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Space and Internal Security -
Developing a Concept for the Use of Space Assets to Assure a Secure Europe

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Executive Summary

The Context for Space and Internal Security

Space applications for security are being increasingly recognised. The use of space for internal security, understood to comprise the handling of civilian threats and terrorism, remains, however, neglected. There is a general lack of awareness of the potential of space applications for internal security. Consequently, questions and challenges such as the coordination of national and European efforts, the optimum institutional set up, the interface with defence related aspects of security and the most efficient procedures for using space applications, still need to be clarified. The lack of a space dimension for internal security is particularly striking, given that the European Security Strategy (ESS) identifies organised crime and terrorism as two of the five key threats Europe is facing.

In the post-Cold War era Europe's security environment has changed. Large-scale military aggression against EU territory has become unlikely while new threats such as terrorism, illegal migration and organised crime have become the main sources of anxiety for both citizens and policy-makers. Thus security is being re-defined moving away from the classical security perception that identified actor-focused threats to territory, towards "functional security" in which the new threats are mainly structural threats that are not "actor/agency"-focused. With the changed concept of security the borderline between defence and security is becoming increasingly thinner and thinner. The concept of "homeland security" is an American concept largely unfamiliar to the Europeans. While U.S. "homeland security" has been driven by the counterterrorist agenda, European efforts to protect society have largely been derived from civilian emergency response communities working with domestic law enforcement agencies. Thus the concept of "homeland security" resonates mainly at the country level in Europe, while Europeans rather refer to "internal security" to encompass the full range of domestic security threats.

In recent years several steps have been taken to increase the role of space in counter-terrorism and organised crime. In this regard the Space and Security Panel of Experts (SPASEC) in 2004 reviewed, inter alia, the role of space in meeting the objectives of the fight against terrorism and organised crime. Its subsequent Report (March 2005) for the first time considered the support space-based applications could provide to counter-terrorism operations. The three year long "Preparatory Action for Security Research (PASR)" (2004-2007) addressed "protection against terrorism" as one of five main areas. PASR's main objective was the development of a full-fledged European civil security research programme to be implemented as part of the 7th Framework Programme for Research and Technological Development (FP 7), the first EU research programme to include security. The 6th Framework Programme for Research and Technological Development (FP 6) had already included space as one of the thematic areas. The preparation of both PASR and the FP 7 Security theme was supported by high level strategy groups: the Group of Personalities (GoP) for Security Research and the European Security Research Advisory Board (ESRAB), whose strategic advice shaped the scope and implementation of these programmes. On the basis of the recommendations of the European Security Research Advisory Board (ESRAB), the European Security and Research Innovation Forum (ESRIF) was established in 2007, mandated to propose by the end of 2009 a European agenda for research and innovation in the field of security capable of guiding European institutions, governments and the private sector in the coming two decades. Composed of eleven working groups, ESRIF considers, inter alia, security of citizens, infrastructure, borders and the role of space in relation to natural and man-made threats. In addition to these bodies, both Galileo and GMES will contribute to the provision of internal security. European level initiatives are complemented by national ones.

Methodology and Goals of the Study

Given the lack of any EU-level policy document setting out the strategy for the provision of internal security, the study follows the analytical structure of the issues areas identified in the U.S. National Strategy for Homeland Security. Thus, it looks at intelligence, critical infrastructure protection, transportation and border security and emergency management. At the same time it distinguishes between three phases of incident management for each of these areas:

- Phase 1: pre-attack including prediction and detection and preventive measures,
- Phase 2: attack, including alerting,
- Phase 3: post-attack, including emergency management and relief.

Desktop research on publicly available sources was complemented with expert interviews and an expert workshop, which was held under the auspices of the Czech EU Council Presidency on 28 May 2009.

The study distinguishes between technological capabilities and limitations and concrete research and development as well as describing implemented projects at both the EU and the national level. At the same time, it identifies the main actors and their mission areas so as to give an overview of the existing institutional set-up and possible gaps and areas for improvement. Given the disregard and lack of awareness of the added value of space applications for the provision of internal security, corresponding questions and challenges that still need to be clarified include: the coordination of national and European efforts, the optimum institutional set up, the interface to defence related aspects of security and the most efficient procedures for using space applications. Thus the study aims to:

- identify the threats and vulnerabilities that need to be planned against (Chapter 2)
- identify and assess the potential of space assets and space-based applications in the provision of homeland security (Chapter 3)
- identify and define differences between the U.S. and the European approach and related lessons learned (Chapter 4)
- identify current actors and indicate future needs and possible space-based solutions (Chapter 4)
- clarify corresponding questions and challenges such as (Chapter 4 & 5)
    - The coordination of national and European efforts
    - The optimum institutional set up
    - The interface with defence related aspects of security
    - The most efficient procedure for using space applications
- raise awareness among principle actors (Chapter 5 & 6)
- encourage the authorities responsible for internal security to be more active in promoting the role of space applications (Chapter 5 & 6)
- establish a dialogue between space and non-space actors (Chapter 5 & 6)

Key Findings in the Context of a European Approach to Internal Security

Detailed findings related to the aims of the study are laid out in the corresponding chapters. Only overarching policy-related recommendations are presented here.

The pillar structure leads to the fragmentation of the European approach to the provision of internal security by artificially dividing security into civil security (Pillar I), military / external security (Pillar II) and Justice and Home Affairs (JHA) (Pillar III).

While most official EU documents have realised the urgency of tackling the matter, space as a distinct technology area as well as its added value to numerous operational areas is usually not mentioned.

The EU has only recently started to seriously look into the provision of internal security with research programmes relating to this matter located in the Framework Programmes (FPs) (FP 5 to FP 7). These cover only the research and demonstration levels. As yet, no full-fledged operational services to be derived from this have been put into practice.

Projects included in the FPs largely focus on Phase 1, the prevention phase. Those projects are difficult to assess regarding the actors involved, the mission areas as well as the users and operators. Information is rather basic and does not provide insights to the actors involved, which would be necessary to identify their potential and make them part of a more comprehensive approach.

Often the target sector is limited to natural disasters such as floods and fires, while the same applications would actually also be useful for other internal security incidents such as terrorist attacks. Many projects look similar in nature and the relationship between them, if any, is not clear. With many projects having overlapping objectives the added-value of the single projects is not transparent. While most projects claim to be user-driven, user involvement is still limited given the lack of understanding of the added value of space applications and rigid cultural/ traditional working structures.

With regard to information dissemination, the websites of the single projects are fragmented, adding to the difficulties of assessing their value. As a consequence, hardly more than a simple introduction or factsheet presentation is given, with most FP5 and FP6 websites being no longer operational. Information can thus not be accessed and assessed. Evaluation of the projects for the purpose of clarifying the EU approach to internal security remains difficult if not impossible. Actors remain limited to their specific mission areas such as natural disasters or CFSP and cannot engage in emergency management following e.g. a terrorist incident.

Recommendations

While primarily tackling the lack of strategic approaches to the use of space applications for internal security, this study goes beyond the mere production of concrete recommendations in this context and provides a building block towards the development of a comprehensive joint European concept for internal security. The specific space-related recommendations can be grouped into three categories, thereby not only substantiating areas that require action but also identifying addressees for immediate action and future steps.

1. Formulating a Common European Approach to Internal Security through an Internal Security Strategy

There is a strong need for further coordination (1) between EU and national approaches and activities, (2) within the different single mission areas and (3) in the overarching area of internal security at the community level. A common European approach must be adopted.

Guidance could be provided through the introduction of a European Internal Security Strategy that should stress the importance of the use of technology - in particular space applications - for the provision of internal security while at the same time providing for better coordination among projects and institutional actors. Such a comprehensive top-down policy approach would be the most efficient way of ensuring that space applications receive the attention of decision-makers, leading ultimately to their implementation as tools in this area.

The current and upcoming EU Presidencies should actively promote this matter and finally move forward to establish an all-encompassing EU approach to internal security. The Commission as the “initiator” has to provide the policy background for an integrated approach in the single areas, but the future EU Presidencies should engage in discussions related to the introduction of an EU Internal Security Strategy. The European Parliament should aim to act as a forum to bring together expertise and get involved through recommendations to the Council regarding the formulation of an Internal Security Strategy.

A comprehensive approach to security should cover all critical mission areas both individually and as a whole in order to account for a cascading effect. The EU has to move beyond the narrow interpretation of internal security as natural disasters and include organised crime and terrorism in threat scenarios to be planned against. The current security scenario is dominated by a combination of asymmetric threats, suggesting the need for a comprehensive approach rather than the traditional separation of civil and military capabilities.

There is thus a need to move away from the separation of civil and military applications particularly in the field of technology and instead to integrate both through the use of dual technologies. The European Defence Agency (EDA) has already taken some initial steps to integrate civil and defence capabilities. These approaches have to be acknowledged by the other institutional actors as well, leading to a re-definition of the EU approach.

In this context, the missions of institutional actors such as the European Union Satellite Centre (EUSC) need to be reviewed to establish a broader field of action taking into account their expertise. Given its international standing, the EU has a responsibility to play an active role in world affairs by taking the lead in developing a

comprehensive approach linking the external and internal dimensions of security and combining the use of civil and military means so as to achieve full exploitation of potential synergies between defence, security and civil research.

2. Issues of Governance and Organisation

The main challenge is to bring together the different instruments and capacities that have been created in different periods of time, within different institutional structures and with different rationales. The institutional set-up mustbe reviewed to take into account the changing security context since the Cold-War and the new possible threat scenarios.

One single actor should thus pool resources and projects and provide the oversight needed. If such a body is not founded, Europe risks delay and continued problematic approaches to internal security, comparable to those in the U.S. which is experiencing a variety of problems due to the fragmented responsibilities of multiple agencies.

Coordination should go beyond mere institutional coordination and lead to cooperation between the industry owning and operating existing assets and Member States. Increased dialogue between the actors involved is needed immediately. Efforts to introduce multi-functional missions have to be supported and fostered. In line with the principle of subsidiarity, an enhanced coordination and communication mechanism between supranational instruments, national capabilities and regional needs has to be created. Particularly in the maritime domain, efforts need to be pooled and coordinated by a single actor. The European Maritime Safety Agency (EMSA) could undertake this.

Commercial applications of space drive many of the systems developments. Thus many different users (commercial, internal security, defence) will be using the same satellites for similar types of applications. The provision of internal security with space applications can thus not be looked at in isolation. It should be part of a larger attempt to coordinate applications development.

With defence and internal security users using the exact same capabilities, close cooperation between these communities has to be fostered. Internal security users could also benefit from the experience of the defence community in making use of satellite applications for reconstruction and security efforts.

Europe should also seek close cooperation with other countries, particularly the U.S. and the relevant international organisations in the security field (e.g. NATO, OSCE) as to consider their positions when adopting a European approach.

3. The Role of Space in an EU Comprehensive Internal Security Strategy

ESA and EC funded projects need to move beyond the demonstrations phase. Thus ESA and the EC have to ensure that the actors involved properly deal with the transition from demonstration to operation. ESA and the European Commission also have to ensure that all critical mission areas in the provision of internal security are covered equally and that the agenda does not get “hijacked” by events such as the piracy attacks off the Somalian coast which moved maritime security up the agenda while fluvial security and efforts in land border security areless discussed. Existing projects have to be re-evaluated so as to include current trends such as the increased integration of energy and information in the area of critical infrastructure.

The adoption of marketable solutions and the introduction of a user-exchange mechanism should improve the current communication problem between users and lead to an improved understanding of the added value of space applications for the provision of internal security through effective outreach activities. The industry should involve itself further in outreach activities, moving beyond mere demonstrations as to enlarge the group of addressees. Space-applications shall be presented as innovative solutions in times of financial and economic crisis, in particular, as laid out in the Lead Market Initiative.

Results from demonstrations and FP research should be seriously considered by ESA when planning and deciding on new satellite missions. User requirements conveyed to ESA should provide the building blocks for new satellite projects. Such projects should be outlined in such a way that they encompass as many opportunities as possible for applications related to internal security. This principle should be applied to single satellite missions as well as satellite constellations or the combined use of satellites with different purposes. Increased cooperation between the different projects of the FP programme relating to space is of utmost importance in the context of formulating user requirements. On the other hand, users also have to be educated as to demanding certain services.

Initiatives like GIANUS (Global Integrated Architecture for Innovative Utilisation of

Space for Security) and programmes such as the Integrated Applications Promotion (IAP), which was launched by the last ESA Council at Ministerial level, are means to provide frameworks for accomplishing interactions with users in order to establish user requirements as a basis for new satellite missions. These are taking a promising direction for establishing a focused dialogue between the users and ESA. Cooperation in this context has to be increased with users in the field of internal security demandingcertain requirements and ESA engaging in this dialogue. The user-exchange mechanism proposed in this report would provide for formalisation of such a process.

1. Introduction

1.1. The Setting

The contributions of space applications to security are being increasingly recognised, but the use of space for internal security, understood to comprise the handling of civilian threats and terrorism, remains neglectedThere is a general lack of awareness of the potential of space applications for internal security. Consequently, questions and challenges such as the coordination of national and European efforts, the optimum institutional set up, the interface with defence related aspects of security and the most efficient procedures for using space applications, still need to be clarified.

In the post-Cold War era Europe's security environment has changed. Large-scale military aggression against EU territory has become unlikely while new threats such as terrorism, illegal migration and organised crime have become the main sources of anxiety for both citizens and policy-makers.[1] In facing this, Europe must strike a balance between protecting its citizens and remaining committed to an open and liberal society.

The lack of a space dimension for internal security is particularly striking, given that the European Security Strategy (ESS) identifies organised crime and terrorism as two of the five key threats facing Europe.In recent years several steps have been taken to increase the role of space in counter-terrorism and organised crime. In this regard the Space and Security Panel of Experts (SPASEC) in 2004 was given the mandate inter alia to review the role of space in meeting the objectives of the fight against terrorism and organised crime. Its subsequent SPASEC-Report of March 2005 for the first time considered the support space-based applications could make to counter-terrorism operations. The three year long "Preparatory Action" (2004-2007) addressed "protection against terrorism" as one of five main areas. PASR's main objective was the development of a full-fledged European civil security research programme to be implemented as part of the 7th Framework Programme for Research and Technological Development (FP 7), the EU's main research funding programme for the period 2007-2013. FP 7 was the first EU research programme to include security. The 6th Research Framework Programme had already included space as one of the thematic areas. Additionally, apart from security, the Preparatory Action as well as the 6th Framework Programme for Research and Technological Development (FP 6) and FP 7 included projects related to emergency management and maritime surveillance.

The preparation of both the PASR and the FP 7 Security theme was supported by high level strategy groups: the Group of Personalities (GoP) for Security Research and the European Security Research Advisory Board (ESRAB), whose strategic advice shaped the scope and implementation of these programmes. The Group of Personalities (GOP) in the field of security research recognised space as a "force enabler". The GOP also concluded that security and civil applications are increasingly forming a continuum with challenges inside and outside of the EU often being similar. Across this continuum, applications in one area can often be transformed into applications in another area. Space is a prime example for such developments.[2] On the basis of recommendations of the European Security Research Advisory Board (ESRAB) the European Security and Research Innovation Forum (ESRIF)[3] was established in 2007. It is mandated to propose by the end of 2009 a European agenda for research and innovation in the field of security capable of guiding European institutions, governments and the private sector in the coming two decades. Composed of eleven working groups, ESRIF considers, inter alia, security of citizens, infrastructure, borders and the role of space

[1] Group of Personalities in the field of Security Research. "Research for a Secure Europe: Report of the Group of Personalities in the field of Security Research." Luxembourg: Group of Personalities, 2004. 9.

[2] Ibid.12.

[3] European Security Research and Innovation Forum (ESRIF) "European Security Research and Innovation in Support of European Security Policies. Intermediate Report." Luxembourg: Office for Official Publications of the European Communities, 2008. <http://www.esrif.eu/documents/intermediate_report.pdf>.

in relation to natural and man-made threats. European level initiatives are complemented by national ones.

Moreover, the EU is developing its own operational capability through the GMES initiative in the form of a series of research and development projects funded by the European Commission and ESA in the domains of maritime surveillance, humanitarian relief, early warning and prevention of conflict.

Most recently, on 29 May 2009, the Sixth Space Council under the Czech Presidency endorsed common orientations for the EU and ESA on a number of key subjects, such as the mobilisation of existing innovation support mechanisms at European, national and regional levels and the consideration of new instruments to ensure cross-fertilisation between the space and non-space sectors. While it touched upon issues related to internal security by emphasising, inter alia, the need to define adequate conditions for services such as the European Geostationary Navigation Overlay Service (EGNOS); the European Navigation Service (GALILEO) and the Global Monitoring for the Environment and Security (GMES) flagship project, and by highlighting the potential of satellite communications services and the growing institutional demand for satcom in European programmes and policies, internal security did not have a strong voice in the preparation of the Sixth Space Council. Structurally, however, internal security would be in a much better position to be a partner in GMES than military defence because it is of an essentially peaceful, civilian character. And GMES is "a civilian system under civilian control"[4]. With the fight against the global economic crisis there may be a window of opportunity for mobilising public funds in support of useful, innovative projects, such as e internal security, which is very high on the agenda.

1.2. Institutional Development

For a long time the debate dealing with security issues in the context of European integration tended to focus only on traditional external and particularly military security issues. The relatively slow development of internal security at the European level in general explains the relatively late development and lack of awareness of the value of space for internal security. Only during the 1990s did internal security start to occupy an increasingly prominent place on the agenda of the EU: Justice and Home Affairs (JHA) emerged as an established area of activity only after the entry into force of the Treaty on the European Union (TEU) in November 1993. Before that time, national governments monopolised internal security and retained it as a closely guarded issue of sovereignty.

The process continued with the creation of an Area of Freedom, Security and Justice as part of the Treaty of Amsterdam. This led in 1999 to a dedicated special European Council summit at Tampere to map out an agenda of future issues. A host of legislative measures have accompanied these developments and a new Directorate General (DG) has been established in the European Commission. It was initially developed as an area of intergovernmental activity within the third pillar of the EU. Since the Treaty of Amsterdam, JHA has taken on a hybrid nature as some issues have become "communitarised" such as immigration and external border controls, while police and judicial cooperation in criminal matters have remained intergovernmental. The evolution of the EU's approach to internal security policy has come about in response to certain threat perceptions. Transnational organised crime and illegal immigration have provided the rationale for building a European governance of internal security.[5] The ambition of the EU is to provide a higher degree of security than individual nation-states can provide.

Moreover, information derived from satellite imagery has only recently started to enter the civilian/diplomatic decision-making process due to the confinement of the technology to military and intelligence institutions. Furthermore, specialised staff needed for the extraction of information from satellite imagery as well as satellite data collection and IT infrastructure remain expensive and are thus largely provided through specialised institutions that are dependent on central governments. Also the availability of new imagery requiring innovation usually provided through research communities has been slow, as research communities have been addressing scientific

[4] Brandtner, Thomas. "Space and Internal Security: A European Task." Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at/images/stories/dokumente/Conference2009/brandtner.pdf>.

[5] Mitsilegas, Valsamis, Jörg Monar and Wyn Rees. The European Union and Internal Security – Guardian of the People? Houndmills: Palgrave Macmillan, 2003. 1-2.

technological improvements for external relations services.[6]

Several challenges in the provision of internal security persist. The EU's pillar structure artificially divides security into civil security (Pillar I), military / external security (Pillar II) and JHA (Pillar III), which results in a huge financial impact.[7] EU Member States have on many occasions confirmed that a more coherent and integrated approach to crisis management is needed. The ESS adopted by the European Council in December 2003, for example, explicitly stated that greater coherence is needed to improve effectiveness in the EU's crisis management. In this regard, the main challenge is to bring together the different instruments and capacities that have been created in different periods of time, within different institutional structures and with different rationales. Greater coherence requires complementarities in the use of EU instruments (civilian and military) and better co-ordination between them. Such structural deficiencies at the institutional and political level hinder Europe in the exploitation of its scientific, technological and industrial strength.[8]

1.3. Approach of the Study

Given recent developments in creating a space dimension for the provision of internal security, as well as the persisting challenges in doing so, this report aims at providing an overview of the various projects in the realm of internal security. By indicating existing measures and shortcomings, the major objective is to develop a concept for the provision of internal security in Europe.

The first chapter provides the background, defining the terms "security", "homeland security" and "internal security" as well as the particular areas under analysis. The second chapter discusses current technological capabilities and limitations. The third chapter provides an overview of the U.S. approach to the topic, thereby evaluating this experience with an eye to possible lessons learned for Europe. The fourth chapter gives a detailed overview of existing measures in Europe. By referring to the main policy documents, the actors involved as well as the existing projects, this chapter points to the gaps in the use of space applications for the provision of internal security in Europe. These will then be outlined in the very last chapter. Each chapter will present a comparative matrix visually highlighting the limitations and indicating fields for further action.

In the framework of this study a workshop was organised by the European Space Policy Institute (ESPI) under the auspices of the Czech EU Council Presidency and took place on 28 May 2009 in ESPI's premises in Vienna (Austria). It contributed to the analysis and findings of this study. During the workshop, invited experts from critical mission areas ranging from critical infrastructure, port and maritime security, the transportation sector to the field of border control, discussed current projects at both the European and the national levels and indicated future needs. These experts from critical mission areas were paired with experts from the space field, thereby initiating a dialogue between the space community and the internal security community. This approachfacilitated the definition of needs and the identification of space-based solutions. Through the systematic approach applied and the fruitful discussions among more than thirty high-ranking participants, the workshop aimed at developing a comprehensive concept for the use of space assets to assure a secure Europe including the identification of needed technological developments, applications and regulations. The morning session on the protection of critical mission areas was complemented by the afternoon session on policy perspectives, putting the discussion in the context of both the ESA and the nation states. The workshop ended with a roundtable composed of representatives of the EU Council Secretariat, the European space industry association Eurospace and ESA, among others.

[6] "Geodata and Crisis Early Warning (2006-2007)." Information Support For Effective and Rapid External Action (ISFEREA) 3 Mar 2009 <http://isferea.jrc.ec.europa.eu/Activities/ProjectPortfolio/Pages/GeoCrew.aspx>.
[7] Molard, Bernard. "Let's be serious about Space for Defence & Security." International Space University Annual Symposium. International Space University, Strasbourg, France, 18-20 Feb. 2009.
[8] Group of Personalities in the field of Security Research. op. cit. 9.



Speakers at the Workshop on Space and Internal Security of 28 May 2009 at ESPI. From left to right: Giovanni Cannizzaro (Telespazio), Erwin Duhamel (Head Strategic Security and Partnerships European Space Agency), Alfredo Roma (Member of ESPI Advisory Board and Consultant for the Aerospace Industry), Aurélie Bressollette (EADS Astrium), Alessandra Ussorio (European Union Satellite Centre), Thomas Brandtner (Head of Unit DG C II, Competitiveness, Lisbon Strategy, Industrial Policy, General Secretariat of the Council of the European Union),Mari Kalliala (Head, Analytical Planning Section, Risk Analysis Unit FRONTEX), Nina-Louisa Remuß (ESPI Project Manager) , Bruno Gagnou (Sales Director Business Unit Navigation ThalesAlenia Space), Kai-Uwe Schrogl (ESPI Director), Petr Havlik (Head Czech EU Council Presidency Team, UN, UNODC, UNIDO), Bernard Molard (Eurospace - Chairman, Security and Defence Workgroup), Marcelo Masera (Joint Research Centre)

Both desktop research and the workshop led to the identification of a number of issues with regard to adapting the current institutional and regulatory framework. The broader use of space applications and services thus creates related requirements that have to be taken up by the relevant legislative and executive bodies on the European as well as member state level. In particular, the incoherence and complexity of the areas of authority in the European institutions for the provision of internal security must be overcome, and the fight against terrorism and organised crime has to be established as subjects that cut across institutions: The EU's pillar structure artificially divides security into civil security (Pillar I), military / external security (Pillar II) and Justice and Home Affairs (JHA) (Pillar III) which results in a huge financial impact. The re-arrangement of the institutional set-up should provide for one actor to take a coordinating role for internal security at EU level, thereby monitoring efforts at national level. The European Union Satellite Centre should be supporting decision-making by the European Union in the area of internal security, but this requires an extension of its mission so as to include internal security threats rather than merely mandating it to support decision-making in the ESDP and CFSP. Civil and military capabilities should be integrated. Islands of data among European countries as well as actors and mission areas need to be overcome. Moreover, in order to provide overall strategic direction, an Internal Security Strategy should be agreed upon as an addition to the European Security Strategy. This should provide for a comprehensive security system integrating all critical mission areas in one European approachthat avoids providing internal security at the cost of limiting civil liberties. Outreach activities, particularly the involvement of users, need to be formalised by establishing a user-exchange mechanism, including the promotion of best practices.

2. The Policy Area of Internal Security

In order to identify the value added that space applications can provide for securing Europe, there is first a need for some conceptual clarification. Definitional explanations are important as the provision of security depends on issue framing.[9] In this regard, this chapter elucidates the development of the concept of "security" and particularly the meaning of the terms "homeland security" and "internal security". It is complemented by a section identifying what needs to be secured. Risk assessment can provide one answer to that question. By putting forward selected statistics, each section attempts to show the dependency of Europe on each critical mission area. Furthermore, terrorist motivations for attacking a specific critical mission area as well as past terrorist incidents are elaborated upon. Additionally, this chapter seeks to provide an overview of the methodology chosen for this study. In this regard, it gives an overview of the three phases of the crisis cycle and elucidates the chosen critical mission areas.

2.1. "Homeland Security" or "Internal Security"? Clarifying the Concept

Any study focussing on homeland security faces definitional problems at the outset. In this regard it is important to first of all understand what "security" is in order to then identify what "homeland security" and "internal security" means.

With the fall of the Iron Curtain as well as events such as 9/11, the Madrid bombings and the tsunami disaster, a re-thinking of the traditional perception of "threats" and "security" occurred.[10] Prior to that, the so-called classical security perception identified threats as actor focused and the classical threat was an armed attack by another state.[11] "Security" thus aimed at preserving the quality of life of a certain population as well as the government's possibility to govern. As economic integration and military rationality proceeded, this purely territorial perception was overtaken and security became part of an enlarged concept in which defence played only a minor role.[12] Consequently, "functional security" as a concept came to oppose traditional territorial security.[13] Therein new threats are mainly structural threats that are not "actor/agency-focused and the greatest threat in today's world is instability" rather than an attack by another state. Human suffering in the 'near abroad' resulting from a natural disaster or man-made disaster bears the risk "of spilling over into our national security system".[14]

Given the interconnectivity of European countries and the transnational character of the threats today's world is facing, EU involvement is becoming inevitable. With the changed concept of security the borderline between defence and security is increasingly becoming thinner and thinner (see figure 1-3[15]).

[9] Sundelius, Bengt. "From National to Total Defense to Embedded Societal Security." Protecting the Homeland: European Approaches to Societal Security – Implications for the United States. Eds. Daniel Hamilton, Bengt Sundelius and Jesper Grönvall. Washington D.C.: Center for Transatlantic Relations, 2005. 3.

[10] Varwick, Johannes and Woyke, Wichard. Die Zukunft der NATO. Transatlantische Sicherheit im Wandel. 2nd Edition. Augsburg: Leske + Budrich, 2000. 127.

[11] Sundelius, Bengt. "Disruption - Functional Security for the EU." Disasters, Diseases, Disruptions: A new D-drive for the EU. Chaillot Paper No. 83. Ed. Antonio Missiroli. Paris: Institute for Security Studies, 2005. 68.

[12] Varwick, Johannes and Woyke, Wichard. NATO 2000. Transatlantische Sicherheit im Wandel. Augsburg: Leske + Budrich, 1999. 30-1. This is in line with Emerson's definition of security, claiming that: "The security nexus involves an interconnected sets of rules": democracy, human rights, economic and political stability and multidimensional crisis management. cf. Emerson, Michael. Redrawing the Map of Europe. Basingstoke: Palgrave, 1998. 28.

[13] Sundelius, Bengt. op. cit. 68.

[14] Tanner, Fred. "Conflict Management and European Security: The Problem of Collective Solidarity." Publication for Workshop 4: Toward the 21st Century: Trends in the Post-Cold War International Security Policy. 1st Conference of the PfP Consortium of Defense Academies and Security Studies Institutes, 19-21 October 1998, Kongresshaus Zürich: The Geneva Center for Security Policy, Switzerland. <http://www.isn.ethz.ch/3isf/Online_Publications/WS4/Tanner.htm>. 4.

[15] The three figures are based on a similar account by Bernard, Molard. "Yes…We Can!" Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009.

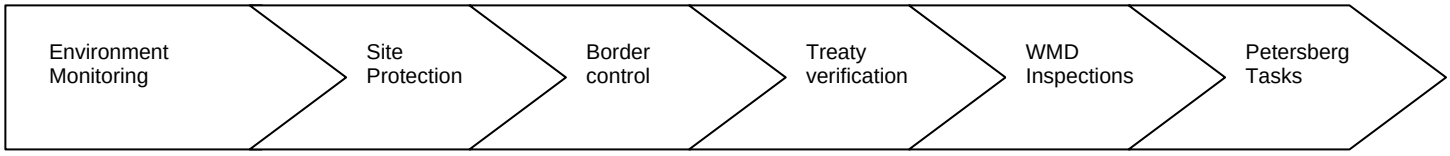


Figure 1: From Environment to Security

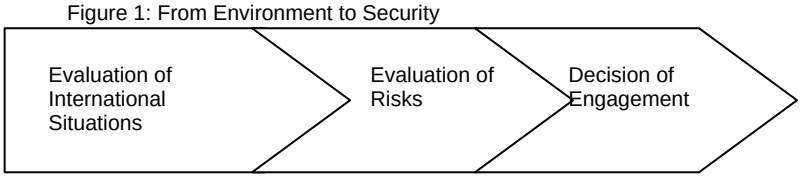


Figure 2: From Security to Defence

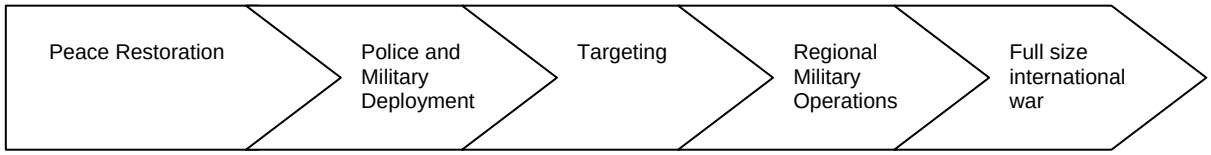


Figure 3: From Defence to Military Operations

The concept of "homeland security" is an American concept largely unfamiliar to the Europeans. While prior to 9/11 the U.S. army defined it as

> those active and passive measures taken to protect the population, area, and infrastructure of the United States, its possessions, and territories by: deterring, defending against, and mitigating the effects of threats, disasters, and attacks; supporting civil authorities in crisis and consequence management; and, helping to ensure the availability, integrity, survivability, and adequacy of critical national assets,[16]

after 9/11 it came to include

> the prevention, pre-emption, and deterrence of, and defence against aggression targeted at U.S. territory, sovereignty, domestic population, and infrastructure, as well as the management of the consequences of such aggression; and other domestic civil support.[17]

In this regard, the concept of "security" was enlarged so as to include threats to the security of critical functions of society Homeland Security thus is the "systematic attempt to reduce society's vulnerabilities".[18]

Regardless of the U.S. emergency planners' experiences with all kinds of hazards since 2001, homeland security came to be strongly associated with counter-terrorism.[19] The U.S. National Strategy for Homeland Security[20] defines it as "a concerted national effort to prevent terrorist attacks within the United States, reduce America's visibility to terrorism, and minimize the damage and recover from attacks that do occur".

Rather than referring to homeland security, Nordic countries point to "societal security", an all hazards approach including the entire spectrum of threats that might confront society (i.e. from environmental threats to natural disasters).[21] With Nordic countries feeling arguably less threatened by global terrorism than both the U.S. and the rest of Europe, this difference in concepts can be ascribed to diverging threat perceptions. This approach is closer to "human security" types of theories, or critical theories in security.

While U.S. homeland security has been driven by the counterterrorist agenda, European efforts to protect society have largely been derived from civilian emergency response communities working with domestic law enforcement agencies.[22] The concept of homeland security is

<http://www.espi.or.at/index.php?option=com_content&task=view&id=364&Itemid=37>.

[16] Brennan, Richard. Protecting the Homeland – Insights from Army Wargames. Santa Monica: RAND Corporation, 2003. 17.

[17] Ibid.

[18] Clarke, Jackie. "The United States, Europe, and Homeland Security: Seeing Soft Security Concerns through a Counterterrorist Lens." European Security 13 (2004): 117-138.

[19] Ibid.

[20] Office for Homeland Security. National Strategy for Homeland Security. Washington D.C.: Office of Homeland Security, Feb. 2002. <http://www.whitehouse.gov/homeland/book/>.

[21] cf. Hamilton, Daniel S, Bengt Sundelius and Jesper Grönvall. Protecting the Homeland: European Approaches to Societal Security – Implications for the United States. Washington D.C.: Center for Transatlantic Relations, 2005.

[22] Hamilton, Daniel S. "Introduction: Transforming Homeland Security: A Road Map for the Transatlantic Alliance." Transforming Homeland Security: U.S. and

relatively new at the EU level. For many, the phrase "homeland" resonates most strongly at the country level. Thus, Europeans refrain from using the term "homeland security" as such and prefer to use terms such as "internal security", "resi-lience", "societal security", "civil protection" or "collaborative security" when referring to intra- EU security issues.[23] Internal security is thus often used to group security threats such as terrorism, organised crime and illegal immigration.

The Bush administration presented counter-terrorism as a "fight" or "war". Such a fra-ming justifies the use of military methods. Europeans see it above all as a matter of law enforcement. Thus, while Europeans refer to the "progressive reduction of a continuing threat", the Bush administration followed the objective of "the elimination of Al-Qaeda as an organisation". President Obama refrained from talking of a broader "war on terror" and suggested distinguishing his campaign against Al-Qaeda from his wider attitude towards the Muslim world in which he promised "to initiate a new partnership based on mutual respect and mutual interest" instead.[24] Given this shift in U.S. counter-terrorism policy under Obama, there is a chance for a new transatlantic understanding on the nature of the fight against terrorism.

Given the differences in the framing of the issue, responses and preventive measures vary on both sides of the Atlantic.

2.2 Risk Assessment

Risk Assessment, through determining both threat and vulnerability, proves invaluable when identifying what to secure as well as a basis for risk reduction and emergency management. The first step to the reduction of the risk is to analyse the threat and the related level of exposure, i.e. the potential probability and severity of the threat.

In order to identify the threat it is necessary to distinguish potential targets from the total range of possible targets; the ideology of terrorist groups has to be considered followed by an analysis of the operational objectives, the perceived capabilities, the targets most researched (i.e. familiarity) and, (if possible) past targets. Indications can also be derived from the symbolic relevance of certain targets. Having conducted a final potential high-risk target selection it is important to determine what equipment, training and planning is necessary to reduce both the terrorist threat as well as the vulnerability to it. Threat and risk assessments enable the determination of how to prioritise scarce resources across various threats and risks in terms of probability and consequences.

After having conducted a vulnerability analysis and having identified ways to reduce this vulnerability, the provision of awareness training and public education is necessary. Communities need to develop and maintain strong partnerships with mutual responsive agencies and identify key personnel. Ways to use existing conventional equipment to manage terrorist activities should be investigated. Appropriate plans to achieve full readiness over time should be agreed upon. Preparedness needs to be clearly defined, i.e. there should be a measurable demonstrated capacity by

> communities, States, and private sector entities throughout the country to respond to acute threats with well planned, well coordinated and effective efforts by all of the essential participants, including elected officials, police, fire, medical, public health, emergency managers, intelligence, community organizations, the media and the public at large (…) such preparedness requires effective and well-coordinated preventative efforts by the components of the Intelligence Community, law enforcement entities, and a well educated and informed public. These efforts must be sustainable, over the foreseeable future while maintaining a free civil community.[25]

The process needs to be reviewed on a periodic basis. Governments should thus adopt a six-step approach:

> Step 1: Identify potential targets by looking at terrorist groups ideology
> Step 2: Conduct a vulnerability analysis
> Step 3: Identify and recommend ways to reduce the vulnerability of the community
> Step 4: Awareness training and public education
> Step 5: Demonstrations and exercises
> Step 6: Review process regularly

European Approaches. Ed. Esther Brimmer. Washington D.C.: Center for Transatlantic Relations, 2006. x.

[23] Gustav, Lindström. "The EU's Approach to Homeland Security: Balancing Safety and European Ideals." Transforming Homeland Security: U.S. and European Approaches. Ed. Esther Brimmer. Washington D.C.: Center for Transatlantic Relations, 2006. 115.

[24] See examples of speeches by President Obama as quoted in Dworkin, Anthony. "Beyond the War on Terror: Towards a New Transatlantic Framework for Counterterrorism." ECFR Policy Brief. ECFR/13 (May 2009): 4. <http://ecfr.3cdn.net/1e18727eafdddcceb7_81m6ibwez.pdf >.

[25] Jenaway, William F. "Comprehensive All Hazards Preparedness is Critical to Effective Emergency Service Preparedness, Response and Funding." The Homeland Security Review 1 (2005). 2.

2.3. The Three Phases of the Event Cycle

Based on the objectives as identified in the definition of homeland security within the U.S. National Strategy for Homeland Security, one can identify three phases of provision of "homeland security" which are transferrable to the European concept of "internal security":

(1) Phase 1: pre-attack including prediction and detection as well as preventive measures,
(2) Phase 2: during an attack, including alerting
(3) Phase 3: post-attack, including emergency management and relief.

The three Phases are shown in Figure 4.

These will serve as a framework when analysing the contribution of space-based applications to the provision of homeland security in the following chapters.

Phase 1: Prevention and Detection

Prevention includes those measures that are aimed at impeding the occurrence of a disaster or preventing such an event from having a harmful effect on a community. Mitigation comprises all actions designed to reduce the impact of disasters.

In this regard one can distinguish between structural or physical measures and non-structural measures (i.e. urban planning, insurance, legislation, training etc.). Actions related to this phase also include any preparatory action. Preventive measures specifically include risk assessment as well as review and development of emergency plans and procedures including related training and exercises. Public awareness campaigns are also cogitable.[27]

Phase 2: Alert and Immediate Response

Early Warning in the form of alert as well as public warning falls within the immediate phase of the crisis. In this regard it mainly focuses on means of communication. The

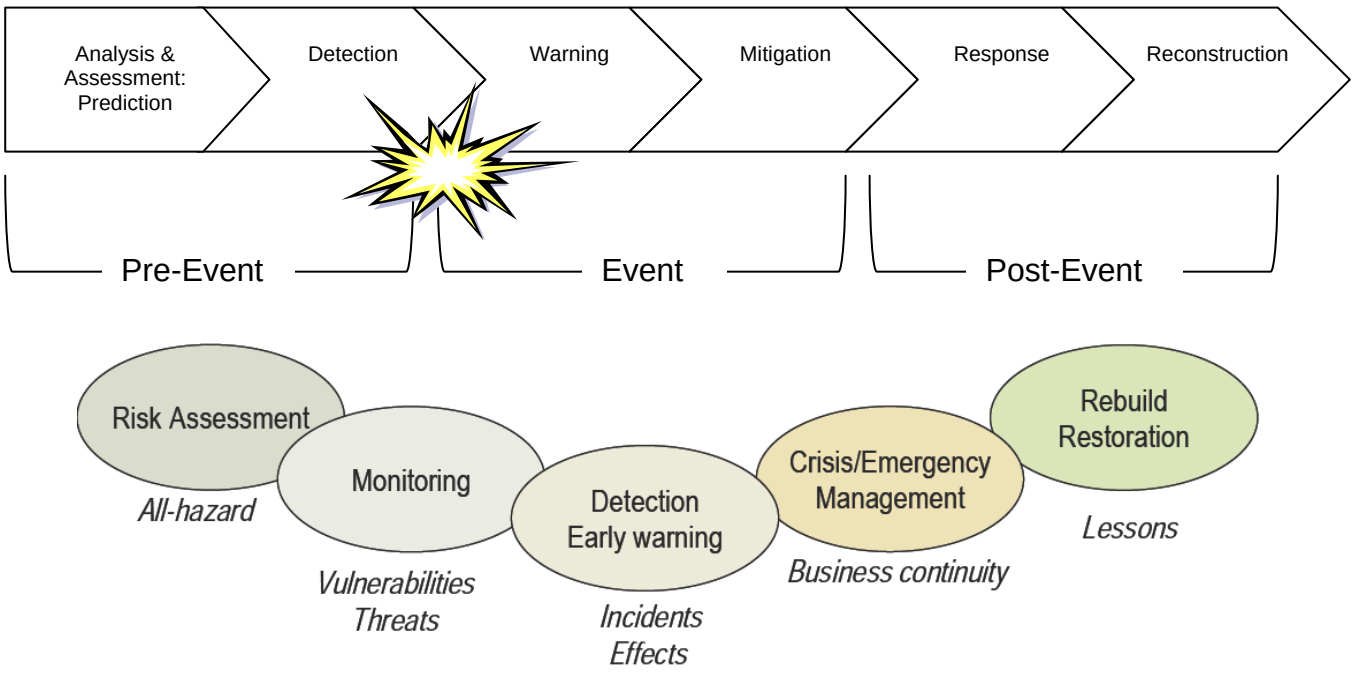


Figure 4: Three Phases of the Event Cycle[26]

[26] Partly based on Marcello, Masera. "Critical Infrastructure Protection Vulnerabilities and Interdependencies: A Vision from the Information and Power Sectors." Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at/index.php?option=com_content&task=view&id=364&Itemid=37>

[27] OASIS FP6 Consortium. "Priority 2.3.2.9 Improving Risk Management Integrated Project." Executive Summary.. OASIS FP 6 Consortium, 2005. <http://www.oasis-fp6.org/documents/OASIS_SP31_RPT_244_lessons_learnt_BAE_2_4_pub.pdf>. 7-9.

immediate response relates to the activities conducted during the impact of a disaster and the short-term aftermath with the main emphasis being placed on the saving of human life. It can also, however, "encompass the protection of assets, the supply of vital goods and services, and protection of the environment".

Phase 3: Relief and Impact Assessment

Phase 3 covers long term reconstruction including recovery or relief, i.e. the "process by which communities return to a normal level of functioning". Apart from restoration, reconstruction and rehabilitation this, in the first instance, includes damage assessment as well as search and rescue operations.[28]

2.4. Threats to Internal Security: The three Critical Mission Areas

In line with the six-step approach presented above, this section gives an overview of the potential target selection and explains the selection of areas under analysis. Through risk and threats assessment, indications of the level of performance demanded from the internal security capabilities are provided. Understanding the threat, its forms and capabilities is vital for the development of measures to secure against potential terrorist attacks. In this regard the U.S.' "National Strategy for Homeland Security" identifies so-called "critical mission areas", like:

- Intelligence and Warning
- Border and Transportation Security
- Domestic Counterterrorism, i.e. redefining national law enforcement mission to focus on the prevention of all terrorist acts
- Protecting Critical Infrastructures and Key Assets
- Defending against Catastrophic Threats, i.e. a coordinated national effort to prepare for,
- prevent, and respond to chemical, biological, radiological, and nuclear terrorist threats to the homeland
- Emergency Preparedness and Response

No internal security strategy for Europe has so far been issued apart from the EU Counter Terrorism Strategy, which does not identify any specific areas of vulnerabilities. The U.S. National Homeland Strategy serves as a starting point for distinguishing policy areas. Pooling together overlapping policy areas, this study identifies three main areas of vulnerabilities, with the potential of risk and vulnerability reduction provided by space assets and space-based applications:

- critical infrastructure (incl. energy, communication, ports and satellite applications)
- transportation (incl. all kinds of cargo), differentiating between land, sea and air transportation
- border security.

In addition to these three critical mission areas, the study identifies space applications for intelligence, early warning and risk management as well as emergency preparedness and management.

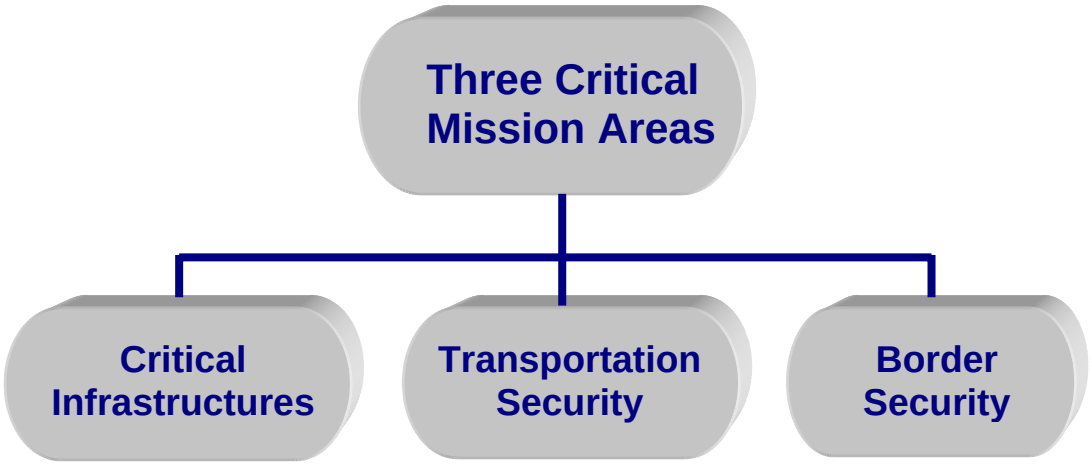


Figure 5: Critical Mission Areas

[28] Ibid.

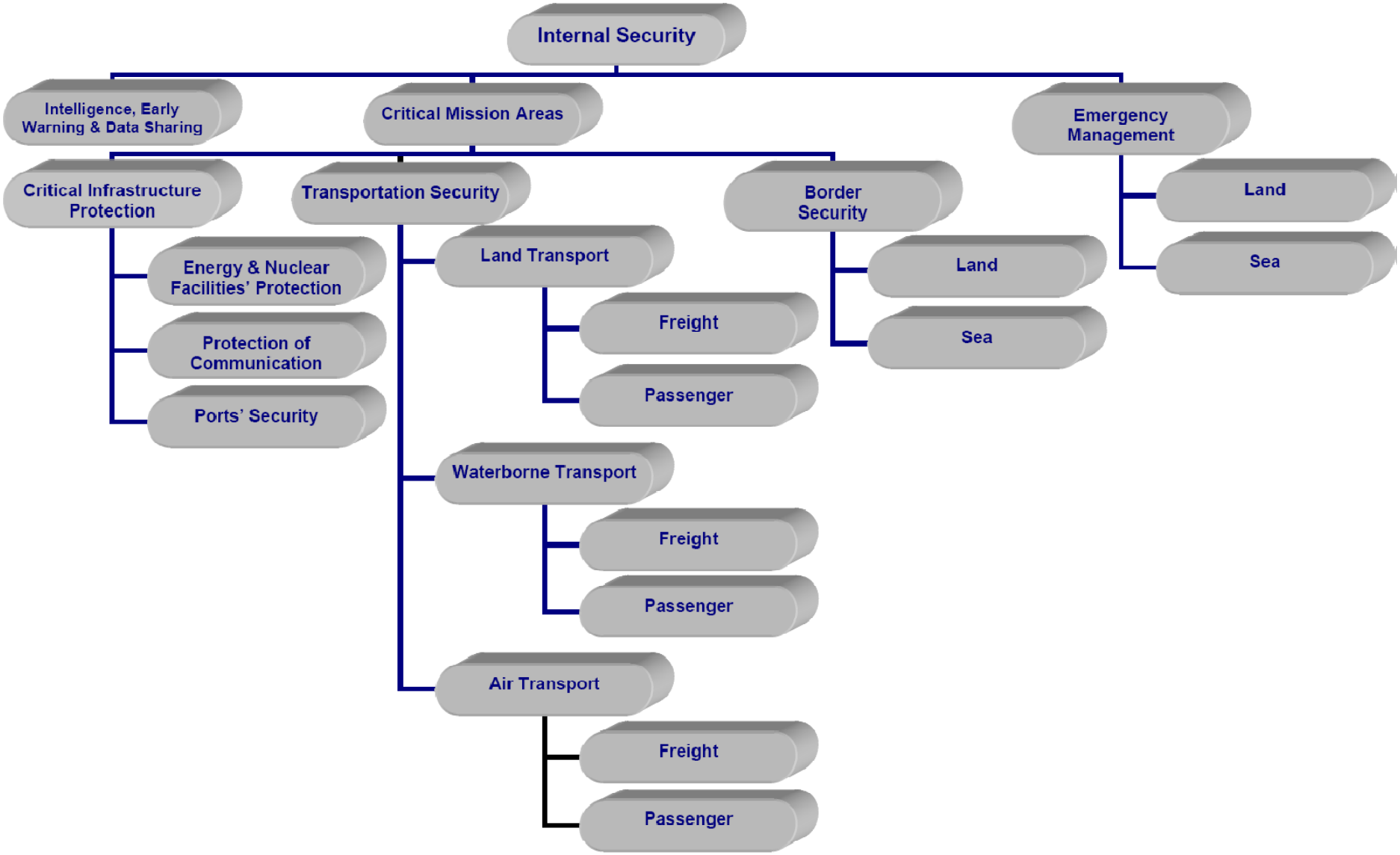


Figure 6: Overview of the Areas under Analysis

As indicated earlier, each of the different areas of vulnerability varies in risk and severity between each other and in different countries and even parts of countries. By looking at motives, preferred tactics and preferred targets it is possible to acquire some indication of the specific vulnerability that in turn indicates which counter-measures are necessary.

2.4.1. Critical Infrastructures

Critical infrastructures (CI) can be defined as "every infrastructure vital to the security, economy and survival of the nation, such as computer networks, energy supplies, and transportation systems"[29] but is also quite often taken to include agriculture and food; water; public health and health care, emergency services, defence industrial base, information technology, telecommunications, energy, transportation systems, banking and finance, chemical facilities, post and shipping, national monuments and icons, dams, government facilities, commercial facilities, nuclear reactors, materials and waste.[30] All of these elements are also intertwined as infrastructures do not exist in isolation, e.g. transport depends on power, telecommunications depends on electricity, and so on. Hence, something happening in one area of infrastructure will result in an effect on another and that effect will in turn result in an effect on the first area. Secondary impacts of an economic or social and political nature can be added to the primary physical impacts. Thus there is a need to consider the system as a whole. Moreover, assets, functions and systems within each CI sector are not of equal importance and different communities will consider them to be of different importance. As the private sector owns and operates many of the critical infrastructures and key assets, the expertise and resources required for emergency planning and protection lie outside of the direct control of the government. Additionally, there appears to be an intractable problem surrounding information exchange between the public and private sectors, meaning that this infrastructure vulnerability will remain a key problem to be exploited.[31] The private sector will however be able to respond much more quickly and effectively to many internal security threats than government agencies can.[32]

The European Union defined "Critical Infrastructures" in its Green Paper on a European Programme for Critical Infrastructure Protection[33], published on 24 November 2005, as including

> those physical resources, services, and information technology facilities, networks and infrastructure assets which, if disrupted or destroyed, would have a serious impact on the health, safety, security or economic well-being of citizens or the effective functioning of governments.

In line with this definition it identifies three types of infrastructure assets:

- Public, private and governmental infrastructure assets and interdependent cyber & physical networks.
- Procedures and, where relevant, individuals that exert control over critical infrastructure functions.
- Objects having cultural or political significance as well as "soft targets" which include mass events (i.e. sports, leisure and cultural).

Furthermore, the Green Paper differentiates between European Critical Infrastructures and national critical infrastructure by providing a specific definition of "European Critical Infrastructures" as

> those assets, systems or parts thereof located in EU member states which are essential for the maintenance of vital societal functions, health, safety, security, economic or social well-being of people (e.g. electricity, gas and oil production, transport and distribution; telecommunication; agriculture; financial and security services; etc.), and the disruption or

[29] Wortzel, Larry M.. "Securing America's Critical Infrastructures: A Top Priority for the Department of Homeland Security" Homeland Security and Terrorism: Readings and Interpretations. Eds. Russel D. Howard, James J. F. Forest & Joanne C. Moore. New York: McGraw-Hill, 2005. 176.

[30] The Executive Office of the President Office of Science and Technology Policy & The Department of Homeland Security Science and Technology Directorate. "National Plan for Research and Development In Support of Critical Infrastructure Protection." Washington D.C.: The Executive Office of the President Office of Science and Technology Policy & The Department of Homeland Security Science and Technology Directorate, 2004. 3-4, cf. for example Commission of the European Communities. Green Paper on a European Programme for Critical Infrastructure Protection. COM (2005) 576 of 17 Nov. 2005. Brussels: European Union.

[31] Bell, Sandra J. "Defending Critical Infrastructure and Systems." Transforming Homeland Security: U.S. and European Approaches. Ed. Esther Brimmer. Washington D.C.: Center for Transatlantic Relations, 2006. 150.

[32] Wortzel, Larry M. op. cit. 179.

[33] Commission of the European Communities. Green Paper on a European Programme for Critical Infrastructure Protection. COM(2005) 576 of 17.11.2005. Brussels: European Union.

destruction of which would have a significant impact on at least two EU member states.[34]

The European Programme on Critical Infrastructure Protection (EPCIP) particularly identifies the following sectors as European Critical Infrastructure sectors:

- energy;
- nuclear industry;
- information, communication technologies, ICT;
- water;
- food;
- health;
- financial;
- transport;
- chemical industry;
- space, and
- research facilities.

This study focuses on energy (electricity, oil, gas and nuclear facilities), ports and communication (including space) facilities only.

The 9/11 terrorist attacks impacted on critical infrastructures directly through their physical effect and indirectly by provoking critical infrastructure users, regulators and owners, to take actions in order to secure the continuous provision of certain critical infrastructures which had not been directly attacked, such as the communication network.

The European Commission in a related Communication[35] on critical infrastructure protection in the fight against terrorism identified an increasing potential for catastrophic terrorist attacks affecting Critical Infrastructures. While the Commission doubts a high number of casualties, it highlights the dangerous consequences for vital infrastructure services as well as possible cascading effects. Moreover, it emphasises that effects will vary depending on the CI attacked.

Risk analysis for CI is difficult with information from a number of sources being needed to conduct threat, incident and vulnerability analysis of Member States' critical infrastructure elements and their dependencies. An alternative and parallel measure is to first explore and understand the motivation for terrorists to attack CI in order to be able to identify preferred targets and tactics. This will enable the possibility of prioritising resources. Drawing on the initially proposed framework of risk assessment, it could be argued that CI has the potential to fall within terrorist target selection when terrorists are motivated by

(1) the effect of disruption of society and economic functions,
(2) an ideological predilection towards CI,
(3) the symbolic relevance of CI,
(4) CI can be taken as a form of organisation building,
(5) familiarity with CI and / or
(6) precedent and adaptation.[36]

Protective measures include a mix of physical security facilities, like electronic monitoring equipment, as well as appropriate training of personnel and an emergency operations plan.

The most prominent example of terrorist impact on CI is 9/11. CI facilities and operations were directly disrupted by the physical impact of the attacks, with the World Trade Center (WTC) and the surrounding area housing a number of key businesses supporting CI. Thus, the destruction of the WTC area caused the disruption of business operations vital to several CI sectors including banking and finance, transportation, and communications. Moreover, the attack on the Pentagon surely had an impact on government continuity which, however, cannot be analysed in detail due to the secrecy with which this national security issue has been handled.[37]

2.4.2. Transportation Security

The transportation sector comprises the highest subdivision in this analysis. In general, one can distinguish between freight (i.e. cargo) transportation and mass surface transportation (i.e. passenger). These two fields can then be further divided into land,

[34] European Council Secretariat. "European Critical Infrastructure." EU Council Factsheet. Luxembourg: Justice and Home Affairs Council, 2008. <http://www.eurunion.org/partner/euusterror/EUCritInfrastructFactsheet-6-5-08.pdf>. Additionally, the Council establishes a four-step-approach in identifying European Critical Infrastructure.

[35] Commission of the European Communities. Communication from the Commission to the Council and the European Parliament: Critical Infrastructure Protection in the Fight Against Terrorism. COM(2004) 702 final of 20 Oct. 2004. Brussels: European Union.

[36] Ackermann, Gary A., Jeffrey M. Bale and Kevin S. Moran. "Assessing the Threat to Critical Infrastructure." Homeland Security Protecting America's Targets: Critical Infrastructure. Ed. James J. F. Forest. Westport: Praeger Security International, 2006 Volume III. 41-5.

[37] Canadian Office of Critical Infrastructure Protection and Emergency Preparedness. "The September 11, 2001 Terrorist Attacks - Critical Infrastructure Protection Lessons Learned." Incident Analysis IA02-001 of 27 Sep. 2002. 3 Feb. 2009 <http://www.publicsafety.gc.ca/prg/em/ccirc/ _fl/ia02-001-eng.pdf>.

sea and air transportation, with land including both road and rail and sea including both ships and ports. Additionally, the freight sector can then be further divided into different transport modalities like containers.

In this regard, supply chain security is even more divided beginning at the manufacturing site with the preparation for transportation and the loading process. Transportation by single mode to the final destination is theoretically thinkable but in practice highly unlikely. Thus freight may be taken to warehouses, storage areas or inland terminals, where they may change the transport modes, involving forwarders, agents or brokers. Thus a detailed analysis of the supply chain makes possible identification of four groups of activities each of which has its own security-relevant characteristics and requirements: (1) preparation of goods for shipment and shipment from the production site; (2) transport of goods; (3) forwarding of goods; (4) warehousing, storage and inland terminal operations.[38] With each individual security measure adding up to the security of the complete chain, an overview for custom authorities becomes difficult.

Ensuring the integrity of a container during its transport from one point to the other is impossible. With sufficient time, opportunity and a remote location, possibilities to open a container and tamper with its contents arise. Ways to do this are even published on the internet. To counter potential threats, the European Union is developing a coordinated approach to maritime, port and container security that combines improved security with the efficient flow of trade and commerce.[39] While security risks relating to terrorism exist in all modes of transport and freight flows, not all are at present covered by security legislation. The security level of various transport modes and freight flows therefore varies significantly. A supply chain perspective has just recently been introduced. While terrorists have so far refrained from attacking cargo, the threat of terrorist attacks on transportation as a whole has highlighted the vulnerabilities of the supply chain.[40]

Land Transportation

The largest share of intra-EU transport is carried by road, which accounts for forty-four percent of freight and around eighty-five percent of passenger transport. Demand factors, such as a reduction in heavy bulk transport and the increasing importance of door-to-door and just-in-time service have undoubtedly contributed to the strong sustained growth of road transport. The share carried by rail is ten percent and six percent, respectively. Rail freight transport has halted its relative decline since 2001 and is on a growth path in a number of Member States.[41]

Transport security has become an issue of international concern, with the risk of a terrorist attack targeting freight transport remaining high and the potential damage in terms of lives and economic activity being unpredictable and incalculable. Security in this regard can be defined as the combination of:

- Vulnerability to attack, which reflects the possibility of a terrorist attacking the transport network successfully, compared to the possibility of protecting it through inherent or managed safeguards.
- Consequences of a successful attack, relating to:
- The possible number of fatalities; and
- The economic impact which is calculated with the following factors:
- The reconstruction costs of the destroyed transport element;
- The disruption time of the transport flow;
- The volume of the transport flow.[42]

While concluding that all transport is at risk, the Commission subdivided transportation security into: (1) Infrastructure risks, i.e. the terrorist has the objective to damage or destroy transport elements in order to disrupt the supply chain. The transport elements are in this case the terrorist's target; (2) supply chain risks, i.e. the terrorist has the objective to misuse the supply chain as a means of

[38] Commission of the European Communities. Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of the Regions: Proposal for a Regulation of the European Parliament and of the Council on enhancing Supply Chain Security. COM(2006) 79 final of 27 Feb. 2006. Brussels: European Union.
[39] Delegation of the European Commission to the United States. "Securing Trade: The EU's Approach to Port and Maritime Container Security." EUInsight 21 (2008).
[40] Commission of the European Communities. COM(2006) 79 final. op. cit.
[41] Commission of the European Communities. White Paper - European Transport Policy for 2010: Time to Decide. COM(2001) 370 final of 12 Sept. 2001. Brussels: European Union.
[42] Commission of the European Communities. Annex to Communication from the Commission to the Council and the European Parliament, the European Economic and Social Committee and the Committee of the Regions on enhancing Supply Chain Security and Proposal for a Regulation of the European Parliament and of the Council on enhancing Supply Chain Security. Impact Assessment. Commission Staff Working Document. SEC(2006) 251 of 27 Feb. 2006. Brussels: European Union.

creating damage or fatalities. The transport elements are in this case not the target but the means and (3) the misuse of the supply chain relates to either cargo or mobile units to conceal and transport various explosives to a location where they are unloaded or detonated, or cargo or mobile units are misused as a weapon.[43] The world's heavy reliance on shipping containers coupled with their high degree of vulnerability means that terrorist attacks using containers would have high economic costs. While containers shipped by land, air and sea share similar weaknesses, there are no consistent security standards.[44] The difficulty in transportation security lies in striking a balance between ensuring security and ensuring a free flow of trade. In the current international environment, there exists the very real risk that terrorist organisations could exploit the mechanisms of international trade.

Contemporary terrorists have made surface transportation a major theatre of operations. In the past, major terrorist campaigns have targeted surface transportation with the IRA, suicide bombers in Israel and terrorists connected with Al-Qaida being at the forefront. While initially commercial planes as symbols of despised nations and policies were their primary targets, terrorists now increasingly turn to surface transportation by attacking trains, buses, stations and depots. One of the reasons for the change in target selection is, the improvement of security in the aviation sector.[45]

Thus, the terrorist threat to surface transportation is real and it will increase as terrorists seek an element of surprise and a way to achieve mass casualties. Surface transportation is often equalled with aviation in terms of security risk and vulnerabilities. This study takes a different point of view in which both surface transportation and aviation security are considered separately as risks and preventive and protective measures are understood to be of a different nature for each sector:

> Commercial aviation is a network, whereas surface transportation is a mosaic: a terrorist bent upon attacking commercial aviation must penetrate security at one commercial airport, once inside the security, however, the terrorist's target range is national; surface transportation offers almost unlimited access points, but once a terrorist is on board, his target range is local.[46]

Surface transportation offers terrorists easy access, little security to penetrate and higher volumes of passengers than aviation. The large crowds of strangers guarantee anonymity for attackers and facilitate their escape. Transportation facilities offer large venues for the spread of biological agents. Historically, attacks on surface transportation have been almost evenly divided between trains and buses (excluding the threat of bombing), with bombing being the most common tactic. The overall impact achieved differs from the one resulting from an attack on aviation: trains and buses cannot be flown into skyscrapers so the casualties that occur in a surface transportation attack are likely to be confined to the passengers themselves.

While some deterrence of terrorist attacks may be possible, prevention is extremely difficult in the absence of screening measures that, given the volume of users, the vast number of access points and the low cost of fares, are unrealistic. Security is thus more about mitigation than prevention. Mitigation can be achieved through station and vehicle design, quick diagnosis of threats, prompt intervention and rapid responses. Hence, surface transportation security has to be more reactive than proactive, aiming at preventing casualties and minimising disruption. In this regard a broad definition of "security" is required, encompassing all efforts to mitigate casualties, damage and disruption and to facilitate rapid restoration of operations. Attributes of effective surface transportation security thus include "flexibility, the ability to increase and decrease efforts according to threat conditions, an emphasis on technology rather than personnel and given limited prevention possibilities, a focus on crisis planning and response training."[47] To properly plan for transportation security, there is a need for a detailed and current knowledge of the assets in each part of the system (e.g stations, control centers and vehicles), their criticality, their vulnerability and the nature and degree of threat posed to each of them. With the threat not being equal everywhere in the country, national threat alerts implying a uniformity of the threat do not seem

[43] Ibid.
[44] Organization for Security and Cooperation in Europe (OSCE). "Containing Terrorism." 25 February 2005. Organization for Security and Cooperation in Europe (OSCE). 21 Jan 2009 <www.osce.org/item/231.html>.
[45] This section draws to a large degree on the related discussion in Jenkins, Brian Michael. "Improving Public Surface Transportation Security: What Do We Do Now?" Homeland Security and Terrorism: Readings and Interpretations. Eds. Russel D. Howard, James J. F. Forest & Joanne C. Moore. McGraw-Hill: New York, 2005.128-39.
[46] Ibid., 136.
[47] Ibid.,137.

appropriate. In order to build an appropriate policy to counter this threat, one should again identify the motivations for engaging in transportation terrorism, preferred tactics and targets.

Waterborne Transportation

> The seas are Europe's lifeblood. Europe's maritime spaces and its coasts are central to its wellbeing and prosperity – they are Europe's trade routes, climate regulator, sources of food, energy and resources, and a favoured site for its citizens' residence and recreation. Our interactions with the sea are more intense, more varied, and create more value for Europe than ever before. Yet the strain is showing. We are at a crossroads in our relationship with the oceans.[48]

Europe depends to a large degree on the maritime sector, with 70.000 kms of coastline along two oceans and four seas: the Atlantic and Arctic Oceans, the Baltic, the North Sea, the Mediterranean, and the Black Sea, This might explain why this sector and the related space applications are the most well developed as will be shown below. The EU has a waterway network of more than 35.000 kms encompassing large to small rivers and canals.[49] The core waterway network with rivers and canals of international importance is formed by more than 12.000 kilometres of interconnected waterways, close to 450 locks and several hundred inland ports and transhipment sites.[50] In general, the system of maritime and port security is perceived to be poorly defended against misuse and terrorism especially due to its global and open nature. Currently, there is no clear set of safety and security criteria from either the U.S. or the European Commission. As a result, individual States take their own uncoordinated initiatives.[51] Historically speaking, shipping has been an important element leading to economic growth and prosperity in Europe. Still today, maritime transport services are essential in helping the European economy and European companies to compete globally.

According to the definition provided by the European Defence Agency, maritime security "encompasses a wide range of policy sectors, information services and user communities, including maritime safety, search and rescue, operational safety for offshore oil and gas production, fisheries protection, marine environment monitoring and protection and navy operations support"[52]. Many of these do not fall within the range of internal security. The European Parliament and the Council rather define it as "the combination of preventive measures intended to protect shipping and port facilities against threats of intentional unlawful acts".[53] With container security being part of supply chain security, which falls within the transportation sector, and ports falling within critical infrastructures, this section focuses on maritime surveillance. That is the "ability to achieve continuity of knowledge on activities in the maritime domain in order to support a timely decision process."[54] Furthermore, one can differentiate between "maritime security", i.e. "the combination of preventive measures intended to protect shipping and port facilities against threats of intentional unlawful acts"[55], "maritime safety", i.e. "to continuously maintain and enhance safety in shipping and the protection of life, health, property and the marine environment, concerning the ships, the crew and the passengers and/or cargo, safety of navigation, environmental safety"[56] and "maritime surveillance, i.e. the ability to achieve continuity of knowledge on activities in the maritime domain in order to support a

[48] Commission of the European Communities. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. An Integrated Maritime Policy for the European Union. COM(2007) 575 final of 10 Oct. 2007. Brussels: European Union.
[49] Inland Navigation Europe. "Facts and Figures." 04 Feb 2008. Inland Navigation Europe 29 July 2009 <http://www.inlandnavigation.org/en/factsandfigures.html>.
[50] Commission of the European Communities. Communication from the Commission on the Promotion of Inland Waterway Transport "NAIDAS" -"An Integrated European Action Programme for Inland Waterway Transport. COM(2006) 6 final of 17 Jan. 2006. Brussels: European Union.
[51] Voort, van de Maarten, Kevin A. O'Brien, Adnan Rahman and Lorenzo Valeri. "Security: Improving the Security of the Global Sea-Container Shipping System." Homeland Security and Terrorism -. Readings and Interpretations. Eds. Russel D. Howard, James J. F. Forest & Joanne C. Moore. McGraw-Hill: New York, 2005. 186.
[52] European Space Agency. "GMES Service Element: Maritime Security". 11 Mar 2009. European Space Agency 24 July 2009 <www.esa.int/esaLP/SEMKCU0DU8E_LP gmes_2.html>; Joint Research Centre. "Maritime Surveillance at the Joint Research Centre." Press Factsheet. Joint Research Centre, 2007. < http://ec.europa.eu/research/press/2007/maritime-briefing/pdf/33-jrc-maritime-surveillance_en.pdf>.
[53] The Commission of the European Communities. Regulation of European Parliament and of the Council on Enhancing Ship and Port Facility Security. (EC) No 725/2004 of 31 March 2004. Brussels: European Union.
[54] Joint Research Centre. Press Factsheet. 2007. op. cit.
[55] The Commission of the European Communities. Regulation (EC) No 725/2004. op. cit.
[56] Commission of the European Communities. Green Paper - Towards a future Maritime Policy for the Union: A European Vision for the Oceans and Seas. COM (2006) 275 final of 7 June 2006. Brussels: European Union.

timely decision process".[57] Consequently "maritime security" in contrast to "maritime safety" does not include the environmental sector but deals solely with counter-terrorism and the fight against organised crime such as piracy.[58]Nonetheless, it "encompasses a wide range of policy sectors, information services and user communities"[59] and thus falls between all three pillars of the European Union. In this regard, maritime safety (for transport) together with the fight against pollution as well as fisheries control fall within the first pillar, intelligence of maritime origin, exercise of sovereignty at sea and military action at sea fall within the scope of the second pillar, and control of external borders, combating drug trafficking and combating clandestine immigration and people trafficking fall within the third pillar. The fight against terrorism is covered by all pillars (Figure 7: Maritime Security and the Three Pillars of the European Union).[60]

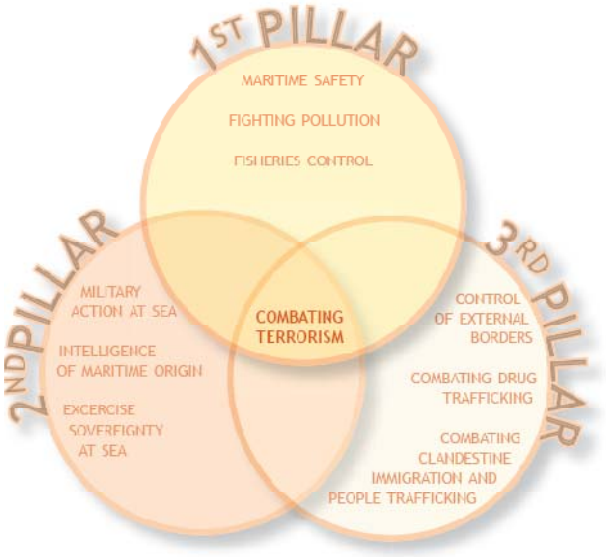


Figure 7: Maritime Security and the Three Pillars of the European Union[61]

Eighty percent of world trade is carried by sea whilst short-sea shipping carries forty percent of intra- European freight.[62] In Europe roughly 3.800 million tonnes are handled annually in the EU-27 ports, with Rotterdam, Antwerp, Hamburg, Marseille and Le Havre being the top five European ports on the basis of gross weight of goods handled.[63] Rotterdam is also Europe's largest container port. About ninety percent of the world's cargo is transported in maritime containers, but only two percent is physically inspected by customs authorities, thus increasing the opportunities for illicit activities such as avoidance of customs duties, circumvention of quotas or smuggling of nuclear materials and weapons.[64] It is widely believed that the only viable way to control containerised cargo is through information-based risk analysis combined with non-intrusive detection technologies. This makes possible pre-selection of high-risk consignments needing physical checks. While customs authorities have been using risk analysis to identify suspicious containers for quite some time, their analysis is based on customs declarations and involves risk indicators, such as the entities involved in the transaction, the type of goods transported and the origin declared. While being a strong tool to detect and intercept suspicious cargo, risk analysis is only as strong as the algorithms and information behind it. It suffers from its reliance on patterns, which can be recognised by terrorists who in turn ensure that their shipments do not match these patterns. Hence, analysis algorithms have to be complicated in order to avoid pattern-recognition. Also, due to difficulties in collecting and analysing the information required, risk analysis currently carried out by customs authorities does not take into account the detailed route followed by specific containers at the global level.[65] Given the great dependency of the European economy on trade, it becomes clear that targeting the transportation sector has an impact on both the European population as well as the economy, i.e. the industrial base of the EU.

When assessing the risks one has to distinguish between two possibilities.

[57] Definition by the European Defence Agency. In Joint Research Centre. Press Factsheet. 2007. op. cit.
[58] Möller, Kay. "Maritime Sicherheit und die Suche nach politischem Einfluss in Südostasien." Berlin: SWP, 2006.
[59] European Space Agency. "GMES Services Element: Maritime Security". op. cit.
[60] Western European Union. Report on the Surveillance of the Maritime and Coastal Areas of European States. Document A/1920 of Dec. 2005. Paris: Western European Union.
[61] .Joint Research Centre. Press Factsheet. 2007. op. cit.
[62] Commission of the European Communities. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions on Strategic Goals and Recommendations for EU's Maritime Transport Policy Until 2018. COM (2009) 8 final of 21 Jan. 2009. Brussels: European Union.
[63] Amerini, Giuliano. "Maritime Transport of Goods – 4th Quarter 2007." Eurostat Data in Focus 44 (2008); Amerini, Giuliano. "Maritime Transport of Goods and Passengers 1997-2006." Eurostat Statistics in Focus 62 (2008).
[64] Joint Research Centre. "Maritime Surveillance ConTraffic: Monitoring Container Traffic." Joint Research Centre, 2008. <http://ipsc.jrc.ec.europa.eu/showdoc.php?doc=promotional_material/JRC42970_contraffic_leaflet.pdf&mime=application/pdf>.
[65] Ibid.

Terrorists can either target the maritime sector itself or abuse it to import or export means to conduct an attack. Given the differences in character, each threat requires different solutions. Sea-related risks and threats falling within the definition of internal security are mainly criminal activities, from the trafficking of human beings and smuggling to terrorism. So far, this has only been done for logistic purposes, i.e. to carry weapons and explosives in large quantities and to smuggle terrorists into Western countries. However, using ships to cause widespread chaos and destruction in a major harbour would also be thinkable. Tackling threats at sea is particularly difficult due to the permeable nature of the frontiers at sea. States cannot respond to these threats, first because there is a power vacuum at sea with most countries having no navy able to react alone and, second, because states can only legally protect vessels carrying their flag in international waters. This requires a coordinated approach at the EU as well as at the international level.[66] One can thus say that maritime security is a global issue with a European dimension. Criminal activities and terrorism could be deterred significantly by a concerted initiative that improves the presence of maritime security forces, enables the boarding of suspicious vessels according to internationally agreed legal rules, and provides important situational data through integrated civil-military surveillance capacities. [67]

Air Transportation

General aviation in turn involves some 15.000 civil aircraft and 30.000 small private planes. Roughly 800 million air passengers are carried annually at EU-27 airports, with the most frequented airports being London/Heathrow with 67.9 million passengers, Paris/Charles de Gaulle, Frankfurt/Main, Madrid/Barajas and Amsterdam/Schiphol.[68]

The existence of a terrorist threat to aviation security is evident since the Pan Am explosion over Lockerbie in 1988. Given that terrorism as a form of warfare is directed against a particular state and airlines are perceived as a nation's flying symbols, aviation is a very attractive target that falls within the ideological scope of most terrorist groups. Hijacking an airliner brings a terrorist three desirable results: First, innocent civilian hostages serve as bargaining counters for political concessions; second, seizing an aircraft shows the government's inability to protect its citizens and, third, it creates a travelling „theatre" which attracts huge publicity. When flying became a mode of mass transportation, aviation became even more vulnerable. Due to the increased airport workforce it became impossible for security personnel to know faces. Aviation security faces five particular threats: (1) conventional hijacking; (2) sabotage bombing; (3) suicide-sabotage bombing; (4) bomb attack on airports and (5) new threats such as Man Portable Air Defence Systems (MANPADs).

Aviation facilities are not only of symbolic significance but also of great functional importance. Their destruction would have widespread economic effects with aviation being a critical part of local and national economies.[69] Commercial planes are not only symbols of nations and policies despised by terrorists but can also serve as portable containers for victims or hostages, a means to escape or a weapon in themselves.

Aviation is a vast and complex system consisting of three main branches, i.e. commercial aviation, general aviation and air cargo, each of which poses separate and complex security challenges. Like mass surface transportation, the difficulty of securing aviation lies in the open character of airports and airline offices that, on the one hand need to be accessible for customers but, on the other hand, have to be secured from the terrorist threat. Additionally, aviation has become increasingly integrated with other transportation modes, especially with light and heavy railways as the landside capacity of airports has become increasingly strained. Moreover, increased linkages between telecommunications and aviation aggravate the security problem, with disruptions of telecommunications networks having profound consequences for global, regional and national aviation systems. Thus, aviation security cannot be addressed independent of telecommunications. Furthermore, the international as well as the national character of aviation complicates the matter. In contrast to transportation security, commercial aviation security emphasises deterrence and prevention:

[66] Commission of the European Communities. COM (2006) 275 final. op. cit.
[67] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.
[68] Fuente Layos, de la, Luis. "Air Passenger Transport in Europe 2007." Eurostat Statistics in Focus 01 (2009).
[69] This section draws to a large degree on the related discussion in Szyliowicz, Joseph S. "Aviation Security: Promise or Reality?" Homeland Security and Terrorism: Readings and Interpretations. Eds. Russel D. Howard, James J. F. Forest and Joanne C. Moore. New York: McGraw-Hill, 2005.

> If the layers of security have been breached, there is seldom much that can be done other than to use blast-resistant cargo containers to provide some protection against bombs smuggled inside checked bags or to implement schemes that override cockpit controls and remotely guide hijacked aircraft to safe landings. Mitigation is not much of an option-an airline crash allows few opportunities for lives to be saved, even with rapid response.[70]

One should further distinguish between commercial and general aviation and air cargo when building a security strategy for aviation. It is often argued that cargo aircraft lack this symbolic resonance of flag-carriers and would not attract the same media attention. Yet, this might be a too easy conclusion. Many of the considerations as put forward for sea cargo are also valid for air cargo, which is often neglected in the debate. With passenger planes carrying large amounts of cargo, cargo security should be allocated the same importance. Cargo planes are just as powerful weapons as loaded passenger planes but not the same security requirements are applied: there are no federal marshals on cargo flights, cockpit doors are weak, if they exist at all and the cargo ramps are insecure in comparison to passenger ramps; thousands of people, many strangersfew if any with background checks, work on loading and unloading air cargo.

While it is important to review transportation security mode by mode, i.e. commercial aviation, ports and container shipping, transport of hazardous cargo, public surface transportation, pipelines, bridges and tunnels as each mode has its own requirement, one needs to avoid missing the broader strategic issues ranging across the entire spectrum of transportation systems. Such an "across-the-board look" could lead to new and innovative security solutions. Furthermore, the nature of the threat faced shows the importance of technology in particular in the counter terrorism strategy for aviation. Given the high chronology of past terrorist incidents in aviation, an examination of statistics of world-wide aviation terrorism can provide a "lessons learned" approach for the design of an effective aviation security system. The establishment of strong national aviation security systems can provide essential building blocks for international cooperation. Both the public and the private sector have to become involved in the implementation process of any counter terrorism policy for aviation. While intelligence proves to be a strong preventive tool for aviation terrorism, a comprehensive counter terrorism strategy in this sector consists of a combination of airport and onboard security as well as integrating international cooperation.

[70] Jenkins, Brian Michael. op. cit. 136.

2.4.3. Border Security

Europe's borders are composed of coastlines, intercoastal waterways, land borders and hundreds of international ports that are frequently crossed by people, cars, trucks, overseas shipments including containers, planes, railcars and ships. The economic value is impressive.[71] Border agencies do not have the resources to examine more than two percent of the cargo that comes into the European Union. They do not have the data to identify what is coming in; they do not possess the analytical capability to discern what to inspect and what to pass through; they do not have the information systems to rapidly update databases and incorporate intelligence from other agencies; and they do not have people to conduct meaningful inspections. While the problem of securing cargo is tackled in the transportation section, ports as points of entry are covered in both the transportation and the critical infrastructure section, coastlines are covered within this section.

Illegal trafficking and other related border security issues have a potential corrosive impact on values and the related threat of corruption of public officials, especially law enforcement officials. There is a higher risk of terrorist incidents due to the ability of human trafficking networks to forge linkages with terrorist organisations. Illegal trafficking of drugs, arms and humans is to a large degree economically motivated and thus the policies designed to address these issues need to be market-oriented solutions tackling the supply and demand aspects of these problems.

[71] Hoffman, Frank. "Border Security: Closing the Ingenuity Gap" Homeland Security and Terrorism -. Readings and Interpretations. Eds. Russel D. Howard, James J. F. Forest & Joanne C. Moore. New York: McGraw-Hill, 2005.

3. The Space Dimension: Technical Possibilities and Limitations

This chapter provides the space dimension to internal security. It gives an overview of the current technically feasible capabilities in all three critical mission areas as well as in the areas of intelligence and emergency management and further qualifies them.

In addition to preventive methods adapted to the characteristics of each of the critical mission areas, terrorism and organised crime can also be countered before reaching European borders through effective preventive measures that do not necessarily depend on the critical mission areas as such (e.g. detection of terrorist facilities etc.). Thus prevention is probably the area where external and internal security most coincide, given that intelligence outside the EU will have an impact on internal security as well.

3.1. Intelligence, Early Warning and Risk Management

The first step in preventing a terrorist attack or to counter organised crime is detection through intelligence. So-called intelligence satellites include all earth-observing sensors as well as the electronic monitoring of telecommunications (COMINT and ELINT).[72]

3.1.1. Earth Observation Imagery

One can further identify image intelligence or IMINT satellites or photo surveillance satellite and signal intelligence, or SIGINT or ferret satellites, which are collectively referred to as surveillance satellites. Image intelligence satellites provide detailed high resolution images and maps of inter alia geographical areas, military installations and activities. The resolution provided by these satellites is of a few centimetres. One can further differentiate between close-look IMINT satellites and area survey IMINT satellites, the former providing high resolution photographs returned to Earth via a re-entry capsule and the latter providing lower resolution photographs transmitted to Earth via radio. Alternatively, IMINT satellites can be categorised according to their wavelength band of operation: (1) Electro-optical satellites and (2) Radar Imaging satellites.[73]

Electro-optical imaging satellites (also known as optical) provide full-spectrum photo-graphic images in the visible and the IR bands and use a CCD camera. Digital information is transmitted to the ground station via electronic communication links. They are able to image heat sources during the night but not objects having normal temperatures. They do not work in cloudy conditions and are able to be camouflaged to only a limited degree.

Radar imaging satellites (also known as SAR) overcome the problem of cloudy conditions. They suffer however from poor resolutions compared to electro-optical satellites and the problem of "backscatter noise", and they are susceptible to active jamming. They usually employ synthetic aperture radar (SAR) to take images in the microwave band. Alternatively, they use Doppler radar technology and Ground Moving Target Indication (GMTI) for detecting ground movement of vehicles.

Both optical and radar imagery cannot only be used as such for the detection of for example terrorist camps but also to detect organised crime through hazard extraction. This is normally done in three stages: (1) change detection; (2) change categorisation and (3) change classification. A database of satellite imagery (both optical and SAR) can serve as a basis for change detection.[74]

[72] European Space Agency. "The European Space Sector in a Global Context - ESA's Annual Analysis 2004." Paris: ESA Policy Department, Paris. 51. <www.esa.int/esapub/br/br242/br242.pdf>.

[73] The description on the workings of reconnaissance satellites is largely based on the related section in: Maini, Anil K. and Varsha Agrawal. "Satellite Technology: Principles and Applications." West Sussex: John Wiley & Sons, Ltd., 2007. 528 -541.

[74] For the exact workflow in change detection refer to Dekker, R.J., I. Lingenfelder, B. Brozek, U. Benz and A.C.

Through fusion of SAR and optical changes, changes can be categorised into priorities:

- highest: coinciding changes (changes that occur in both SAR and optical at the same time)
- high: changes in SAR only
- medium: changes in optical only.

The last stage is change classification based on object size, spectral signature, GIS data and what could possibly be detected using remote sensing images.

Satellite imagery also helps in terms of treaty monitoring and verification through, for example, the detection of underground nuclear tests revealed in satellite radar interferometry.[75] Commercial satellite imaging proves to be especially valuable for intelligence purposes, given its unclassified nature and the lack of legal restrictions.



Figure 8: was shown to the U.N. Security Council during Colin Powell's February 5, 2003 presentation. Powell described it as showing a terrorist poison and explosive factory in Iraq, operated by an Islamic terrorist group, Ansar al-Islam, with ties to Al-Qaeda[76]

3.1.2. Digital Image Processing

Pure imagery data is however of limited use. Derived imagery has to be placed in the relevant context, analysed and assessed. This is what is referred to as GEOspatial INTelligence (GEOINT), combining disciplines such as mapping, charting, imagery analysis and imagery intelligence. GEOINT organises and combines all available data around a particular geographical location and then exploits it in order to prepare products useable by decision makers, intelligence agencies and emergency responders.

In order to compile geospatial information,[77] risk management requires:

1. reference data such as orthoimagery, transportation elevation, political boundaries, property ownership, hydrography and industrial installations
2. data on critical infrastructure such as telecommunications, electrical power systems, gas and oil production, storage and distribution, banking and finance, water supply systems and emergency services
3. residence and employment data related to specific locations such as schools, government facilities, hospitals and population density

The combination of satellite navigation and remote sensing is also increasingly important in the prevention of organised crime.[78] Based on this data risk, mapping can be conducted to determine vulnerabilities and predict potential hazards. In order to prevent unwanted groups such as terrorists from using hazardous material, all potentially dangerous sites must be entered into databases and site analyses performed to evaluate the potential danger of these materials to the civilian population. This is a first step towards improving information to the affected public, drawing up emergency plans and adopting corrective measures.[79] Cooperation between geographic information system (GIS) tools, satellite navigation and remote sensing and their applications in government and private industry is nowadays known as geographic management systems (GMS).[80]

van den Broek. "Object-Based Detection of Hazards to the European Gas Pipeline Network Using SAR Images." 10 Feb. 2009 <http://www.presense.net/pdf/11_Object_based_detection.pdf>.

[75] Liebig, Volker. "Remote Sensing for Civil and Security Applications." Presentation. An AFCEA International Symposium on AEROSPACE TECHNOLOGIES and APPLICATIONS for DUAL USE. Hotel Parco del Principe: Rome, Italy. 13 Sep 2007. <http://www.afcearoma.it/presentazioni/internazionali/AFCEA_VLiebig_Final.pdf>.

[76] Richelson, Jeffery T. "Eyes on Saddam: U.S. Overhead Imagery of Iraq." Briefing book no: 88. 30 Apr. 2003. National Security Archive 24 July 2009 <http://www.gwu.edu/~nsarchiv/NSAEBB/NSAEBB88/>.

[77] Atkinson, M. "The EC FP6 Integrated Project ORCHESTRA: General Introduction: Moving Towards Information interoperability in Risk Management." Presentation. Joint Research Centre, Ispra, Italy. 16 June 2005. <http://www.eu-orchestra.org/docs/200500615_Orchestra_OverviewEndUser.pdf>.

[78] Forest, James K. (Ed.). "Homeland Security Protecting America's Targets". Vol. III: Critical Infrastructure. London: Praeger Security International, 2006. 360-361.

[79] The GMES Working Group on Security. "The Security Dimension of GMES." Position Paper of the GMES Working Group on Security. GMES Working Group on Security, 2003. <http://www.gmes.info/pages-principales/library/reference-documents/?tx_kkdownloader_pi1%5Bpage%5D=2&cHash=ed61e0a4c2>.

[80] Forest, James K. op. cit.

Geographical Information Systems and satellite remote sensing are also used for nuclear safeguards. With reference to the Non-Proliferation Treaty (NPT) satellite imagery provides a useful tool:

- to verify the completeness of the site layout map attached to the declaration;
- to monitor ongoing status and detection of changes at declared sites
- to corroborate information from other sources about undeclared sites
- to provide inspectors with a real site layout for inspection planning and orientation.[81]



Figure 9: A satellite image of Iran's Natanz uranium enrichment facility [82]

The imagery type of use ranges from spatial resolutions of 5-10 metre resolution corroborating open source information to high resolution satellites of 1 metre resolution. Such a system can also be applied as a remote tool for monitoring nuclear facilities, reducing the effort and cost of inspections. Some commercial companies have already started offering 3D Terrain Models for nuclear site observation and monitoring after IAEA freezed a certain nuclear site.

3.1.3. Electronic Monitoring of Telecommunications

Signal intelligence from SIGINT satellites detect transmissions from broadcast communication and non-communication systems such as radar, radio and other electronic systems and are capable of intercepting and tracking mobile phone conversations, radio signals and microwave transmission. They are not capable of intercepting landline communication.

SIGINT is considered to be the most sensitive and important form of intelligence, providing early warning. The main missions carried out by these satellites are:

- interception and decryption of governmental, military and diplomatic communications transmitted by radio
- interception of ESM (electronic support measure) signals characterising the operating modes of higher command organisations
- relay of radio messages from intelligence agents in foreign countries

SIGINT can be further divided into communication intelligence (COMINT) and electronic intelligence (ELINT). COMINT or communication intelligence satellites perform covert interception of foreign communications in order to determine the content of these messages. Given the encryption of these messages, various computer-processing techniques for decryption are needed. This way sensitive data about critical individuals can be obtained. ELINT, electronic intelligence satellites, are used for the analysis of non-communication electronic transmissions. Most common ELINT satellites are designed to receive radio and radar emanations of ships at sea, mobile air defence radar, fixed strategic early warning radar and other components for the purpose of identification, location and signal analysis.[83]

While the added value of geospatial information for military use as well as disaster management has been understood, this is not the case for disaster prevention related to counter terrorism and organised crime. Effective integration and assimilation of data and information in situ and space-based observation is of great importance as governments civil protection bodies and emergency services are becoming increasingly dependent on integrated information systems to aid different phases of disaster risk management.[84]

[81] Poucet, A., Contini, S. and Bellezza, F. "Geographical Information Systems and Satellite Remote Sensing in Nuclear Safeguards." Paper presented at 5th EC-GIS Workshop GIS OF TOMORROW. Stresa, Italy. 28 - 30 June 1999. <http://www.ec-gis.org/Workshops/6ec-gis/papers/contini-nuclear.doc>.

[82] Blair, David. "Iran Could Have Nuclear Bomb by 2010 Study Warns." Telegraph 27 Jan. 2009 <http://www.telegraph.co.uk/news/worldnews/middleeast/iran/4362989/Iran-could-have-a-nuclear-bomb-by-2010-study-warns.html>.

[83] "Signals Intelligence Programs and Activities." 26 Nov. 1997. FAS 25 Aug. 2009 <http://www.fas.org/irp/world/russia/program/sigint.htm>.

[84] European Space Agency & European Commission. "Global Monitoring for Environment and Security: Final Report for the GMES Initial Period (2001-2003)." European

Currently the EU has "islands of data" of different standards and quality. The current data situation in Europe often results in the absence of necessary information due to lack of coordination across borders and between levels of government, and a lack of common standards and their use resulting in incompatible information and information systems, fragmentation of information and redundancy. This situation is worsened by current data policy restrictions such as pricing, copyright, access rights and licensing policy.

With this in mind, in 2001 the European Commission launched an initiative to develop an Infrastructure for Spatial Information in Europe. The INSPIRE initiative aims mainly at developing a legal framework to underpin the creation of a European SDI thereby starting from priorities in the environmental field. While the INSPIRE initiative, which is currently in its transposition phase, aims at solving the standardisation issues, other obstacles to satellite data exchange still persist. In this regard, architecture constraints remain and in particular information exchange in risk management is not yet resolved.

3.1.4. Risk Mapping

To assess the risk resulting from a given hazard, it is necessary to determine the vulnerability of the population and of the infrastructure. Thus background information is needed on:

- basic geographical features
- natural risk zones
- man-made alterations to the landscape
- locations and nature of economic infrastructure (e.g. road and rail, factories, offices)
- locations of Seveso sites and other potential hazards (e.g. chemical plants),
- location and size of population
- location and population density of residential areas.

Combining risk information with other geographical information will allow the identification of potential incompatibilities as a first step towards improving information to the affected public and drawing up emergency plans, improved land use planning, and adoption of corrective measures.[85]



Figure 10: Example of Risk Mapping for Floods[86]

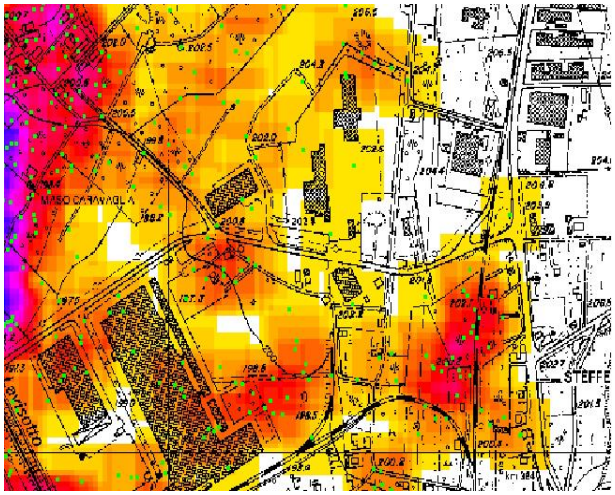


Figure 11: Risk map for unexploded bombs of World War II[87]

Space Agency / European Commission, 2004. <http://www.gmes.info/pages-principales/library/reference-documents/?tx_kkdownloader_pi1%5Bpage%5D=2&cHash=ed61e0a4c2>.

[85] The GMES Working Group on Security. op. cit.

[86] "Flood Risk Mapping and Damage Management." Flood Forecast and Risk Management Europe 24 July 2009 <http://www.floodrisk.eu/Preview/index.php?page=RiskMappingDamageAssessment>.

[87] UXB Trentino Project. "Risk Map of Unexploded Aerial Bombs". UXB-Trentino 27 July 2009 <http://mpa.itc.it/projects/pr-ux-ri-en.html>.

3.2. The three Critical Mission Areas

As will be shown by the following analysis, satellite requirements depend to a large degree on their specific application. For this reason this study looks at the different critical mission areas separately when giving an overview of the current technological capabilities.

3.2.1. Critical Infrastructures

Given the diversity of threats to critical infrastructures, effective protection can only be achieved by integrating various technologies and systems for a common protection goal. Theoretically, satellite imagery can be used for the monitoring of all critical infrastructures. High-resolution imagery has only been available since 2000 and thus EO has only been applied to CIP since the existence of centre-metric images.

The three stage process of change detection, change categorisation and change classification explained earlier can also be used for CIP monitoring. However, given the existing revisit times 24 hour monitoring is currently not possible.

A general first step is the mapping of critical sites including an assessment of the environment, the current situation and the potential risks (cf. 2.1 Intelligence).[89] High resolution EO satellites like QuickBird can take detailed pictures of different critical infrastructures, which provide the basis for more detailed maps. Feature level fusion techniques can be applied for extraction of man-made structures combining SAR and PolSAR data and combining SAR and hyperspectral data and are particularly useful for road detection. A classifier based on polarimetric decomposition methods is useful for detection of large roads in SAR and a specific detector for linear features is more effective for smaller roads. Fusion of E-SAR and hyperspectral is not sufficient for building detection. It is not possible to detect

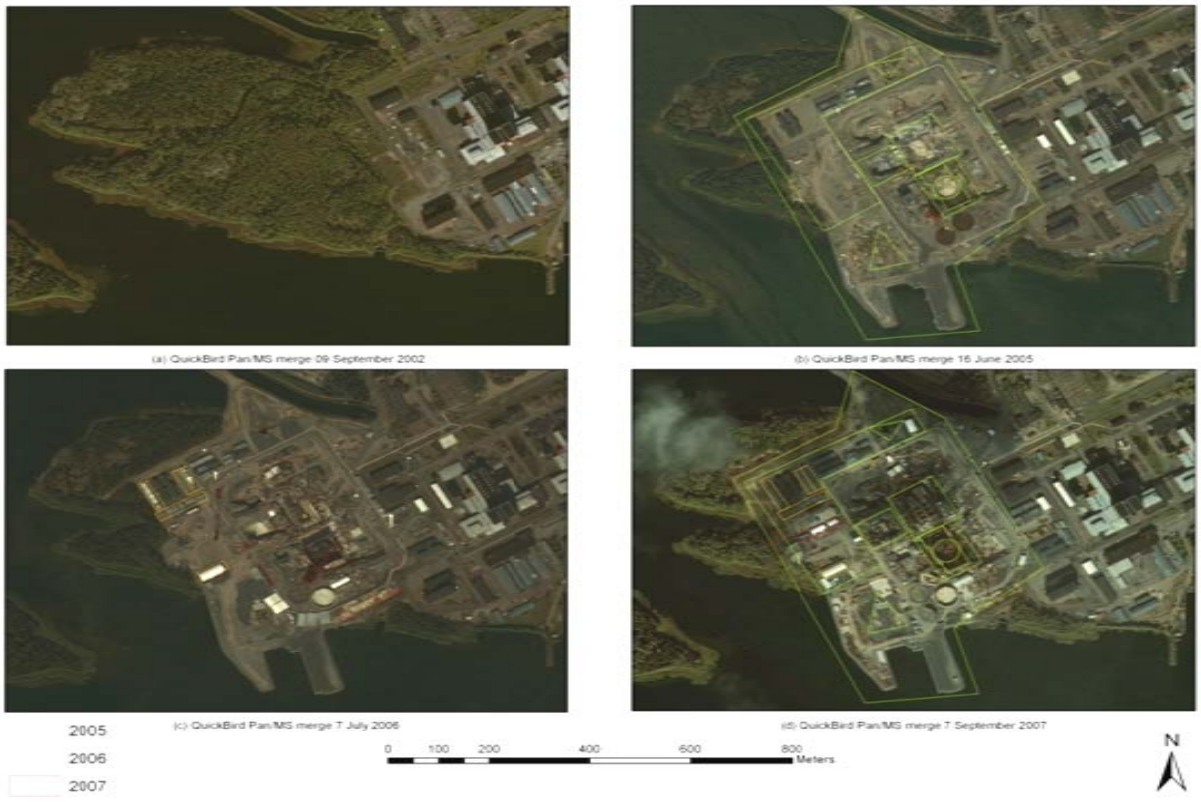


Figure 12: Change Detection in Nuclear Power Plant[88]

[88] Ussorio, Alessandra. "Earth Observation Applications and Critical Infrastructures in the Field of CFSP/ESDP: the Experience of EUSC." Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at/images/stories/dokumente/Conference2009/ussorio.pdf>.

[89] Institute for the Protection and Security of the Citizen & Sensors Radar Technologies and Cybersecurity Unit. "Research Strategy Paper: Emerging Technologies in the Context of Security." Issued in the Framework of Science and Technology Foresight: Joint Research Centre, 2005. <http://serac.jrc.it/index.php?option=com_docman&task=doc_click&Itemid=76&bid=22&limitstart=0&limit=5>.

individual houses in an urban environment using SAR, mainly due to the imaging geometry.[90] Combined with local geospatial information, high resolution satellite imagery can provide updated infrastructure and terrain analysis. 2D modelling and risk mapping can be conducted on this basis, to determine vulnerabilities and for predicting potential hazards.

Based on EO imagery and the acquired maps one can build and maintain a comprehensive assessment of European critical infra-structures with catastrophic potential.[91] In order to prevent unwanted groups such as terrorists from using hazardous material, all potentially dangerous sites must be entered into databases and site analyses must be performed to evaluate the potential danger of these materials to the civilian population. This can be complemented by technologies for the detection of abnormal behaviour. Space applications have the advantage of providing unbiased sources for analysis.

Apart from these general space applications for critical infrastructure protection, their application for more specific protective measures depends on the characteristics of the particular infrastructure.

Energy Facilities

High resolution imagery can provide the basis for putting together a map of any energy facility for risk analysis. High resolution multi-spectral cameras and spectrometerss on board of LEO satellites are suitable tools to help to detect and evaluate industrial and power plant disasters in hidden regions where the disaster should be kept secret.92 Hence, one can distinguish two categories of monitoring of energy facilities: (1) monitoring in cooperation with the owner/operator and (2) monitoring in the absence of any cooperation with the owner/operator. Satellite remote sensing data is also applied for satellite monitoring of nuclear power plants and pipelines. Hazards can be detected through change detection application on SAR and optical images followed by change categorisation and change classification (Figure 12).

Twenty-four hours monitoring of, for example, a nuclear power plant provides data on the normal functioning of the facilities. Any change in this data allows for immediate detection of malicious behaviour or accidents. In situ data can contribute to the further identification of threats. GIS techniques can also be used to develop maps of gas and oil pipelines. Combined with other technologies such as, for example, a hydraulic simulation engine one can develop simulations of the gas flow.[93]

Ports

At a basic functional level, satellite navigation can ensure secure port navigation and satellite imagery can be used for port surveillance. At a more sophisticated level,

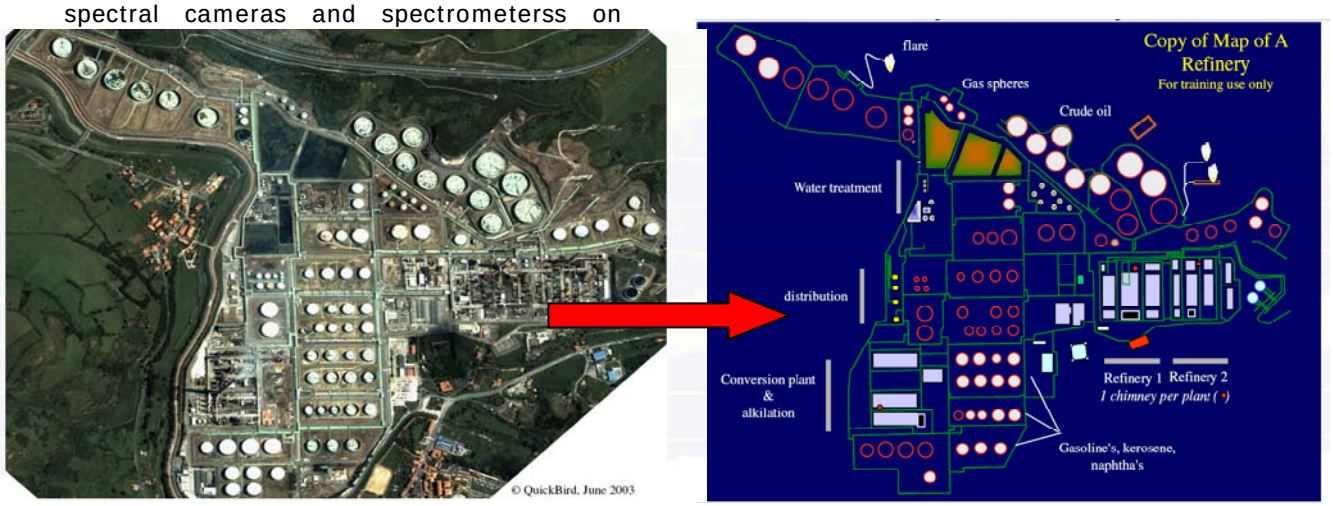


Figure 13: Satellite Imagery of a Refinery and Derived Map

[90] "Man-made Object Classification Using Fused Polarimetric SAR and Hyperspectral Imagery Data (HYSAR)." 2006. Belgium Earth Observation Platform 30 Apr. 2009 <http://eo.belspo.be/Directory/ProjectDetail.aspx?projId=789>.

[91] European Security Research and Advisory Board. "Meeting the Challenge: The European Research Agenda". Luxembourg: Office for Official Publications of the European Communities, 2006.

[92] Sandau, Rainer. International Study on Cost-Effective Earth Observation Missions. London: Taylor & Francis Group, 2006. 82.

[93] Institute for the Protection and Security of the Citizen. "Annual Report 2007." Luxembourg: Office for Official Publications of the European Communities, 2008. 20.

space-based applications can help detect illicit traffic of radioactive material.

> Detection and identification of dangerous radioactive material has to be quick and reliable, without disrupting the normal flow of commerce. Symetrica and Smiths Detection are jointly developing a unique handheld and backpack Human Portable Radiation Detection System (HPRDS) tailored to meet those criteria. The device features space-based radiation detection hardware and signal processing software for use by emergency responders, border patrol agents, customs and coast guard officers and other law enforcement personnel. The detectors can identify and determine the location of incoming radiation and reliably discriminate between normally-occurring radioactive materials and potential threats.[94]

Communication

One can distinguish three different types of communication systems as critical infrastructures: (1) Telecommunication and computer infrastructures, (2) Mobile and Personal Communications Systems and (3) Satellite Equipment and Systems. For secure communication for voice, advanced data, video and multimedia applications several countries rely on military communication satellites. In order to secure mobile communication systems use PMR (Private Mobile Radio) can be used.

While satellite telecommunication technology is perhaps the most effective tool that can uninterruptedly perform its functions during any natural or human induced disaster, space assets can themselves become targets. In view of their vital importance for the functioning of society, space assets themselves can also be considered a critical infrastructurethat needs to be secured. There are three main measures to consider for space system protection:

1. Capabilities to detect space negation attacks are often a pre-condition for successful protection and a pre-requisite for deterrence
2. Physical and electronic means to withstand attacks on ground stations, communication links and satellites
3. Reconstitution and repair mechanisms to recover from space negation attacks.

The main problem in detecting space negation lies in the difficulties in distinguishing between technical failure and deliberate attack.[95] Detection of rocket launches can be conducted through space-based early warning capabilities complemented by sea-based and terrestrial assets performing ballistic missile launch detection and tracking. Most actors additionally have a basic capability to detect a ground-based electronic attack on their space system, such as jamming, by identifying the interference signal of the attacker or detecting the loss of communication with the system under attack. Many actors also have the capability to use multiple sensors to geo-locate the source of jamming signals, which helps to determine if the interference is intentional. It can also be assumed that most actors have some capability to detect spoofing, since basic electronic error code checking routines are relatively simple to implement. Early warning of ground-based electronic attacks remains challenging as direct energy attacks such as laser dazzling or blinding and microwave attacks move at the speed of light making advance early warning very difficult. Space-based anti-satellite weapons can be detected through the tracking of satellite manoeuvres to monitor whether a satellite is in an orbit that could allow it to intercept or attack another satellite. While the ability to constantly monitor all satellites to detect hostile manoeuvres would constitute a significant protection capability, no space actor currently has this ability. Alternatively, sensors on satellites could detect nearby satellites and negation efforts. Nuclear detonation warning sensors on satellites can detect nuclear tests. Direct energy attacks such as laser dazzling or blinding and microwave attacks can be detected either by the loss of a data stream from optical or microwave instrumentation or by thermal sensors. Onboard satellite-specific laser sensors can detect either the key laser frequencies or radiant power.

Physical protection of ground stations is mostly guaranteed within the general boundaries of the relative military capabilities. Physical protection of satellites is more difficult to achieve. It is almost impossible to provide physical hardening against conventional weapons because of the high relative velocities of objects in orbit. Radiation hardening can be used for protection against nuclear attacks.) One option would be the use of radiation-tolerant components and automatic sensors designed to switch off non essential circuits during a nuclear detonation (e.g. photocvoltaic or

[94] Ibid.

[95] Governance Group for the Space Security Index. “Space Security 2008.” Montreal: Project Ploughshares, 2008. 137ff.

solar cells employed as power sources can be replaced by nuclear reactors, thermal-isotopic generators or fused silica-covered radiation-resistant solar cell models built with gallium arsenide. Alternatively, one can think of electromagnetic pulse shielding and scintillation and blackout avoidance. Satellites can be protected against direct energy attacks like laser dazzling or blinding through automated mechanical shutters. All physical protection capabilities are designed to mitigate vulnerabilities but they cannot make satellite systems invulnerable. Thus initiatives to prevent proliferation, effective laws, policies and doctrines, are also needed.

Electronic Protection can be achieved through basic protection capabilities such as:

- data encryption;
- error-protection coding to increase the amount of interference that can be tolerated before communications are disrupted, directional antennas that reduce interception or jamming vulnerabilities, or antennas utilising natural or manmade barriers as protection from line-of–sight electronic attacks;
- shielding and radio emission control measures that reduce the radio energy that can be intercepted for surveillance of jamming purposes; and,
- robust encryption onboard satellites.

Sophisticated electronic protection measures are generally unique to the military communications systems of technologically advanced States and consist of:

- narrow band excision techniques mitigating jamming by using smaller bandwidth;
- burst transmissions and frequency-hopping methods that communicate data in a short series of signals or across a range of radio frequencies to keep adversaries form “locking-on” to signals to jam or intercept them;
- antenna side-lobe reduction designs that mitigate jamming or interception vulnerabilities by providing more focused main communication beams and reducing interference from jamming in the side-lobe regions;
- nulling antenna systems (adaptive interference cancellation) which monitor interference and combine antenna elements designed to nullify or cancel the interference.

Alternatively, quantum communication or optical data communication (i.e. Free Space Optical Laser Communication) can be considered.[96]

Quantum communication is a secure way of transmitting data by means of quantum mechanics. It has the advantage of making it possible to detect an interception of communication. Optical Data Communication uses laser beams to transmit information between two locations. This has the advantage of high security regarding interception: a focused laser beam is hard to intercept without notice / path to quantum cryptography.

Considering space assets as critical infrastructures, the third phase of a crisis is replacement and repairing. Although efforts are underway, no actor currently has a rapid recovery capability. The same holds true for rapid rebuilding. The use of smaller and smaller satellites to provide operational requirements in outer space, as well as the use of defensive clusters of satellites or the inclusion of space-based backup systems, could contribute to space systems’ protection strategies.

3.2.2. Transportation Security

While transportation security can be divided into several transportation areas such as land, sea and air, containers are used for all of these. Technological solutions for ensuring container integrity do not depend on the transportation mode. Electronic systems installed inside the container detect movements within the container, including opening of the door and can be programmed to send a signal to a control centre if the container is opened at position other than its predetermined destination. Such tools involve space-based navigation and communication systems. A distinction can be made between systems giving a time-to-time or a continuous update on the container’s position and systems that signal status-changes, like the opening of the door or movement within the containers. While experience proves that theft is practically non-existent after the system’s application, global coverage is far from being achieved. Due to the high costs, only containers carrying high class loads are currently equipped and protected with these systems. Costs could be reduced by producing larger numbers.

[96] Oerlikon space. “Optical Data Communication: An attractive Space Technology for terrestrial Applications.” Presentation. Space National Information Day. Lucerne, Switzerland, 23 April 2009. <http://www.euresearch.ch/fileadmin/documents/events2009/Buchheim_LCT_FP7Lucerne_OSZ_print.pdf>.

In addition to electronic systems installed inside the container, a transponder can be attached via plug and play to the container, allowing monitoring of the container's route via satellite navigation and communication systems. By monitoring container traffic and systematically gathering and analysing data on global container movements, potentially suspicious and particularly dangerous routes can be identified. Additionally, ground independent and worldwide position determination of containers through satellite navigation becomes possible. Permanent real-time data transfer via communication satellites enables world-wide availability of all data and documents, electronic creation and signing of loading and customs documents by authorised persons, and goods-condition monitoring and coding of all data.

Thus satellite applications prove to be a promising alternative to physical verification, i.e. the actual opening of a container and manual verification of its contents with the bill of lading, a process taking approximately eight hours and leading to significant delays, thereby introducing further uncertainties in the logistics chain, and to X-ray scanners, which produce a fairly accurate image of the container's contents through scanning. In the pas,t scanners have been known to generate such a large amount of taxes from confiscated contraband that they can be regarded as a profitable investment instead of a costly expenditure. However, they require a rather large section of terminal space. Moreover, the contents of cans and lead within a container cannot be verified. The same applies to materials inside boxes or tins. Training for custom officials in interpreting X-ray images is additionally required. It remains problematic however for them to distinguish between e.g. a Weapon of Mass Destruction (WMD), which has been taken apart and is now transported in pieces, and car-parts. Inspection technologies like hyper-spectral imaging used to identify potential terrorist weapons and toxins, which are spin-off technologies from NASA, would be needed in addition to X-ray scanners.

Land Transportation

- Road

The road sector is a major potential market for Global Navigation Satellite System (GNSS) applications with satellite navigation receivers now commonly installed in cars as a key tool providing new services. The main contribution of satellites to transport security through assisted GNSS lies in improved traceability which, inter alia, helps to secure the transport of dangerous goods and improves transport and traffic monitoring as a basis for improved search and rescue and emergency calls. Assisted GNSS is also more immune to jamming and spoofing.[97]

Additionally, radar or imaging techniques such as SAR and along-track SAR interferometry approaches have been proposed for traffic monitoring using SRTM/ X-SAR ATI data. With satellite VHR optical systems having two different sensors on board, multispectral (MS) and panchromatic (PAN) respectively, information is not captured instantaneously but there is a given time lag between the two data collections. By knowing the exact time lag between the two data collections, it is possible to estimate the target's ground velocity with the approximation of the pixel size.[98] This allows for three steps: (1) image-to-image registration between MS and PAN images, (2) precise location of barycentre of targets and (3) estimation of the targets' ground velocity and direction by Image-to-ground transformation. Additionally, satellite navigation contributes to the monitoring of dangerous goods transport. Positioning and tracking applications can be used for fleet management. More reliable traffic management based on satellite navigation applications also contributes to improved road transport security.

- Rail

Currently, the locating of rail vehicles and trains in Europe is done via ground-based solutions, with high equipment and maintenance costs. The use of satellite navigation would reduce costs, allowing cheaper train signalling and traffic management systems whilst simultaneously increasing line capacity and efficiency. Tracking and tracing of trains by satellites would additionally lead to the design of more safety and security related applications such as proximity alarms that trigger reports when two trains on the same track are too close and velocity limit alarms that report when a train is exceeding the allowed speed for that section of the track. These could be useful should terrorists decide to hijack a train and run it into another one or into a crowded

[97] Gagnou, Bruno. "Added Value of Space Solutions for Transportation Security." Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at /images/stories/dokumente/ Conference2009/gagnou.pdf>.

[98] Pesaresi, Martino, Karlheinz Gutjahr and Eldoie Pagot. "Moving Targets Velocity and Direction by Using a Single Optical VHR Satellite Imagery." Presentation. ESA-EUSC 2008: Image Information Mining: Pursuing Automation of Geospatial Intelligence for Environment and Security Conference. ESRIN, Frascati, Italy. 4-6 March 2008. <http://earth.esa.int/rtd/Events/ESA-EUSC_2008/index.html>.

station at high speed. Improved positioning through satellite navigation would allow for fast rescue missions. Such technologies could also be combined with container security plug and play modules for tracking of certain freight transports. Additionally, one could also think of satellite communication for secured internal communication.

Satellite navigation also allows for improved fleet management and goods tracking. Potential deviations from envisaged routes in the case of e.g. hijacking of a train could be quickly identified. It also allows for the monitoring of dangerous goods transport by rail. Furthermore, satellite navigation can also be used to conduct track surveys as a basis for tracking defects in 'active' railways to prevent accidental or intentional derailing of trains. This is achieved through a combination of satellite navigation receivers with inertial navigation systems.[99] Derailing can be caused either by rail defects due to bad maintenance, or intentionally by for example terrorists.

- Urban Public Transportation

Existing applications in the field of public transportation are mainly concerned with fleet management and tracking. Emerging technologies seem rather to focus on permanent monitoring of tracks.

Waterborne Transportation

Satellite observation fits particularly well with the geographic and thematic diversity of maritime activities requiring monitoring and surveillance. The global characteristic of monitoring from space makes space systems particularly attractive for long-term monitoring of a very large geographic area. Satellite observation systems are multipurpose and non intrusive. They can provide additional information that complements other existing coastal vessels monitoring systems (VTS, AIS, etc).[100] These systems work in parallel and provide a unique capability for periodic ocean monitoring.[101]

Satellites can enhance maritime surveillance in order to counter and prevent organised crime like piracy and illegal transport. By extending the surveillance range in comparison to coastal radars, satellite based applications offer a better anticipation of threats. They cover larger zones in one shot and thus allow for an optimisation of the use of resources. Satellite data is not limited to borders but is non-intrusive and continuously available. Three space-based applications enhancing the maritime situation picture can be identified: imagery (optical and SAR in different resolutions); collection of signals both cooperative (e.g. AIS, LRIT) and non-cooperative (e.g. SIGINT); and satellite communication. Through continuous observation, space applications mainly contribute to the observe-and-detect function and allow for tracking and, to only a limited degree, for classification. They are thus used complementary to ground, sea and air assets in order to pinpoint potential threats and abnormal behaviour. Satellite-based applications mainly show "where to look" as a basis for users to activate other systems to confirm or identify threats.[102]

- Space-Based Imagery

Although very-high resolution optical images reveal a lot of information on ships, detection and classification from optical images is much less developed than from radar images. This can be explained by the relative novelty of very-high resolution optical satellite sensors, their limitation to cloud-free conditions, higher costs per km² and the fact that the swath of very-high resolution imagery is relatively small limiting its suitability for ocean surveillance. Ships can be easily detected with the human eye on optical images of a minimum of 2.5 metre resolutions. Ship sizes can be estimated and larger vessel types, e..g. container ship, oil tankers and bulk carriers can be recognised. Both intermediate vessels and small ones (<10 metres) can be detected. Their identification is however more difficult or only possible through guessing based on context information such as, for example, the nearness of a marina indicates that a small target is likely to be a pleasure boat. Guessing is however a risky method when the purpose is to detect threats or illegal activities.

[99] Feuerbacher, Berndt and Heinz, Stower. Eds. Utilization of Space Today and Tomorrow. Berlin/Heidelberg: Springer-Verlag, 2006, 259.

[100] This section is to a large degree based on the findings of the DECLIMS project as presented in: Greidanus, Harm & Naouma Kourti. "Findings of the DECLIMS Project: Detection and Classification of Marine Traffic from Space." Conference Paper. SESAR 2006: Advances in SAR oceanography from ENVISAT and ERS, ESA-ESRIN, Italy. 23-26 Jan 2006. <http://earth.esa.int/workshops /seasar2006/participants/87/paper_SeasarDECLIMS12p3.pdf>. "Results of DECLIMS." Joint Research Centre 24 April 2009 <https://declims.jrc.ec.europa.eu/web/declims/6>.

[101] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.

[102] This section is to a large degree based on the insights provided by the following presentations: Bressollette, Aurélie. "Space Current and Future Contributions to Maritime Security." Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at/images/stories/dokumente /Conference2009/bressollette.pdf>. Gagnou, Bruno. op.cit.

Currently, there are only three partners involved in optical ship detection world wide compared to fifteen in radar. A number of service providers and suppliers offer automatic systems for ship detection in satellite SAR images based on images of RADARSAT, ENVISAT ASAR and ERS-2. For ship detection, resolutions used range from 150 metres (ENVISAT ASAR Wide swath, only for large merchant ships) via 50 metres (RADARSAT ScanSAR) and 25 metres (ENVISAT ASAR Image and Alternate polarisation modes, RADARSAT Standard) to 8 metres (RADARSAT Fine). Several benchmarking tests have been conducted in the past on SAR ship detection with RADARSAT but only few have been done using ENVISAT ASAR. In this regard, the DECLIMS (cf. chapter 4) benchmarking test showed a >85 percent detection rate but tests sets were to small for good statistics. JRC benchmarking tests on delectability of fishing vessels, analysing almost 900 known fishing ship positions in 100 images, indicate a delectability of 80 percent for larger fishing vessels (45 metres on average) in RADARSAT ScanSAR Narrow B images (50 metres resolution) and over 90 percent for somewhat smaller fishing vessels (35 metres on average) in RADARSAT Standard images (25 m resolution) Available experience however indicates comparable performance of ENVISAT ASAR's image mode to RADARSAT's standard mode. ENVISAT's wide swath mode is not suitable for detection of vessels (15 metres resolution). While most systems do provide a length estimate; some provide a width and heading estimate; but only a few provide a speed estimate or target RCS (Radar Cross Section), SAR imagery currently does not allow for vessel type identification. Length estimation has an accuracy of about 15 metres (standard error) under calm sea conditions for slow moving targets with high resolution.

Moving ships show wakes in both radar and optical satellite imagery, which can provide information about speed and direction of the ship.

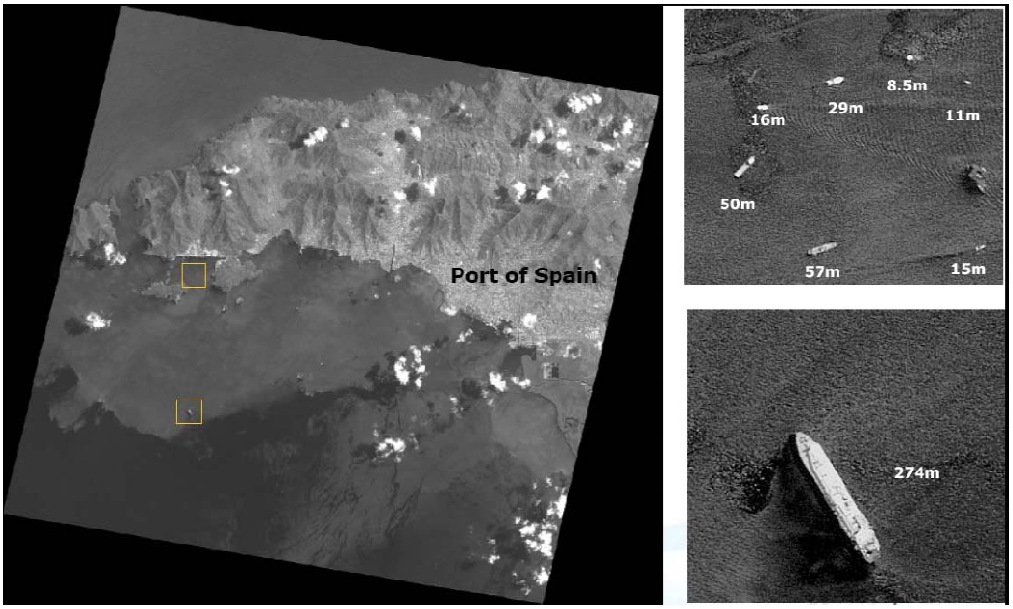


Figure 14: Example of optical imagery capability[103]

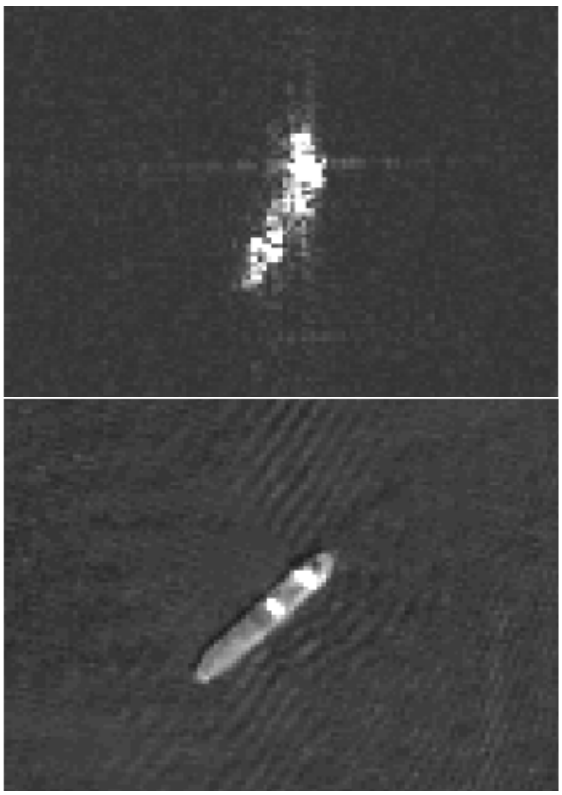


Figure 15: The same 220 m long ship imaged by SPOT-5 (2.5 m resolution, top) and RADARSAT Fine (8 m resolution, bottom)[104]

Wake detection in optical imagery proves difficult as the moving vessel and its wake can be difficult to separate due to similar brightness. Radar imagery too is under-

[103] Ibid.
[104] The DECLIMS Partners. "Detection and Classification of Maritime Traffic from Space". The DECLIMS Partners, 2007. <http://ec.europa.eu/research/press/2007/maritime-briefing/pdf/31-jrc-declims-brochure_en.pdf>.

developed in wake detection. This can be explained by the reliance on RADARSAT (HH) by most operational ship detection applications, where wakes are not prominent.[105]

For counter-terrorism, i.e. vessels intending to attack targets in the EU, satellite imagery is only partially of added value. Assuming that terrorists would use medium to large vessels, these can be detected in SAR imagery but would not be identified as dangerous or suspicious without identification as such from other sources, like intelligence data. As explained earlier, SAR imagery proves particularly valuable at open sea, far away from the coast. Vessel classification from SAR data, however, is still difficult: automatic algorithms do not yet always provide reliable size estimates as explained earlier. In several aspects the implemented, operational capabilities lag behind available theoretical knowledge: e.g. in wake detection and analysis and for automatic vessel detection and classification in optical images.[106]

In the past, terrorists have used large vessels with small, explosive-packed boats, so-called Improvised Explosive Devices (IED), to target other vessels. The challenge is to detect small boats which may be on the water for a very short time in a crowded area, and to recognise their hostile intent. Constituting still a huge challenge for local, high resolution, continuously operating sensors, such a scenario is essentially out of reach for support by imaging satellites. Piracy is similar to this IED scenario in that the threat is formed by small boats that can suddenly appear from a nearby coast. As attacks can be expected further from shore in order to avoid any risk or reaction by local authorities or population, there is more pre-warning time. Thus, it would make sense to look for small boats leaving hot-spot coasts, heading to an intercept course with passing vessels. In order to capture such an event, much higher revisit rates are needed.

Current capabilities in maritime surveillance are based on Earth Observation satellites that have already been operating for some time and whose primary mission is not maritime surveillance. For this reason, they often cannot achieve “best delivery times” as revisiting is not sufficient. In a wide area, ship detection remains limited (< 8-10 metres (optical), < 30 metres (SAR)). As indicated earlier, satellite applications for maritime surveillance cannot be used on their own but should rather be understood as complementary to other data sources. This is a result of the rather limited information on vessel traffic provided by satellite images: Data is available not continuously but only at one instant; within a limited swath; with limited resolution; and with limited detection probability and classification potential. In order to be able to identify a potentially dangerous unknown ship from satellite SAR imagery, in near-real time, with detection and false alarm rates that are sufficient for some users and scenarios but need to be improved for others, resulting detections have to be fused with vessel traffic data from other sources (e.g. AIS, VMS, coastal radar, airborne patrol etc.). This is increasingly being done.

Given that radar (at medium resolution) is most suitable for ship detection but does not allow for ship classification, which in turn has to be derived from (very) high resolution optical imagery, a combination of radar (for day and night all-weather imagery) and optical (for large swath and high resolution) is needed for maritime surveillance. Requirements for future sensors differ according to the application. Most scenarios require the combination of a wide swath (several 100 km) with the capacity to detect also relatively small boats (tens of metres). In some cases even smaller vessels need to be detected.[107] Apart from detection problems, challenges like the extension of the surveillance range and non-intrusive and continuous availability persist. Limits set on the volume of operations based on space assets are quickly reached under the current regime in Europe. The availability of space systems and their capacity to be replaced or augmented for operational purposes remains weak and hardly reactive. Thus, there is a need for a dedicated maritime surveillance system that is better adapted to maritime surveillance in terms of surveillance, detection, classification and tracking. Given the need for a comprehensive maritime picture, which can only be derived on the basis of a variety of data sources like intelligence, conventional monitoring systems (coastal radars, AIS, VMS), aircraft patrolling

[105] Ship wakes are a rarity in these RADARSAT images, whereas in ERS images (VV, steep incidence) they are more the rule. It must be concluded that operational capability at present lags behind theoretical capability in this area.

[106] For a more detailed analysis on technicalities related to false alarm rates and other common problems refer to: Greidanus, Harm & Naouma Kourti. op. cit.

[107] A discussion on different scenarios and their requirements can be found in the related DECLIMS document: Greidanus, Harm. “Requirements for Future Systems.”. D8 of 26 Apr. 2007. DECLIMS. <https://declims.jrc.ec.europa.eu/c/document_library/get_file?p_l_id=1782&folderId=2482&name=DLFE-188.pdf>.

and EO satellites (both optical and radar), there is a need for an integrated approach. Better shared resources between the Member States and the EU can be envisaged[108] to improve the situation. Furthermore, outreach activities would help to counter the lack of knowledge, misconceptions and the "community gap" which make space tools not fully accepted by users.

- Satellite Navigation

While being used in every phase of marine navigation: ocean, coastal, port approach and port manoeuvres, under all weather conditions, satellite navigation advances transport safety rather than security.[109] It can also be used for the tracking of containers.

- Space-Based Collection of Signals

By picking up navigation signals, ship tracking and positioning becomes possible.

The International Maritime Organization's (IMO) International Convention for the Safety of Life at Sea (SOLAS) requires Automatic Identification Systems (AIS) to be fitted aboard internal ships of three-hundred or more tonnes as well as all passenger ships regardless of their size. An Automatic Identification System (AIS) provides the means for ships to electronically exchange ship data like identification, position, course and speed with nearby ships and Vessel Traffic Services (VTS) and is used by ships and VTS for identification and location of vessels. By integrating a standardised VHF transceiver system with an electronic navigation system like a GPS receiver it enables tracking and monitoring of vessel movements. The AIS is capable of handling well over four-thousand-five-hundred reports per minute and updates every two seconds.[110]

There are currently two types of shipboard AIS available. Class A AIS using Self Organizing Time Division Multiple Access (SOTDMA) data communication technology (for larger ships in line with SOLAS) is mandatory for all ships of three-hundred gross tonnage and all ships engaging on international voyages, cargo ships of five-hundred gross tonnage and upwards not engaged on international voyages, some search and rescue aircraft and passenger ships irrespective of size.[111] Class B AIS, using Carrier Sense Time Division Multiple Access (CSTDMA) technology is a voluntary AIS service for pleasure vessels and small craft.[112] With the system becoming more popular, AIS receivers and display systems are becoming a common tool for local-area awareness on board ships. Current AIS reception is limited to coastal antennas' range of 50 to 100 km.

While the signal was originally picked up by coastal receivers, the idea is to pick it up from space (AIS-S). This way the application range of AIS would be extended. Trials with the ESSAIM constellation have already proven this possibility. Exploiting AIS data collected from satellites can make worldwide maritime traffic surveillance a reality. For commercial operators it provides the advantage of making tracking of own ships easier, leading to potential for comparative advantage. The benefits of a space based AIS system are manifold, offering worldwide coverage (including the strategic maritime polar areas), high Quality of Service (rich information provided by AIS message) and reliable positioning information. It improves internal security mainly through its application to cooperative and semi-cooperative ships, allowing for more accurate vessel traffic management which is needed for transport security but would also support search and rescue operations.
While AIS is an existing anti-collision maritime system with 30 nautical metres terrestrial transponders which is deployed to more than 70 000 ships, space-based AIS is still not a worldwide reality. It still faces problems in particularly crowded areas. Orbcomm (6 satellites) and COMDEV (1 test satellite) are the first two operators of AIS-equipped satellites. Trials are ongoing. Problems with the existing AIS system and its signals being picked up by space evolve mainly due to the systems' optimisation for terrestrial use. With space-based receivers being capable of receiving messages from multiple cells the problem of message collisions arises. In areas with low ship density, the probability of message collisions is low and AIS provides reasonable performance. Long collection times can help improve the overall detection statistics and the chances of a clear message being received. Alternatively the antenna footprint

[108] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.
[109] Feuerbacher, Berndt and Heinz, Stower. op. cit. 259.
[110] U.S. Departement of Homeland Security. "What is the Automatic Identification System (AIS)?" 27 July 2009 <http://www.navcen.uscg.gov/enav/ais/default.htm>.
[111] Cain, J.S. and E. Meger. "Space-Based AIS: Contributing to Global Safety and Security." Presentation. ISU Annual Symposium. International Space University, Strasbourg, France. 18-20 Feb. 2008.
[112] For further information related to technology refer to U.S. Department of Homeland Security. "How AIS works." <http://www.navcen.uscg.gov/enav/ais/how_ais_works.htm>.

can be reduced in order to increase the number of ship detections through reducing the cells in view at any point in time. This however increases the time to provide global coverage while reducing the amount of time that any single cell is under observation. Both the Canadian COMDEV Ltd. and ThalesAlenia Space are currently working on a solution to that.

Both AIS-S and terrestrial systems are needed for complete maritime domain awareness for two reasons. First, terrestrial AIS systems have the benefit of continuous coverage and detection rates that approach one hundred percent close to shore but a very limited range and high cost per square mile covered.Secondly, AIS from space has the advantage of providing complete global coverage with comparable average detection performance as well as low cost per square mile covered however faces the disadvantages of lower detection rates close to shore stations and only periodic vessel refresh..

COM DEV Ltd. is using an AIS radio to address the collision problem in order to achieve a much higher level of ship detection per spacecraft pass. The result is a large improvement in the number of detections versus the commercial receiver approach. ThalesAlenia Space (TAS) proposed instead to dedicate specific AIS channels to a space-based link and to optimise the transmission protocol (periodicity, message, synchronisation). Apart from the collision problem, AIS-S requires several other regulatory arrangements: once AIS-S data is captured by a satellite, cost-effective transport of the data down to the earth and then to a central repository must be guaranteed. The data must be processed, archived, routed, tagged, quality checked, filtered and then distributed to authorised users. A comprehensive approach to AIS-S has to be adopted.[114]

AIS is mainly used for collision avoidance but also provides traffic awareness for Vessel Traffic Services (VTS). VTS is a marine traffic monitoring system established by harbour or port authorities similar to air traffic control. Radar, closed-circuit television, and VHF radiotelephony complement AIS to monitor vessel movements and provide navigational

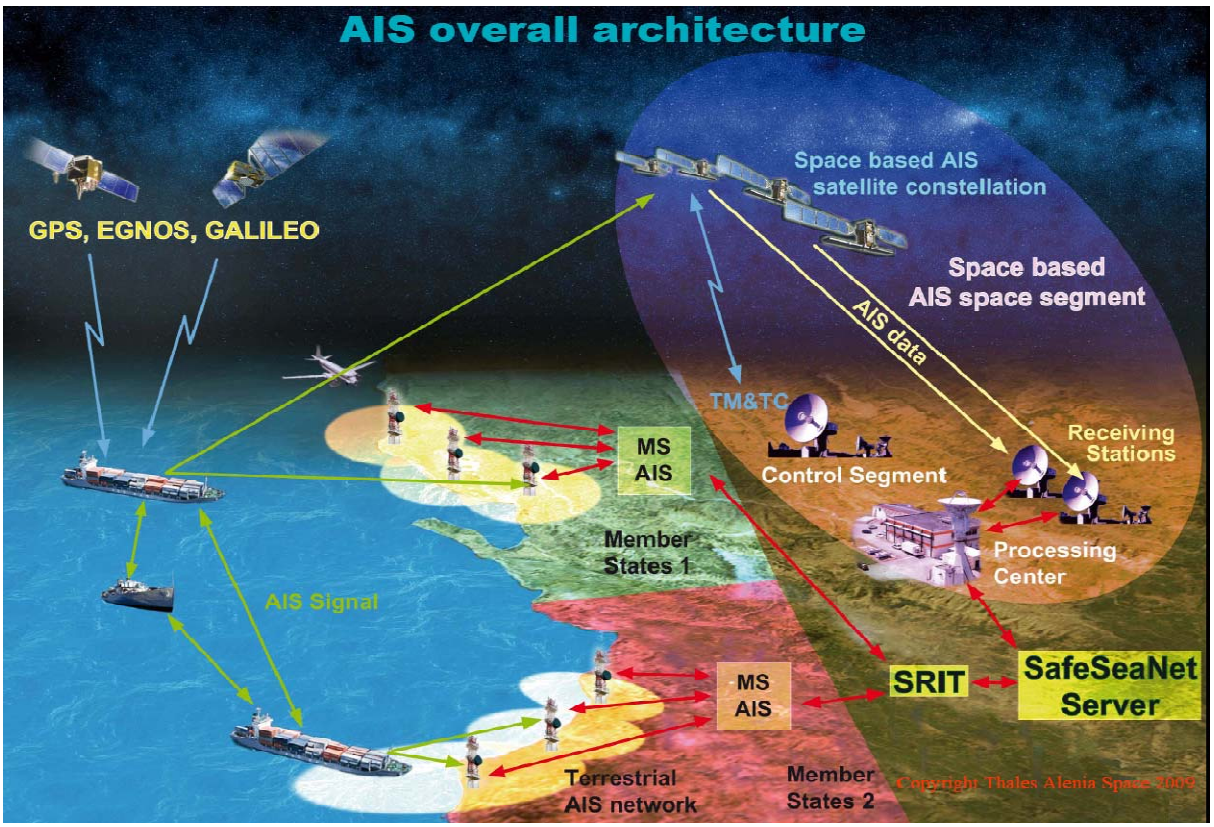


Figure 16: The AIS overall architecture[113]

[113] Gagnou, Bruno. op. cit.

[114] This section is mainly based on Cain, J.S. and E. Meger. Cain, J.S. and E. Meger. op. cit.

safety in a limited geographical area. In addition to providing a comprehensive traffic image, a VTS should have information about all participating vessels and their intentions in order for developing situations to be evaluated and respondedto. The Vessel Monitoring Service (VMS) which is primarily used for fisheries monitoring is also able to detect cooperative sea traffic.

Additionally, AIS is used for coordination resources for marine search and rescue operations, which require exact position and navigation status of ships in the vicinity of the ship or person in distress. The AIS standard envisions the possible use of SAR aircraft, including messages for aircrafts to report positions. To aid SAR aircrafts in locating people in distress, a standard for an AIS Search and Rescue Transmitter (AIS-SART) is currently being developed. Several demonstrations have proved its functions.

In addition to AIS, the IMO adopted a resolution in May 2006 requiring the establishment of an international system for the Long Range Identification and Tracking of ships (LRIT), amending chapter V of SOLAS. The so-called LRIT regulation will apply to all passenger ships including high speed craft, cargo ships including high speed craft of three-hundred gross tonnage and higher and mobile offshore drilling units. These ships must automatically report their position to their flag administration at least four times a day. Contracting governments having a legitimate interest under the regulation may additionally request information about vessels. LRIT is a global standard for long range reporting accepted by nearly all governments. The EU's ambition for "LRIT" of ships heading towards Europe requires long-range detection over large maritime zones, which would not be feasible by means other than satellites.[115] Providing additional information (e.g. course and speed), AIS-S is understood as supporting and complementing LRIT. LRIT is only sent to specific recipients for confidential treatment.[116] AIS-S is considered as providing an independent surveillance capability that operates regardless of the status of the LRIT deployment in a particular country. Thus both systems seem to be needed in providing mutually supporting value to the market.

Moreover several reporting regimes like port notifications, hazardous material reporting (HAZMAT) and the West European Tanker Reporting System (WETREP) can be identified.A further distinction can be made between multinational (WETREP), single state (Italian ARES (Automated Search and Rescue System) or bilateral reporting systems sometimes covering international straits (e.g. CALDOVREP for the Dover Strait, GOFREP for the Gulf of Finland). Different reporting systems can be mutually exclusive [117]

Air Tansportation

Generally speaking, air transportation and particularly aviation security is the sector that probably received most attention from the tragic 9/11 event. However, the use of satellite applications for aviation security is not easily accessible through publicly available sources.

- Space-Based Imagery

Satellite Earth Observation (both optical and radar) combined with change detection techniques can help to detect malicious behaviour around airports. [118]

- Satellite Navigation

In aeronautical applications, satellite positioning and timing services have long been an additional means of navigation, providing supplementary services for many flight phases in both private as well as commercial air transportation. Post 9/11, particularly in the U.S., research has been conducted on technological solutions to improve aviation security. In this regard, satellite systems offer the possibility of audio-video monitoring equipment, which broadcasts cockpit communications based on a "silent alarm" and transmits this to the ground. The "silent alarm" could be triggered by authorised aircraft personnel, an air marshal or a ground controller, or by anomalous aircraft performances. The system could be further extended so as to include back-up transponders with aircraft identification, altitude, speed and location information being transmitted to the ground. The very existence of such a system might prevent terrorist incidents and goes further than previously existing flight control systems which could only help explain a crash after it was too late to stop it.

[115] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.
[116] European Commission/ Joint Research Centre, Ispra. Integrated Maritime Policy for the EU: Working Document III - On Maritime Surveillance Systems. 14 June 2008. Ispra. <http://ec.europa.eu/maritimeaffairs/pdf/maritime_policy_action/maritime-surveillance_en.pdf>.
[117] European Commission/ Joint Research Centre, Ispra. op. cit.
[118] Schreier, Gunter. and Peter Reinartz. "Supporting Civil Security with Earth Observation in GMES: DLR Technology and Projects." Presentation. ISU Annual Symposium. International Space University, Strasbourg, France. 18-20 Feb. 2008.

Air traffic control relies mainly on satellite navigation. In combination with the above mentioned applications, it can help to track aircraft in emergency. The accuracy and integrity of specific navigation systems will help to optimise the use of existing airports.[119]

- Satellite Communication

Air transportation also depends on safe and secure air-to-air, air-to ground and ground-to-air communications. Much of the secure communication used by airborne systems depends on the use of military and commercial satellite communication: satellites can provide encrypted voice, data, teletype or fax communications and are interoperable between the services. However, commercial aircraft communication mostly relies on commercial satellite communication, which is currently not protected against jamming or spoofing or other forms of satellite interference.[120]

3.2.3. Border Security

To maintain secure borders, around-the-clock surveillance is required. Monitoring what flows across borders and approaches a country's shores requires constant vigilance in good weather and bad.

Land Borders

EO together with change detection techniques is the main service to be used for land border monitoring and land border management.121 Satellite imagery technology can map land borders and has the potential to detect illicit behaviour such as a tunnel under the border by detecting earth disturbances or unusual vegetation patterns.[122] SAR sensors make this service functional in all weather conditions, day and night.

Earth Observation can also be used to detect the location of movements of individuals, vehicles and / or biological and chemical substances crossing unregulated land borders and, when required, track and trace their movements thereafter. Detection, identification and authentication are conducted through the use of EO imagery combined with hyper-spectral / multi-spectral processing, SAR/ ISAR equipment and non-co-operative target recognition. Satellite surveillance and in situ data fused with data from intelligence provides for a complete system that monitors immigration flows.[123]

Based on such monitoring, immigration routes can be identified and changes in crossings in different seasons can be identified. This way forecasting as a basis for decision-making and planning of operational procedures for intervention becomes possible. Given that EO has only been applied to civil work since 1979 and high resolution imagery has existed only since 2000, technological development is not advanced enough for immediate alert and detection. EO imagery allows for planning but does not allow for on-the-spot detection of illegal migrants and immediate action.

Satellite navigation can also be used for tracking (stolen) vehicles or people. An Israeli SME developed an integrated hardware and software platform called Stolen Vehicle Recovery (SVR) system. It includes location, built-in security systems, real-time alarms, web-based vehicle tracking and direct contact with security agents.[124]

Sea borders

There is growing interest in the use of satellite SAR imagery for maritime border control. Recently completed studies coordinated by FRONTEX highlighted the potential role of satellite based surveillance and airborne patrols for maritime areas outside the coverage of coastal surveillance systems (e.g. open waters and territorial waters of third party states). In particular, the Integrated Surveillance System for Europe's southern maritime borders as requested by the European Council intends to

[119] "Satellite Navigation Applications." 19 Jan 2007. Summaries of EU Legislation 29 July 2009 <http://europa.eu/legislation_summaries/transport/intelligent_transport_navigation_by_satellite/l24463_en.htm>.
[120] For protective measures for satellites refer to: Remuss, Nina-Louisa. "The Need to Counter Space Terrorism – A European Perspective." ESPI Perspective 17. Vienna: European Space Policy Institute.
[121] EOMag. "WALPHOT uses TerraSAR-X data for land border monitoring." 19 June 2009. European Association of Remote Sensing Companies (EARSC) 27 July 2009 <http://www.eomag.eu/articles/813/walphot-uses-terrasar-x-data-for-land-border-monitoring>.
[122] "Commercial Satellite Imaging for Homeland Security: Emerging Market." Homeland Security Research Corporation 24 Apr. 2009 <http://www.homelandsecurityresearch.net/ 2008/04/03/commercial-satellite-imaging-for-homeland-security-emerging-market>.
[123] European Security Research and Advisory Board. op. cit.
[124] "S3 PAMS Profile." Safety and Security Systems in Europe 27 July 2009 <http://www.safety-and-security.de/technologies-from-the-enterprise-europe-network/07%20IL%20ILMA%200IOY>.

integrate satellite based surveillance with conventional vessel tracking systems.[125]

By picking up navigation signals, ship tracking and positioning becomes possible. AIS-S allows for worldwide maritime traffic surveillance to become a reality. It improves internal security through its application to cooperative and semi-cooperative ships, allowing for more accurate vessel traffic management needed for border protection Non-cooperative signals are picked up by SIGINT satellites, shedding light on the "bad guys".

Optical images in combination with change detection can show groups of people accumulating at the coast, giving an indication of the possibility of a vessel with illegal migrants shortly leaving the coast (Figure 17). Satellite capabilities then depend on the destination (the distance) and the size of the vessel. Relatively large ships can be detected via SAR imagery based on previous suspicious indications. Without alert by, inter alia, intelligence, recognition of the vessel as carrying illegal immigrants would be impossible. As explained earlier, SAR imagery proves particularly valuable at open sea, far away from the coast. Illegal migration with small vessels is more challenging for satellite imaging. Small vessels are far more difficult to track and difficult to re-acquire in any subsequent satellite image. The largest pressure of illegal overseas immigration is across narrow straits (such as the Gibraltar Strait, the Greek-Turkish waters, West Africa-Canaries at the point of closest approach, and Italy-Albania). Given the very short distances, crossings can be made by very small ships (down to 6 metres) in a very short amount of time. Thus reaction time needs to be immediate. This is out of reach for any satellite imaging system expected in the near future. EO satellites can however help to detect and observe ship building factories inland or detect any abnormal concentration of ships.[126]

Often illegal goods like drugs, small arms and light weapons (SALW) and weapons of mass destruction (WMD) or stolen cars are hidden behind innocent cargo, so that the ship seems to be a legitimate vessel. Thus, as in the case of illegal migration, smuggling can only be countered when ships are indicated as suspect based on intelligence reports or upon detection. Large vessels can then be detected and tracked via SAR imagery, medium size vessels are difficult to re-acquire and small vessels are impossible to detect. Large vessels used for smuggling goods sometimes anchor off the coast to transfer the load to other vessels. Satellite imagery can detect such a vessel and identify it as suspect if it is found to stay at an offshore location without reason and if smaller vessels are found moving between it and the coast (Figure 19). In order to detect the smaller vessels real-time availability is needed. A high value of small-volume goods like narcotics is usually carried by very small and fast boats. The high vessel speed, requiring fast reaction time, makes boats too difficult to detect by satellite imagery. Given the infrequent update rate, crossings are most likely missed.



Figure 17: Coastal Monitoring showing illegal migrants gathering for departure[127]

[125] This section draws to a large degree on the insights provided by Gagnou, Bruno. op. cit.; Bressollette, Aurélie. op. cit.; Greidanus, Harm. op. cit.

[126] Bressollette, Aurélie. op. cit.

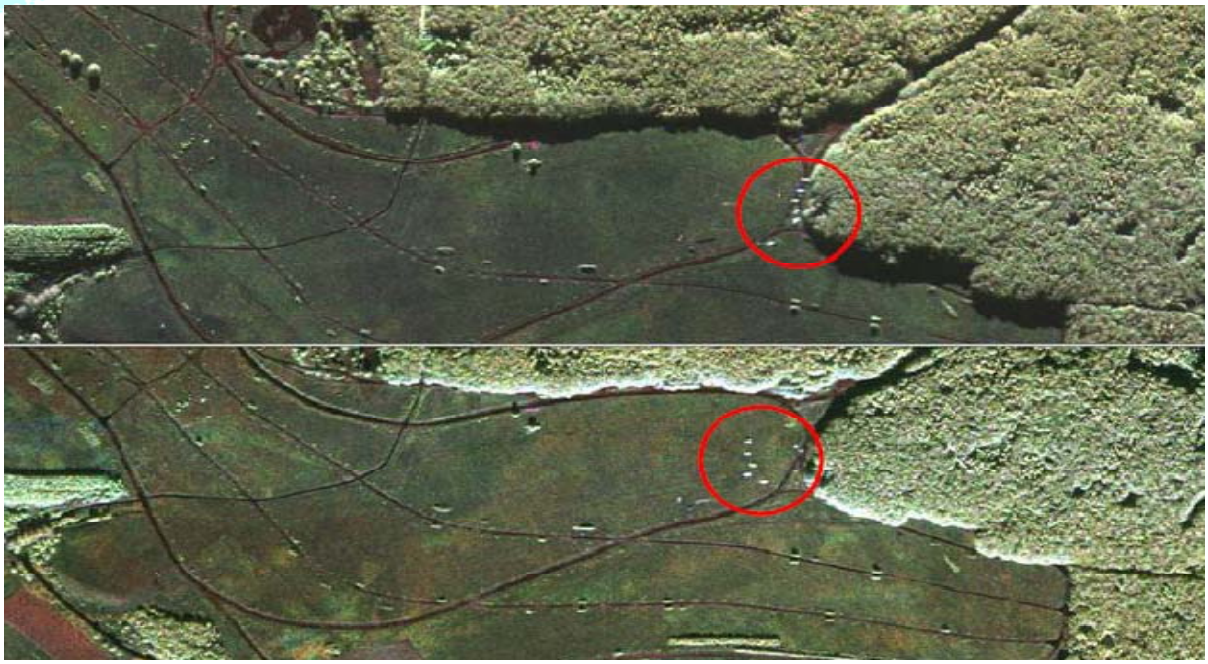


Figure 18: Terrasar-X image showing vehicle movement[128]

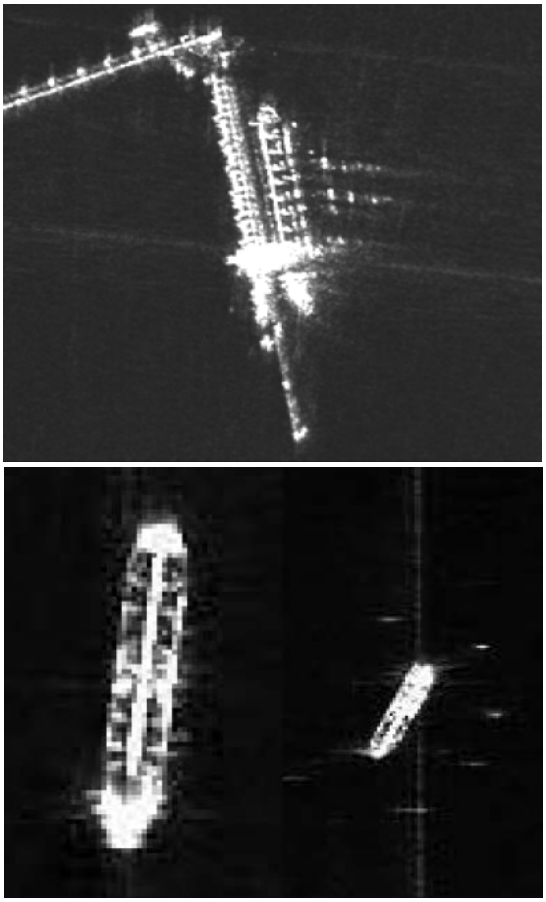


Figure 19 and Figure 20: X-Band SAR showing vessel structure and enabling identification, high resolution X-band enabling detection of small vessels close to larger vessel for monitoring of illegal transfer activities[129]

[127] Ibid.
[128] Beer, Thomas. "The European Space Agency: A new Actor in Security and Defence." Presentation. NATO Space Workshop. Kalkar, Germany. 22 Apr. 2008.
[129] Ibid.

3.3. Preparedness for Emergency Management

Disaster and emergency management can be improved in all crisis phases by involving space applications. Contingency planning depends on satellite based data. Risk mapping shows particularly attractive targets. Geographic information systems combined with a wide range of attribute data, such as data on population density, age distribution, health, infrastructure, land use, etc. are extremely useful to planners and decision-makers for both prevention and mitigation of a major disaster. Disaster management planning and particularly demonstrations benefit from the use of satellite imagery based maps. Standardised mapping - an essential element for planning crisis management – is not readily available in up-to-date digital form for many of the EU's areas of interest. As the updating of traditional hardcopy maps is very costly and requires highly trained cartographers, photo-maps based on commercial satellite imagery provide a cost-effective and relatively fast alternative.[130]

Space applications contribute in a crucial way to the management of crises through satellite images and navigation, providing information to assess the situation, indicate risks, damages and needs. Satellite communication

[130] European Space Agency/European Commission. op. cit. 7-8.

can be used for supporting response forces in the field particularly when terrestrial systems have been attacked as well.

One can differentiate between disasters at sea and land. While the former require techniques as described in the section on waterborne transport, the latter are mostly based on EO derived rapid mapping.

Reaction time for disasters at sea is even smaller than in other cases. In coastal and major inland waterways, coast guards and other maritime safety agencies are called upon, dispatching rescue forces, conducting search and rescue missions and providing initial response to national security and environmental threats. In addition to these, as explained earlier, an AIS Search and Rescue Transmitter (AIS-SART) is currently being developed to aid SAR aircrafts in locating people in distress. There are also several emergency reporting systems relying on SAR information. Additionally, the integration of satellite navigation receivers in cell phones supports location-based services and emergency calls.[131]

The International Charter on Space and Major Disasters (hereinafter referred to as The Charter), initiated by ESA and CNES together with the Canadian Space Agency (CSA) following the UNISPACE III conference of July 1999 (operational since 1 November 2000),[132] provides for the requesting party to obtain easy and free of charge access to satellite data (Earth Observation resources), with top priority in satellite tasking. Representing a joint effort by global space agencies to put resources at the service of rescue authorities responding to major natural or man-made disasters, the civil protection authorities of the affected country (or the EU Civil Protection Unit) can invoke The Charter in case of major crises. Triggering The Charteris however limited to crises linked to natural or technological disasters.[133] Thus, in addition to technical limitations like insufficient revisit times and the limited coverage and a lack of services for data interpretation, The Charter cannot be triggered in the aftermath of a terrorist attack. Providing a good starting point, The Charter needs to be extended to include other forms of disasters needing satellite imagery for emergency management. In the long run, The Charter could be replaced by a more user-driven system using a GMES infrastructure.[134]

Rapid Mapping Products are based on radar archive images and radar event images. After a process of radiometric and geometric corrections as well as mosaicking and orthorectification, change detection is possible. After merging of optical and radar images a rapid mapping product can be put together.[135]

Rapid mapping based on EO satellite imagery as provided for by The Charter can be used for a variety of different services in the aftermath of a crisis: Based on up-to-date maps, situation and impact assessment becomes easier. Satellite-based maps may also provide the geographic context, showing critical infrastructures and population density in related areas. Further damage may be limited based on this information.

Users have identified rapid mapping as “a valuable tool concerning civil protection issues, complementing other types of information (field, aerial)”.[136] They have also indicated that the overall time for the information delivery has to be improved. They require the first rapid mapping product within a maximum 24 hours of their request followed by one every six hours thereafter. Future EO systems for emergency situations should thus be characterised by an improved revisit frequency (twice a day / every three hours) and an improved spatial resolution (sub-metric).[137] While space-based information is already used for disaster management, it has an even greater potential, which should be further exploited. Many applications have been developed for very particular users only, but can in fact be of great value to an even larger user group.

3.4. Comparative Matrix

The following comparative matrix shows the technical applicability of space applications for the provision of internal security. It also

[131] Lockheed Martin. “Maritime Safety, Security & Surveillance: Integrated Systems for Monitoring Ports, Waterways and Coastlines.” New York: Lockeed Martin, 2009. <http://www.lockheedmartin.com/data/assets/7457.pdf>.
[132] “Charter activation brings space dimension to European Emergency Exercise.” 12 Apr 2005. European Space Agency 9 Feb 2009 <www.esa.int/esaEO/SEM041W797E_environment_2.html>
[133] International Charter Space and Major Disasters. Rev.3 (25/4/2000) of 25 Apr. 2000. <http://www.disasters charter.org/charter_e.html>.
[134] The GMES Working Group on Security. op. cit.
[135] Tholey, Nadine. “Rappid Mapping: A GMES Service for Emergency Management.” Presentation. ISU Annual Symposium. International Space University, Strasbourg, France. 18-20 Feb. 2008.
[136] Ibid.
[137] Ibid.

indicates their limitations. Grey cells indicate the non-applicability of a certain phase for a mission area. The pre-attack phase for emergency management should be understood as preparedness for emergency management.

Event Cycle		Technical Capabilities for Space Applications for Internal Security						
			Critical Mission Areas					
		Intelligence, Early Warning, Data-Sharing (WMD, terrorist facilities)	Critical Infrastructure	Transportation			Border Security	Emergency Management
				Land	Sea	Air		
Pre-Attack	Prediction / Prevention	– SIGINT – COMINT – ELINT – GEOINT – IMINT (optical and radar) – Risk mapping / vulnerability assessment based on EO imagery (population density etc.) – GEOspatial intelligence (GEOINT), geographic information system (GIS) tools, geographic management system (GMS) needed for context as a basis of analysis – Treaty verification and non-proliferation through satellite imagery – Risk mapping	– Risk mapping: mapping of critical sites including assessment of environment, current situation and potential risks – EO imagery provides basis for maps of critical structures – Risk mapping as a basis for database of critical sites	– Assisted GNSS for improved traceability – tracking and tracing of cars, busses, trains based on satellite navigation – Radar and imaging techniques for traffic monitoring – through monitoring of container traffic identification of dangerous routes – monitoring of dangerous goods transport – secure satellite communication for inter-train communication	– continuous maritime surveillance (MarSur) – through monitoring of container traffic identification of dangerous routes – tracking and tracing of ships (depending on vessel size, speed and length of travel route) – AIS-S can make worldwide maritime traffic surveillance a reality – Vessel Traffic Service (VTS): provides comprehensive traffic image on all participating vessels and their intentions	– monitoring of airports as critical infrastructures – through monitoring of container traffic identification of dangerous routes – secure satellite communication for air-to-ground communication – audio-video monitoring broadcasting cockpit communications based on a “silent alarm”	– mapping of land borders based on EO imagery – trade: worldwide availability of all data and documents, electronic creation and signing of loading and customs documents via communications satellites – land border surveillance: satellite imagery provides proof of evidence (shows passage ways in different seasons) as a basis for restructuring teams, resources and technical capabilities. It also shows current weaknesses in the existing set-up of land border surveillance.	– Repository of satellite imagery and map data – Use of satellite imagery based maps for disaster management planning
	Detection	– SIGINT – COMINT – ELINT – GEOINT – IMINT (optical and radar) for change detection – Database of satellite imagery (from long-term monitoring) serves as basis for change detection – Three step approach: (1) change detection; (2) change categorisation and (3) change classification – GEOspatial intelligence (GEOINT), geographic information system (GIS) tools, geographic management system (GMS) needed for context as a basis of analysis – Treaty verification and non-proliferation through satellite imagery	– Monitoring provides for data on normal functioning which can be continuously compared to detect malicious interferences – Three step approach: (1) change detection; (2) change categorisation and (3) change classification	– Assisted GNSS for improved traceability – tracking and tracing of cars, busses, trains based on satellite navigation – Radar and imaging techniques for traffic monitoring – electronic systems inside the container monitoring movements and interference connected with an alarm and a monitoring centre – transponders for tracking and tracing of the container – monitoring of dangerous goods transport – identification of deviations from envisaged routes through satellite navigation (detection of hijacking) – immediate detection of de-railing through continuous monitoring	– electronic systems inside the container monitoring movements and interference connected with an alarm and a monitoring centre – transponders for tracking and tracing of the container – ship detection and identification depending on size and intensity of ship traffic based on optical and radar imagery for detection of illicit events – SAR imagery allows for near-real time identification of a potentially dangerous unknown ship – Detection and false alarm rates are sufficient only for some users and scenarios but need to be improved for others THUS fusion with vessel traffic data from other sources (AIS, VMS, coastal radar, airborne patrol) needed – AIS-S can make worldwide maritime traffic surveillance a reality: allows for application to cooperative and semi-cooperative ships for more accurate vessel traffic management – Vessel Traffic Service (VTS): provides comprehensive traffic image on all participating vessels and their intentions	– Monitoring of airports as critical infrastructures: Three step approach: (1) change detection; (2) change categorisation and (3) change classification – electronic systems inside the container monitoring movements and interference connected with an alarm and a monitoring centre – transponders for tracking and tracing of the container	– satellite imagery can detect illicit behaviour by detecting earth disturbances or unusual vegetation patterns – monitoring allows for detection of and changes in crossings (not real-time)	

Attack	Alert	– identification of threat on the basis of intelligence source	– monitoring centres allow for spread of message based on alert by sensor through satellite communication, positioning based on satellite navigation	– electronic systems inside the container monitoring movements and interference connected with an alarm and a monitoring centre	– electronic systems inside the container monitoring movements and interference connected with an alarm and a monitoring centre	– electronic systems inside the container monitoring movements and interference connected with an alarm and a monitoring centre – audio-video monitoring broadcasting cockpit communications based on a "silent alarm" – back-up transponders with aircraft identification, altitude, speed and location information being transmitted to the ground	– land borders: on-spot detection hardly possible: low-revisit time does allow for planning but not for on-spot detection – sea borders: SIGINT allows for identification of "bad guys" – sea borders: satellite imagery and change detection techniques allow for detection of people accumulating at coast for departure; provides basis for where to look for boats with immigrants	– monitoring centres allow for spread of message based on alert by sensor through satellite communication, positioning based on satellite navigation
	Immediate Response	– intelligence (GEOINT, GIS, GMS) needed as a context for the immediate response	– impact assessment for emergency brigades – Rapid mapping: maps for planning of emergency brigades work	– improved positioning through satellite navigation allows for faster search and rescue missions – impact assessment for emergency brigades – Rapid mapping: maps for planning of emergency brigades work	– AIS Search and Rescue Transmitter (AIS-SART) to aid SAR aircrafts in locating people in distress – improved positioning through satellite navigation allows for faster search and rescue missions	– back-up transponders with aircraft identification, altitude, speed and location information being transmitted to the ground	– land borders: on-spot detection hardly possible: low-revisit time does allow for planning but not for on-spot detection	– at sea: AIS Search and Rescue Transmitter (AIS-SART) to aid SAR aircrafts in locating people in distress – rapid mapping: Situation and damage assessment; provision of geographical context for security provisions and humanitarian aid – satellite imagery based maps as a basis for determination of rescue routes' etc. – space applications contribute in a crucial way to the management of crises through satellite images and navigation, providing information to assess the situation, indicate risks, damages and needs – advantage of continuous working even when terrestrial systems have been attacked as well – *The International Charter*
Post-Attack	Relief / Impact Assessment	– intelligence (GEOINT, GIS, GMS) needed as a basis for impact assessment and reconstruction	– Impact assessment based on satellite imagery – Risk mapping can provide insights for reconstruction	– Impact assessment through EO imagery	– risk mapping can allow for detection of dangerous routes	– improved search and rescue based on improved positioning based on satellite navigation	– land borders: constant monitoring allows for detection of routes and changes in pathways for allocation of staff – sea borders: smugglers or illegal migrants at sea can only be identified upon prior identification by intelligence source: need to know where to look for illicit behaviour	– rapid mapping: situation and damage assessment; provision of geographical context for provision of security and humanitarian aid – Impact assessment based on satellite imagery – Risk mapping can provide insights for reconstruction – *The International Charter*
Limitations		– pure imagery data is of limited use; need for placing in relevant context, analysis and assessment – GEOspatial intelligence (GEOINT), geographic information system (GIS) tools, geographic management system (GMS) needed for context as a basis of analysis – Effective integration and assimilation of data and information from in situ and space-based observation necessary: need for common data standards and quality; EU currently has "islands of data"	– In situ data needed for threat identification and risk mapping – Satellites can themselves be considered critical infrastructures and thus are vulnerable too	– Use of space applications limited to fleet management and improved use of resources rather than security	– Optical imagery is limited to cloud free conditions – Swath of very-high resolution imagery is relatively small limiting its suitability for ocean surveillance – Optical imagery does not allow for identification of intermediate and mall size vessels – SAR imagery does not allow for vessel type identification – Speed and direction based on wake detection is currently practically impossible – While tracking and detection of large and medium size vessels is possible, they currently cannot be identified as suspicious, threatening or conducting any other illicit mission; this needs to be done upon indication by the intelligence – Small vessels cannot be detected due to insufficient revisit time – Challenge is to detect small boats which may be on the water for a very short time in a crowded area and to recognise hostile intent – Data is available only at one instant, within a limited swath, with limited resolution and with limited detection probability and classification potential – Detection and false alarm rates are sufficient only for some users and scenarios but need to be improved for others – AIS-S is still not a world-wide reality – Persistence of message collisions problems in AIS-S	– Current use of space applications for air transportation focussed on air traffic management and safety rather than security	– Smuggling/Illegal Immigration and Terrorism at sea: While tracking and detection of large and medium size vessels is possible, they currently cannot be identified as suspicious, threatening or conducting any other illicit mission; this needs to be done upon indication by the intelligence – Small vessels cannot be detected due to insufficient ?? – EO can be used for planning and allocation of staff for land border security but given the existing technological limitations it does not allow for immediate action – It remains difficult to convince users of the added value of satellite imagery for land border surveillance. Border brigade at Eastern borders showed army culture and traditional structures which were hard to overcome. – Limitation in revisit time results in applicability of satellite applications for planning but not on-spot detection	– The International Charter can currently only be triggered in case of natural or technological disaster – *Technological limitations of EO resources of the International Charter: insufficient revisit time and limited coverage*

Figure 21: Comparative Matrix on the Technical Capabilities of Space Applications for Internal Security

Matrix Key
Technical capabilities
Potential capabilities / capabilities with potential applicability

4. Internal Security and Space - A Comprehensive Status Report

The main objective of this chapter is to provide an overview of current measures related to space and internal security. It thus elucidates the main policy documents and highlights the different actors in this field. It also provides an overview of European projects related to space and internal security, thereby evaluating each single project with regard to the European Union's overall approach to internal security.

The European Union is characterised by open borders as well as freedom of movement and transport. It has a very high population density, a road network of nearly 4.3 million km, a railroad network of nearly 155,000 km and an extended network of oil and gas pipelines. Nearly five million commercial trucks move on the European road network each day and there are over 41,000 landings and take-offs at European airports per day. There are over 1,100 container and passenger ports. In Rotterdam, the largest EU container port, nearly 10,000 containers leave the port every day. These figures demonstrate the enormous challenges the EU faces in developing an internal security profile covering all critical mission areas.[138]

For a long time counter-terrorism was considered part of judicial and domestic policy within the EU and only came to be understood as a cross-policy task in the aftermath of 9/11 and the attacks in Madrid and London. The European approach can be divided into the nation states level and the European Union level, which includes EU agencies. The EU seems to have started to seriously look into the provision of internal security only recently with research programmes relating to this matter being included in the Framework Programmes (FPs) (FP 5 to FP 7). There is neither a European "Department of Homeland Security" nor a "Directorate General (DG) for Homeland or Internal Security" in the EU.[139] The EU, in line with the principle of subsidiarity, seems to aim at complementing security policies already existing at a national level. Thus, its contributions take the form of initiatives enhancing "cross-border cooperation (…), coordination support, information sharing, and collaboration with national and international partners".[140] The EU's approach to internal security is not centralised in a single agency, institution or Directorate General but is rather spread across the EU's three pillars and consequently also across the respective institutions and agencies. In this regard, the Commission's efforts to improve critical infrastructure protection and the Council's introduction of a Counter-Terrorism Coordinator facilitating policy coordination in the fight against terrorism are relevant examples. This will become even more striking when looking in detail at projects in the critical mission areas. The EU's approach is functional and reactive in nature. It has evolved gradually and sporadically as policy initiatives in specific areas have progressively come to form part of a nascent EU approach. Fragmentation among different policies (horizontal), different bodies and actors in the three pillars (institutional) as well as between the EU and its member states (vertical) has resulted in incoherence and the absence of a holistic and integrated approach to the matter. Moreover, it can be seen that the EU focuses on the fight against terrorism as well as emergency management in the field of internal security, thereby putting less effort on the issue of organised crime.

With European States having already developed their own policies in the field of

[138] Gustenau, Gustav E. "The Concept of Homeland Security in the European Union and in Austria: A Challenge for the Austrian EU Presidency." Transforming Homeland Security: U.S. and European Approaches. Ed. Esther Brimmer. Washington D.C.: Center for Transatlantic Relations, 2006. 60. Similar Figures can also be found in the European Commission Directorate-General for Energy and Transport. EU Energy and Transport Figures Statistical Pocket Book 2009. Belgium: European Communities, 2009. <http://ec.europa.eu/energy/publications/statistics/doc/2009_energy_transport_figures.pdf#pagemode=bookmarks>.

[139] Bendiek, Annegret. "EU Strategy on Counter-Terrorism: Steps towards a Coherent Network Policy." SWP Research Paper. Berlin: German Institute for International and Security Affairs, 2006. 5-7.

[140] Some of the classifications in this section draw to a large degree on the related discussion by Gustav, Lindström. op. cit. 116-7.

internal security, the question arises as to how a common European Union approach could provide added value. In this regard, it seems important to clarify institutional competences and decide on guiding principles.

4.1. The Domestic Threat Assessment

Threat perceptions in the U.S. and Europe vary. Generally speaking, Europeans feel less threatened than Americans. A survey on transatlantic trends, conducted by the German Marshal Fund in May and June 2005, supports this argument. Thus, seventy-one percent of Americans think it is likely that they will be personally affected by international terrorism while only fifty-three percent of their European counterparts feel the same way. Europeans, by contrast, feel more likely to be affected by global warming (seventy-three percent versus sixty-four percent of Americans).[141]

Possible Explanations for Diverging Threat Perceptions

This difference can be explained by reference to different national experiences with terrorism which form the context for understanding terrorism today. Generally speaking, a distinction can be made between internal (domestic) and external (international) terrorism. European countries do not all see terrorism in the same way; but they share a broadly similar outlook that is distinctly different from the U.S. vision after 9/11. Differences are rooted in culture, ideology and geopolitics. Americans see terrorism primarily as an external threat, while for European societies it has a large internal dimension.[142] Europe's domestic terrorism as exemplified by the IRA and ETA, differs greatly from the terrorism experienced by the U.S. on 9/11. While U.S. officials were surprised by a catastrophic incident, Europeans have long been taunted by domestic terrorists.

In Europe the threat of terrorism was traditionally perceived as latent, except for countries that have historically experienced domestic terrorism. Since the attacks on Madrid and London, this threat perception has changed. Terrorism in Europe is one of five key threats mentioned by Javier Solana in the European Security Strategy: (1) terrorism, (2) proliferation of weapons of mass destruction, (3) regional conflict, (4) state failure and (5) organised crime. The focus on these five threats can be seen as the basis for the European concept of internal security.[143]

The recent FP 7 FORESEC project report showed that societies in EU Member States feel relatively safe from external threats and do not perceive international security challenges to be among their priority concerns. It also showed that there are important distinctions to be made: When asked whether certain issues such as terrorism and non-proliferation represent a security risk a high percentage of EU citizens will say "yes", while when asked whether they will be directly affected by these challenges the number goes down and when asked about their personal priority concerns these challenges drop to single digits in most cases.[144]

In a Eurobarometer survey of November 2002, a year after 9/11, terrorism was the threat most feared by EU citizens (eighty percent), followed by organised crime that was ranked as the second highest threat (seventy-five percent). This has been underlined by a more recent Eurobarometer survey of June 2008 on the European Union's role in Justice, Freedom and Security policy areas, in which the fight against terrorism (fifty percent) and the fight against organised crime (forty-eight percent) are the priorities respondents cited most frequently.[145] Similar results were found by FORESEC. Accordingly, organized crime remained the second most often cited threat – with a highly elevated sense of concern amongst citizens and governments alike as to the threat posed from organized crime. Transnational terrorist networks were seen as the next most immediate hard security threat.

However, while almost all strategic documents and outlooks list it as a major threat, at a public level the actual sense of

[141] Föhrenbach, Gerd. "Transatlantic Homeland Security and the Challenge of Diverging Risk Perceptions." Transforming Homeland Security: U.S. and European Approaches. Ed. Esther Brimmer. Washington D.C.: Center for Transatlantic Relations, 2006. 46.
[142] Dworkin. Anthony. op. cit.
[143] Föhrenbach, Gerd. op. cit. 46.
[144] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. "FORESEC-Europe's Envolving Security: Drivers, Trends and Scenarios Deliverable D 2.2 Synthesis Report." FP 7 FORESEC, 2008. 8. <FORESEC-Europe's Envolving Security: Drivers, Trends and Scenarios Deliverable D 2.2 Synthesis Report>.
[145] Eurobarometer. "The Role of the European Union in Justice, Freedom and Security Policy Areas." Special 290 (2008) 21. <http://ec.europa.eu/public_opinion/archives/ebs/ebs_266_en.pdf>.

threat is relatively low. This sense is best exemplified by Swedes, who fear terrorism when they are travelling abroad but not at home, adding an additional geographical element to the threat perception.[146] Among the twenty-seven EU Member States one can identify twenty-seven different interests and threat perceptions, addressed by different security structures operated by individual Member States. Individual EU Member States' threat perception and perceived urgency may vary depending on national history, previous experiences with terrorism and organized crime, national demographics, specific national characteristics (i.e. political strengths and weaknesses), the importance of bilateral relations as well as historical legacies (i.e. mistrust), budget constraints, intricacies of each country's national security system, technological development as well as strategic choices.[147]

In this regard, the most stringent "homeland security policy" can be found in the UK, followed by The Netherlands, whose open society was shaken by extremism and terrorism as a result of the murders of Pim Fortuyn, a maverick politician, in 2002 and Theo van Gogh, a film-maker, in 2004. The Scandinavian or Nordic countries of Denmark, Sweden and Norway (not an EU member) as indicated earlier started to understand security in terms of civilian crisis management. Europe faces the challenge of finding a unifying policy both on the EU as well as on the national level. European unity itself has its limitations: In this regard, both the Madrid and London bombings have been perceived as attacks on Spain and the UK respectively. Heightened national risk perception translates into heightened EU risk perception to only a limited degree.[148]

4.2. Existing Documents and Strategies

The priorities of the EU in the fight against terrorism are summarised in the "Plan of Action on Combating Terrorism".[149] In the Laeken European Council of December 2001, EU Heads of State and Government agreed on the introduction of a European Arrest Warrant, Joint Investigation teams and a definition of terrorism to be applied in all EU Member States. With the attacks in Madrid creating a new momentum, EU Member States were able to agree on a "Declaration on Combating terrorism" aimed at strengthening cross border cooperation in fighting terrorism, sharing of intelligence and preventing the financing of terrorism. A Counter-Terrorism Coordinator has been appointed. The European Council also underlined the importance of existing bodies such as Europol and Eurojust. In this regard, the Europol Counter-Terrorist Task Force has been reactivated. In order to strengthen "an area of freedom, security and justice", the EU agreed on the Hague Programme in November 2004. In December 2005 the Justice and Home Affairs Council adopted the European Union Counter-Terrorism Strategy[150] which revolves around four strands of work: prevention, protection, pursuit and response. It commits the EU to "combat terrorism globally while respecting human rights and makes Europe safer, allowing its citizens to live in an area of freedom, security and justice."[151] In addition the European Neighbourhood Policy (ENP) proposed by the Commission in 2003 and endorsed by the Council in 2004 provides for an overall strategy towards neighbouring countries.[152] Further, the European Defence

[146] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. op. cit. 9.
[147] Cardona, Meliton. "The European Response to Terrorism." Western Responses to Terrorism. Eds. Crelinsten Ronald D. and Alex P. Schmid. London: Frank Cass, 1993. 249; Crelinsten Ronald D. and Alex P. Schmid. Western Responses to Terrorism. London: Frank Cass, 1993. 308-9; Linde, Erik J. G. van de, et al. Quick Scan of Post 9/11 National Counter Terrorism Policymaking and Implementation in Selected European Countries. Leiden: RAND Europe, 2002. 4 & 13; Hippel, Karin von. Europe Confronts Terrorism. New York: Palgrave Macmillian, 2005. 9 & 11 & 266; Grabbe, Heather. "Breaking new ground in Internal Security." Europe After September 11th. Eds. Edward Bannerman et al. London: Centre for European Reform, 2001. 67; Adrich, Richard J. and Wyn Rees. "Contending Cultures of Counterterrorism: Transatlantic Divergence or Convergence?" International Affairs 81, 5 (2005): 907.
[148] Föhrenbach, Gerd. op. cit. 49.

[149] For a more detailed summary on the EU's actions in the fight against terrorism cf. European Commission's Specialised Documentation Centre. 27 July 2009 <http://ec.europa.eu/justice_home/fsj/terrorism/strategies/fsj_terrorism_strategies_en.htm>.
[150] Council of the European Union. The European Union Counter-Terrorism Strategy. 14469/4/05 REV 4 of 30 Nov. 2005. Brussels: European Union. <http://register.consilium.europa.eu/pdf/en/05/st14/st14469-re04.en05.pdf>. A summary on the related latest developments can also be found here: European Commission Freedom, Security, Justice. "The European Union's Counter-Terrorism Strategy." 21 Sept. 2005. <http://ec.europa.eu/justice_home/fsj/terrorism/strategies/fsj_terrorism_strategies_counter_en.htm>. The Commission activities in the fight against terrorism are also listed in MEMO/05/332 of 21 Sept. 2005.
[151] Council of the European Union. The European Union Counter-Terrorism Strategy. op. cit. 6.
[152] For the current status and country reports cf. European Commission. European Neighbourhood Policy-Reference Documents. 23 Apr. 2009 European Commission 29 July 2009 <http://ec.europa.eu/world/enp/documents_en.htm>.

Agency has presented strategy documents on the development of a European Defence Technological and Industrial Base (EDTIB) and initial thoughts on a European Defence Research and Technology Strategy.[153]

In addition to the body of documents related to counter terrorism, the EU has also established several strategies related to organised crime and illegal migration as well as human trafficking. The Vienna Action Plan of 1999 establishes the fight against organised crime as an EU priority and foresees and integrated approach. The "Millennium Strategy" called upon the Commission to develop policy initiatives to combat organised crime. The Hague Programme of 2004 defined the objectives for 2004-2009 in the areas of freedom, security and justice and the subsequent Commission Communication complemented the priorities with concrete actions.[154]

In 2004 the Report of the Groups of Personalities in the field of Security Research[155] advised the European Security Research programme (ESRP) to take advantage of the duality of technologies and the growing overlap of security functions in order to bridge the gap between civil and defence research in support of a comprehensive approach to security. In this regard the importance of funding research activities targeted at the development of systems and products that are useful for, inter alia, the protection of Member States' territory, sovereignty, domestic population and critical infrastructure against transnational threats was stressed.

While the potential of Space for defence and security-related issues had already been recognised it was only then that the added value of space assets for the provision of internal security was first mentioned:

> security without the support of technology is impossible. It provides us with information about threats, helps us to build effective protection against them and, if necessary, enables us to neutralize them. In other words: technology is a key 'force enabler' for a more secure Europe. (…) As a result, the technology base for defence, security and civil applications increasingly forms a continuum. Across this continuum, applications in one area can often be transformed into applications in another area. There is an increasing overlap of functions and capabilities required for military and non-military security purposes (such as is found between border police, coast guard and emergency response teams) that often allows the use of the same technology for the development of both security and defence applications. Space technologies are a perfect illustration of this: a decision as to whether global positioning or earth observation systems, for example, are to be used for defence and security purposes is primarily political in character, not technological.[156] [emphasis added]

The report thus suggested a capability related approach as a basis for a European Security Research Programme. It should thus identify the threats, the missions required to tackle these threats, the capabilities needed to accomplish these missions and the technologies or combination of technologies that can provide the necessary capabilities. With an ever-changing strategic environment, however, threats are of a changing nature.

Moreover, there is no single user and thus needs and capabilities are difficult to define. The report comes to the conclusion that while single threats seem to require different specificities, different threats actually require the same missions The report goes even further in concluding that a capability-related approach can help to identify more specific technologies and applications. In this regard the report shows the multi-functionality of capabilities and technologies, both within the spectrum of internal security missions and across the boundary between external and internal security. Intelligence and secure communications, in particular, are crucial for all missions and actors. Surveillance is needed for the protection of national borders and for crisis management operations abroad. In both cases, the means used to fulfil the capability can often be the same with applications being adapted to the specific needs of the user but the basic technology staying the same. This will be exemplified at a later stage when looking at the critical mission areas in detail.

The report was followed by a Commission Communication on "The Implementation of the Preparatory Action on the Enhancement of the European Industrial Potential in the

For a foreign policy analysis on the ENP cf. Lynch, Dov. "In Search for EU Foreign Policy." 29 July 2009 <http://www.fes.de/aktuell/focus_europa/6/Docs/FES_Lynch.pdf>.

[153] Antonenko, Oskana, Bastian Giergerich, Nigel Inkster, Raffaello Pantucci and Rahul Roy-Chaudhury. op. cit.

[154] "The Fight against Organised Crime." Apr. 2008. European Commission Website 29 July 2009 <http://ec.europa.eu/justice_home/fsj/crime/fsj_crime_intro_en.htm>.

[155] Group of Personalities in the field of Security Research. op. cit.

[156] Ibid. 12.

Field of Security Research - Towards a Programme to Advance European Security through Research and Technology".[157] The Preparatory Action[158] addressed five particular areas including "protection against terrorism" and "the enhancement of crisis management". From 2004 to 2006 it funded research and technology and supported mission-oriented projects through calls for proposals and public procurement. It can be seen as the predecessor of the FP 7 Programme. It aimed at preparing the groundwork for a successful Security Research Programme. In particular its objective was to identify the optimal EU added value and priorities to be addressed in such a programme. While all five project areas (improving situation awareness, optimising security and protection of networked systems, protection against terrorism, enhancing crisis management, achieving interoperability and integrated systems for information and communication) fall within the possible realm of space technology, only one project description explicitly mentions the use of space-based assets.[159]

Also in 2004 (June) the Panel of Experts on Space and Security (SPASEC) was given the mandate to review, inter alia, the role of space in meeting the objectives in the fight against terrorism. The resulting SPASEC Report considered the support of space-based applications in counter-terrorism operations and showed that space technologies can play a key role in assisting the police, emergency response services, armed forces and agencies undertaking humanitarian relief efforts to respond more effectively to e.g. terrorist attacks.

Based on the Preparatory Action, the Framework Programme contributes to competitiveness in specific technological fields.[160] In this regard the FP 7 covers internal security in two out of seven clusters, which focus on space and security respectively. Given that the objective of the security theme is, inter alia, to develop technologies and knowledge needed to ensure the security of citizens from threats such as terrorism and (organised) crime, most of the activities are congruent[161] with the critical mission areas identified as a theoretical framework by this study. The EU Member States have earmarked a total of € 1.4 billion for funding this theme over the duration of FP 7.

Practically the most important European existing measure is probably the flagship programme Global Monitoring for Environment and Security (GMES).[162] It was launched in May 1998 in Baveno and adopted by the ESA and EU Councils respectively in June and November 2001. It is a European initiative for the implementation of information services dealing with environment and security. GMES will address three main themes, one of which is "contributing to the security of citizens"[163], by providing the information support to decision-makers and to operational actors such as civil protection teams and NGOs.

The benefits of GMES are therefore threefold:

- An improvement in monitoring capability leading to better data and information access, and effort on characterising environmental events and issues.
- A better understanding of events based on a more comprehensive integrated service bringing together diverse but complementary data sources and stakeholders
- An improved capability for forecasting and prediction.[164]

[157] Commission of the European Communities. Commission Communication On the implementation of the Preparatory Action on the Enhancement of the European Industrial Potential in the Field of Security Research, Towards a Programme to Advance European Security Through Research and Technology. COM (2004) 72 final of 3 Feb 2004. Brussels: European Union.

[158] Commission of the European Communities. Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of Regions - Promoting Decent Work for all: The EU Contribution to the Implementation of the Decent Work Agenda in the World. COM(2004) 249 final of 24 May 2006. Brussels: European Union.

[159] cf. related section on "optimising security and protection of networked systems" in Commission of the European Communities. COM(2004) 72 final. 8

[160] Group of Personalities in the field of Security Research. op. cit. 14.

[161] i.e. increasing the security of infrastructures, Intelligent surveillance and border security; Restoring security and safety in case of crisis

[162] Annex IV of the Commission of the European Communities. Commission Staff Working Document Accompanying the Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of Regions. Global Monitoring for Environment and Security (GMES) We Care for a Better Planet. SEC(2008)2808 of 12 Nov. 2008. Brussels: European Union.

[163] For a detailed discussion on the security theme refer to The GMES Working Group on Security. op. cit.

[164] This section is to a large degree based on the related discussion in European Space Agency/European Commission. op. cit.

The GMES initiative is based on four inter-related components: services, observations from space, in-situ (including airborne) observations and data integration and information management capacity, which should be addressed collectively. This data will be coordinated, analysed and prepared for end-users. Apart from the Sentinel 1-5 missions, GMES will also draw on national, Eumetsat and third party missions (e.g. CosmoSkymed, SPOT, Pleiades, Jason-2, Radarsat, TopSat, Terrasar-X, Rapideye, UK-DMC, METOP). It can be divided into three core services (land, marine, emergency response) and two pilot services (atmosphere and security).[165]

Generally speaking, GMES services include: prevention, monitoring and risk assessment, organisation and distribution of aid, impact assessment, protection against emerging security threats such as proliferation of WMD and trafficking and border surveillance.[166] Based on the idea of maximising the use of existing European capacities, the EU concentrates its actions on filling the gaps in full cognisance of the principle of subsidiarity. Thus GMES is a partnership involving different actors: The European Commission and ESA fund several projects that contribute to the development of the security dimension of GMES.

The EU Satellite Centre (EUSC) is operational in similar and related areas and is therefore closely involved in current GMES developments. In addition, the Commission's Joint Research Centre is engaged in GMES developments. In a similar vein, the agencies of the EC, in particular the European Environmental Agency, the European Maritime Safety Agency (EMSA) and the European Agency for the Management of Operational Cooperation at the External Borders of the Member States (FRONTEX), are also involved in the development of GMES. This user-driven approach is based on a wide consultation process with stakeholders. In this regard each service was initiated by a thematic workshop gathering users of the future service. In the security theme this meant setting-up expert groups with the tasks of describing scope, architecture and implementation plans for each service including the necessary infrastructural requirements. With Satellite imagery playing an increasing role in disaster management, it is ever more striking that the Commission, the Council and ESA all have separate imagery databases and imagery analysis and photo-mapping facilities. Albeit working in separate subject areas, resources among the institutions should be pooled. GMES should facilitate the easy access to the multitude of databases or meta-databases relevant to crisis management.[167] At present the development of GMES' services is channelled through demonstrations within the EC's FP 6 and FP 7 for space and security.

The recently published European Space Policy Progress Report[168] mentions elements for a European Strategy for International Relations in Space. In this regard it does not explicitly mention the use of space applications for counter-terrorism or crisis management resulting from a terrorist attack. It just mentions Galileo and EGNOS as "European investments in essential European infrastructure for crucial applications such as border control, transport management and logistics, financial operations and the surveillance of critical energy and communications infrastructure" and calls for better coordination between institutional actors.

On the basis of a recommendation by the European Security Research Advisory Board (ESRAB), the EU Member States and the Commission set up the European Security Research and Innovation Forum (ESRIF)[169] in September 2007, which is tasked to present by the end of 2009 a widely-shared perspective on European security research needs and priorities. Its members represent three different interest groups ("stakeholders")

> (1) Those using and applying the achievements of security research: European, national and regional authorities, police, fire brigades, all kinds of emergency organisations and first responders, private and public operators of critical infrastructure etc. ("demand side");
> (2) Those performing security research and turning its outcome into technologies and products: universities, research establishments, industry, including SMEs ("supply side");

[165] Commission of the European Communities. Communication from the Commission to the European Parliament and Council - Global Monitoring for Environment and Security (GMES): Establishing a GMES Capacity by 2008 - (Action Plan (2004-2008)). COM(2004) 65 final of 3 Feb 2004. Brussels: European Union.
[166] A more detailed analysis of the services will follow in the related section on critical mission areas.
[167] European Space Agency/European Commission. op. cit. 7-8.
[168] Commission of the European Communities. Commission Working Document European Space Policy Progress Report. COM(2008) 561 final of 11 Sept. 2008. Brussels: European Union.
[169] European Security Research and Innovation Forum (ESRIF). op. cit

> (3) And representatives of the citizens, often non governmental organisations or special think tanks, that are affected by both potential security incidents as well as the efforts to ensure their security ("civil society").[170]:

ESRIF is particularly tasked with helping to enhance the security of European citizens. Specifically, ESRIF addresses:

- mid-term and long-term security perspectives (up to 20 years),
- European, national and regional perspectives,
- bringing the demand and supply sides together to focus on requirements for research
- societal and technological aspects of security research,
- the promotion of innovation to lay the foundation for a European Security Market that exploits economies of scale at European level
- guidance for all stakeholders who prepare security-relevant research programmes in Europe.

One of its eleven working groups focuses on situational awareness and the role of space. In completing its activities by the end of 2009, ESRIF will draft detailed recommendations for security-related research and innovation in Europe.

4.3. The Current Institutional Set-Up

The provision of internal security is internally distributed across different authorities and operators, each with their respective roles, capabilities and resources.[171] With transport policy falling under the first pillar, i.e. the community pillar, and border control falling mainly under the third pillar, i.e. the intergovernmental pillar, responsibilities as well as decision-making structures vary.

4.3.1. European Union Institutions

European Parliament (EP)

The European Parliament, as the only elected body within the EU's institutional setting, is often consulted by the Council in matters related to security and crisis management. The EP can provide recommendations or ask for clarifications from the Council. Together with the Council, the EP enjoys the budgetary authority of the European Union. None of the most relevant committees: the Civil Liberties Committee, Justice and Home Affairs (LIBE) and the Foreign Affairs Committee, (AFET) cover internal security explicitly. The Foreign Affairs subcommittee on Security and Defence (SEDE) is mandated to be responsible for CFSP and ESDP only.

European Commission (EC)

The Commission enjoys a monopoly in initiating legislation. It also monitors how this legislation is implemented once it has been adopted by the EU Council of Ministers. It delivers opinions on any EU matter and is obliged to publish an annual report on the activities of the EU, which give the Commission scope to influence policy debates or steer the EU in specific directions. It is supranational in character, being composed of 27 Commissioners one from each Member State, each heading a Directorate-General. Over time the Commission has become increasingly accountable to the European Parliament.[172] The Commission ensures and coordinates the consistency of the EU's policies.[173] Given the limited competencies of the EU in Justice and Home Affairs and the Common Foreign and Security Policy, the power of the European Commission is limited compared to other policy areas. Some of its Directorates-General however are involved in the provision and development of internal security.

Directorate-General for Justice, Freedom and Security (DG JLS)

Within the area of Justice, Freedom and Security the European Commission shares its right to make legislative proposals with the Member States. DG JLS is mandated to ensure that the whole European Union is an area of freedom, security and justice. Its specific tasks and responsibilities are laid down in the Treaty of Rome,[174] the Treaty of Amsterdam and the conclusions of the European Council meeting in Tampere. Already from the beginning the European Commission has been involved in the

[170] Ibid.
[171] Group of Personalities in the field of Security Research. op. cit. 20.
[172] Peterson, John. "The College of Commissioners." The Institutions of the European Union. Eds. John Peterson and Shackleton Micheal. Oxford: Oxford University Press, 2006. 96.
[173] Regelsberger, Elfriede. "The Institutional Setup and Functioning of EPC/CFSP." Foreign Policy of the European Union: From EPC to CFSP and Beyond. Eds. Elfriede Regelsberger, Schoutheete de Tervarent, Philippe and Wessels, Wolfgang. USA: Rienner Publishers Inc, 1998. 100.
[174] Cf. Part Two, Articles 17-22; Part Three, Title III, Articles 39-47.

discussions to bring justice, freedom and security matters within the ambit of the European Union and to set up a small task force for justice and home affairs when the Maastricht Treaty was signed in 1992. This was expanded into a full Directorate-General in October 1999.[175] Apart from safeguarding civil liberties, DG JLS is engaging in an industrial study on national border surveillance systems, secure communications and pre-frontier intelligence in the context of fostering integrated border management and the development of the European Border Surveillance System (EUROSUR). Together with FRONTEX, DG JLS is responsible for the coordination of the EUROSUR programme. It analyses the development of this programme and provides recommendations to the European Commission.[176] DG JLS is also involved in the EU action against trafficking in human beings.[177]

Directorate General Joint Research Centre (JRC)

The Joint Research Centre is a European Commission service dedicated to improving "Europe's capacity to prevent, forecast and deal with natural hazards, such as floods and drought, and technological disasters". Counter-terrorism and the fight against organised crime is very closely related to these dedicated mission areas if not exactly included in them. The JRC is subdivided into specialised institutes and units such as the Institute for the Protection and Security of the Citizen (IPSC) and the Traceability and Vulnerability Assessment (TRVA) unit, which follow specific projects in line with their specialisations and focus. For its work, the JRC can draw on state-of-the art facilities such as its European Laboratory for Structural Assessment (ELSA) used to simulate the earthquake loading of full-scale buildings. It can be said to be a precursor as it has already understood the added value of space applications in disaster management.[178]

The JRC activities can be divided into institutional actions and ongoing and completed collaborative research projects falling within FP 6 or FP 7 respectively. In this regard one can identify a strong focus on maritime surveillance and container security issues. In addition, JRC is engaged in practical and scientific studies to improve knowledge and techniques to provide information related to civil protection (natural and man-made hazards) and foreign policy (e.g. humanitarian and development aid, de-mining, conflict prevention and crisis management). A cooperation agreement has been signed with EUSC.[179] The JRC

> has a leading position in satellite ship surveillance and oil spill detection; this knowledge (algorithms, software, systems) will be further developed (…). Since some operational tasks related to maritime surveillance have become the remit of EU Agencies, notably EMSA, FRONTEX, CFCA and EUSC, the action will concentrate on R & D support for these customers.[180]

JRC has adopted a consultancy evaluation role. While most of its maritime research activities have the objective of countering maritime pollution, those research projects often also have relevance for internal security: The JRC develops, tests and demonstrates technologies related for tamper-proofing of containers, tracking and tracing of shipments (Vessel Detection System and Vessel Monitoring System) and remotely operated ship inspections as well as issues such as container integrity. freight logistics (i.e. container traffic monitoring), anti-fraud and counter-terrorism. Its Institute for the Protection and Security of the Citizen (IPSC) is also strongly involved in emergency response with satellite applications. In the maritime border domain, the JRC also cooperates with FRONTEX, DG Justice Liberty Security (DG JLS), DG Mare and DG Enterprise (DG ENTR).

The Institute for the Protection and Security of the Citizen (IPSC) is also conducting

[175] Commission of the European Communities Directorate-General for Justice, Freedom and Security. "Justice, Freedom and Security from a European Commission Perspective." 27 July 2009 <http://ec.europa.eu/dgs/justice_home/index_en.htm>.
[176] Commission of the European Communities Directorate-General for Justice, Freedom and Security. "Inter-service Consultation on a draft Communications from the Commission on Examining the Creation of a European Border Surveillance System (EUROSUR). Work Programme. JLS.B1 D(2007) 16488 of 11 Dec. 2007. Brussels: European Union. <http://www.statewatch.org /news/2008/feb/eu-com-inter-service-eurosur.pdf>.
[177] cf. "EU action against Illegal Trafficking." 1 July 2009. European Commission 29 July 2009 <http://ec.europa.eu/justice_home/fsj/crime/trafficking/fsj_crime_human_trafficking_en.htm>.
[178] Ehrlich, Daniele. "Disaster Management at the JRC: The Use of Space Products." Presentation. The United Nations International Workshop on the Use of Space Technology for Disaster Management. United Nations, Munich, Germany. 18-22 Oct. 2004. <http://www.zki.caf.dlr.de/media/download/events/2004/unoosa_workshop_presentations/04_pres_session02_chair-bessis/08_UNOOSA-DLR_Ehrlich_JRC.ppt>.
[179] The GMES Working Group on Security. op. cit.
[180].Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.

Vulnerability Assessment in Transport and Distribution System (VATDIS Action) in which it reviews and develops methods for the identification, protection and security assessment of critical infrastructures with respect, primarily, to the threat posed by terrorism. Two specific groups in Ispra (Italy) and Petten (The Netherlands) specialise in critical infrastructure protection and occasionally conduct demonstrations to improve both the understanding of the dependence on certain infrastructures and to develop back-up systems. This includes mainly analysis of the threat itself, development of scenarios relating these threats to the vulnerabilities of the infrastructure, assessment of the consequences of attacks and development of protection measures. Methodological developments are supported by specific studies related to the threat associated with the transportation of dangerous substances. The JRC contributed decisively to the development of the criteria and of the guidelines that will be used by Member States to identify European Critical Infrastructure (ECI) in the context of Directive 2008/114/EC on the identification of CI. It is currently focussing its efforts on modelling and simulation of critical infrastructure, their failures and cascading effects, in support of the recent European Directive on European Critical Infrastructure, which does not directly involve the use of space applications. Among the major achievements is the development of a gas pipeline model to support critical European energy infrastructure assessment.

The JRC's institutional action on Information Support for Effective and Rapid External Action (ISFEREA) focuses on the external dimension of crisis management.It includes space technologies in crisis response and monitoring for security thereby providing sound political assessment and decision-making for external security and external relations (i.e. specifically addresses regional security issues in addition to crises, conflict and instability in third countries). Some of its actions such as map production, automatic image understanding and scientific support as well as spatial data infrastructure design also have implications for internal security. ISFEREA is also involved in the GMES projects focusing on security aspects. It contributed towards shaping the fast track GMES emergency response service. It also developed the geo-spatial database, collated global datasets, developed local geo-spatial data sets from VHR earth observation data and has made available online geo-spatial layers over crisis areas.[181] ISFEREA maintains a Digital Map Archive (DMA) as part of the JRC Institutional Action on Crisis Monitoring and Response Technologies (CRITECH).182 ISFEREA also provides geospatial information and analysis in support of crisis management (e.g. damage assessment, reconstruction monitoring etc.).

Directorate-General for Transport and Energy (DG TREN)

The European Commission's Directorate-General for Energy and Transport manages work in these two fields and aims to ensure to take advantage of all possible synergies. In this context it is also responsible for providing policy guidelines for both fields which among other things cover security-related aspects. DG TREN is working on the technical aspects of port security with DG JRC.[183]

Directorate-General for Enterprise and Industry (DG ENTR)

The Directorate-General for Enterprise and Industry works to ensure that EU policies contribute to the sustainable competitiveness of EU enterprises and facilitate job creation and sustainable economic growth. To reach this objective, DG ENTR has identified three action areas:

1. Competitiveness, industrial policy, innovation and entrepreneurship
2. Internal Market for goods and sectoral policies
3. Space and Security

With regard to the third action area, it has identified the following objectives:

- to move to coordinated space activities at EU and international level, so that the competitiveness of the sector is enhanced;
- to bring the Global Monitoring for the Environment and Security services into operation;

[181] "Digital Map Archive (2000-ongoing)". ISFEREA - Geo-Spatial Information Analysis for Global Security and Stability 3 Mar. 2009. <http://isferea.jrc.ec.europa.eu/Activities/ProjectPortfolio/Pages/DMA.aspx>.
[182] "Institutional Action – Crisis Monitoring and Response Technologies (CRITECH)." Institute for the Protection and Security of the Citizen – Joint Research Centre 3 Mar 2009 <http://ipsc.jrc.ec.europa.eu/showaction.php?id=15>.
[183] For more information on DG TREN's mission cf Ruete, Matthias. "Mission of the Directorate-General for Energy and Transport." Directorate General for Energy and Transport 29 July 2009 <http://ec.europa.eu/dgs/energy_transport/wcm/mission_dg_tren.pdf>.

- to strengthen the technological base of the space industry through actions carried out under the FP 7;
- to increase the technological basis for security in Europe so that higher security levels are possible.

In this context, DG ENTR is responsible for the "Space" theme of FP 7 as providedby Decision No.1982/2006/EC of the European Parliament and of the Council of 18 December 2006 concerning the FP 7 of the European Community for research, technological development and demonstrations activities (2007-2013). The objective of the FP 7 space theme is to support a European Space Policy focusing on applications such as GMES (Global Monitoring for Environment and Security) that has benefits for citizens and also on research and development in areas related to the competitiveness of the European space industry.[184]

Humanitarian Aid Office (DG ECHO)

The Humanitarian Aid Office (ECHO) is the service of the European Commission responsible for managing humanitarian aid funds given by the Community. It is mandated to provide emergency assistance and relief to the victims of natural disasters or armed conflict outside the European Union. The aid is intended to go directly to those in distress, irrespective of race, religion or political convictions. Around 80 per cent of the aid goes to conflict-driven crises. ECHO's task is to ensure goods and services get to crisis zones fast. Goods may include essential supplies, specific foodstuffs, medical equipment, medicines and fuel. Services may include medical teams, water purification teams and logistical support.

Goods and services reach disaster areas via ECHO partners. Among its tasks are:

- carrying out feasibility studies for its humanitarian operations
- monitoring humanitarian projects and setting up coordination arrangements.
- promoting and co-ordinating disaster prevention measures by training specialists,
- strengthening institutions and running pilot micro-projects.
- giving its partners technical assistance.
- raising public awareness about humanitarian issues in Europe and elsewhere, through actions carried out directly or by call for proposals by ECHO in order to promote the awareness of humanitarian aid.

In terms of disaster types, the main areas of interest are earthquakes, tropical storms, floods, droughts, volcanoes and conflicts. Looking at the three phases, it provides:

- early warning: methodologies and alert tools for rapid onset disasters (earthquakes, tropical storms, flash floods, epidemics)
- rapid damage assessment: methodologies for decision support tools (earthquakes, tropical storms, floods)
- long-term planning instruments (vulnerability analysis).

Considering both its mandate and its tasks, ECHO does not directly fall within the realm of the provision of internal security. However, many of the methodologies it provides with regard to crisis management as such are also applicable to the provision of internal security.[185] DG ECHO is one of the users of GMES.

Directorate-General for Maritime Affairs and Fisheries (DG MARE)

DG MARE is mandated to "steer, in close relationship with stakeholders at regional and European level, the development and implementation of the Integrated Maritime Policy and to manage the Common Fisheries Policy with a view to promote the sustainable development of maritime activities as well as the sustainable exploitation of fisheries resources within and beyond Community waters." In the context of providing for an integrated maritime policy, DG MARE has been involved in pilot projects and preparatory actions with regard to AIS-S with ESA, EMODNET and the integration of maritime surveillance in some sea basins.[186]

EC Maritime Affairs Task Force

The task force was created within the framework of the overall strategic objectives of the EC for 2005-2009 to initiate "an all-embracing maritime policy" pursuant to the "aim of launching a wide consultation on a future Maritime Policy for the Union". The

[184] "Grants Work Programme 2008 – General Publicity." DG Enterprise 29 July 2009 <http://ec.europa.eu/enterprise/contracts-grants/calls-for-proposals/grants-work-programme-2008/index_en.htm>.

[185] The GMES Working Group on Security. op. cit.

[186] For the current status of these projects cf. "Calls for Proposals and Tenders." European Commission Maritime Affairs 30 July 2009 <http://ec.europa.eu/maritimeaffairs/call_proposals_tenders_en.html>.

task force interacted with the Member State Experts Group, Inter-service Group and stakeholders for an integrated analysis (science and research, safety and security, maritime sectors competitiveness, jobs, human welfare, marine environment, climate change, and maritime heritage) in order to deliver the Maritime Policy Green Paper on 7 June 2006, followed by an Action Plan presented on 10 October 2007 by the European Commission. Its Maritime Data Expert Group[187] has been involved together with the Commission in the basic design principles of the European Marine Observation and Data Network (EMODNET).[188]

Civil Protection Unit (CPU)

At community level the Civil Protection Unit (CPU) in the DG for Environment has been involved for years in improving responses to crisis management in Europe. Together with civil protection authorities within the Member States and ESA Member States, the EU Civil Protection Unit is involved in risk mapping, early warning and crisis management with an explicit applicability of terrorist incidents.[189] The EU Civil Protection Unit can invoke the International Charter "Space and Major Disasters" in order to obtain easy and free of charge access to satellite data, with top priority in satellite tasking.[190] Together with the European Union Satellite Centre (EUSC) and JRC's Institute for the Protection and Security of the Citizen, CPU is one of the main actors in the field of emergency response with satellite applications at the European level. EUSC in comparison also has access to military satellite data and is the only EU actor providing GEOspatial information (GEOINT). Additionally, the Regional Service of Image Treatment and Remote Sensing (SERTIT) works closely with the UN in providing satellite data in case of International Charter reaction but SERTIT is limited to natural hazards rather than internal security.

Apart from the Civil Protection Unit and the related civil protection authorities within Member States of the EU and ESA, DG ECHO is involved in risk mapping, early warning and crisis management. ECHO's mandate is however to provide aid externally and not within the EU. In such a decentralised environment, interoperability of communication and information systems and the links between different networks are crucial. All relevant security services should be able to exchange information rapidly and securely. Areas of competency should be better coordinated and combined into a common situation picture. An agreement on common standards in early stages would facilitate interoperability and information security.[191]

Joint Situation Centre (SitCen)

The Joint Situation Centre of the Council's Secretariat was created to produce intelligence that no national agency is willing to produce or better where a single country's support would not be acceptable for political reasons. Its analysts primarily support the High Representative, COREPER II and other parts of the Council such as the Political and Security Committee (PSC) and the Working Party on Terrorism (COTER) in the second pillar, and the Article 36 Committee (CATS) and the Terrorism Working Group in the third.[192] By compiling national reports and complementing these with information from the EU and to a significant extent from open sources it produces intelligence that is customised for the European Council and the High Representative. SitCen has a Civilian Intelligence Cell (CIC) in general and a Counter Terrorism (CT) cell in particular.

Counter-Terrorism Coordinator

In the wake of the Madrid Bombings the European Council agreed to establish a EU Counter-terrorism Coordinator in March 2004 and appointed Gijs de Vries to this position. His main tasks are to coordinate the work of the Council of the EU in combating terrorism and to maintain an overview of all the instruments at the Union's disposal. He shall also closely monitor the implementation of the EU Action Plan on Combating Terrorism and secure the visibility of the Union's policies in the fight against terrorism. He should not be taken as a European "Minister for Homeland Security" available to the U.S. Secretary of Homeland Security. The EU Coordinator for Counterterrorism

[187] For more information refer to "Marine Observation and Data Expert Group." Register of Expert Groups 30 July 2009 <http://ec.europa.eu/transparency/regexpert/detail.cfm?ref=2146&l=M>.
[188] "European Marine Observation and Data Network (EMODNET)." European Commission Maritime Affairs 30 July 2009 <http://ec.europa.eu/maritimeaffairs/emodnet_en.html>.
[189] The GMES Working Group on Security. op. cit.
[190] European Space Agency/European Commission. op. cit. 7.
[191] .Group of Personalities in the field of Security Research. op. cit. 20.
[192] Müller-Wille, Björn. "The Effect of International Terrorism on EU Intelligence Co-operation." JCMS 46.1 (2008): 49-73. 49.

has neither line of authority over Commission bureaucrats or member state agencies, nor a significant budget to promote harmonisation of policies, procedures, standards, or equipment, which vary widely across member states. He cannot prescribe; he can only persuade. He reports to the High Representative for Foreign and Security Policy in the European Council, and thus is of a lower level than the U.S. Secretary, and works out of the European Council rather than the European Commission, and so only has a small staff at his disposal.[193]

4.3.2. European Union Agencies

European Union Satellite Centre (EUSC)

Among the lessons learned initiatives from the Chernobyl accident was the establishment of the European Union Satellite Centre.[194] It provides products or services to Member States, the Commission, third States having agreed to the provisions set out in the Annex on the association with the Centre's activities and international organisations whose request is of relevance to CFSP. EUSC is tasked to "support the decision-making of the European Union in the field of the CFSP, in particular of the ESDP, including European Union crisis management operations, by providing, as appropriate, products resulting from the analysis of satellite imagery and collateral data, including aerial imagery, and related services".[195] It is thus not directly missioned to support the provision of internal security.

EUSC, being dedicated to the support of CFSP and ESDP in particular, undertakes the production of imagery analysis reports and geospatial products such as geographic information systems (GIS). The Centre has access to a wide range of commercial imagery including high-resolution imagery (0.6 meters) and there are plans for itto have access to some national satellite systems in the future, which would further enhance the support that the EUSC gives at present.[196] In line with the increased dependency on critical infrastructure and energy as identified by the European Security Strategy, the European Union Satellite Centre is conducting security analyses of oil and gas infrastructure. The European Union Satellite Centre had been tasked by the OHQ in support of the ATALANTA NAVFOR Operation,[197] which aims at helping deter, prevent and repress acts of piracy and armed robbery off the coast of Somalia, to analyse imagery and report on key ports and specific coastal areas with the objective of gaining understanding of the current infrastructure and service ability and to give an outline of possible access/egress routes important to the operation.

European Maritime Safety Agency (EMSA)

Within the field of maritime transport policy the European Maritime Safety Agency (EMSA) established in 2002 (operational since 2003) provides technical assistance to the Commission in order to improve ships' identification and traffic monitoring in EU waters.[198] EMSA was not given tasks and responsibilities in respect of maritime security until 2004,[199] when it also received the operational responsibility for the community vessel traffic monitoring and information systems (SafeSeaNet (SSN), SafeSeaNet Tracking Information Relay and Exchange System (STIRES), European Union Long Range Identification and Tracking of ships Data Centre (EU LRIT Data Centre)). EMSA is also developing a 24/7 Maritime Support Service as a means to support Member States in their request for maritime data. EMSA cooperates closely with national administrations and provides technical input to discussions within the International Maritime Organisation (IMO), as well as to the development of Galileo and its application to maritime navigation.[200] Based on a Council Resolution adopted in October 2007 that decidedto set up an EU Long Range Identification and Tracking of Ships (LRIT) Data Centre (EU LRIT DC), EMSA is developing, operating and in charge of the maintenance of the EU LRIT DC. It is a combined effort of the European Flag States to be managed by the European Commission in cooperation with Member States through EMSA. After successful developmental testing, the EU LRIT DC began production on 1 June 2009.

[193] European Space Agency. Annual Analysis 2004. op. cit. 17-18.
[194] Molard, Bernard. op. cit.
[195] Council of the European Union. Corrigendum to Council Joint Action 2006/998/CFSP of 21 December 2006 amending Joint Action 2001/555/ CFSP on the establishment of a European Union Satellite Centre. Council Joint Action 2006/998/CFSP of 21 December 2006. Brussels: European Union. Art. 2.
[196] The GMES Working Group on Security. op. cit.
[197] "EU NAVFOR Somalia. European Security and Defence Policy." Council of the European Union 30 July 2009 <http://www.consilium.europa.eu/showPage.aspx?id=1518&lang=en>.
[198] Commission of the European Communities. Commission Staff Working Document – An integrated Maritime Policy for the European Union. SEC(2007)1278. Brussels: European Union.
[199] Following the coming into force of. Regulation (EC) No 724/2004 of 31 March 2004. Amending Regulation (EC) No 1406/2002 establishing a European Maritime Safety Agency.
[200] Western European Union. Document A/1920. op. cit.

- EU Long Range Identification and Tracking of Ships Data Centre (EU LRIT DC)

The EU LRIT DC has been operational since 30 June 2009. When all Member States' ships are phased in by the end of 2009, it will track around 10,000 ships, which will generate a minimum of 40,000 position reports per day. At present, there are 32 Member States, EFTA countries and Overseas Territories participating in the EU LRIT DC. Third countries might join in the future. The EU LRIT DC covers an estimated twenty to twenty-five percent of the world fleet subject to LRIT. In addition to tracking EU-flagged ships, the EU LRIT DC also provides Member States, on request, with the LRIT information of any third country vessel bound to, or sailing within, EU waters. So it is possible to track any ship within a 1,000 nautical mile zone of a participating State's coastline, no matter what flag the ship is flying. To support the work of the competent maritime authorities of Member States, EMSA has set up a permanent monitoring function (Maritime Support Services) to ensure the continuity, quality and reliability of the information exchanged through the new EU LRIT DC.[201]

European Agency for the Management of Operational Cooperation at the External Borders (FRONTEX)

The European Agency for the Management of Operational Cooperation at the External Borders (FRONTEX), established in October 2004, is coordinating the EU's external border activities. Its mission is the "coordination of intelligence driven operational cooperation at EU level to strengthen security at external borders". Among its main tasks are to carry out risk analyses and follow-up on the development of research relevant for the control and surveillance of external borders. FRONTEX assists the competent services of the member states responsible for implementing the Schengen *acquis* on control of persons at the external borders. In this regard it also cooperates with Europol, other international organisations and third countries. In particular, Frontex' Risk analysis Unit is tasked to

> produce appropriate, accurate and timely intelligence products which provide the foundation for Frontex operational activities, as well as to keep all principal customers (i.e. the Member States and the Schengen Associated Countries as well as the Council and the Commission) informed of the current situation regarding illegal migration at the external borders.[202]

FRONTEX, within the framework of European external relations policy, establishes technical working arrangements for joint operations with relevant third countries and invites them to participate in operational activities where appropriate. It is also providing action plans on different migratory routes using recommendations of the International Labour Organization (ILO).[203] Together with EUSC, FRONTEX' Risk Analysis Unit launched a training programme related to GIS called FronBac whose main objective is to develop analytical skills in FRONTEX and the Member States.

In the maritime area, FRONTEX cooperates with EMSA, the Joint Research Centre (JRC) and in the past with two centres for the control of maritime borders: (1) the Western Sea Borders Centre (WSBC), located in Madrid, with responsibility for the Baltic and the Western Mediterranean and (2) the Eastern Sea Borders Centre (ESBC), situated in Piraeus (Greece), which had responsibility for the Eastern Mediterranean. The latter two are no longer operational and FRONTEX thus has one Sea Borders Sector within its Joint Operations Units as part of the Operations Division.

The EU has already undertaken multinational maritime border control operations in the Mediterranean (Operation Guanarteme, Operation Magallanes, Operation Alhambra, Operation Dolphin, Operation triton, Operation Fer.I.A.S.), including inspections of ports. FRONTEX is also involved in the FP 6 project LIMES which inter aliadeals with land and sea border surveillance. FRONTEX is developing a permanent regional border security concept, the European Patrols Network (EPN), which is based on the preceding studies MEDSEA and BORTEC, and has the objective of integrating the EPN into a European Surveillance System (EUROSUR).

European Defence Agency (EDA)

[201] European Maritime Safety Agency. "The EU LRIT Data Centre is in Production." Press Release. 3 June 2009. <http://www.emsa.europa.eu/Docs/other/press_release_03-06-2009.pdf>.

[202] Kalliala, Mari. "Situation Regarding Illegal Migration at the EU External Borders." Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at/images/stories/dokumente/Conference2009/kalliala.pdf>.

[203] Commission of the European Communities. Communication from the Commission to the Council Reinforcing the management of the European Union's Southern Maritime Borders. COM (2006) 733 final of 30 Nov. 2006. Brussels: European Union.

Established in 2004, the European Defence Agency's mission is to "support the Member States and the Council in their effort to improve European defence capabilities in the field of crisis management and to sustain the European Security and Defence Policy as it stands now and develops in the future". Its functions and tasks mainly relate to improving Europe's defence performance by promoting coherence.[204]

EDA has identified twenty-four action areas which have been grouped into two priority sections of twelve topics each. Among Priority 1 areas are "Intelligence, Surveillance, Target Acquisition and Reconnaissance Architecture", "Medical Support" as well as "Network Enabled Capability" and under Priority 2 areas which are space related are "Information Management", "Open Source Intelligence (OSINT)- military implications", "Space", "IDR Sensors and Collectors", "Electromagenetic Spectrum Management" and "Wide-area Maritime Surveillance".[205] Similar issue areas can also be found within EDA's Capability Development Plan (CDP).
Initiated by the Western European Armaments Group (WEAG) in 2004, a Research Technology Programme on Microsatellite Cluster Technology was transferred to EDA in 2006 and completed successfully in May 2007. It particularly focussed on the application of SAR and ELINT satellite clusters for high resolution imaging of targets and change detection, moving target indication and geolocation and classification of objects based on detection of radar emissions.

Among the ongoing projects in the realm of imagery and reconnaissance, EDA is involved with MUSIS by:

- informing MUSIS Contributing Members about its own activities and standardisation activities in particular in the ISR (Intelligence, Surveillance and Reconnaissance) area to support MUSIS nations in considering coordination between the respective initiatives
- promoting standards based on results coming from the MUSIS Federating Common Programme
- based on conditions and timeframe agreed by the MUSIS Contributing Members, assisting in identifying other Member States interested in the MUSIS programme
- according to the objective of supporting the ESDP, informing the MUSIS Contributing Members of the EU needs for space-based images.

EDA is also responsible for ensuring

- liaison with the European Commission for R&T projects in FP 7 that could be of interest for MUSIS, i.e. hyperspectral technology
- liaison with the European Commission for ESDP related projects like GMES (Global Monitoring for Environment and Security), EDRS (European Data Relay Satellite), etc. in order to support MUSIS nations in considering appropriate synergies
- liaison with appropriate EU bodies regarding:
- establishment of conditions of their access to the MUSIS capacity
- access to the MUSIS capacity for ESDP operations according to the agreed conditions

EDA is also conducting the EDA PT MARSUR project which aims at having naval forces identify operational requirements for using space assets that should be taken into account when developing GMES. JRC is in liaison with PT MARSUR in this context.

With regard to Satellite Communication, in the short term EDA will establish an EU Satcom procurement cell.I In the long term it will look into new research and technology activities in this field (Milsatcom 2020). Some of the topics that have been identified as areas that need action are related to the protection of satellites from anti-jamming, anti-piracy and anti-spoofing (MilsatCom-C1), conventional weapons (MilsatCom-C4) and to develop optical communication (MilsatCom-D2). Transition from need for action areas to actual projects depends on Member States' interest.

EDA engaged in the so-called "Structured Dialogue on Space" with the European Commission, the Council General Secretariat, the European Space Agency and Member States, aimed at raising awareness about programmes and identifying opportunities for the complementary development of space based assets for respective user communities. In this context, valuable interfaces have been created with other key players. The European Commission's activities within the FP 7 in areas such as

[204] "Background." 3 May 2005. European Defence Agency 20 June 2009 <http://www.eda.europa.eu/genericitem.aspx?area=Background&id=122>.
[205] The Overview of EDA's activities related to space is partially based on the insights given by De Vries, Jan Sjoerd. "Persistent Intelligence supporting Knowledge-based EU Operations." Presentation. Space Systems and Security. Dorint Hotel, Potsdam, Germany. 25-27 May 2009.

Space and Security research are of high interest to EDA participating Member States. In this context the cooperative European Framework Initiative was formed in April/May 2009, creating joint programmes and committees that might build the bridge between FP 7 and FP 8.

EDA is also involved in developing the defence-related requirements for Space Situational Awareness and other ESA projects. Regarding research and technology, EDA has formed a Joint Task Force together with the European Commission and ESA on critical technologies for European non-dependence. It is also exploring future activities in support of space, system of systemsand simulation/modelling. With a view to a more structured approach, on 18 May 2009 Ministers of Defence tasked the Agency to propose a framework for maximising such complementarities and synergy between defence and civilian security-related activities and to propose the content of a coordinated programme.

Equally EDA investigates possible synergies and military requirements as to the GMES programme. Regular exchange is also taking place with the European Space Agency, whose space-related technology and application programmes bear a similar potential for coordination and synergies. Common areas of interest currently range from the definition of military requirements for security-relevant programmes such as "Space Situational Awareness" and possible interest in the European Data Relay Satellite System to the conduct of coordinated feasibility studies regarding the demonstration of Unmanned Aerial Systems (UAS) Command and Control over Satellites. The "Structured Dialogue" needs to be further put into practice and the development of a more comprehensive European agenda fostered.[206]

Apart from these EDA initiated projects (Ad Hoc Cat. A projects), EDA is conducting projects initiated by Member States (Ad Hoc Cat. B projects).[207] Among the latter, two projects are currently in the definitional phase: (1) a demonstrationof a Maritime Surveillance Network Architecture, initiated by Germany and the Netherlands, and the role of space for land and maritime surveillance (initiated by Belgium, Finland, Norway). In contrast to many other actors, EDA focuses on user needs. The need for networked data systems to share, fuse, analyse and exploit available data led EDA and EUSC to conduct a series of trials between February and September 2008 with the objective of demonstrating how imagery from diverse sources and in various formats could be fused to a single ground station. This Tactical Imagery Exploitation System (TIES) will improve the interoperability between intelligence, surveillance and reconnaissance assets and ground terminals.[208]

In general, EDA aims at providing coherence across all efforts and actors and acts as a "systems integrator". It provides analysis, injects proposals, monitors and evaluates. It also forms a channel and instrument for political will. Its concrete areas of action depend however on decisions and funding of the Member States as EDA is inter-governmental in character.

4.3.3. Other European Agencies and Institutions

The European Space Agency (ESA)

In general under European Space Policy, the European Space Agency (ESA) is responsible for implementing space capabilities that respond to EU policy needs. In this regard, the EU is cooperating strongly with ESA in the development of several security related service elements of its flagship programme Global Monitoring of Environment and Security (GMES). ESA is, however, also undertaking its own projects in the area of internal security provision. It particularly engages in the maritime area as well as in emergency management. ESA has already been actively involved in the maritime field for over the past twenty-five years.[209] In emergency management it focuses on floods and fires with its projects RISK-EOS and RESPOND. A structural dialogue among European institutional actors including the European Commission, the General Secretariat of the Council, the European Defence Agency, ESA and the Member States was set-up during the fifth Space Council with the aim of achieving a substantial increase in the coordination of space, security and defence related activities. ESA supports the identification of user

[206] Trioulaire, Denis and Michael Simm. "European Cooperation on Space Matters." EDA Bulletin. Issue 12. June 2009. 17.
[207] Starlinger FFG. Presentation. Space and Security – Aktueller Status und Entwicklungen. Österreichische Forschungsförderungsgesellschaft (FFG), Vienna, Austria. 15 May 2009.
[208] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. "FORESEC Deliverable D 2.3 Global Trends and Actors Shaping European Security." FORESEC. 7-8. <http://www.foresec.eu/wp2_docs/GlobalTrendsAndActors.pdf>.
[209] "EC unveils new EU Maritime Policy." European Space Agency. 12 Oct. 2007. European Space Agency 16 Mar 2009 <www.esa.int/esaCP/SEM6DE2PL7F_index_2.html>.

requirements to be presented and discussed within the structured dialogue among European institutional actors.

ESA is also working on the European Data Relay System (EDRS) that contributes to the improved provision of internal security through improved delays of Earth observation images. ESA is also currently working on European Space Situation Awareness that, inter alia, will improve the detection of negation of space systems. Additionally, it is currently in the process of developing a Global Integrated Network for Innovative Utilisation of Space for Security (GIANUS), aimed at improving European preparedness for a variety of threats through the development of integrated space infrastructure as part of a European security and crisis management system in coordination with already existing initiatives.

Similarly, ESA has recently introduced the Integrated Applications Promotion (IAP) programme, which is ARTES 20 (user-driven applications), and the ARTES 3-4 Telecommunications Applications element (product-driven applications), dedicated to development, implementation and pilot operations, utilising not only Telecommunications satellites but also combining the use of different types of space assets, including Earth Observation and Navigation, as well as Human Spaceflight technologies.

Advisory Council for Aeronautics Research in Europe (ACARE)

The Advisory Council for Aeronautics Research in Europe was launched at the Paris Airshow in June 2001. It comprises about forty members, including representation from the Member States, the Commission and stakeholders, including manufacturing industry, airlines, airports, service providers, regulators, research establishments and academia. Its main focus is to establish and carry forward a Strategic Research Agenda (SRA) that will influence all European stakeholders in the planning of research programmes, particularly national and EU programmes, in line with the Vision 2020 and the goals it identifies.[210] Currently, among other things, it is looking into the use of highly sophisticated fully automated systems to detect sudden off track manoeuvres of airplanes in flight, on the ground and remote control systems to prevent and minimise hijacking, and automatic surveillance for critical circumstances alerts. Research is planned to be conducted in both cargo security and passenger security. Although the Addendum to the SRA does not explicitly mention the use of space systems for these security applications, it does express the need for systems other than ground-based to enhance aviation security.[211] The European Aeronautics Vision of 2020 envisages sophisticated ground and satellite-based communication, surveillance and navigation system to enhance airport security by 2020.[212] These security measures are most likely to be supervised and controlled by the Joint Aviation Authorities (JAA), which is currently responsible for Europe's aviation security.[213]

4.4. Measures for Intelligence, Early Warning and Risk Management

Following the attacks in New York, Madrid and London, intelligence and security agencies in Europe and around the world concurred on the need to work more closely together in response to the new international terrorist threat. Within the EU, in contrast to the U.S., only a small number of voices was raised in favour of creating a new, potent European intelligence agency.[214]

In the context of the provision of intelligence, GEOspatial information (GEOINT) is increasingly being used by civilian and private sector organisations. Projects are either run on a purely national basis or organised along the lines of a limited multinational cooperation of two to three countries. Some are also managed under PPP schemes.

[210] Advisory Council for Aeronautics Research in Europe. "European Aeronautics: A Vision for 2020 – Meeting Society's Needs and Winning Global Leadership." Report of the Group of Personalities. Luxemburg: Office for Official Publications of the European Communities, 2001. <http://www.acare4europe.com/docs/Vision%202020.pdf>.

[211] Advisory Council for Aeronautics Research in Europe. "2008 Addendum to the Strategic Research Agenda." Advisory Council for Aeronautics Research in Europe, 2008. <http://www.acare4europe.com/docs/ACARE_2008_Addendum.pdf>.

[212] Advisory Council for Aeronautics Research in Europe. "European Aeronautics: A Vision for 2020 – Meeting Society's Needs and Winning Global Leadership." Report of the Group of Personalities. Luxemburg: Office for Official Publications of the European Communities, 2001. 14. <http://www.acare4europe.com/docs/Vision%202020.pdf>.

[213] Ibid. 16.

[214] Müller-Wille, Björn. op. cit. 59.

National Responses

On the Member States' level, France seems to be the precursor in this field with several dedicated missions to signal intelligence and early warning. CERISE is intended to eavesdrop on weak HF communications while Clementine, Essaim and the upcoming ELISA and SPIRALE are devoted to technology demonstration for electronic intelligence. French President Sarkozy has insisted on developing a new strategic function "Knowledge and Anticipation" through space means. In the recently published White Paper on Defence he underlined the importance of the strategic use of outer space. France will develop a significant capability in ELINT (Electronic Intelligence) satellites based on the experimental satellites in this field.[215] Britain's technological demonstrator, TopSat, is also used for intelligence purposes. It provides for land-use monitoring and its maximum resolution of 2.8 meters is good enough to identify individual houses or vehicles. The Italian Space Agency (ASI) is also working on a technological demonstrator called PRISMA (PRecursore IperSpettrale of the application mission) to be launched in 2010. Neither of these has been explicitly mentioned to be used for counter-terrorism purposes. Sweden (SVEA) and Spain (SEOSAT) are both developing a Signal Intelligence satellite to be launched in 2010.

Apart from specifically devoted signal intelligence and reconnaissance missions, systems such as the German SAR-Lupe and the Italian COSMO-Skymed (Constellation of Small Satellites for Mediterranean Basin Observation) prove Europe's high competence in radar imagery for Earth Observation. COSMO-Skymed consists of four identical LEO satellites providing global coverage independent of weather and day time. They are characterised by short revisit and response times and a high volume of high resolution images daily. The COSMO-Skymed system architecture strongly pursues the goal of conceiving design solutions for sustaining Multi-Mission/Multi-Sensor (MM/MS) capabilities.

Optical capabilities are again solely in the hands of France, operating missions like SPOT or HELIOS and the earlier described Pleiades. CNES's SPOT (Satellite pour l'Observation de la Terre) satellites are characterised by a maximum resolution of 2.5 meters and offer intelligence services. Their imagery data is also used by EUSC. HELIOS is a signal intelligence technology demonstrator which is mainly used for military intelligence by the French, the Italian and the Spanish armed forces. Its imagery data is also used by EUSC. The German DLR's TerraSAR-X offers a maximum resolution of 1 to 16 meters depending on the image size and is used for intelligence purposes. Germany is planning a twin asset called TanDEM-X to be launched in October 2009. The RapidEye constellation that was launched last year offers a maximum resolution of 6.5 meters. It is not primarily intended for intelligence purposes but will potentially contribute to GMES. Among its prime field of application is disaster management.

Italy is also working on a technology demonstrator called PRISMA, an earth observation system with innovative electro-optical instrumentation combining a hyperspectral sensor with a panchromatic, medium-resolution camera. In addition to the classical capability of observation based on the recognition of the geometrical characteristics of the scene, it has the advantage of being able to determine the chemical-physical composition of objects present on the scene through hyperspectral sensors. Primarily intended for applications in the field of environmental monitoring, resource management, crop classification, pollution control, etc., it will also be applied to the field of national security.

Generally speaking the best analysis needs to be based on a combination of radar and optical satellite imagery. Thus, several agreements on a bilateral or multilateral basis ensure data exchange. In this regard, the Schwerin agreement ensures exchange between optical French Helios and German Radar SAR-Lupe. The imagery data of the Italian COSMO-Skymed is used by the EUSC and will, together with Pleiades, form the dual-use ORFEO (Optical and Radar Federated Earth Observation) Programme. Moreover, the COSMO-Skymed imagery data will contribute to GMES. There are also various projects related to the improvement of specific techniques related to SAR imagery such as the Belgium ASARTECH and PolInSAR projects.

The French government has repeatedly been advocating the development of a European signal intelligence capability based on its technology demonstrators. In this regard France in 2008 started working with Belgium, France, Germany, Greece, Italy and Spain (plus Sweden as an observer country) on the ultra-high resolution European system for space-based surveillance and reconnaissance called MUSIS (Multinational Space-based Imaging system), which will achieve

[215] European Space Agency/European Commission. op. cit. 19.

operational capability in 2015. It ensures continuity of services from the current HELIOS II, German SAR Lupe and Italian Cosmo-Skymed system from 2015 onwards. MUSIS will complement GMES (environment and security) in dealing with security and defence. MUSIS is now in its definition phase under the responsibility of CNES (Centre National d'Etudes Spatiales). The overall MUSIS programme is managed by the French Ministry of Defence together with its international partners. While the space segments will remain national, constant efforts are being deployed in order to make the MUSIS ground segment more European and more interoperable.

Three satellites are planned to be launched on orbits with two different altitudes. The overall design is thereby derived from the Pleiades system. Pleiades is a new generation dual system for optical observation of the Earth, characterised by high resolution and great manoeuvrability. The programme is being developed in cooperation with Austria, Belgium and Sweden. The first satellites are planned to be launched in early 2010. EDA has recently approved MUSIS as a Category B programme, aiming at federating nationally provided space components through a generic user ground segment. Potential synergies with Europe's Global Monitoring for Environment and Security (GMES) initiative will be assessed.[216]

European Reponses

International and European initiatives, such as Global Earth Observation System of Systems (GEOSS) and Global Monitoring for Environment and Security (GMES), aim at finding a viable way to provide cost-effective solutions for future EO systems, providing users with simple access to Multi-Mission/Multi-Sensor (MM/MS) Capabilities. Another multilateral programme in the Earth Observation sector is the Common Requirements for a Global European Earth Observation System by Satellites/Besoin Opérationnel Commun (BOC), which was signed by Belgium, France, Germany, Italy and Spain in 2002 with Greece joining later on in 2003. It aims at harmonising the operational requirements and national Earth observation programmes of participating States and identifying requirements for building an independent European military satellite Earth observation system as a first step towards creating a European-level strategic imagery capability.[217] Initiated by the European Defence Agency (EDA) in cooperation with EUSC, the Tactical Imagery Exploitation System (TIES) aims at compensating for the lack of interoperability between European satellite imagery systems and ground systems through the creation of a capacity for receiving imagery data in different formats provided by different satellite systems and feeding them into one intelligence product.

As indicated above, the main GEOINT provider at European Union level is the European Union's Satellite Centre. It combines imagery, collateral information, imagery intelligence and geospatioal information with intelligence sources so as to provide a Common Operating Picture (COP). In this regard the timely, relevant and accurate provision of geospatial intelligence is of great importance for planning, decision-making and operational use. EUSC offers a geospatial intelligence database for stakeholders and authorised users and creates tailored geospatial intelligence services and solutions aimed at meeting specific customer needs. EUSC's mission statement however limits the support of EUSC to decision-making in the field of the Common Foreign and Security Policy and in

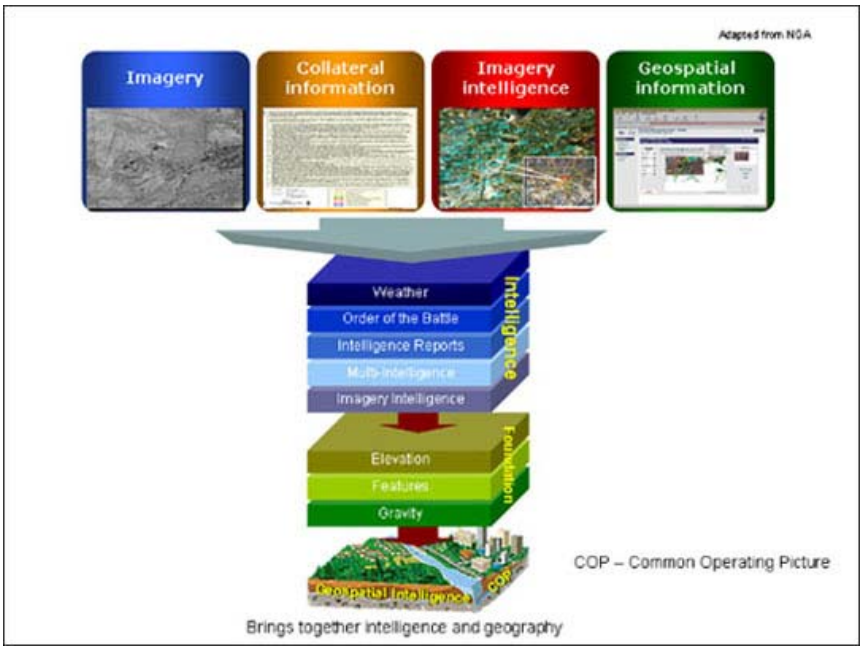


Figure 22: EUSC's GEOINT Working Structure

[216] Vries de, Jan Sjoerd. op. cit.

[217] Gleason, Michael P.. "European Union Space Initiatives: The Political Will For Increasing European Space Power." Astropolitics, 4:7 (2006): 7-41.

particular the European Security and Defence Policy. While explicitly including crisis management operations, EUSC's analysis of satellite imagery and collateral data as well as related services are thus not primarily used for the provision of internal security.

Together with FRONTEX, EUSC is also providing training on GIS applications as part of a training programme called FronBac aiming at developing analytical skills in FRONTEX and among the Member States. The emphasis of this programme lies on border security.

Apart from the GEOINT application of EUSC, the European Union has developed several research projects as part of its Framework Programmes, which aim at improving the sharing of intelligence at the EU level. These can be divided into four categories:

1. projects aiming at countering fragmentation and developing a European Spatial Data Infrastructure (ESDI)

2. projects focussing on treaty and land monitoring as part of non-proliferation and the detection of clandestine weapon development[218]

3. projects aiming at external security which might have a positive impact on internal security as well. Similarly, there are projects that explicitly address disasters as distinct from counter-terrorism or organised crime but their results might still be relevant to internal security

4. projects primarily aimed at improving external security but having implications for internal security as well.

Category 1: Data Harmonisation and Sharing

FP 6 project HUMBOLDT[219] was aimed at the implementation of a European Spatial Data Infrastructure (ESDI) which would integrate the diversity of spatial data available for a multitude of European organisations. Specifically it aimed to provide the necessary software modules in order for different organisations to share their spatial information in an effective and sustainable way. In a similar vein the FP 6 project ORCHESTRA (Open Architecture and Spatial Data Infrastructure for Risk Management) aimed to establish an open service-oriented architecture (SOA) based on common standards for sharing relevant information for both disaster prevention and emergency management. Such a SOA would improve the geospatial situation in Europe through enhanced data-sharing and end-user accessibility.

The FP 6 project WIN (Wide information network for risk management integrated project) followed the objective of integrating all existing reference results or initiatives to contribute to the design, the development, and the validation of what could be referred to as a "European Risk Management information infrastructure". This information infrastructure will be a major element of the future overall European Spatial Data Infrastructure (ESDI). Complementing

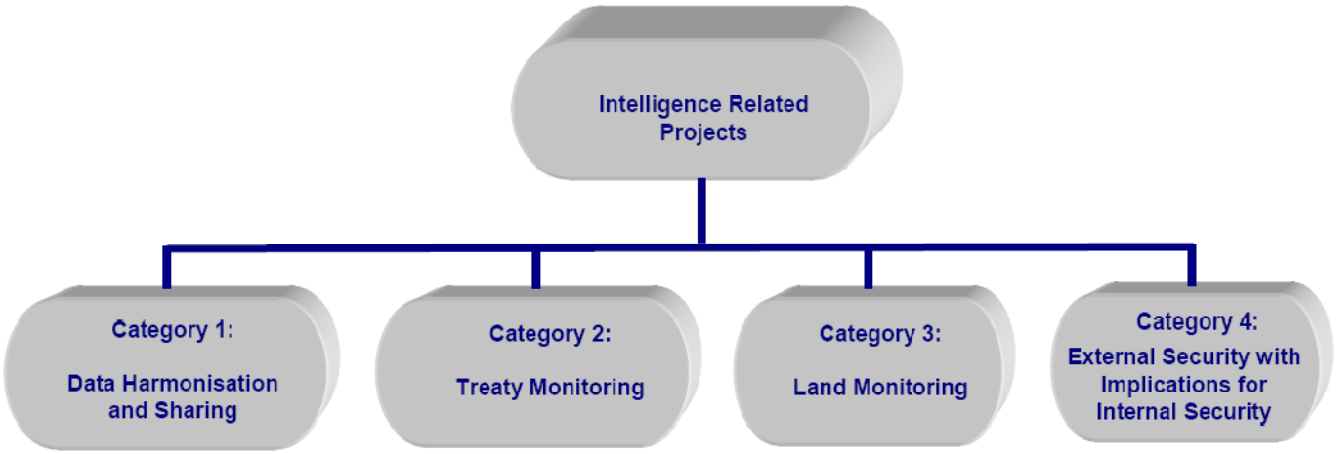


Figure 23: Categories of Projects for Intelligence Gathering

[218] Most of the projects do not explicitly mention their applicability for counter-terrorism or organised crime.

[219] Information on the specifics of each project can be found in the Annex in alphabetical order. The respective websites of each project as well as the CORDIS website served as the main source of information for this.

ORCHESTRA and WIN, which serve the prevention and early warning phase of the disaster cycle, the FP 6 project OASIS (Open Advanced System for dISaster and emergency management) aimed at improving the spatial data situation in the response phase (cf. related section on Emergency Management). The differences among ORCHESTRA, OASIS and WIN are depicted in the figure below.

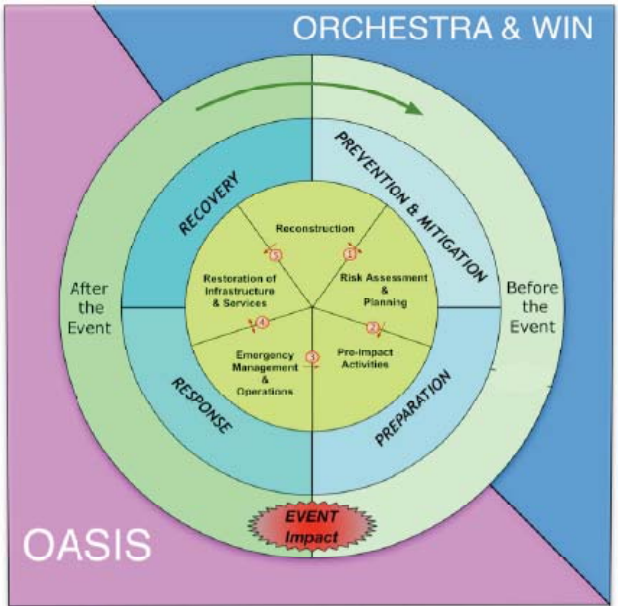


Figure 24: ORCHESTRA, OASIS and WIN Contribution to the Disaster and Emergency Cycle[220]

Category 2: Treaty Monitoring

The FP 6 project Global Monitoring for Security and Stability (GMOSS) focused on research on specific science and technology needed to provide, inter alia, effective monitoring of international treaties protecting against the proliferation of weapons of mass destruction. GMOSS contributed to the security dimension of GMES through the integration of remote sensing data analysis, social science and the analysis of key threats and risks. It advances the field of Earth Observation for civil security applications through joint research and training activities, sharing resources and infrastructure and exchange of expertise and staff. The FP 6 project LIMES (Land and Sea Monitoring for Environment and Security) supports Non-Proliferation Treaty (NPT) monitoring with the objective of improving information products to respond to the needs of users such as the IAEA.

Category 3: Land Monitoring

LIMES is divided into three clusters one of which deals with land surveillance. It mainly applies solutions based on EO technology and on its integration with satellite telecommunication and navigation capacities where relevant. Similar to the LIMES FP 6 project, HAWKEYE evaluated the benefits of thermal infra red hyper spectral imaging (TIHSI) technologies for the detection of clandestine weapon development or activities putting at risk civil security linking airborne or space borne systems. It particularly aimed at identifying potential applications of the developed technology for non-proliferation enforcement and internal security. Experimental results through field and airborne experiments were used for a preliminary evaluation of space borne solutions.

Through evolution in geospatial data processing, GMES will provide more and more tailored intelligence to the police forces thus improving detection of suspicious or criminal behaviour and thereby making better use of police resources. Processing of geospatial images could detect routes used for trafficking (human beings, drugs, weapons), terrorist safe houses, illegal cultivation for drug production but also more indirect traces of criminal activity. It will also provide assistance to simulation and planning.

The FP 6 project PREVIEW researched into risk maps and early warning systems aiming at better anticipating short-term risks. The FP 6 project TANGO (Telecommunications Advanced Networks for GMES operations) explored synergies between Earth Observation and satellite communication in GMES, thereby advancing early warning for all GMES services. Building on satellite navigation, the FP 7 project LOTUS (Localisation of Threat substances in Urban Society) is developing a new anti-terrorism tool in the form of an integrated surveillance system for continuous chemical background monitoring.

GIANUS will provide the greatest improvements. Providing very high resolution (below 1 meter) in a fast response time (less than eight hours), it aims at independent European access to intelligence for crisis management. GIANUS will complement MUSIS. It is currently in the preparatory phase (2009 – 2011) with the technology development phase starting in 2011 and system development starting with the new

[220] Atkinson, M. op. cit.

EU financial perspectives in 2014[221]. EDA took over the Research Technology Programme on Microsatellite Cluster Technology from the Western European Armaments Group (WEAG) in 2006 (initiated in 2004) and in May 2007 completed its research on SAR and ELINT satellite clusters for high resolution imaging of targets and change detection, moving target indication and geolocation and classification of objects based on detection of radar emissions.

Category 4: External Security Projects relating to Internal Security

While the FP 7 project G-MOSAIC (GMES services for management of operations, situation awareness and intelligence for regional crises) primarily aims at EU security related to "out of EU Area Crisis" for inter alia peacekeeping and peace-building, some of its final applications also advance the field of internal security. In this regard its services also offer solutions to intelligence and early warning in particular in the case of proliferation. Thus G-MOSAIC, which is mainly intended for RELEX-related entities, should not be overlooked when outlining internal security related projects. As it is used by intelligence centres, it definitely also advances the field of internal security. Its main advantage is to offer services in all three phases of the disaster and emergency cycle thereby providing for an all-encompassing solution without artificially dividing crisis management.

4.5. Activities in the three Critical Mission Areas

Several initiatives at both the national and European level aim at improving the provision of internal security through space applications. At a general level, the FP 7 project FORESEC is a European wide participatory foresight project on Europe's evolving security, its drivers, trends and scenarios. In particular, the PASR project ASTRO+ aims to show the potential of using space-based assets to support security operations through demonstrating to users the unique advantages of space in specific situations and contributing to establish a research and technology basis by networking users, institutions, industry and research to enhance European competitiveness. Building on the results of SENTRE222 and STACCATO[223], the FP 7 project STRAW aims at providing a European Service of Technology Watch (TW) on security technologies, which will also include space applications.

STRAW not only advises potential end-users about the fundamental technologies and existing technology gaps but also brings together the defence and security research industry for developing new civil applications. The main output will be a web-based IT system with a TW list and an interface for entering data on user requirements. The project HIDE (Homeland Security Biometric Identification & Personal Detection Ethics), promoted by the European Commission (EC) and coordinated by the Centre for Science, Society and Citizenship, aims to establish a platform devoted to monitoring the ethical and privacy implications of biometrics and personal detection technologies including GIS technologies and other space based applications.

Apart from these holistic projects, most FP projects tend to focus on one specific mission area. The stage of development in the different areas is different both at the policy level (documents) as well as at the level of practical applications (R&D projects, implementation).

4.5.1. Critical Infrastructures

As earlier described (cf. chapter 1), the protection of critical infrastructures depends on the definition of critical, which, in turn depends on the degree of dependency. Even though different players and policy documents define differentsectors as critical, dependency is still unknown, underestimated or neglected. The FORESEC study concluded that while the increasing vulnerability of critical infrastructures is a common concern across EU Member States, threat assessment again varies among Member States, with

[221] Duhamel, Erwin. "Space-Based Solutions for Alert, Crisis Management and Relief. European Space Agency." Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at/images/stories/dokumente/Conference2009/duhamel.pdf>.

[222] On a more general level the PASR project SENTRE aimed at developing a strategic research plan for European security by establishing and consulting a network of users and technology experts at national and European levels as to identify European/common capability needs and research priorities for comprehensive security missions.

[223] The PASR project STACCATO aimed at proposing methods and solutions for the creation of a security market and a structured supply chain in Europe. In line with ESRAB's (European Security Research Advisory Board) recommendations, it goes beyond research needs and gap analysis already undertaken through efforts supported by PASR, by identifying implementation measures.

Swedes, for example, expressing great concern about potential cascading effects from bad infrastructures in neighbouring States they rely upon for trade and raw materials. Countries still connected into aging Russian systems show a growing concern about the lack of investment in renovation. The lack of effective connectivity amongst European States is generally seen as a potential weakness and greater coordination is understood as essential in protecting critical national infrastructures. While some steps have been taken towards European coordination, the potential risk that infrastructure failure in one nation might have a cascading catastrophic effect upon neighbours has highlighted the need for further European cooperation.[224]

The European Security Strategy characterises the post Cold War environment as one of "increasingly open borders" with an "increased European dependence – and so vulnerability – on an interconnected infrastructure in transport, energy, information and other fields". Following the Madrid terrorist attack of March 2004, the European Council asked the Council to prepare an overall strategy to enhance the protection of critical infrastructures in the European Union. In this regard, in October 2004 the Commission adopted a Communication on "Critical Infrastructure Protection in the Fight against Terrorism", putting forward suggestions on what would enhance European prevention, preparedness and response to terrorist attacks involving critical infrastructures. In the following,the Council reached political agreement on a Directive on the Identification and Designation of European Critical Infrastructure (ECI) and on the assessment of the need to improve its protection. The Directive establishes the necessary procedure for the identification and designation of ECI and a common approach to the assessment of the needs to improve the protection of such infrastructure in order to contribute to the protection of people. It concentrates on the energy and transport sectors.

In 2005, the Commission created a Critical Infrastructure Warning Information Net-work (CIWIIN), bringing together Member State CIP specialists to assist the Commission in drawing up a programme to facilitate exchange of information on shared threats and vulnerabilities and appropriate counter-measures and strategies. The U.S. has a similar system known as the Critical infrastructure Warning Information Network (CWIN), which has been operational since 2003.[225] The completion of the development and deployment of CIWIIN is envisaged in the 2009 annual work programme adopted under Council Decision No 2007/124/EC, EURATOM.[226] In December 2006, the Commission presented a Communication on a European Programme for Critical Infrastructure Protection (EPCIP), which sets out the principles, processes and instruments proposed to implement the EPCIP.[227] It recognises that the security and economy of the European Union as well as the well-being of its citizens depend on certain infrastructures and the services they provide. Pursuant to this, the Commission organised two seminars involving both Member States and industry and decided to put forward a Green Paper outlining the options for EPCIP.

The third Annual Work Programme adopted under Council Decision No 2007/124/EC, establishing the Specific Programme "Prevention, Preparedness and Consequence Management of Terrorism and other Security related Risks for the Period 2007-2013 as part of the General Programme on "Security and Safeguarding Liberties"[228] offers a comprehensive framework and contributes to the development of the EPCIP.[229] Additionally, the Green Paper on a European Programme for Critical Infrastructure Protection (EPCIP) has laid down as the main objective of EU policy in Critical Infrastructure Protection (CIP)[230] the ability to prepare for,

[224] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. "FORESEC-Europe's Envolving Security: Drivers, Trends and Scenarios Deliverable D 2.2 Synthesis Report." FP 7 FORESEC, 2008. 8. <FORESEC-Europe's Envolving Security: Drivers, Trends and Scenarios Deliverable D 2.2 Synthesis Report>. 11-5.

[225] Euractiv. "Critical Infrastructure." 7 June 2005. Euractive 11 Mar 2009 <http://www.euractiv.com/en/security/critical-infrastructure/article-140597>.

[226] Euratom. "ANNEX Annual Work Programme 2009 – Prevention, Preparedness and consequence management of terrorism and other security related risks." <http://ec.europa.eu/justice_home/funding/cips/doc/awp_cips_2009_en.pdf>.

[227] European Council Secretariat. op. cit.

[228] The framework programme aims at ensuring an effective operational cooperation in the fight against crime and terrorism and strengthening their prevention. It consists of two financial instruments encompassing the following specific programmes: Prevention of and fight against crime and Prevention, Preparedness and Consequence Management of Terrorism and other Security related risks. Both specific programmes are implemented via annual work programmes adopted by the Commission each year. Cf. "Security and Safeguarding Liberties." DG Justice, Freedom and Security 22 July 2009 <http://ec.europa.eu/justice_home/funding/intro/fundin_security_en.htm>.

[229] Euratom. op. cit.

[230] Commission of the European Communities. COM (2005) 576 final. op. cit.

protect against, mitigate, respond to, and recover from critical infrastructure disruptions or destruction.

Finally, Council Directive 2008/114/EC on the identification and designation of European critical infrastructure and the assessment of the need to improve their protection filled the gaps in the existing structures by providing definitions long yearned for. By defining European Critical Infrastructures (ECI), it laid down, inter alia, the difference between a national and a European critical infrastructure. In particular it establishes CI as part of the internal market. The JRC has contributed decisively to the development of the criteria and of the guidelines that will be used by Member States to identify European Critical Infrastructures through the organisation of five workshops which led to the establishment of a common procedure for the identification and designation of ECI and of a common approach to the assessment of the needs to improve their protection. The third annual work programme 2009[231] adopted under Council Decision No 2007/124/EC, EURATOM establishing the Specific Programme "Prevention, Preparedness and Consequence Management of Terrorism and other Security related Risks" for the Period 2007-2013 offers a comprehensive framework and has contributed to the development of the EPCIP by funding, in particular, actions based on an all-hazards approach with priority to countering threats from terrorism.

While all of these documents advance the field of CIP through defining threats and vulnerabilities and putting CIP in the context of the fight against terrorism as well as other man made disasters, none of them propose the incorporation in CIP of space applications. The policy level thus does not explicitly recognise the potential of space applications for the provision of internal security.

National Responses

- Energy Facilities

While most of the EO missions (such as the German TerraSar-X or the UK's Topsat), can (depending on the capabilities) be used for geospatial information on critical infrastructure for risk and vulnerabilities or impact assessment, there do not seem to be any national projects explicitly covering CIP. Belgium has conducted a research project (HYSAR) on fusing polarimetric SAR and hyperspectral data to classify man-made objects in urban and industrial scenes. Detected images include inter alia industrial buildings and nuclear power plants.

A Greek Small and Medium Size Enterprise (SME), which is active in the area of Space Applications and Information Communication Technologies, has developed an innovative pipeline surveillance system using remote sensing and earth observation techniques. The system involves image operation in different spectral bands (visible, infrared, thermal), lineament analysis and geostructural investigations so as to assure safety of pipeline facilities.232

- Ports

The two hanseatic cities Ventspils (Latvia) and Bremen (Germany) have initiated the SMARTRegions (Space and Security Knowledge for Maritime Regional Development) project. Specifically, the project looks at the potential for satellite-based technology application fields like port/ transport/ trade (port security, movement of goods, container security), maritime security / safety (detection of ship-source pollution, ship tracking/AIS, sensitivity mapping for coastal protection) and coastal and city environment. The SMARTRegion project also attempts cooperation with regional initiatives conducted in parallel such as the GMES-initiative in Bremen and several regional initiatives in Ventspills. A pilot project is currently being developed in Bremen by OHB to prove the feasibility, applicability, value-adding and high quality of space technology-based services.

- Communication

As described earlier, satellites themselves can be considered as critical infrastructures.[233] While both the U.S. and Russia have developed space-based early warning capabilities to detect rocket launches, France is the only European country with two early warning demonstrator satellites (Spirale 1 and Spirale 2). Those can be complemented in their space negation detection efforts by sea-based and terrestrial assets performing ballistic missile launch detection. In Europe only France (i.e. Monge tracking ship with ARMOR radars to track ballistic missiles) and the UK (i.e. space surveillance site with a Large Phased-Array Radar) employ such radars. Alternative to employing electronic protection capabilities for the up-and-down links between satellites and ground stations, laser communication systems provide a degree of immunity from conventional jamming techniques. France's Astrium

[231] Euratom. op. cit.

[232] "S3 PAMS Profile." op. cit.

[233] A detailed description of the policy measures to be taken can be found in Remuss, Nina-Louisa. op. cit.

satellites and its subsidiary Tesat Spacecom in Germany launched a civil-military laser communication programme in 2007. Austria has conducted a research and development project on quantum communications ("Austrian QUANTUM") providing for secured exchange of information between satellites and spacecraft and ground stations. This project became the basis for a still ongoing reflection process among several European countries with ESA.[234]

Both France and the UK maintain hardening to protect their military satellites against nuclear attacks in a high altitude. Satellites can be protected against direct energy attacks through protective measures like automated mechanical shutters or the release of smoke. Only France and Germany have reconnaissance satellites that might employ such capabilities. No single European country currently possesses rapid rebuilding capacities. The UK is looking into the development of advanced microsatellites.

France is pursuing a research and development project (T2L2) on using optical signals to compare atomic clocks for, for example, GPS and Two-Way Satellite Time and Frequency Transfer (TWSTFT). While its security implications are not obvious, one can assume that results could lead to preventive measures against any illicit attempt relying on the limitations of current technological support of the banking system. Through its "Applied Remote Sensing Cluster" the German Aerospace Center is conducting, inter alia, risk and vulnerability assessments of urban areas.

European Responses

In the area of CIP one can identify two categories of projects:

- projects tackling one particular critical infrastructure such as pipelines or ports, and
- projects of a more general character aimed at improving CIP in general.

Category 1: Projects Aiming at Improving the Security of a Particular Critical Infrastructure

- Energy Facilities

Falling under the scope of the first category, the FP 5 project PRESENSE (Pipeline Remote SENsing for Safety and the Environment) uses SAR images for object-based detection of hazards to the European gas pipeline network.[235] Its main objective is to develop a new form of pipeline management system based upon remote satellite surveillance. It offers a hazard detection system based on a threefold architecture composed of (1) change detection based on SAR and optical images, (2) change categorisation, i.e. the fusion of SAR and optical changes and the categorisation of changes into priorities, and (3) change classification, i.e. the classification of categorised changes into hazards (e.g. digger, excavation, tree etc.).

The FP 6 project LIMES focuses on closing research gaps in order to improve EO and other space technologies for critical infrastructure monitoring and aims at setting up a framework for cooperation between

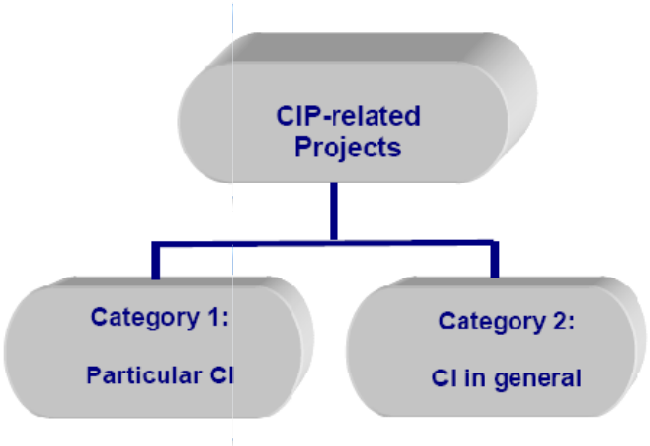


Figure 25: Categories of Projects for Critical Infrastructure Protection

[234] "Un pas de plus vers la communication quantique spatiale". BE Autriche 23 April 2009. <http://www.bulletins-electroniques.com/actualites/053/53978.htm>.

[235] Dekker, R. J., I. Lingenrfelder, B. Brozek, U. Benz and A. C. Van den Broek. op. cit.

Member States' bodies and the EU. During the first project cycle, LIMES developed a set of services for critical infrastructure surveillance during a demonstration in Spain for gasification plants and pipelines in collaboration with ENAGAS. These combine local geospatial information and very high resolution satellite imagery in order to provide updated infrastructure and terrain analysis for the benefit of the security of premises, population, land uses and economic activities. They also include 2D modelling of critical infrastructures. During the second project cycle, LIMES continued testing of change detection techniques so as to develop new algorithms with the difference of using a non-cooperative nuclear power plant.[236] A third step would be to monitor power plants in critical geographical areas as conducted by GMOSS. LIMES is also conducting research on pipeline monitoring. It currently focuses on the accident in the pipeline network of Moscow in 2008. LIMES discovered that accidents such as these have such a huge impact that they can actually be detected by meteorological satellites with low resolutions although the actual extent and impact is only detectible in high resolution imagery. This is however limited given the low revisit time. Thus LIMES is currently conducting archive research in order to advance the field of on-spot detection.

The JRC pursues an institutional action called Security of Critical Networked Infrastructures (SCNI), which aims at gathering, exchanging and generating security-relevant data in order to develop

(1) models and methodologies for the assessment of networked systems, their weaknesses, potential malicious actions that can be deployed for attacking them, and the effectiveness of countermeasures;
(2) experimental security facilities;
(3) tools for data sharing and information exchange, for the construction of assurance cases for infrastructure systems, and
(4) criteria for the identification and designation of critical information infrastructures and ICT systems.

While currently there is no use of space data in the framework of CIP, a new GIS tool for energy infrastructure monitoring, which includes the use of satellite imagery, is under development. JRC is currently conducting an internal project based primarily on open-source information supplemented with commercially limited to basic maps and geographic/topological data called the CEI-GIS project (Critical European Infrastructures – Geographic Information System) that is aimed at creating an interactive Geographic Information System supporting

- the representation of the CEI over the European territory;
- the modelling and simulation of security relevant behaviours of the CEI;
- the storage and visualisation of the appropriate information, mainly focusing on security scenarios;
- further risk analysis elaborations and models.

It builds upon a database (SDE geo database) bringing together all the relevant information related to European critical infrastructures including social, administrative, organisational, technical and geographic data derived from both public and private information supplied by infrastructure operators. Through a web application, qualified EC users can continuously update the stored data (ArcGIS Server Manager). The map service draws on satellite imagery, the Euroglobal Map (EGM)[237], the GISCO database (Geographic Information System for the European Commission), the Teleatlas – MultiNet for territorial information, the Platts commercial database containing updated information regarding the European Electric Grid (including generating stations, substations, cross border sections, transmission lines), NUTS codes (Nomenclature of Territorial Units for statistics) as its standard location code, EUROSTAT tables for statistical data at NUTS level for analysing power loads and the impact of potential power failures and consumption and production data, annual statistics at national or regional level; specific data related to cross border sections, and

[236] The Treaty establishing the European Atomic Energy Community (EURATOM) calls upon its signatories to create the conditions of safety necessary to eliminate hazards to the life and health of the people. Among the tasks of EURATOM it mentions the establishment of "uniform safety standards to protect the health of workers and the general public and ensure that they are applied." Over the years nuclear energy issues notably operational safety of nuclear facilities has grown ever more important. "Treaty establishing the European Atomic Energy Community (EURATOM)." Summaries of EU legislation. 22 July 2009 <http://europa.eu/legislation_summaries/institutional_affairs/treaties/treaties_euratom_en.html>.

[237] The EGM is a pan-European digital data base and mechanisms. It builds upon e work already done at the national level in almost 40 European countries, by harmonising the national datasets into one harmonised and seamless pan-European database. "EuroGlobalMap." GIS-Centras. 30 July 2009 <http://www.gis-centras.lt/gisweb/index.php?pageid=298>.

general reliability statistics of the Union for the Co-ordination of Transmission of Electricity (UCTE). Users can navigate through the CEI database, view maps or simply read information about projects and follow developments and analysis via an intranet site.

By building on previous work undertaken within the JRC, a methodology was developed from 2007-2008 to help identify elements of a network that may be considered to be critical with the objective of developing a gas pipeline model to support critical European energy infrastructure assessment as part of the VATDIS action (Vulnerability Assessment in Transport and Distribution Systems). In this regard a gas pipeline network model using open-source data for the Czech Republic, Slovakia and Hungary was assembled, consisting of

- a Geographical Information System (GIS) to provide a background capability for mapping pipelines and facilities making use of satellite imagery
- a powerful hydraulic simulation engine of the gas flowing in the system including the capability to include system components such as compressor stations, storage fields and pipeline furniture.[238]

Beginning as a VATDIS ACTION at the Institute for the Protection and Security of the Citizen in Italy, the model has now been continued by the sister institute on energy in Petten. Two labs in ISPRA conduct research on the possible effects of failures of critical infrastructures to advance the European understanding of "critical".

Furthermore, the JRC supports the implementation the EPCIP focusing on chemical plants by working on a Major Accident Hazards Bureau (MAHB), aiming at contributing to consistent and efficient control of industrial hazards through the development of tools and methods for risk assessment and management, monitoring accidents and incidents through an accident reporting system (accident data base), analysing their cause, drawing lessons for the competent authorities and mainstreaming them into the risk management process. It supports the SEVESEO directive[240] and occasionally uses space data. It provides a Community Documentation Centre on Industrial Risk as a service, aiming at facilitating the exchange of information between the EU Member States on the control of major hazard industrial activities.

In line with the increased dependency on critical infrastructures and energy as identified by the European Security Strategy, the European Union Satellite Centre is conducting security analysis of oil and gas infrastructures**.** In this context it conducted a risk analysis of the Baku-Tbilisi-Ceyhan (BTC) pipeline in Turkey. Based on satellite imagery it was able to provide a detailed 3D model of the pump stations. By combining data such

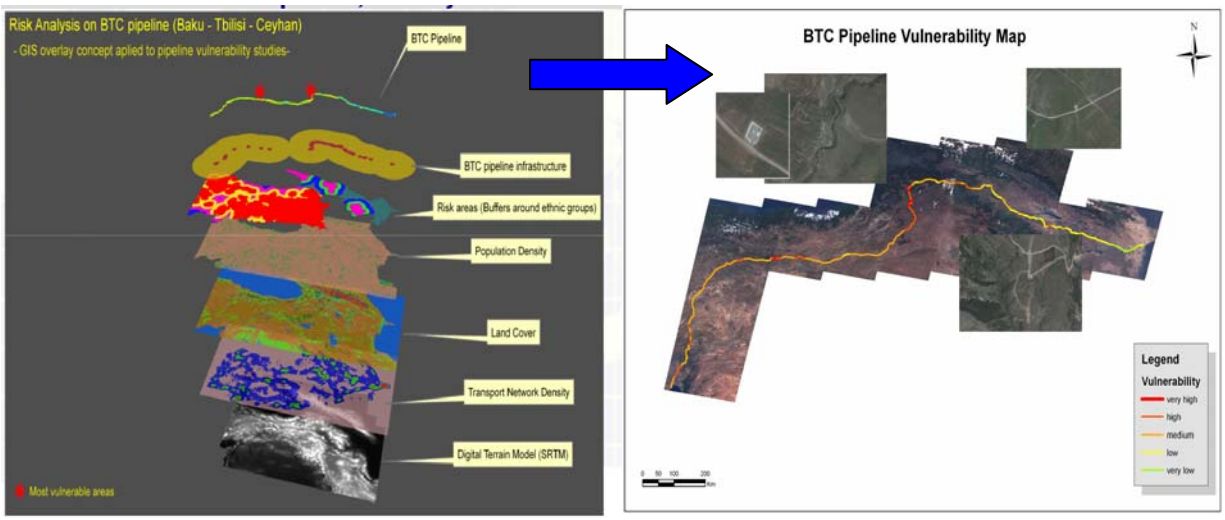


Figure 26: Example of a GIS Analysis as a Basis for a Vulnerability Map239

[238] Institute for the Protection and Security of the Citizen. "Annual Report 2007." Luxembourg: Office for Official Publications of the European Communities, 2008. 20.
[239] Ussorio, Alessandra. op. cit.

[240] Meuleman, Koen, Filip Lefebre, Eric Gontier, Els Knaeps, Sindy Sterckx and Marc Paganini. "SEVESEO: EO-Based Services in Support to Industrial and Technological Risk Management." Paper presented at Envisat Symposium 2007. Montreux, Switzerland. 23–27 April 2007. <http://envisat.esa.int/envisatsymposium/proceedings/posters/3P9/506999me.pdf>.

as digital terrain modelling, transport network density, land cover, population density and risk areas (e.g. buffers around ethnic groups) with the BTC pipeline infrastructure, a vulnerability map could be composed.

The EUSC also provides site analysis on the basis of visual interpretation of satellite imagery. The EUSC conducts site monitoring drawing on Quickbird, Ikonos and Cosmo-SkyMed images, among others. Change detection capabilities help to detect interferences with nuclear power plants for treaty monitoring.

- Ports

Considering ports as a critical infrastructure, the FP 7 project SECTRONIC (Security System for maritime infrastructures, ports and coastal zones) also deals with the protection of critical infrastructures. Specifically it offers a 24-hours small area surveillance system, which improves both the detection of terrorist activity or organised crime and improves situational awareness during a crisis.

The OHQ in support of the ATALANTA NAVFOR Operation[241] that aims to help deter, prevent and repress acts of piracy and armed robbery off the coast of Somalia has tasked the European Union Satellite Centre to analyse imagery and report on key ports and specific coastal areas with the objective of gaining understanding of the current infrastructure and service ability and to give an outline of possible access/egress routes important to the operation.

The JRC is currently also involved in the FP 7 project VESPO (VEssel Surveillance and POrt security),[242] aiming at improving vessel detection through the development of algorithms in satellite radar images for improved port security. Additionally, it is supporting DG TREN to provide guidelines for the most efficient application of the new port security regime, thereby focusing on the technologies necessary to fulfil the functional security requirements in the technical domain.

In a second project cycle, LIMES also focuses on the monitoring of ports as critical infrastructures building the interface between ocean and cities, often handling critical substances and being industrial in character. LIMES will now conduct different demonstrations in order to gain data on port monitoring. One demonstration will include a terrorist attack. The FP 7 project SOBCAH will provide an efficient, real-time, user-friendly, highly automated surveillance system to be deployed in one of the larger European Ports. The FP 6 project EFFORTS ("Effective Operations in Ports") dealt among other things with the use of satellite navigation for safe navigation in ports.

EADS Defence & Security is developing an intelligent system (Integrated MARitime SECurity System) for border security which integrates cameras, radar, UAVs and satellites for border surveillance and security and also aims at improving the protection of seaports.[243]

- Communication

As described earlier, satellites themselves can be considered as a critical infrastructure, which is in line with the current work of the JRC.[244] While both the U.S. and Russia have developed space-based early warning capabilities to detect rocket launches, there is currently no European early warning capacity. Space-based ASATs can be detected through tracking of satellite manoeuvres to monitor whether a satellite is in an orbit that could allow it to intercept or attack another satellite. Both the U.S. and Russia have a limited ability to do this through their space surveillance capabilities. The introduction of a European Space Situational Awareness (SSA) programme[245] will improve Europe's detection capabilities of space-based ASATs. Through tracking satellite manoeuvres it enables monitoring of whether a satellite is in an orbit that could allow it to intercept or attack another satellite. While the ability to constantly monitor all satellites to detect hostile manoeuvres would constitute a significant protection capability, no space actor currently has this ability. The European SSA will also help to detect high altitude nuclear detonations, thereby identifying non-compliance with relevant international treaties and recommendations.

[241] "EU NAVFOR Somalia. European Security and Defence Policy." op. cit.
[242] "Institutional Action -Vessel Surveillance and Port Security (VESPO)." Institute for the Protection and Security of the Citizen. 30 July 2009 <http://ipsc.jrc.ec.europa.eu/showaction.php?id=28>.
[243] EADS. "The Step Beyond – Sicherheit in einer veränderten Welt." Issue 1.2009. 50-1.
[244] A detailed description of the policy measures to be taken can be found in Remuss, Nina-Louisa. op. cit.
[245] Rathgeber, Wolfgang. "Europe's way to Space Situational Awareness (SSA)." ESPI Report 10. Vienna: European Space Policy Institute; Rathgeber, Wolfgang. "Space Situational Awareness (SSA) for Europe. A First Important Step." ESPI Perspectives 16. Vienna. European Space Policy Institute.

Having acknowledged a potential cascading effect resulting from interrupted satellite navigation, Galileo's Public Regulated Services (PRS) which are designed for use by governmental bodies, are expected to withstand any jamming attempts and provide continuity through robustness. PRS is protected by each Member State against abuse by hostile groups. Thus Galileo's infrastructures and signal located in national territory are monitored and detection of interferences and their sources will be guaranteed by European and National Security Centres.[246]

In order to secure communication between satellites and their ground stations, ESA, like the U.S., is working on including laser terminals for laser communication systems onboard of the Sentinel series (Earth Observation) of the GMES programme. ESA has also since 2002 supported research in the field of quantum communications. On the basis of this, a European research consortium submitted the mission proposal Space-QUEST ("QUantum Entanglement for Space ExperimenTs") in 2004 to the European Life and Physical Sciences in Space Programme of ESA, aiming at a quantum communications space-to-ground experiment from the International Space Station. In 2007 several European countries including Austria, France and Italy were involved in a reflection process with ESA to test quantum communication between a satellite and a ground station at approximately 800 km distance. The decision should be taken in 2009.[247] Apart from ESA, EDA is also looking into securing satellite signals. It is currently offering a series of projects related to the protection of satellites from anti-jamming, anti-piracy and anti-spoofing (MilsatCom-C1), conventional weapons (MilsatCom-C4), and the development of optical communication (MilsatCom-D2). Those are however still in the initial phase with only one country having issued a letter of intent.

There is currently no rapid recovery plan for European satellites.[248] USAF's Operationally Responsive Space programme, by contrast, is aimed at rapid design, build, test and responsive launches. Europe is also looking for its own approach to responsive space as an element of GIANUS. With satellite missions being currently fragmented and shattered over different players in different mission areas, Europe needs to improve its information chain to get the right information to the right people when needed.[249]

Category 2: Projects covering CIP in general

The FP 6 project GMOSS conducted research on specific science and technology needed to provide better monitoring of infrastructures and provided specific proposals for remote sensing data analysis in this context. It also contributed to the awareness of the security dimension of GMES. The Commission intends to undertake the establishment of a voluntary EU reference network for critical infrastructure (ERN-CIP) through administrative arrangements with the JRC in 2009.[250] During the preparatory phase the JRC is mainly concerned with analysing the experimental security requirements at the EU level, reviewing the current EU-wide security-related test, trial and evaluation capabilities and capacities, proposing a roadmap for the implementation of the reference network and the development of the capabilities needed and proposing setting-up the management structure for the reference network (2009-2010). During the execution of implementation it deals with the design and development of experiments for the testing of critical infrastructure-related commercial detection and protection equipment as well as for testing the vulnerability of critical infrastructure systems. The ERN-CIP will contain real world data, data from desktop simulations as well as data from experimental security test platforms and ranges, and aims to support industry and governments during the whole lifecycle. Once established, all Member States will have access to the testing facilities of the ERN-CIP and will be notified of the execution of security experiments. VATDIS (Vulnerability Assessment in

[246] Roma, Alfredo. "A Case Study: SYNTHESIS—Using the European Satellite Infrastructures for National Internal Security." Presentation. Space and Internal Security Workshop. European Space Policy Institute, Vienna, Austria. 28 May 2009. <http://www.espi.or.at/images/stories/dokumente/Conference2009/roma.pdf>.

[247] "Un pas de plus vers la communication quantique spatiale." op. cit.; Perdigues-Armengol, Josep Maria, Bernhard Furch, Clovis Jacinto de Matos, Olivier Minster, Luigi Cacciapuoti, Martin Pfennigbauer, Markus Aspelmeyer, Thomas Jennewein, Rupert Ursin, Tobias Schmitt-Manderbach, Guy Baister, John Rarity (8), Walter Leeb), Cesare Barbieri, Harald Weinfurter, Anton Zeilinger. "Quantum Communication at ESA: Towards a Space Experiment on the ISS." Proceedings of the 58th International Astronautical Congress. Ed. IAC/IAA, 24-28 Sept. 2007, Hyderbad, India. <homepage.univie.ac.at/rupert.ursin/php/?download=2007-ESTEC_IAC07_Paper.pdf>; Perdigues-Armengola, Josep Maria, Zoran Sodnik and Clovis de Matos. "Leap Ahead in Space Communications – Quantum Technologies for Space Systems." European Space Agency Bulletin.13 (2009). <http://www.esa.int/esapub/bulletin/bulletin137/bul137e_perdigues_armengol.pdf>.

[248] Governance Group for the Space Security Index. op. cit.

[249] Erwin, Duhamel. op. cit.

[250] Euratom. op. cit.; Masera, Marcelo. op. cit.

Transport Distribution Systems) also contributed significantly to the progress made during 2007 on a proposal of a Directive on the "Identification and Designation of European Critical Infrastructures and the Assessment of the Need to Improve their Protection".

JRC's VATDIS focuses on the review and development of methods for the identification, protection and security assessment of European Critical Infrastructures with respect, primarily, to the threat posed by terrorism. These methods include analysis of the threat itself, development of scenarios relating these threats to the vulnerabilities of the infrastructures, assessment of the consequences of the attacks, and development of protection measures. A method based on the spatial distribution of urban vulnerability was developed to support the identification of particularly threatened zones thereby determining vulnerability in terms of casualties that would result if a well-defined target were attacked by a particular means of transport. For a given urban area, only its land-use and a list of possible targets with their spatial locations are required. Each target is classified into a specific target type (e.g. a government building, a school). Collateral casualties around targets are also estimated using land-use information. The method was validated against historical events (e.g. Oklahoma City attack and Enschede accident) and demonstrated on an anonymous large-sized European city in a GIS environment.

Additionally, JRC is developing a web service for mapping as a preparatory service for risk management in the context of GMES. It mainly consists of a pilot web service featuring geo-spatial interoperability in full compliancy with open standards, covering a trans-boundary area of Italy and France that provides support for the emergency management of different risks: floods, quakes and transport of dangerous goods. The services allow representation of the zones impacted by hazards, routing of rescue teams and identification of near-by civil protection resources.

The FP 6 project VASTS (Vulnerability Assessment and Surface Transport Security) focused on the review, comparison and development of methods for the security management of European Critical Infrastructures with respect, primarily, to the threats posed by terrorism, sabotage and other negative intentional acts. These methods include analysis of the terrorist threat itself, development of the scenarios relating these threats to the vulnerabilities of the infrastructures, assessment of the consequences of the terrorist attacks, and development of defence layers and protection measures to deter, detect, prevent and mitigate such actions. Methodological developments were supported by more technical applications particularly in the Surface Transport Sector (cf. Land transportation).[251]

4.5.2. Transportation Security

The transport sector is characterised by its internationalism. Thus an all-embracing transportation policy needs to take into account both the external dimension as well as the internal one. Europe's prosperity depends to a large degree on its flourishing transportation sector. It is thus of great importance to not endanger the economy by introducing too strict protection for the sake of security. As highlighted earlie,r transport policy covers both mass and surface transportation of passengers as well as freight transport. Moreover, it can be divided into sea, land and air transportation. All require different security applications resulting from the different vulnerabilities and threats. Member States have a common interest in developing an international regime offering high levels of safety and security. According to the European transport White Paper the EU is a "world leader in transport sector regulation". A coherent and integrated approach to all levels of transport policy has still to be found.

Land Transportation

While since 2002 elements of EU security legislation related to aviation, maritime transport and ports have been implemented, the entire land transport supply chain has not yet been covered.[252] Already in 2003 the Commission had emphasised the need for enhanced security in land freight transport.[253] Until now, multiple sectoral policies exist at various levels of government (EU, national, regional and local).[254] The supply chain is

[251] "FP 6 VASTS (Vulnerability Assessment and Surface Transport Security)." Institute for the Protection and Security of the Citizen 16 May 2009 <http://projects-2006.jrc.ec.europa.eu/show.gx?_app.page=show-printable.html&Object.object_id=PROJECTS000000000001 73F1>.

[252] Commission of the European Communities. SEC(2006) 251. op. cit.

[253] Commission of the European Communities. COM(2006) 79 final. op. cit.

[254] Commission of the European Communities. Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of the Regions Towards a

defined as comprising all the transport and transport-related operations and processes beginning at the production site and ending at the cargo's point of destination. There are currently no rules in place for the European land transport supply chain in its entirety.
While maritime and air transportation have a great number of similarities with both markets being characterised by a limited number of operators who operate in geographically confined and defined controllable areas, the land transport supply chain as a whole has quite different dimensions. Additionally, operators in the land transport sector are not yet used to security arrangements, with which operators in the other two sectors are already acquainted. The land transportation sector is largely non-transparent with more than half a million companies involved in transport and related services ranging from major multinational companies to tiny services rooted in different cultures and business settings. Current initiatives focus on containerised intermodal transport leaving other cargo transport entities uncovered. Given these different characteristics of each of the operators it seems impossible to establish a single all-embracing security framework comparable to the air and maritime sector for the land transport supply chain.

Considering it more realistic to establish a framework of minimum security requirements, the Commission proposed in February 2006 a regulation on enhancing the entire supply chain[255], by introducing minimum European standards for freight operators. The related Communication SEC(2006)251 explicitly refers to the potential of space and particularly Galileo for transport security. Accordingly, a more flexible "policy tool box" is needed in order to counter the sustained terrorist threat. The Communication refers to the use of space technology in the form of communication and navigation in air traffic management as a positive examplethat should lead to an increased use of the related technology for road and maritime security as well. However, given protests by operators and business associations who claimed that the increased costs could not be accepted due to only minor security benefits, the Commission decided to put the proposal on hold in December 2006.[256] In December 2008 the Commission published an "Action Plan for the Deployment of Intelligent Transport Systems in Europe", a major initiative towards an Intelligent Transport System (ITS). This detailed roadmap for ITS development and deployment in Europe has recently (March 2009[257]) been welcomed by the Council. The Action Plan draws on a series of ongoing European Commission initiatives such as the Action Plan on Freight Transport Logistics, the Action Plan on Urban Mobility, Galileo deployment, the Greening Transport Package, the i2010 initiative on Intelligent Cars, eSafety, the FP 7, eCall, European Technology Platforms and their strategic research agendas, CARS 21, and aims at complementing existing work and maximising synergies.[258] It proposes a directive laying down the framework for the deployment of Intelligent Transport Systems (ITS) in the field of road transport and for interfaces with other transport modes and explicitly emphasises the use of positioning services and satellite-based infrastructures. The ITS concept can be extended to encompass other supply-chain activities such as the exchange of content-related data for regulatory or commercial purposes, using innovative technologies such as radio frequency identification (RFID) and building on applications of the EGNOS/Galileo satellite positioning system. In the future this may lead to a concept of "Intelligent Cargo", meaning that goods become self-, context- and location-aware as well as connected to a wide range of information services.[259] The success of this initiative remains to be seen.

Waterborne Transportation

Looking at the transport sector as a whole, the maritime sector seems to be the most developed. It is definitely the highest on the agenda of EU policy makers. In this regard one can identify an institutional drive coming from the European Union in the form of the previously discussed initiative for an

future Maritime Policy for the Union: A European Vision for the Oceans and Seas. COM(2006) 275 final of 7 June 2006. Brussels: European Union.
[255] Commission of the European Communities. COM(2006) 79 final. op. cit.

[256] EurActive. "Transport Security." 17 Aug. 2004. EurActive11 Mar 2009 <http://www.euractiv.com/en/transport/transport-security/article-117548>.
[257] Cf Council of the European Union. Council Conclusions on the Commission Communication "Action Plan for the Deployment of Intelligent Transport Systems in Europe". 2935th Transport, Telecommunications and Energy Council meeting of 23 April 2009. Brussels: European Union. <http://www.consilium.europa.eu/uedocs/cms_data/docs/pressdata/en/trans/106964.pdf>.
[258] Commission of the European Communities. Communication from the Commission - Action Plan for the Deployment of Intelligent Transport Systems in Europe. COM (2008) 0886 final of 20 Mar. 2009. Brussels: European Union.
[259] Ibid. Art. 6.2.

integrated maritime policy. The establishment of dedicated agencies has further added to this institutional drive. Moreover, the EU has been given an impetus by other actors like the IMO, which issued an LRIT regulation recently. Additionally, some other national coastal surveillance programmes such as the U.S. Deep Water programme, Qatar's National Border Shield and the Australian CoastWatch programme that have recently emerged might have triggered some EU policies. Recent increases in piracy incidents probably added to the perceived urgency leading to an emphasis on actions to be taken in this area.

Other than the smaller number of operators in the field of freight transport, the maritime sector is particularly complex given the large number of issues at stake. Pre-9/11 the majority of terrorist surveillance and response measures were a result of individual action at Member State level. The International Maritime Organisation (IMO) took the window of opportunity created by 9/11 and established obligatory security standards for ships and ports, which were integrated by the Commission into a binding regulation in March 2004. Only then did the European Parliament and the Council agree to measures at Community level.[260] Based on the IMO standards each ship intending to enter a Member State's port must provide 24-hour advance information concerning ship and cargo safety. Given the proximity of ports to cities, terrorist attack could easily trigger a "knock-on effect on the surrounding industry", harming people in the neighbouring population. While the European shipping industry has made substantial efforts to implement the mandatory security measures adopted in 2002 by the IMO and introduced into EU law in 2004, the terrorist threat is not decreasing and the maritime transport security situation seems to be deteriorating with piracy and armed robbery at sea increasing.[261] In 2002, Directive 303/59/EC of the European Parliament and the Council established a Community Vessel Traffic Monitoring and Information System (VTMIS) with the objective of enhancing safety. It covers all ships of 300 gross tonnage and upwards, regardless of their freight, except for: warships, fishing vessels, traditional ships and recreational craft less than 45 metres in length, and bunkers below 5 000 tons.

Operators of ships bound for a European port must notify information like ship identification, total number of persons on board, port of destination, estimated time of arrival) to the port authority at least twenty-four hours in advance, where this is feasible. Each ship built on or after 1 July 2002 calling at a European port must be fitted with an AIS as well as a voyage data recorder (VDR, "black box").[262] Cooperation with the U.S. has been agreed upon through the Container Security Initiative (CSI), launched in September 2004. Under the CSI containers heading to the U.S. are pre-selected for inspection according to specified risk assessment criteria. This aims to improve cargo security on a reciprocal basis for both partners, while ensuring equal treatment of U.S. and EU ports and operators. Furthermore, it fosters exchange of information and best practices and the establishment of common standards for risk assessment, inspection and screening methods.[263]

The European Parliament and the Council issued a Directive[264] in September 2005 on the harmonisation of a River Information Service (RIS) on inland waterways in the Community. It aims at transforming inland waterway transport into a transparent, reliable, flexible and easy to access transport mode. For the purposes of RIS for which exact positioning is required, the use of satellite positioning is recommended. RIS will provide traffic information and traffic management service through AIS and satellite navigation for high accuracy of data on ship movements thus enabling improved tracking.[265]

In 2006 the EU introduced a Green Paper entitled "Towards a future Maritime Policy for the Union: A European vision for the oceans and the seas"[266], which declared "the particular need for an all-embracing maritime policy". The Green Paper emphasised that such a maritime policy needs to provide answers to questions related to decision-making as well as the added value of an EU

[260] The Commission of the European Communities. Regulation (EC) No 725/2004. op. cit.
[261].The Commission of the European Communities. COM (2009) 8 final. op. cit.

[262] "Maritime safety: Community monitoring, control and information system for maritime traffic." Summaries of EU legislations 30 July 2009 <http://europa.eu/scadplus/leg/en/lvb/l24243.htm>.
[263] Delegation of the European Commission to the United States. op. cit.
[264] European Parliament and Council. On harmonised river information services (RIS) on inland waterways in the Community. DIRECTIVE 2005/44/EC of 7 Sept. 2005. Brussels: European Union.
[265] "Inland waterways: River information services." Summary of EU legislations. 30 July 2009 <http://europa.eu/legislation_summaries/transport/waterborne_transport/l24239_en.htm>.
[266] Commission of the European Communities. COM(2006) 275 final. op. cit.

policy in the context of existing national, local and private initiatives in the field. Given the complexity of the maritime transport sector, i.e. short sea shipping differing from deep sea shipping and its competition with land transport as well as the differing externalities, the EU launched a wide ranging study assessing trends and shipping scenarios over the period 2008-2018 with the aim of preparing a European strategy for maritime transport. Additionally, the Commission initiated a consultation process with stakeholders on the concept of a European Space for Maritime Transport without barriers, and implementing options including a related conference that took place in 2008.

The European Commission presented a Maritime Policy Action Plan[267] for an "Integrated European Maritime Policy" in October 2007, enumerating a set of actions to be taken as a first step towards the implementation of the new policy. The subsequent Commission Communication entitled "An Integrated Maritime Policy for the EU"[268], the so-called blue book, additionally emphasised the need to establish a network for maritime surveillance. It specifically mentioned vessel tracking and e-navigation including satellite monitoring and long range identification and tracking (LRIT) as an invaluable tool. In the context of the development of a roadmap facilitating maritime spatial planning in 2008, it emphasised the importance of building on, inter alia, the GMES initiative. It called upon the EU to take steps in 2008 towards a European Marine Observation and Satellite Network and to promote the multi-dimensional mapping of Member States' waters in order to improve access to high quality data building upon GEOSS and GMES.

In June 2008 the Commission presented a working document on the present status of maritime surveillance systems and practices in EU Member States with a focus on data sharing aspects.[269] In October 2008 the Commission published a non-paper on the current state of the art of surveillance, monitoring, tracking, identification and reporting systems put in place by EU Member States and EU Agencies, thereby identifying the main challenges ahead and indicating a set of next steps in view of the establishment of a work plan foreseen by the Action Plan.[270] In the Transportation Strategy of January 2009,[271] the Commission called for an integrated information management system interfacing with the e-Freight, e-Customs and the Intelligent Transport systems in order to enhance supply chain security as a whole. In this context, it particularly emphasised fusion of broadband applications with other on-board and land-based monitoring and communication systems like AIS, L RIT, SafeSeaNet[272] or CleanSeaNet, or those that are being developed, such as Galileo and GMES. It also emphasised the need to promote the creation of a platform to ensure the convergence of sea-, land- and space-based technologies, the integrity of applications and appropriate management and control of information on a "need-to-know" basis. This will improve the integration of shipping in the various logistic chains and simplify paperwork on board ships.[273]

Given the initiatives already conducted, the challenge is to establish a comprehensive framework of security measures covering all phases of the disaster and emergency cycle, thereby establishing a "genuine 'security culture'" [274] that does not compromise rapidness of trade or quality of life of seafarers and passengers. In January 2009, the Commission proposed an action plan with a view to establish a European maritime transport space without barriers.[275] At the same time the Commission also presented a comprehensive ten year strategy plan[276] to

[267] Commission of the European Communities. SEC(2007)1278. op. cit.
[268] Commission of the European Communities. COM(2007) 575 final. op. cit.
[269] European Commission/ Joint Research Centre, Ispra. op. cit.
[270] Commission of the European Communities. Non-Paper on Maritime Surveillance. SEC(2008)2337 of 13.10.08. Brussels: European Union. <http://ec.europa.eu/maritimeaffairs/pdf/maritime_surveillance_non_paper.pdf>.
[271] The Commission of the European Communities. COM (2009) 8 final. op. cit.
[272] SafeSeaNet is a European Platform for Maritime Data Exchange (4 types: ship, port, alert, hazmat) between Member States' maritime authorities in form of a network/Internet solution based on the concept of a distributed database.
[273] Commission of the European Communities. SEC(2007) 1278. op. cit.
[274] The Commission of the European Communities. COM (2009) 8 final. op. cit.
[275] Commission of the European Communities. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. Communication and Action Plan with a View to Establishing a European Maritime Transport Space without Barriers. COM(2009) 10 final of 21 Jan. 2009. Brussels: European Union.
[276] Commission of the European Communities. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. Strategic goals and recommendations for the EU's

promote safe, secure, clean and efficient shipping. In this context it explicitly mentioned the promotion of better use of information and communication technologies such as satellite broadband communications.

With national projects for improving monitoring of maritime traffic being currently brought forward, there is a need to also develop maritime situation monitoring capabilities using European space-based means. Given the fragmentation across various actors in the field, several observers are calling for more coordination. In this regard the EU seems to be the perfect candidate: having all "pieces of the jigsaw in its own box", it just needs to put them together and develop a comprehensive maritime security policy in order to create a so-called "enhanced maritime recognised picture". The same idea has also been put forward by the blue book as discussed earlier. It is of utmost important to integrate the isolated programmes and projects in progress in order to avoid unnecessary duplication. By doing so, the EU would be in a position to create a 24/7 maritime surveillance system by combining existing surveillance measures. Sensors such as coastal radars, Vessel Tracking Systems (VTS) and Automatic Identification Systems (AIS) and other satellite technology under the responsibility of the police, the intelligence services, coast guards and navies need to be pooled so as to contribute to comprehensive maritime situation awareness.[277] Some observers[278] have also suggested the establishment of a European coordinating body guaranteeing the smooth operation of trade. While the European Maritime Safety Agency (EMSA) constantly monitors the level of maritime safety in order to maintain a high level, there is no such body for the transportation sector as a whole.

Air Transportation

The recently conducted FORESEC study concluded that common airspace technologies needed to limit the risk of accidents or misunderstanding in an increasingly open sky would also improve European level cross-border counter-terrorism and organised crime efforts. Increasing investment in common airspace technologies will increase safety, but will also lead to greater protection from outside airborne threats.[279] Given the global character of the transportation sector in general, aviation is predisposed to international cooperation. In the past, however, international organisations in the field continuously faced problems because of the lackof any enforcement mechanisms.

Born in 1945 when the Chicago Convention was ratified, the International Civil Aviation Organisation (ICAO) – a UN body - is a means of securing the highest possible degree of uniformity in regulations, standards, procedures and organisation regarding civil aviation matters.[280] Its Annex 17[281] is considered the "cornerstone of aviation security" and remains the basis of today's international security regulations. Designed to balance the needs of civil aviation with political and economic considerations, it mainly includes details on requirements for a valid national aviation security programme. Through a tiered committee structure ICAO develops and publishes international standards (mandatory rules) and recommends practices that its Member States are urged to implement but cannot be enforced. Annex 17, a security manual containing operating guidelines and training programmes, should enable any State to develop its own efficient response to terrorism aimed against civil aviation. Additionally, the Tokyo, Hague and Montreal Conventions, which are also ICAO products, provide the basis for international law in respect of acts of unlawful interference.

Another attempt at international cooperation in aviation security is conducted by the International Air Traffic Association (IATA) that was first established in 1919 and subsequently reborn during the Chicago Conference in 1944. Since 1945, when the Canadian Parliament acting on behalf of the world's governments gave the association its current Charter, it has followed its mandate "to promote safe, regular and economical air transportation for the benefit of the peoples of the world, to foster air commerce and to

Maritime Transport Policy until 2018. COM (2009) 8 final of 21 Jan. 2009. Brussels: European Union.

[277] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.

[278] Cf. for example. Voort, van de Maarten, Kevin A. O'Brien, Adnan Rahman and Lorenzo Valeri. op. cit. 186.

[279] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. op. cit. 13-5.

[280] This section is to a large degree based on the related overview in Tony Ellerbeck and Mike Mason. "Analysis of the Terrorist Threat to All-Cargo Aircraft – A European Perspective." Lancaster: CdiSS, 2005.19. 30 Nov. 2006 <http://www.cdiss.org/default.aspx?fa=publicationlist>; Rodney Wallis. "The Role of International Aviation Organisations in Enhancing Security." Terrorism and Political Violence 10 (1998): 84-95; EurActive. "Transport Security." op. cit.

[281] For more detailed information on Annex 17 consider: "Annex 17 to the Convention on International Civil Aviation Security - Safeguarding International Civil Aviation against Acts of Unlawful Interference." ICAO 30 July 2009 <http ://www.icao.int/eshop/pub/anx_info/an17_info_en.pdf>.

study the problems connected therewith".[282] With the upsurge of terrorism in aviation during the 1960s, IATA established the Security Advisory Committee (SAC). Unlike ICAO, IATA has the mandate to intervene during on-going acts of unlawful interference and in post-incident investigations. This way it can on the one hand identify where and how a security breach occurred and on the other hand it is able to apply lessons learned in the future. In practice, IATA established a set of minimum criteria for securing airports against acts of terrorism and launched a complementary programme of surveys to test airport defence. While applying the same criteria for assessing all airports, it also takes into account local conditions, the political environment and the fiscal possibilities. Locations to be surveyed by an international IATA team are either nominated by airlines or airport authorities volunteer themselves. In 1988 IATA initiated the creation of an international force to take control of any future hijack and for hijackers to be tried by a new international court. Hence, IATA's role in this web of international organisations in civil aviation is that of an initiator.

When the Council of Europe in 1951 looked into the possibility of closer cooperation of European states it decided to complement IATA with the European Civil Aviation Conference (ECAC), which was working through ICAO's good offices. ECAC was officially founded in December 1955. With its mandate to "promote the continued development of a safe, efficient and sustainable European air transportation system"[283], ECAC seeks to harmonise civil aviation policies and practices among its Member States. Compared to other regional organisations ECAC is rather powerful, using its collective voice in the ICAO forum to promote policies and also having been adopted as the European Union's aviation security "standards manual". Due to the rather similar experiences of ECAC's Member States, it can adopt procedures well in advance of those established in ICAC. The combined output of the various sub-groups appears in the form of a manual of recommendations and resolutions known as DOC. No.30. While initially this was largely based on ICAO's Annex 17, policies adopted by ECAC in Doc. 30 are now being included in ICAO Annex 17. Given the lack of enforcement power, these only serve as guidelines and provide valuable data when advising developing countries on security methodology.

Within the European Union aviation security moved up the political agenda as a consequence of 9/11. Whereas before that, each Member State had its own national rules on air safety, by adopting Regulation 2320/2002 the EU launched its first common rules in 2002, based on the standards contained in ICAO Annex 17, recommendations of ECAC and various Commission proposals. Regulation 2320/2002 made baggage matching mandatory from 31 December 2003 and laid down detailed provisions on: access to sensitive areas of airports and aircrafts; passenger screening and baggage handling; control of cargo and mail; staff screening and training; and classification of weapons and other items prohibited on board planes or in airports.It also established an Aviation Security Regulatory Committee (AVSEC) to ensure technical adoption of the Annex of the regulation was introduced. It is chaired by the Commission and consists of experts representing all Member States. Regulation 2320/2002 furthermore requires the establishment of National Aviation Security Programmes and National Quality Control Programmes needed to ensure proper implementation of Community and national standards. The EU is given the authority to inspect airports to enforce compliance with this new regulation. Thus, unannounced inspections of the security procedures of every commercial airport within the EU at any time and issuing of a report listing the deficiencies has become possible.

Regulation 849/2004 amended Regulation 2320/2002 in areas where clarification and necessary corrections were needed.[284] The Annex to Regulation 2320/2002 and its implementing Regulation[285] include detailed

[282] Rodney, Wallis. op. cit. 88.
[283] Tony Ellerbeck and Mike Mason. op. cit. 20.

[284] DG Energy and Transport. "Aviation Security in the Community after Implementation of Regulation 2320/2002 of European Parliament and Council on Common Rules in the Field of Civil Aviation Security. Executive Brief." Press Kit 2004. 1-5. <http://ec.europa.eu/dgs/energy_transport /security/aviation/doc/aviation_en.pdf>.
[285] Regulation 2320/2002 was followed by Regulation 622/2003 containing the operational standards needed to ensure harmonized implementation of the standards set force in Regulation 2320/2002. The consequent Regulation 1217/2003 introduced the requirement for widespread auditing to ensure compliance with regard to the EU airport inspections. It also provides detailed requirements on how MS shall effectively monitor proper implementation of Community and national Aviation standards. Regulation 1486/2003 sets forth the standards under which the Commission shall undertake its inspections of Member States' application of Community standards in compliance with Regulation 2320/2002. Regulation 68/2004 and Regulation 1138/2004 provided additional implementing details conformed with requirements set in Regulation 2320/2002. Tony Ellerbeck and Mike Mason. op. cit.; DG Energy and Transport. op. cit.

rules in the following areas which are subject to community rules:

> Airport Security, including access control and 100 % staff screening; Aircraft Security, including aircraft inspections and protection of aircraft when in and out of service; Passenger and Cabin Baggage Screening; Hold Baggage Screening and Protection; Cargo, Courier and Express Parcels, including detailed rules on handling, screening and protection of cargo; Company Mail and Materials; Public Mail; Air Carrier Catering, Stores and Supplies; Air Carrier Cleaning, Stores and Supplies; General Aviation; Staff Recruitment and Training and Equipment Standards.[286]

This means that EU airports, service providers, catering, cleaning and cargo parties as well as all airliners both EU and foreign departing or transiting any EU airport are subject to Regulation 2320/2002. Thus it provides a comprehensive and advanced base for EU aviation security standards, thereby establishing the first supranational airport security regime in history.[287] Both the European Parliament and Council appreciated the Commission's proposal to amend Regulation 2320/2002 by introducing air marshals.[288]

In 2006 the Commission concluded that "the protection of European citizens against terrorist attacks is essentially a state responsibility"[289]. Given the enforcement power of directly applicable regulations of the European Court of Justice, Regulation 2320/2002 also establishes international (at least regional) standardsthat could not be enforced through ICAO.[290] Problematic, however, are the individual amendments by some Member States that threaten to erode the common standards.

Moreover, as a consequence of 9/11 one can observe increasing cooperation between the U.S. and the EU regarding mainly air travel as exemplified by the Passenger-Name-Record (PNR). While the Council, in order to comply with U.S. anti-terror requirements, initially agreed to the PNR thereby "forcing European airlines to pass on information about passengers on planes flying to or through the US - including credit card details, email addresses, telephone numbers and hotel or car reservations - to US security authorities", the EP appealed in September 2004 to the ECJ for the annulment of the agreement, arguing that it did not provide sufficient privacy protection for European passengers. Following the ECJ ruling on 30 May 2006,[291] declaring the Council's decision to be invalid given the "inappropriate legal basis" and giving the EU and the U.S. until 30 September 2006 to renegotiate an interim agreement which was reached on 5 October 2006. Accordingly, all data on European passengers can be passed on to all U.S. counter-terrorism agencies.[292] Supporters of the PNR argue that information sharing and intelligence gathering should be one of the most important lessons from 9/11 and that protection of privacy should not reflexively block the development of new screening tools.[293] Critics, while acknowledging the necessity of cooperation, recall the importance of not infringing on fundamental rights[294] and point to the inferior data protection of the American system.[295] The PNR Agreement between the EU and the U.S. was signed in July 2007.[296]

None of the above-mentioned international or European documents related to aviation security explicitly mentions the use of space applications in the provision of internal security in air transportation. The Hague Convention refers to "preventive security

[286] DG Energy and Transport. op. cit. 6-10.
[287] Hainmüller, Jens and Jan M. Lemnitzer. "Why do Europeans Fly Safer? The Politics of Airport Security in Europe and the U.S." Terrorism and Political Violence 15.4 (2003): 28.
[288] EurActive. "Transport Security." op. cit.. & EurActive. "Council agrees to Armed Air Marshals." 13 Oct. 2006. EurActive 7 Dec. 2006 <http://www.euractiv.com/en/transport/council-agrees-armed-air-marshals/article-158771>.
[289] This is in line with both AEA (Association of European Airlines) & ACI (Airports Council International Europe), which claim that aviation security is a national security duty and that governments should therefore help to finance the costs linked with it. EurActive. "Transport Security." op. cit.
[290] Wilkinson, Paul. "Enhancing Global Aviation Security." Terrorism and Political Violence 10 (1998). 165.
[291] Case C-318/04 (2004/ C 228/67) published in the Official Journal of the European Union of 11.9.2004.
[292] EurActive. "Transport Security." op. cit.
[293] Chertoff, Michael. "A Tool We Need to Stop the Next Airliner Plot." Washington Post 29 August 2006. 7 Dec. 2006. <www.washingtonpost.com/wp-dyn/content/article/2006/08/28/AR2006082800849_pf.html>.
[294] EurActive. "ECJ puts end to EU Air Passenger Data Transfers to US." 31 May 2006. EurActive. 1 Dec. 2006 <http://www.euractiv.com/en/security/ecj-puts-eu-air-passenger-data-transfers-us/article-155680>.
[295] EurActive. "EU-US Passenger-Data Deal 'Dodges Real Issue." 6 Oct. 2006. EurActive 6 Dec. 2006 <http://www.euractiv.com/en/transport/eu-use-passenger-data-deal-dodges-real-issue/article-158548?_print>.
[296] Council of the European Union. On the signing, on behalf of the European Union, of an Agreement between the European Union and the United States of America on the processing and transfer of Passenger Name Record (PNR) data by air carriers to the United States Department of Homeland Security (DHS) (2007 PNR Agreement). Council Decision 2007/551/CFSP/JHA of 23 July 2007. Brussels: European Union.

measures for use against acts of unlawful seizure of civil aircraft, sabotage, and armed attacks against civil aviation and its facilities" but does not identify space applications explicitly. Similarly, the Chicago Convention mentions the importance of air navigation facilities without referring to space applications.

Following a Communication[297] from the Commission in December 1999, a High Level Group was set up within DG TREN by the civil and military authorities responsible for air traffic control in Member States. EUROCONTROL contributed its technical expertise to the work. The Commission also involved social partners, relevant industrial groups (service providers, airlines, airports, equipment manufacturers) and passenger associations in this work through an Industrial and Social Group. The High Level Group report suggested, inter alia, "promoting the introduction of new technologies, with absolute priority for safety, and improving interoperability between systems and technologies"[298]. This report provided the basis for the establishment of the Single Sky I (SES I) regulation which was established on 10 March 2004 by the European Parliament and Council.[299] Its main objective was the enhancement of existing safety standards and overall efficiency for general air traffic in Europe. Its aim was to establish a harmonised regulatory framework for the creation of the single European sky by 31 December 2004. SES I mentions the use of space technology for aviation management and security, particularly in providing support for air navigation services for all phases of the flight. The single European sky regulation is in line with the Chicago convention. At the time of SES I, the greatest problem in air traffic management was congestion in the air leading to delays. Thus these topics together with safety related issues became the focus of SES I.

Single Sky II (SES II) was a follow up to Single Sky I because of various changes within the aviation management sector. Whilst safety and capacity are still major issues, the picture has become more varied with a greater emphasis on the environment and, more recently due to the fuel price crisis, on cost efficiency. Additionally, the regulatory approach has been changed due to requests from Member States and stakeholders for a less prescriptive approach ("better regulation").[300] SES II aims at the implementation of a European Air Traffic Management (ATM) Master plan in order to be more efficient in the planning of air traffic. In order to implement this master plan, the SESAR research project was established. The single European Sky II package was adopted by the European Parliament on 25 March 2009.

The SESAR project (formerly known as SESAME) is the European air traffic control infrastructure modernisation programme. SESAR[301] aims at developing the new generation air traffic management system capable of ensuring the safety and fluidity of air transportation worldwide over the next 30 years.[302] The first phase of SESAR, called the definition phase, which ended in 2008, delivered an ATM master plan defining the content, the development and the deployment plans of the next generation of ATM systems. Led by Eurocontrol and co-funded by the European Commission, the work was executed by a consortium of representatives of all air transportation stakeholders. SESAR can be seen as the technological or industrial complement to the SES legislation. Space-related examples of key SESAR technologies include: high capacity digital and voice telecommunications between ground and air, a secured system-wide telecommunication network and Active satellite navigation (GALILEO) for all flight phases (take-off/cruise/landing).[303]

[297] Commission of the European Communities. Communication from the Commission to the Council and the European Parliament - The creation of the single European sky. COM(1999)614 of 1 Dec. 1999. Brussels: European Union.

[298] "Single European Sky I." 16 Dec. 2008. European Commission Transport 30 July 2009 <http://ec.europa.eu/transport/air/single_european_sky/ses_1_en.htm>.

[299] European Parliament and Council. On the organisation and use of the airspace in the single European sky. REGULATION (EC) No 551/2004 of 10 March 2004. Brussels: European Union.

[300] "Single European Sky II." 12 May 2009. European Commission Transport 30 July 2009 <http://ec.europa.eu/transport/air/single_european_sky/ses_2_en.htm>.

[301] Council of the European Union. COUNCIL DECISION endorsing the European Air Traffic Management Master Plan of the Single European Sky ATM Research (SESAR) project. Council Decision (2009/320/EC) of 31 Mar. 2009. Brussels: European Union.

[302] "What is the SESAR project." 31 March 2009. European Commission Transport 30 July 2009 <http://ec.europa.eu/transport/air/sesar/sesar_en.htm>.

[303] DG TREN. "SESAR - The EU Air Traffic Management modernisation Programme." Presentation. Fifth Community Aeronautical Days - Aeronautics Days 2006. Vienna, Austria. 19.6. - 21.6.2006. <http://www.dglr.de/veranstaltungen/extern/aerodays2006/sessions/H_Sessions/H3/H31.pdf>; DG TREN. "The SESAR programme: Making air travel safer, cheaper and more efficient." MEMO Nov. 2005. <http://ec.europa.eu/transport/air_portal/sesame/doc/2005_11_memo_sesar_en.pdf>; Ludwig, K.-P. and J. Federhen. "Satellites in Air Traffic Management ." Presentation. Sächsisches GIS-Forum. 25 Feb. 2008.

National Responses

- Land Transportation

Among the national projects dealing with land transportation one can distinguish two categories: projects dealing with 'pure' traffic management and projects dealing with the transport of dangerous goods.

Italy is researching into employment of satellite navigation for an Italian vehicle traffic management system. While its main fields of application are related to efficient use of roads and tolling, its monitoring and control application will considerably improve first aid and assistance on roads. It does not have immediate or explicit reference to counter terrorism or organised crime but rather has the potential to be applicable to emergency management on roads. In a similar vein, Germany's Photogrammetry and Image Analysis Unit is developing a processor for interferometric radar data from the TerraSAR-X satellite for traffic monitoring. Both projects are intended to become Europe-wide applications with Italy hoping to produce European standards and Germany aiming at integrating these processors into ESA and EU GMES projects. According to its Vision 2025, the UK is working on an intelligent transport structure but details are so far not available.304

France (TransControl), Austria (EMOGES) and Italy (Safe Transport of Dangerous Goods) are working on projects related to innovative monitoring and inspection application for the safe transport of dangerous goods. Austria is particularly looking into integrating satellite navigation with complementary technologies for areas where satellite navigation systems cannot be guaranteed (e.g. high mountain roads). The Italian project offers a more all-encompassing approach by looking at the transport planning stage, the transportation phase as well as emergency management. It offers continuous monitoring and control of vehicle position as well as an alert function in case of changes in the route or changing conditions of the load.

France is conducting a more general research and development project (TOCHA) which aims at developing a continuous operational tracking tool for all kinds of tracking units (e.g. objects, persons, vehicles). It combines different localisation techniques and offers a web-service to localise the tracking unit. The Austrian GaWaLoc project is examining both the commercial and technical aspects of implementing a wagon tracking system using Galileo. France is also conducting a project called MOBISIC as part of the competitiveness cluster SYSTEM@TIC, aimed at developing and testing a mobile plug and play security system using satellite navigation for localisation to be used indoors and outdoors. While it could be deployed during any social event (political presentation, soccer match etc.), it could also be used after a bomb attack close to an underground station.

- Waterborne Transportation

As explained earlier (cf. chapter 2), maritime surveillance can be conducted with space-based imagery. An example of radar imagery is Gemany's TerraSAR-X satellite which provides day and night all-weather images. It has a resolution of up to 1 meter and a large area coverage of 100 km * 1.500 km. First deliverables are available within 1 hour after downlink; near-real time improvement is planned. For optical imagery it draws on inter alia SPOT 2,4 and 5, Formosat-2 and Kompsat-2, Ikonos, EROS and Quickbird which are all commercialised by SPOT IMAGE. Offering medium to high resolution (10 meters to 1 meter) they provide the capabilities for small vessel detection (ca. 10 meters) and large vessel classification. SPOT IMAGE and SPOT Infoterra Group control a constellation of satellites, combining optical and radar capabilities. They offer near real-time access of 2 hours (i.e. imagery acquisition, image production, intelligence extraction and transmission). The Italian COSMO-SkyMED radar imagery constellation is also contributing to maritime surveillance.

Apart from these imagery capabilities, several European countries are pursuing R&D projects aimed at integrating satellite navigation with other technologies so as to get the most all-encompassing picture. Norway is working on a national maritime programme (SatHav) for using satellite Earth observation data in marine mapping, monitoring and warning. The French project GALILEOCEAN is aimed at optimising Galileo for maritime positioning. The final outcome will be a prototype receiver using a simulation platform. While focusing on land transportation the previously mentioned Italian project on safe transport of dangerous goods includes the transportation of dangerous goods along roads and inland

<http://www.gdi-sachsen.de/skripte/gis09/GIS-Forum%20Pres_SatellitesInATM_25022009.pdf>; Eurocontrol. "SESAR in Brief." Eurocontrol, 2008. <http://www.eurocontrol.int/sesar/gallery/content/public/docs/sesar%20in%20brief_2008_final%20low.pdf>; "SESAR." EUROCONTROL 30 July 2009 <http://www.eurocontrol.int/sesar/public/subsite_homepage/homepage.html>.

[304] Gunner, Tom. Ed. "Vision 2025 – A World of Opportunities for UK Space Technology." UKSpace, 2008. 15.

waters, and tackles the intermodal connections with ports.

Additionally, Italy is conducting the project Sicurezza nel Trasporto Marittimo (Safety in Sea Traffic), which develops and tests innovative advanced satellite navigation technologies supporting efficient implementation of Sea Roads. It also aims at reducing the need forinterventions at sea for rescuing pleasure boaters through preventive actions and targeting as far as possible the search operations.

With current AIS reception being limited to the range of coastal antennas, trials have been conducted with the ESSAIM constellation to pick up the signals from space so as to provide a space-based AIS. CNES and the French Ministry of Defence began a study on maritime security based on space means in early 2008 with the objective of improving tracking of ships through space-based AIS. Following some definition studies to be performed through bilateral cooperation, the programme could be proposed as a European service (possibly of GMES).305 CNES is also working together with NASA on a radar ocean surveillance successor for TOPEX-Poseidon and Jason-1. Jason-2 is a four party Programme of EUMETSAT, NOAA, CNES and NASA.306

The British Sci-Tech System developed a system monitoring the position of people over board in realtime through GNSS. This Person Overboard (POB) system will improve search and rescue missions at sea.307 Germany and Poland (Maritime University Czczecin and LOPOS Technologies) together developed a design for an advanced IT infrastructure in the Berlin-Schwedt-Szczecin-Baltic Sea corridor in which the waterway Havel-Odra and the parallel rail and road lanes may be monitored, synchronised and controlled by an EGNOS-Galileo-based multimodal traffic system. Single vehicles will be connected interactively by wireless broadband communication with the traffic control centre. Such an Active Vessel Traffic Management Information System (A-VTMIS) allows for synchronisation of the land-based transloading capacities at the ports. Controlling with high accuracy ship's positions, sailing, speed and course and identifying these ships unambiguously in real-time enables delivery of the complete traffic image to service ships such as the police.308

In December 2004 EADS was tasked by Portugal's Instituto dos Transportes e Portuário Marítimos (IPTM) to lead a Portuguese-German consortium in the introduction of a Vessel Traffic System (VTS) to monitor traffic off the coasts of Portugal.309
CNES is currently conducting a maritime surveillance project called S MAR. The SMARTRegions (Space and Security Knowledge for Maritime Regional Development) project of the two Hanseatic cities Ventspils (Latvia) and Bremen (Germany) looks at potential mutual satellite-based technology application fields such as port/ transport/ trade (port security, movement of goods, container security), maritime security / safety (detection of ship-source pollution, ship tracking/AIS, sensitivity mapping for coastal protection) and coastal and city environment.

Commission research has shown that VMS operational data sharing in general is relatively far advanced considering integration but at the same time quite restricted in any sharing outside the fisheries sector. National and regional sharing of AIS data is developing fast.310

- Air Transportation

The FORESEC Study showed that greater investment in space-based technologies for aviation security, in terms of either intelligence assets or defensive tools was seen as a crucial asset for Europe to invest in. France and the UK in particular shared this perspective.311 Publicly available research showed very few projects that include space applications for aviation security. Belgium has conducted a research project (HYSAR) on fusing polarimetric SAR and hyperspectral data to classify man-made objects in urban and industrial scenes. Detected images include airports and could provide a basis for contingency planning.

[305] Brudieu, P. "CNES and French MoD: A Common Approach for Security and Defence in Space." Presentation. ISU Annual Symposium. International Space University, Strasbourg, France. 18-20 Feb. 2008. <http://www.isunet.edu/index2.php?option=com_docman&task=doc_view&gid=737&Itemid=26>.
[306] European Space Agency/European Commission. op. cit. 19.
[307] "Leben retten mit Hilfe von Satellitendaten." 20 Jan. 2009. European Space Agency 21 Jan. 2009 <www.esa.int/esaCP/SEMVRVVPXPF_Germany_2.html>.
[308] Stateczny, A. "An Idea of Future Oriented Active VTMIS." Geophysical Research Abstracts 7 (2005). <http://www.cosis.net/abstracts/EGU05/02443/EGU05-J-02443.pdf>.
[309] Perrimond, Guy. "EADS: A World Leader in Maritime Surveillance." TTU International 232 (2009). 1.
[310] European Commission/ Joint Research Centre, Ispra. op. cit.
[311] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. op. cit. 13-5.

European response

9/11 had probably the greatest impact on the transport sector due to the direct involvement of the aviation sector. The terrorist incidents that followed in London and Madrid added to this and showed the vulnerability of the mass transportation sector.

- Land Transportation

As the land transportation sector is underdeveloped in terms of satellite applications for transportation, security projects can be categorised as:

1. satellite applications for road transportation security;
2. satellite applications for rail transportation security and
3. mass transportation.

Category 1: Satellite Applications for Road Transportation Security

Space applications can be seen as one of the main contributors to transport security within Europe in general.[312]
FP 4 TRACAR 1 and TRACAR 2 TR 1059 and TR 4026 are aimed at encouraging users of transport to improve the management of their intermodal cargo movement by automating aspects of tracking, tracing and monitoring from end-to-end of the supply chain using the combination of tag, reader and short range radio technology with GPS positioning. While part 1 of the project involved the transport of frozen cargo by road and rail, part II dealt with road and sea demonstratorsfor both dry and frozen cargo.
Similarly, FP 4 MULTITRACK TR1039 addressed the problem of the intermodal logistics chain using powerful Geographic Information Systems. In a similar vein FP 4 MAGNET A TR 1034 and FP 4 MAGNET B TR 1035 aimed at developing GNSS 1 user segment prototypes and at assessing their performance in terms of availability, accuracy, integrity and continuity of service through field trials involving civil, land, maritime and aviation users in order to achieve a multimodal core module common to all user segments.

Currently Assisted GNSS and EGNOS are the main space applications improving land transportation security at the European level. Assisted GNSS mainly contributes to transportation security through improved

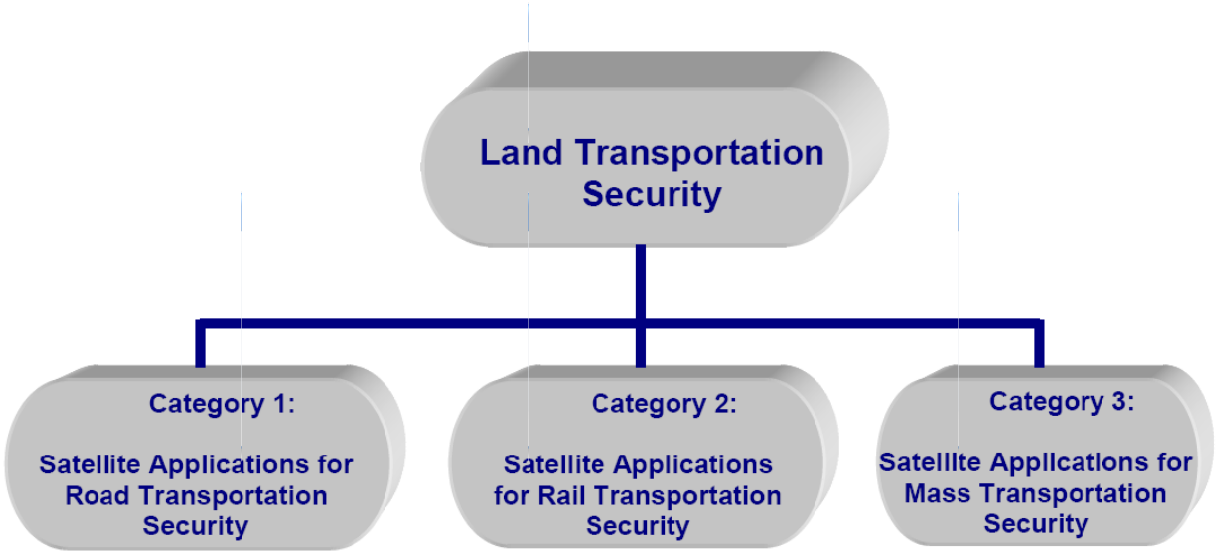


Figure 27: Categories of Projects for Land Transportation Security

[312] Gagnou, Bruno. op. cit.

traffic management. It provides exact positioning of all vehicles even in an urban environment, which improves traceability for dangerous goods transportation. Additionally, it enables faster sending of rescue teams with exact positioning. The system is resistant to fraud and jamming.[313] This situation should be considerably improved with the implementation of Galileo and the GMES service element on Land Monitoring (LMCS), addressing a wide range of resources and policies (inter alia transport).

Galileo will provide only limited improvements for some road applications through for example the Advanced Driver Assistance Systems (ADAS). While primarily improving traffic management and safety of the driver through e.g. road preview, path prediction etc. it also improves situational awareness, which could be exploited for security-related applications for road transport. It will also improve search and rescue on roads by providing continuous monitoring with a vehicle accuracy of well under 1 metre.[314]

As described earlier, GMES will improve police missions through more effective operation. In this regard it will improve planning and conduct of protection missions (e.g. transportation of politicians, VIPs etc). 3D surface elevation models coupled with relevant processing techniques could help to choose the safest route, identifying vulnerabilities in the chosen itinerary. Based on the preliminary requirements and demonstrations of services (e.g. EC FP projects) the current priorities for development appear to be: All weather imaging capability at high and medium resolution for land in cloudy regions and during night coupled with radar interferometry capability for detection of small (millimetre or sub-millimetre level) ground movements, with the appropriate frequencies and operating modes required to support the GMES services. This is needed in support of disaster management, urban management and transportation security.[315]

In the past, the OSCE organised a workshop and simultaneous exercise to demonstrate the need for global cooperation and attention to the issue of container security. Taking place in Vienna (Austria) it brought together 170 participants from across the OSCE region and some of its partners. The simulation exercise illustrated the role of different actors in the container transport chain and how they might react to unfolding events in a threat scenario.[316] While it tackled the topic of "international measures taken to enhance security and the use of technology" it did not directly involve the use of space applications.

JRC's VATDIS (Vulnerability Assessment in Transport Distribution Systems) defines surface transportation as a critical infrastructure. Focusing on the review and development of methods for the identification, protection and security assessment of European Critical Infrastructures with respect, primarily, to the threat posed by terrorism, methodological developments are supported by specific studies related to the threat associated with the surface transport of dangerous substances, how this type of transport can be misused in terrorist actions and the impact of these actions, and how on-board communication and spatial geo-reference equipment can support real-time traceability, alarm management and rapid emergency response. It develops urban vulnerability maps against terrorist attacks using freight vehicles loaded with explosives.

The FP 6 project VASTS takes a similar approach and also considers surface transportation as part of CIP. It focuses on the review, comparison and development of methods for the security management of European Critical Infrastructures with respect, primarily, to the threat posed by terrorism, sabotage and other negative intentional acts. Methodological developments have been supported by more technical applications in the Surface Transport Sector, including specific studies related to:

- the threat associated with the surface transport of dangerous substances, how this type of transport can be misused in terrorist actions and the consequence assessment of these actions, and how on-board communication and spatial geo-reference equipment (also exploiting the forthcoming European global satellite navigation system Galileo) can support real-time traceability, alarm management and rapid emergency response;

[313] Ibid.
[314] "Satellite Navigation Applications." op. cit.
[315] European Space Agency/European Commission. op. cit. 19-20.
[316] Organization for Security and Cooperation in Europe (OSCE). op. cit. also refer to Organization for Security and Cooperation in Europe (OSCE). Enhancing Container Security. DECISION No. 9/04 of 7 Dec. 2004. Sofia: OSCE. <http://www.osce.org/documents/mcs/2004/12/3912_en.pdf>.

- the development of improved functional and security requirements for a new generation of Digital Tachograph equipment and for the conceptual design of a Universal On-Board Platform for road vehicles that provides security measures along with other safety and business functions;

- the threat of terrorist actions targeting passenger transport and the technical requirements for the monitoring and surveillance of public transport infrastructure;

- the security needs at national, regional and local levels of a Trans- European Corridor, with an emphasis on consequence analysis, emergency preparedness, exploitation of satellite information and in-situ data (GMES), and the establishment of a structured dialogue with all stakeholders.[317]

Belgium, Finland and Norway are currently in the process of formulating a project including a demonstration on the role of space for land and maritime surveillance (Belgium, Finland, Norway) as part of a country-initiated EDA project.

Category 2: Satellite Applications for Rail Transportation Security

The railway network is gradually introducing the European Rail Traffic Management System (ERTMS) and Telematics Applications for Freight (TAF-TSI)[318]. The European Rail Traffic Management System (ERTMS) is becoming the European standard for train control, signalling and traffic management. Two layers of ERTMS rely on satellite navigation: the European Train Control System (ETCS), dealing with train control and protection, and the European Traffic Management Layer (ETML) dealing with non safety–of-life related aspects such as traffic management and regulation.

Communication does not utilise any space applications but relies on GSM-R, a radio system. The decision to use such a system was already taken in the early 1990s when no satellite communication was able to fulfil railway requirements for operational use such as availability and tunnel coverage.[319] The main objective of the ERTMS is to harmonise the currently existing national train control and command systems in Europe to increase European railway's competitiveness. ETML will rely on satellite navigation for positioning of trains and wagons. Improved positioning will advance the transportation security of dangerous goods transports and search and rescue missions after a disaster.[320]

Galileo will additionally improve the safety of train speed control and operating systems.[321]

The Galileo Joint Undertaking is piloting a number of projects in this area, which can be grouped in several themes:

- positioning and tracking applications (cf Galileo) for fleet management and mobile resources integrated managements
- containers: positioning and trackings
- security at terminals, warehouses and distribution centres
- protection of rail and road infrastructure, railway protection and detection of missing parts

The EUREKA project "Logchain Translog Safety" has developed a satellite-based system for monitoring the transportation of hazardous rail freight between Europe and Asia based on a combination of geostationary and orbiting satellites into a single antenna toensure a fast response mechanism during the transportation of potentially hazardous materials. The status of railway vehicles is gathered by sensors, which then transfer the data to the transceiver, and via satellite to the control centre, which can be situated anywhere based on needs. Another advantage is its independence from terrestrial GSM communication, which is not always available along railway lines.[322]

Most projects related to the security of rail transportation were covered by FP 5. They mostly deal with emerging technologies like supervision, permanent monitoring of tracks to prevent interference and misuse, and potential technologies like energy-saving drive control, rail control and monitoring. The FP 5 project LOCOPROL was aimed at developing and demonstrating a GNSS based fail-safe train location system at the core of a

[317] "FP 6 VASTS (Vulnerability Assessment and Surface Transport Security)." op. cit.
[318] A database for freight transportation data.
[319] According to Senechal, Hélène. "Statement regarding railway communication via satellite." Via email to author. 17 June 2009.
[320] ERTMS Website. 30 July 2009 <http://www.ertms.com>; "The ERTMS in 10 Questions." MEMO/05/235 of 4 July 2005. Brussels: European Union. <http://europa.eu/rapid/pressReleasesAction.do?reference=MEMO/05/235&format=PDF&aged=0&language=EN&guiLanguage=en>.
[321] "Satellite Navigation Applications." op. cit.
[322] "EUREKA project presents monitoring system for hazardous rail freight." Times Higher Education 26 July 2005. <http://www.timeshighereducation.co.uk/story.asp?storyCode=197556&sectioncode=26>.

train's protection, control and command system. It particularly focussed on three sub-systems: (1) fail-safe track mapping, (2) fail-safe interlocking functions and (3) satellite positioning of the train.

The FP 5 project GADEROS demonstrated Galileo's added value for Railway Operation Systems while FP 6 project INTEGRAIL looked at reliable train positioning using GNSS and EGNOS. GADEROS particularly researched the integration of GNSS within the European Railway Traffic Management System (ERTMS)/ European Train Control System (ETCS). INTEGRAIL particularly researched the development of an intelligent system for monitoring and control.

The ESA funded ECO-RAIL project was one of several GNSS_1 Rail User Navigation Equipment projects dealing with the implementation of satellite navigation into the railway domain to demonstrate the feasibility and benefits of GNSS. In particular, it researched the use of GNSS and EGNOS for autonomous and automated level crossings. FP 5 project Rune looked at train sensor enhancement through the application of GNSS. Building on LOCOPROL, the ESA funded LOCOLOC intends to develop an innovative, cost-effective satellite based train location system (based on GNSS) as a primary component of a train protection, control and command system thereby complementing LOCOPROL.[323]

Category 3: Satellite Applications for Mass Transportation Security

The use of Galileo for mass transportation will improve search and rescue in the aftermath of an incident. The Commission has started consultations with the public transport sector on how to factor in security constraints from the start of the planning and design stages of installations and networks.[324] Most strikingly, some sectors of the transportation sector are less covered by inter alia FP projects than others. The third call for security as part of the FP 7 will also include the topic of mass transportation (SEC-2010.2.1-1 Security of mass transportation) with the objective of "planning, (re)design, and (re)engineering of urban areas to make them less vulnerable and more resilient to security threats".

- Waterborne Transportation

Existing projects demonstrate a fragmented and uncoordinated European approach, even though EU stakeholders, EU Member States and industries participating in the debate are convinced that "inter-pillar" and "inter-agency" cooperation should take place.[325] It is possible to group projects according to the institutions conducting them, i.e. EU FP projects, ESA funded service elements or pure ESA projects, JRC projects and projects conducted by industrial consortia. Alternatively, one can group them according to the main objectives, i.e. (1) technology development or benchmarking of existing capabilities; (2) GMES or Galileo related projects and (3) projects related to

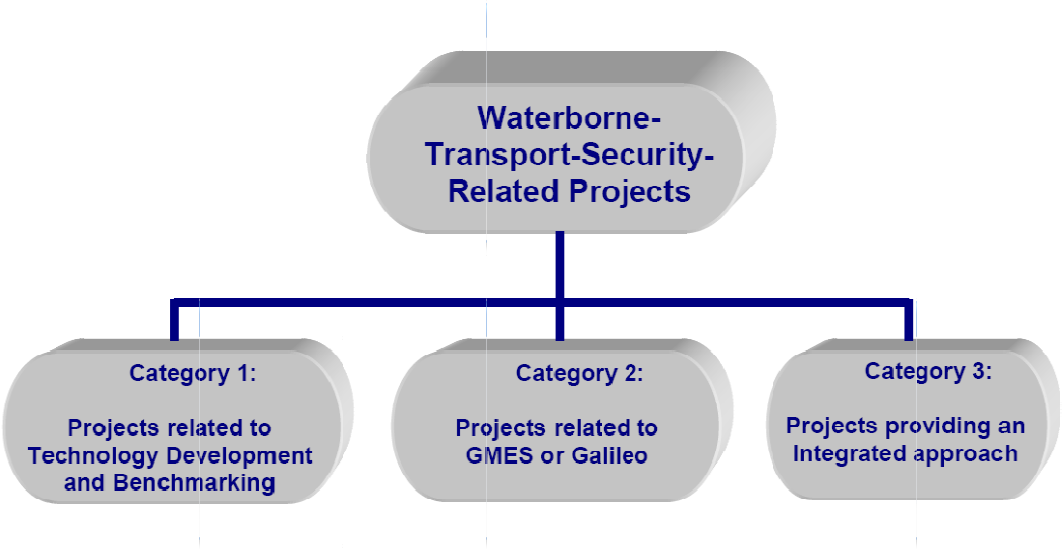


Figure 28: Categories of Projects for Waterborne Transportation Security

[323] "Navigation Projects." 8 Feb. 2006. European Space Agency 30 July 2009 <http://www.esa.int/esaNA/SEMHCYW4QWD_index_0.html >.

[324] Euractive. "Transport Security." op. cit.
[325] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.

coordination and the improvement of the entire EU approach. Most projects do not have a particular relevance for transportation but rather cover vessel tracking and maritime surveillance as a whole, so that they also are applicable to the area of sea borders.

Category 1: Projects related to Technology Development and Benchmarking

The FP 5 project DECLIMS conducted specific research with the aim of benchmarking both optical and radar sensors for vessel detection. Its results give a good overview of the current capabilities and provide a basis for future missions. EDA and NATO engaged in some of their own research and development programmes on new Earth Observation sensors and system architecture. CNES and ESA are currently conducting a study on a possible European space-based maritime surveillance system.

JRC in collaboration with the European Anti-Fraud Office (OLAF) is developing the project ConTraffic, monitoring container traffic. The ConTraffic System has become one pillar of the JRC's institutional action MASURE. Through ConTraffic, systematic gathering and analysis of data on global maritime container movements and subsequent identification of potentially suspicious consignments may become possible. The project is carried out in the framework of mutual assistance between EU customs. ConTraffic directly tackles the issue of supply chain security by giving custom authorities information on the whole transport route rather than just the start and end points. In addition to the ConTraffic project, several industrial consortia are conducting research and development projects for container and supply chain security.

In this regard Bosch is working on a transport supply chain security initiative. Like ConTraffic it covers the whole transport chain but additionally also monitors interference with the container as such through a plug and play container security box (CSB) which is connected to a 24/7 – 365 days alarm management coordination centre. Thus it is not only tracking the container through its entire route but also securing its content.

The industrial consortium project SeCureSystem (EADS astrium, EUROGAT, hellmann worldwide logistics, MSCGATE) is also aimed at developing smart and secure logistics services. In addition to offering ground-independent and worldwide position determination of containers through GPS, SeCureSystem offers permanent real-time data transfer via communications satellites enabling world-wide availability of all data and documents, electronic creation and signing of loading and customs documents by authorised persons, and goods-condition monitoring and coding of all data. ESA's ERS satellites have been used for testing and demonstrating the feasibility of using EO satellite data in different maritime policy areas. Apart from supporting development in environmental applications, the ERS mission also advanced coastal bathymetry mapping and vessel detection. Processing near-real time products from ENVISAT's Advanced Synthetic Aperture Radar (ASAR) ESA is able to create new maps revealing European shipping routes as never before.[326] EUSC is analysing satellite imagery to help deter, prevent and repress acts of piracy as part of the ATALANTA NAVFOR Operation.[327]

ESA is conducting a GMES service Element project called MARISS, which is investigating the added value from combining information derived from satellite based Earth Observation with conventional data streams such as AIS, coastal radar, maritime patrol aircraft and intelligence sources for maritime border control and maritime situation awareness.[328] It focuses on (1) near real time tactical monitoring of vessel movements and (2) strategic monitoring for anomaly detection and threat characterisation. It thus researches

- the combination of non-cooperative wide area detection (based on EO imagery) with more limited area cooperative identification systems, ensuring that any tampering (whether accidental or deliberate) with identification systems does not result in a particular vessel being invisible to maritime surveillance authorities.[329]

- the combined coverage of as wide an area as possible using satellite and airborne surveillance systems with coastal systems, meaning that limited temporal coverage of open waters by satellite and aircraft can be combined with regular temporal coverage of limited area coastal waters by AIS and coastal radar resulting in a significant

[326] "ESA map reveals European shipping routes like never before." News 22 May 2009. European Space Agency 29 May 2009 <www.esa.int/esaEO/SEMBDI0OWUF_index_2.html>.
[327] "EU NAVFOR Somalia. European Security and Defence Policy." op. cit.
[328] "European Maritime Security Services (MARISS)." 30 July 2009 <http://www.gmes-mariss.com/sections/theproject/objectives.html>.
[329] Ibid.

improvement to the overall recognised maritime picture.[330]

ESA is also exploring the feasibility of a system to pick up VHF signals emitted from ships' AIS from space. The European Commission, through the JRC, in close collaboration with the European Space Agency (ESA) is working to establish whether and how such a system could become operational, in which case it would considerably increase the added value of AIS data. Commission services are supposed to report on the feasibility project of picking up AIS signals from space by the end of 2009.[331]

ThalesAlenia Space (TAS) is already working on an AIS-S system, which will be capable of detecting an average of more than ninety percent of ships worldwide[332] (including dense maritime areas) in less than two hours.[333]

The JRC is also working on a Vessel Detection System (VDS) correlating VDS with AIS and VMS in real-time]. It is aimed at fisheries policy control only. EMSA's Vessel Monitoring System is a quite similar service based on satellite services, which however solely covers fishery. Similarly, EMSA's CleanSeaNet initiative only covers oil spill detection. EMSA through the CleanSeaNet service is cooperating with the International Charter (Space and Major Disaster) for rapid access to satellite imagery.

In 2002 the Commission initiated a quite similar project named SafeSeaNet (SSN), capable of tracking and following ships and obtaining information on their cargoes. This community vessel traffic monitoring and information system aimed at facilitating identification of vessels and the measures to be taken by the authorities after an incident or accident. Operational responsibilities were transferred from the Commission to EMSA in 2004. While SSN so far does not include space-based applications, EMSA will examine the possibility of integrating other applications and functionalities into SafeSeaNet (e.g. waste notification messages). Alternatively, SSN might become part of a larger system in support of the development of a comprehensive EU vessel traffic monitoring and information system.

EMSA cooperates closely with national administrations[334] and provides technical input to discussions within the International Maritime Organisation (IMO) as well as to the development of Galileo and its application to maritime navigation.[335] Based on Directive 2002/49 establishing a Community Vessel Traffic Monitoring and Information System, EMSA and SafeSeaNet, EMSA has together with SAAB conducted a study[336] on SafeSeaNet Tracking Information Relay and Exchange System (STIRES) aimed at facilitating relaying and exchanging information between the EU Member States, Norway and Iceland. It relies on the existing AIS systems of the EU Member States and/or regional AIS servers and the existing SSN core system, the EU LRIT Data Centre and the EMSA Geodatabase system.

Operational since 30 June 2009, the EU LRIT DC provides Member States, on request, with the LRIT information of any third country vessel bound to, or sailing within, EU waters. Drawing on LRIT messages picked up by satellites (e.g. Inmarsat and Iridium) it is possible to track any ship within a 1,000 nautical mile zone of a participating state's coastline independent of the flag of the ship. The EU LRIT DC tracking data can be used for search and rescue missions or other security related positioning requests.[337] While EMSA's current applications and services tend to operate independently, there are naturally synergies to be drawn across various applications and integration will probably proceed gradually.

The JRC has also developed an Eclipse RCP-based tool for Search Unidentified Marine Object (SUO) which is used in collaboration with national agencies to police fisheries. By law, fishing vessels are required to retrieve their location from GPS and report it to authorities every two hours. SUMO takes that data and compares it to high-resolution satellite imagery. This way it is possible to determine whether the activities on the ocean match with the positions reported. While it is primarily used for policing of fisheries, it could in the future be used against any

[330] Ibid.
[331] Commission of the European Communities. SEC(2008)2337. op. cit.
[332] Performances provided with a fleet distribution estimated for the next 15 years.
[333] Gagnou, Bruno. op. cit.

[334] Western European Union. Document A/1920. op. cit.
[335] Ibid.
[336] "SafeSeaNet Tracking Information Relay and Exchange System (STIRES)." European Maritime Safety Agency Extranet. 30 July 2009 <https://extranet.emsa.europa.eu/index.php?option=com_content&task=view&id=94&Itemid=152>.
[337] European Maritime Safety Agency. "The EU LRIT Data Centre is in Production." op. cit.

tampering thus preventing fake VMS positions reporting.[338]

Category 2: Projects related to GMES or Galileo

Directive 2005/44/EC on the River Information Service (RIS) on inland waterways in the community recommends the use of satellite navigation and positioning for vessel tracking and tracing in inland waterways transport. Moreover, the efficiency, safety and optimisation of marine transportation can benefit from satellite navigation. Galileo will thus provide benefits for safety applications, safety improvements and automatic identification systems (AISs). It can also be used for port approaches.[339] In this context, the FP5 project Nauplios was aimed at improving the Galileo services to be established in the area of maritime safety and security through improved maritime surveillance. Particularly, it demonstrated the added value of Galileo's positioning and SAR services for maritime transport surveillance and related services with regard to sea traffic management, the transportation of dangerous goods, search and rescue and piracy and terrorist incidents. Additionally, the JRC is conducting a competitive activity called MARUSE, aiming at including GNSS, EGNOS and Galileo in the maritime domain, thereby bringing together small and medium size enterprises, industry, key service providers and users.

Likewise several FP 6 projects conducted research and development for the Marine Core Service (MCS) of GMES, which seeks to establish ocean forecasting and monitoring allowing for detection and evaluation of anomalies in maritime traffic. The FP 6 project MERSEA (Marine Environment and Security for the European Area) was aimed at developing an ocean monitoring system envisioned as an operational networking acquiring data from EO satellites, in-situ ocean observing networks and weather prediction agencies.

The FP 6 project LIMES (Land and Sea Monitoring for Environment and Security) like the previously discussed Nauplios project, focuses on surveillance and protection of maritime transport for sensitive cargo. It is among the more transparent and often cited projects. Having already been referred to in the section on Early Warning, Critical Infrastructure and Land transportation, it can be seen that it actually covers most of the critical mission areas discussed in this study. LIMES is particularly interesting as it is the only project that explicitly mentions the development of a business plan indicating future developments.

The ERCS project SAFER covers, inter alia, the GMES's Marine Core Service (MCS), does however focus on humanitarian crises rather than terrorist incidents or piracy. FP 7 project MyOcean aims at providing the best possible information available on oceans and is working on the combination of space and in situ observation and their assimilation in numerical models. The JRC is also working on the FP 7 project VESPO (Vessel Surveillance and POrt security), aimed at examining scientific and technical issues related to maritime safety and security specifically with respect to maritime surveillance.

With AIS depending on satellite navigation, Galileo will improve AIS reliability thereby contributing to increased safety and vessel traceability. The high accuracy and signal availability of a combined Galileo/GPS receiver are ideal for navigation in the open sea. The integrity information contained in the Galileo signal adds confidence in the calculated position of a vessel. Galileo will be an additional means of implementing the regulations on Automatic Identification Systems (AIS) and vessel traffic management systems to increase navigation safety and collision prevention in line with IMO requirements.[340]

Category 3: Projects providing an Integrated approach

Satellite-based systems for maritime surveillance have already been developed but are successfully operational only for certain sectors (e.g. Vessel Monitoring System (VMS) for fisheries control, CleanSeaNet for oil spill detection etc.). This situation should be considerably improved when GMES becomes operational. The goal is not to create an additional surveillance system but rather to set up an interface and integrate existing systems across sectors and borders.[341] The JRC institutional action on maritime surveillance (MASURE) aims at developing and providing scientific expertise that is

[338] Joint Research Centre. "European Commission Helps National Agencies Monitor Fishing Activity using RCP." 2007. 30 July 2009 <http://www.eclipse.org/community/casestudies/jrcfinal.pdf>.
[339] "Satellite Navigation Applications." op. cit.
[340] Galileo Joint Undertaking. "Galileo Application Maritime." October 2002. 30 July 2009 <http://ec.europa.eu/transport/galileo/doc/galileo_application_maritime.pdf>; Feuerbacher, Berndt and Heinz, Stower. op. cit. 260.
[341] Commission of the European Communities. SEC(2008)2337. op. cit.

essential for policy makers particularly with regard to the desirability and feasibility of an integrated plan for EU-wide maritime surveillance (in line with the Green Paper) and fits with the GMES objectives but is not primarily aimed at technology development for GMES. MASURE started in 2007 to cover different activities related to monitoring maritime traffic, container traffic and port security aiming at developing a system which can provide a complete picture of vessel traffic. Additionally, it foresees the exploration of the use of new sensors for maritime surveillance. MASURE aims to develop tools for data fusion and integration of the difference sources of vessel traffic information. Technology transfer to other agencies is planned.

GMES will contribute significantly to the surveillance of activities at sea.[342] In the long-term GMES requires a comprehensive and complementary set of operational space missions providing permanent and continuous observing capabilities at all levels (global, regional and local). Based on the preliminary requirements and demonstrations of services (e.g. ESA GSE, EC FP 6 projects) the current priorities for development appear to be:

- A radar satellite providing high resolution imagery for continuity with ERS and ENVISATclass radars, with a medium resolution mode with the widest possible coverage for marine surveillance. Co-ordination will be sought with the Canadian Radarsat mission to increase observation frequency and geographical coverage.

- Multi-spectral optical imaging satellites at two spatial resolutions: High resolution for local and regional operational monitoring applications (continuity of SPOT and Landsat classes); Medium resolution for global applications (continuity of ENVISAT-global imaging and SPOT-Vegetation -classes), with multi-spectral capabilities and optimised for vegetation, cloud & aerosol and ocean colour.[343]

FP 6 project MarNIS, involving port authorities, ship owners, harbour masters, pilots and search and rescue authorities, focusses on improving exchange of information and aims to develop Maritime Navigation and Information Services on a pan–European basis.[344] MarNIS is directly linked with the Commission Initiative to develop a future maritime policy as put forward in the Green Paper. It is also contributing to the development of E-Maritime, focusing on improved exchange of information and provision of services and the required infrastructure to meet the requirements placed at both the authority and business level. The stakeholders may include on the one hand the ship itself, together with the ship owner, operator and agent, and on the other hand shore-based entities including maritime authorities (e.g. Search and Rescue (SAR), coastal and port), related authorities (e.g. customs and immigration) and commercial parties within the port sector.[345]

FP 7 project MyOcean, a joint EC and ESA initiative, is a first step towards the implementation of the Marine Core Service. Aiming at providing the best possible information available on oceans it is working on the combination of space and in situ observation and their assimilation in numerical models. Its main objective is to set-up an integrated pan-European capacity for ocean monitoring and forecasting. It particularly aims at integrating Member States' capacities and procedures as to avoid resource duplications and define European procedures, organisations as well as a European architecture.

Similarly, the FP 7 project OPERAMAR (An InterOPERAble Approach to European Union MARitime Security) is meant to provide the foundations for a pan-European Maritime Security Awareness by addressing the insufficient interoperability of European and national assets with a view to fuse data into a common comprehensive picture, thereby overcoming the division into national and EU initiatives. Similar to OPERAMAR but not yet as concrete, the Commission in 2007 called for the establishment of a European Marine Observation and Data Network[346] (EMODNET) integrating existing but fragmented initiatives in order to facilitate access to primary data for public authorities, maritime services, related industries and researchers. As part of this initiative the European Commission, with the support of a specially constituted advisory group, will prepare in 2009 an EU action plan to make progress in the area on the basis of

[342] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.
[343] European Space Agency/European Commission. op. cit.19-20.
[344] Commission of the European Communities. SEC(2008)2337. op. cit.
[345] "MarNIS - Maritime Navigation and Information Services." Maritime Navigation Information Service 30 July 2009 <http://www.marnis.org/home.asp>.
[346] Commission of the European Communities. SEC(2007)1278. op. cit.

a road map published in April 2009[347]. This roadmap will provide an overview of the main data and information service categories to be covered as well as their sources and uses. It also includes examples of benefits and added value of better integration. This will be one of the cornerstones of the integrated maritime policy. EMODNET will contribute to INSPIRE with respect to data sharing and marine standards. Parameters made available through EMODNET will facilitate the GMES's marine core service, through:

- collecting data once for using it many times
- developing standards across disciplines as well as within them
- processing and validating data at different levels. Structures are already developing at national level but infrastructure at sea-basin and European level is needed
- building on existing efforts where data communities have already organised themselves
- developing a decision-making process for priorities that is user-driven
- accompanying data with statements on ownership, accuracy and precision.[348]

The "proof of concept" of EMODNET is being tested through preparatory actions. The projects will identify the main challenges in moving from an ur-EMODNET to an operational EMODNET. An impact assessment to be conducted in 2009 will assess options for moving towards a definitive EMODNET, both in the intermediate period 2011-2013 and in the long term after 2014.[349] The ur-EMODNET will be operational throughout 2010 and 2011, collecting feedback from users on fitness for purpose and indicating how the definitive EMODNET might be set up. The FP 6 project SeaDataNet constructs a standardised system for managing the large and diverse data sets collected by oceanographic fleets and new automatic observation systems, thereby networking and enhancing currently existing infrastructures. It is developing and implementing a Common Data Index (CDI) service, giving users detailed insights into the availability and geographical spreading of marine data. SeaDataNet provides an efficient Pan-European distributed marine data management infrastructure, in agreement with the principles of the European INSPIRE directive and in support of the development of the European Marine Observation and Data Network as foreseen in the new European Marine Strategy. It mainly covers multidisciplinary datasets relevant to the monitoring of ocean state and health (e.g. distribution of heat and salt concentrations, sea level, currents but also data related to living marine resources and ecosystems such as nutrients and oxygen), which are of limited use for internal security purposes.[350]

Additionally, the JRC institutional action MASURE addresses maritime surveillance in an integrated manner. By taking into account the many actors and their remits it aims at developing tools for data fusion and integration in order to subsequently use the developed integrated tools for risk assessment studies. By combining existing approaches such as GIS,and the SDI initiative INSPIRE, risk assessment will be improved. The JRC has set up a European Group of Experts on satellites which conducts R&D activities relating to pollution.[351]

Among EDA's Cat B projects, a demonstration on a Maritime Surveillance Network Architecture (Germany and the Netherlands) is currently in the drafting process.

Additionally, there is an FP 6 project called DRAGONESS, with the objective of harmonising European and Chinese marine monitoring for Environment and Security Systems. It does not fall in either of the categoriesof projects identified above but should not be kept out of sight. As part of its objectives it aims at creating an inventory of Chinese and European capacities for marine monitoring for environment and security. It is related to GMES in terms of harmonising methods in the frame of international programmes.

- Air Transportation

In line with the objective of the SES I, both FP 4 AATMS TR 1001 and FP 4 EOLIA TR 1021 aim at improving air traffic management through improved communication and navigation by satellites. The latter also improves surveillance, which might be applicable to the threat of terrorism.

[347] cf. Commission of the European Communities. Commission Staff Working Document - Building a European marine knowledge infrastructure: Roadmap for a European Marine Observation and Data Network. SEC(2009) 499 final of 7 April 2009. Brussels: European Communities.
[348] Ibid.
[349] "European Marine Observation and Data Network (EMODNET)." op. cit.
[350] Pan-European infrastructure for Ocean& Marine Data Management. "SeaDataNet." 30 July 2009 <http://www.seadatanet.org/>.
[351] Ferraro, Guido and Harm Greidanus. "Maritime surveillance at JRC: MASURE action." Presentation. 19 June 2007. Joint Research Centre. 30 July 2009 <https://masure.jrc.ec.europa.eu/c/document_library/get_file?p_l_id=17836&folderId=124742&name=DLFE-19380.ppt&targetExtension=pdf>.

The FP 4 ProATN TR 1042 aims at improving aviation security by developing a prototype aeronautical telecommunication network.

Galileo will support the SESAR joint undertaking, implementing the legal framework as laid down in the SES I and SES II.[352] In line with SESAR, ESA conducted a project on satellite communication systems that best answersuser requirements and end-to-end concept of operations so as to include satellite communication infrastructure in the SESAT ATM Master Plan. This is part of its general ARTES programme (Advanced Research in Telecommunications Systems) enabling European and Canadian industry to explore, through research and development (R&D) activities, innovative concepts to produce leading-edge SATCOM products and services. Iris, element 10 of the ARTES programme, aims to develop a new Air-Ground Communication system for Air Traffic Management. It is the satellite-based solution for the Single European Sky ATM Research (SESAR) programme. By 2020 it will contribute to the modernisation of air traffic management by providing digital datalinks to cockpit crews in continental and oceanic airspace.

During the first phase ARTES-10 Iris had the objective of analysing and defining a satellite system study in order to provide ESA and SESAR through analysis with satellite communication options. This was conducted through several sub-projects like AVISAT; PHOENIX, ICOS and SAMARA. The project Final Review was completed in 2009 and the project finalised.[353] Phase Two of the programme focuses on the development of a new satellite-based communication system including a new satellite communication standard. Its performance will be validated end-to-end with a subset of the infrastructure required for the future European ATM System and Phase Three will support in-orbit verification and certification of the pre-operational system, technical support to deployment of the full system, and preliminary work leading to an enhanced future role for satellites.[354]

The FP 6 project ANASTASIA (Airborne New and Advanced Satellite Techniques & Technologies in A System Integrated Approach) aims to carry out research, evaluation and cost benefit analysis of space based technologies such as satellite communications and satellite navigation new technologies and new avionics architectures suitable for aircraft operation in the future satellite-based European Air Traffic Management environment. The future needs of surveillance will be consolidated with these requirements and key technology prototypes from communication and navigation.

No evidence could be found that Europe, like the U.S., is thinking about satellite-based plane cabin monitoring. The FP 6 project NEWSKY aims to develop an innovative networking concept to integrate different existing and emerging link technologies into a single, global network for a secure, seamless and robustly redundant communication system. In this context it also aims at integrating satellite- communication.

The FP 6 project SAFEE ("Security of Aircraft in the Future European Environment") was to develop a set of advanced on-board security functions in order to allow the crew to handle in-flight security incidents. SAFEE builds on an Onboard Threat Detection System (OTDS) aiming at detecting unauthorised access, dangerous materials, or suspicious human activities through evaluation and correlating of data from several kinds of sensors. The alert is forwarded to a Threat Assessment and Response Management System (TARMS). Output is sent to the crew and in some cases even to ground actors. When TARMS concludes that the cockpit crew is no longer in control of the aircraft it will protect the flight path by initiating an automated manoeuvre through the EAS (Emergency Avoidance System). SAFEE also deals with securing SATCOM in air flight management and secures data flows between the aircraft and the ground through detection of disrupted data due to manipulation or consistency loss.

Compared to ports, airports are not considered critical infrastructures but rather form part of the transport sector. Thus different actors deal with this area, which might provide an explanation for the lower number of projects in this sector.

4.5.3. Border Security

As a result of Europe's geography, the major component of its external borders is maritime, giving rise to extensive monitoring and surveillance requirements. Over the last decade, the threat from illegal trafficking of persons, weapons, drugs and contraband has

[352] "Satellite Navigation Applications." op. cit.
[353] "Artes Elements." 9 Mar. 2009. Telecommunications and Integrated Applications - European Space Agency 30 July 2009 <http://telecom.esa.int/telecom/www/object/index.cfm?fobjectid=28910>.
[354] "Artes 10 – IRIS Programme." 14 Apr. 2009. Telecommunications and Integrated Applications - European Space Agency 30 July 2009 <http://telecom.esa.int/telecom/www/object/index.cfm?fobjectid=28507&fcategory=226>.

steadily increased as a consequence of the EU's enlargement and the abolition of internal border controls. Effective monitoring of vessel movements in European coastal waters and nearby high seas for timely detection of illegal activity is therefore of great importance for the security of Europe's citizens and for the safety of lives in Europe's maritime domain.

In spite of the restrictive immigration policies that have been in place since the 1970s in most Member States, large numbers of legal and illegal migrants have continued to come to the EU. Similar to the other critical mission areas the main challenge in the area of border security is the establishment of integrated border management. Since the Tampere Programme of 1999, the management of the external borders has been one of the cornerstones of the establishment of the European Union as an area of freedom, security and justice. The Commission set out priorities for the development of integrated management of the external borders, focusing on:

- a common body of legislation,
- operational cooperation between Member States and
- solidarity between Member States and the Community.[355]

These goals have now broadly been reached with the entry into force of the Schengen Borders Code, the establishment of the European Agency for the Management of Operational Cooperation at the External Borders of the Member States of the European Union (FRONTEX) and the adoption of the External Borders Fund which entered into force in 2007.

Building upon the Tampere Programme, the Hague Programme of 2004 focuses on so-called second generation measures strengthening the management of the external borders. It set the objectives for strengthening freedom, security and justice in the EU for the period 2005-2010. In 2005, the European Council agreed on a Global Approach to Migration, including the management of the southern maritime external borders. In 2006 the Commission in its Communication to the Council and the European Parliament entitled "The Global Approach to Migration one year on: Towards a Comprehensive European Migration Policy"[356] proposed a two-pronged approach for the control of maritime borders, setting out complementary measures which can be implemented separately:

- Operational measures to fight illegal immigration, protect refugees and reinforce control and surveillance of the external maritime border to be implemented immediately, and
- Building on the existing relations and practical cooperation already established with the third countries, pursuing and strengthening our dialogue and cooperation with third countries on these operational measures in the context of the Association Agreements and ENP Action Plans as well as in the context of the Cotonou Agreement.[357]

Among the priorities, the Commission mentioned the need to increase mutual trust and the exchange of information among Member States. The main tools for return and readmission are voluntary return programmes and plans for enforced return.358

In February 2002, the EU Council of Ministers adopted a comprehensive action plan on illegal immigration. Building upon the Return Action Programme agreed in 2002, on 16 December 2008 the EP and the Council adopted a Directive359 laying down common standards and procedure in Member States for returning illegally staying third country nationals. Additionally, readmission agreements have been concluded with several third countries.[360]

In 2006 the Commission issued a Communication on "Reinforcing the

[355] Commission of the European Communities. Communication from the Commission to the Council and the European Parliament - Towards integrated management of the external borders of the Member States of the European Union. COM(2002) 233 of 7 May 2002. Brussels: European Union.

[356] Commission of the European Communities. Communication from the Commission to the Council and the European Parliament - The Global Approach to Migration one year on: Towards a comprehensive European migration policy. COM(2006) 735 final of 30 Nov. 2006. Brussels: European Union,

[357] Commission of the European Communities. COM (2006) 733 final. op. cit.

[358] Immigration and trafficking of human beings are considered seperately within the EU policy areas. For more information on the EU action against trafficking in human beings consider: "EU action against trafficking of human beings." European Commission Justice and Home Affairs 30 July 2009 <http://ec.europa.eu/justice_home/fsj/crime /trafficking/fsj_crime_human_trafficking_en.htm>.

[359] European Parliament and Council. On common standards and procedures in Member States for returning illegally staying third-country nationals. DIRECTIVE 2008/115/EC of 16 December 2008. Brussels: European Union.

[360] For more information cf. "Towards a common European Union immigration policy". European Commission Justice and Home Affairs. 30 July 2009 <http://ec.europa.eu/ justic_home/fsj/immigration/fsj_immigration_intro_en.htm>.

management of the European Union's maritime borders"[361], which recommends, inter alia,:

- the maximisation of the capacity of FRONTEX
- setting-up a permanent coastal patrol network based on the MEDSEA study presented by FRONTEX
- the creation of a European Surveillance System for Borders (EUROSUR).

In 2007 the Commission issued a Working Document on Maritime Surveillance Systems[362], summarising the information on existing systems at EU level thereby focusing on the data sharing aspects. It discussed systems like VMS, AIS, VTS, LRIT, several special reporting regimes, GMDSS and SSAS, explicitly pointing out what information is transmitted, when and to whom, for each of the systems under discussion. It concluded that VMS is relatively far advanced in operational data sharing between countries, but at the same time quite restricted in any sharing outside the fisheries sector. National and regional sharing of AIS data is developing fast. Europe-wide sharing of vessel traffic data is progressing under SafeSeaNet based on the Community vessel traffic monitoring and information system directive of 2002. The degree of integration and cooperation between surveillance systems and authorities in the southern EU countries in the framework of border security varies widely between almost non-existent cooperation in some countries, via different authorities using the same surveillance system, to relatively advanced integrated systems to which several authorities contribute.

On the basis of this research, the Commission in 2007 proposed the development of a European Border Surveillance System (EUROSUR) supporting the Member States in reaching full situational awareness at their external borders and in increasing the reaction capability of their law enforcement authorities. It would help prevent unauthorised border crossings, reduce the number of illegal immigrants losing their life at sea as well as contribute to the prevention of cross border crime. In this regard, the Commission mentions the significant contribution space applications offer for surveillance at sea.

The GALILEO system will provide an advanced technological platform for the development of satellite-based surveillance applications. If surveillance and monitoring systems can be made more efficient and more compatible, ultimately allowing government agencies in the Member States access to a fully integrated, single picture with complete information on ship movement and activities, all government activities concerned will benefit. This would also help the Member States to carry out search and rescue activities more efficiently, thus preventing loss of life at sea.[363] Additionally, EUROSUR would draw on radar satellites at European level, building on GMES.[364] EUROSUR could be set up in three parallel on-going phases:

> Phase 1: interlinking and streamlining existing surveillance systems and mechanisms at Member States' level
>
> Phase 2: making better use of surveillance tools at land and sea borders at EU level; by using the EU programmes for research and development, the technical performance of surveillance tools and sensors (satellites) shall be improved (taking into account developments in the framework of GMES)
>
> Phase 3: the objective of this phase is to create a common information sharing environment for the EU maritime domain by integrating all existing reporting and monitoring systems in sea areas under the jurisdiction of the Member States and in adjacent high seas into a broader network, thus allowing border control authorities to take advantage of the combined use of these various systems.[365]

Once implemented, EUROSUR would constitute a decisive step in the further gradual establishment of a common European integrated border management system.[366]

In 2008 the Commission issued a Communication on "Preparing the next step in border management in the European

[361] Commission of the European Communities. COM (2006) 733 final. op. cit.
[362] European Commission/ Joint Research Centre, Ispra. op. cit.
[363] Commission of the European Communities. SEC(2007)1278. op. cit.
[364] Chapman, John. "The Questionmarks over Europe's Maritime Security." 11 Mar 2009. Security and Defence Agenda Discussion Paper. 24 July 2009 <http://www.securitydefenceagenda.org/Portals/7/Reports/2007/Final_Discussion_Paper.pdf>.
[365] Commission of the European Communities. COM (2006) 733 final. op. cit.
[366] Commission of the European Communities. Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of the Regions Examining the creation of a European Border Surveillance System (EUROSUR). COM(2008) 68 final of 13 Feb 2008. Brussels: European Union.

Union"[367] in which it notes a series of problems in the existing border management and points to new measures in order to reach the objective of developing an integrated border management strategy. While drawing on specific technologies for border management, the Commission does not mention space applications.

Also in 2008 the Commission issued a Communication examining the creation of a European border surveillance system (EUROSUR).[368] Therein the Commission proposes to implement EUROSUR in three phases:

- Phase 1: Interconnect and rationalise border surveillance systems at national level
- Phase 2: Improve the performance of surveillance tools at EU level
- Phase 3: Creation of a common monitoring and information-sharing environment for the EU maritime domain.[369]

In 2008 the Commission published a Non-Paper on maritime surveillance.[370] Accordingly, space is an important element in an envisaged integrated network. It recognises the existence of satellite-based systems for sectoral purposes (e.g. Vessel Monitoring System (VMS) for fisheries control, CleanSeaNet for oil spill detection etc.). By giving an overview of GMES and surveillance, it also recognises the attempts of several FP 6 and FP 7 projects to develop tools for a pan-European Maritime Surveillance capacity. In this context the EU's main objective is not to create an additional surveillance system but rather to integrate maritime policy and set-up interfaces and integrate existing systems across sectors and borders.

The FORESEC study concluded that a sense of greater investment was required to improve border security. Improving border checks and strengthening Europe's borders through increased investment in both personnel capacity and technological capabilities were seen as key in countering transnational terrorism and organised crime. It has also been shown that existing institutions could be put to better value. Apart from the strong overlap with the maritime critical mission area, border security is also the area with the strongest "external dimension" of internal security.[371]

The main routes of illegal migration to the EU have been identified as the West Africa route, the Western Mediterranean route, the Central Mediterranean route and the Eastern Mediterranean route. The West African route saw a sharp decrease of illegal migrants arriving on the Canary Islands (minus twenty-six percent between 2007 and 2008) due to a combination of mainly four factors: (1) the implementation of an effective return agreement, (2) increased patrolling, (3) collaboration with local police in the countries of embarkation (Mauritania, Senegal) and (4) information campaigns in the country of origin (Senegal) warning of the dangers of illegal sea crossing.

The Western Mediterranean route showed a decrease in detections of West Africans. However, the situation with Moroccans remains stable. The Central Mediterranean route showed a large increase of migrant reports in Lampedusa (plus one hundred percent) and Malta (plus fifty percent). Migrants mainly come from East Africa (Somalia) and Tunisia, with most migrants departing from Libya. There is no displacement between the West African route (Canary Islands) and the Central Mediterranean route (Lampedusa, Malta), but a possible displacement for West African nationals from the Western Mediterranean route (Ceuta and Melilla) to the Central Mediterranean route. Along the Eastern Mediterranean route, illegal migrants transit through Turkey. They were mostly from Afghanistan and Iraq. Detections were mostly made along the maritime border separating Greece and Turkey (sometimes a distance of only a few hundred metres). It is relative less important than the previously described routes. Most detections have been reported along the EU border with Ukraine. Ukraine serves as country of origin and country of transit (two percent of all detections). Air routes for illegal migration are mostly used by nationals from Latin America.[372]

[367] Commission of the European Communities. Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of the Regions Preparing the next steps in border management in the European Union. COM(2008) 69 final of 13 Feb. 2008. Brussels: European Union.
[368] Commission of the European Communities. COM(2008) 68 final. op. cit.
[369] "European external border surveillance system (EUROSUR)." Summary of Legislations. 10 July 2009 <http://europa.eu/legislation_summaries/justice_freedom_security/free_movement_of_persons_asylum_immigration/l14579_en.htm>.
[370] Commission of the European Communities. SEC(2008)2337. op. cit.
[371] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. op. cit. 11.
[372] Kalliala, Mari. op. cit.

The current economic crisis will also have an impact on the outlook for 2009 and 2010. Migration to the EU is primarily driven by job opportunities. Thus a lower volume of job opportunities in the EU is likely to reduce the flow of illegal migrants. However legal migrants might lose their jobs and rather than return to their home countries are likely to slip into illegality and add to the number of illegal migrants in the EU. Member States are likely to react to the crisis with protectionist measures such as strengthened border controls, which is likely to increase the number of detections of illegal border crossing, despite a decreasing flow.[373]

National Responses

Spain's SEOSAT intelligence satellite will be used for land border control. It provides cloud-free land optical images for border control for Spanish government authorities for territory surveillance and commercial users. France is conducting an R&D project called MODENA on a marine environment modelling platform for simulating maritime surveillance systems. It aims at integrating numerous methods and various technologies for ocean surveillance to counter, inter alia, illegal immigration and all types of trafficking. Additionally, France's GALILEOCEAN project is contributing to border security through ensuring the optimisation of the use of Galileo for coastal surveillance. It is also contributing to search and rescue operations at sea. Offering maritime surveillance capabilities, the Italian COSMO-SkyMed constellation, Germany's TerraSAR-X and SPOT 2,4 and 5, Formosat-2 and Kompsat-2, Ikonos, EROS and Quickbird (which are all commercialised by Spot Image) all contribute to monitoring of sea borders. TerraSAR-X also has the potential to be used for AIS. TerraSAR-X, ENVISAT and ALOS provide coastal surveillance capabilities. The Canadian RADARSAT-2 SAR-satellite, commercialised by SpotImage is also used for maritime surveillance in the context of border security. The future Sentinel satellites will be employed for a similar purpose as part of GMES.

National actors have also shown interest in the Land and Sea Integrated Monitoring for European Security. In the context of the FP 6 project LIMES, the French Navy was involved in a demonstration that took place in the Caribbean in June 2008 with the objective of fighting drug smuggling. Other navies (Spain, Italy) are together with FRONTEX involved in activities that will be demonstrated in the Atlantic, the North Sea and the Mediterranean Sea. The purpose of the French exercise was to provide satellite imagery (optical and radar) of vessel traffic in distant maritime areas. It focused on three main requirements specific to the job at hand:

- observe a large area on a regular basis and generate subsequent situation of vessels at sea
- observe hot spots on reduced timescales
- sort through a broad range of observation targets, and quickly process and deliver vessel detection information.

The French Navy was the end-user. As many as five satellites were used. Imagery was of two types – very high resolution for coastal and island zones, and high resolution for larger, high sea areas.[374] The service providers produced intelligence and provided vessel traffic monitoring with the aim of delivering final products to support the monitoring, detecting and tracking of suspicious ships.[375] User-feedback showed that two hours after the satellite overpass, pre-processes imagery could already be provided. However, limitations as to too low revisit time and optical imagery in cloudy conditions as well as ships smaller than 30 meters not being visible were underlined. The French Navy also stressed that a 2-hour delay required for processing and transmission can be significant if you need to locate a ship in order to prepare an immediate intervention at sea.

As many Member States are upgrading their surveillance infrastructure and deployment of AIS and coastal surveillance radar, there is growing interest in integrating additional surveillance capabilities at the EU level.[376]

European Response

Given the different actors engaging in the field of border security, one can see an emphasis on institutional actions by agencies themselves rather than only FP projects. One can categorise these in three blocks:

[373] Ibid.

[374] "PORTRAITS OF GMES USERS - Intergovernmental Panel on Climate Change MEDSUN service
French Navy in The Caribbean Polar swell alert." Window on GMES. Paris: Graham Deane, Infoterra Ltd, Co-ordinator of the *BOSS4GMES* project. 24. <http://www.moon-oceanforecasting.eu/files/BOSS4GMS3-UK-BAT-final.pdf>.

[375] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit..

[376] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. op. cit. 11.

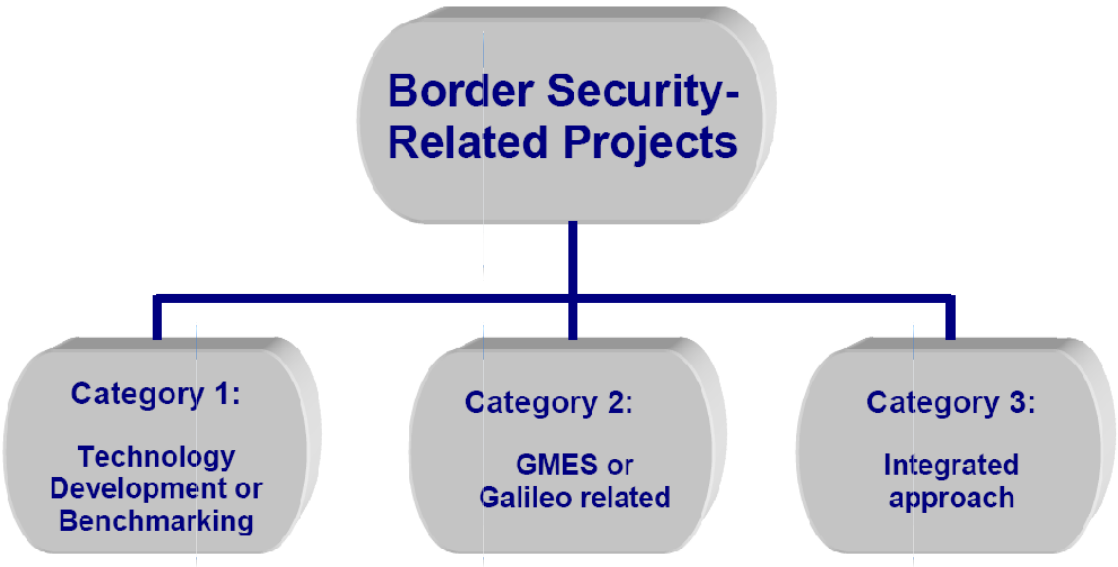


Figure 29: Categories of Projects for Border Security

- projects dealing with technological development
- GMES related projects
- Projects countering fragmentation

Category 1: Projects dealing with Technological Development

The FP 6 project GMOSS, which has already been partially described in the section on critical infrastructures, also conducted an analysis of the potential of remote sensing data for monitoring of borders and the migration of peoples, as well as benchmarking of recent data analysis and visualisation concepts and techniques.

The JRC is developing ConTraffic, which systematically gathers and analyses data on global maritime container movements in the framework of mutual assistance between EU customs authorities. Additionally, in the context of the FP 6 project LIMES, the JRC conducted a thirty day long demonstration in Caribbean waters together with EADS Astrium, EUSC, SPOT Image, Nev@antropic and the French Navy on data provision in an ongoing anti-drug and counter piracy operation of the French Navy.[377] In this demonstration Astrium conducted pre-operational information gathering from different sources for improved intelligence.

It was concluded that satellites allow threat anticipation for vessel route forecasting. There are two initiatives in Europe (by ESA and CNES) for the development of AIS-S. The ESA project (on-going phase-A) is led by OHB. The CNES initiative is more ambitious aiming at the development of a constellation of MSS (Maritime Security and Safety) with the AIS being one of the sensors. There are

two parallel contracts for this phase A from CNES, with ThalesAlenia Space (TAS) leading one of them.

In parallel, TAS is following several comparatively small contracts funded by CNES on this topic in order to study further the problem of data collision for really dense areas, or to support the French position towards regulatory bodies. AIS-S for Europe will provide a promising system for improving maritime surveillance for cooperative and semi-cooperative ships and thus improve border protection and terrorism prevention as well as supporting search and rescue operations.[378] The JRC has adopted a consultative role in maritime border surveillance. Thus it is supporting DG MARE's pilot projects and preparatory actions (.e.g AIS-S with ESA), DG JLS' industrial study on national border surveillance systems known as EUROSURE, DG ENTR research in GMES for border security, FRONTEX EPN, VDS test campaigns, EDA (liaison with PT MARSUR), and DG TREN technical aspects for port security.

The FP 7 project SOBCAH will provide an efficient, real-time, user-friendly, highly automated surveillance system to be deployed in one of the larger European ports, i.e. an area exposed to terrorist attacks, covering all kinds of border security threats.

The FP 7 project VESPO[379] aims at improving vessel detection through the development of algorithms in satellite radar images for

[377] Greidanus, Harm. "JRC R&D activities related to border surveillance." FP 7 Demo Program Workshop, Brussels, 12 March 2009.

[378] Gagnou, Bruno. op. cit.

[379] "Institutional Action -Vessel Surveillance and Port Security (VESPO)." op. cit.

improved port security and might in this way also advance border security.

Category 2: GMES or Galileo related Projects

Several FP 6 and FP 7 projects have and are aimed at developing tools for pan-European maritime surveillance. In the context of GMES, LIMES and MARISS focus on the use of technology for detection and deterrence of illegal activities. Their primary objective is to show the added value of earth observation (both optical and radar) from space combined with other technologies for monitoring vessels.[380] Within its maritime surveillance cluster, FP 6 project LIMES aims at supplementing existing surveillance measures like coastal radar and airborne and ship-based surveillance with satellite sensors. Its main objective is to monitor both vessels and cargo movements for reasons of maritime safety, policing and border security.

LIMES land surveillance cluster has also conducted land border monitoring together with FRONTEX. While the derived EO imagery was able to show migration routes, user involvement proved to be difficult. Given the effective field brigades at the Eastern European Borders, who were mainly educated and formed by the local armies, introduction to EO as an effective tool proved difficult. It was also shown that RS for land border surveillance is mainly useful for planning but does not allow for immediate so-called onspot detection of illegal crossings.

ESA is additionally funding MARISS (MARitime Security Service project), a GMES service element for vessel detection, a Vessel Tracking System and an Automatic Identification System with Earth Observation satellite data.[381] While MARISS explicitly mentions collaboration with other FP projects in the field of maritime security, the difference between MARISS and LIMES is not that obvious. Both seem to conduct vessel tracking. LIMES is focussed on improving the performance characterisation of the different satellite radar systems and their various operating modes with respect to the detection and identification of types of vessel and the impact of different environmental and viewing conditions. Thus newly-derived algorithms can provide an improved basis for integrating the different data streams within any future follow-on activity to MARISS.

LIMES is not only tackling maritime border security but also developing services for land border management together with FRONTEX and relevant border guards. In LIMES, Astrium deployed an operational satellite information platform for combating drug trafficking over Caribbean waters with the objective of anticipating a next-generation integrated system. It provided a high value added service from satellite tasking down to ship detection reports.

Similarly to the intelligence area, border security is a prime example of the overlapping character of both internal and external security. Thus, one can identify projects explicitly aimed at external security but which are also of added value for internal security. One of these is G-MOSAIC. The FP 7 project G-MOSAIC primarily provides the EU with intelligence data to be applied in crisis management and rapid interventions but also for early warning and crisis prevention. In this context it offers monitoring of border crossing and related infrastructure along the border as part of its service chain.

Category 3: Projects related to an Integrated Approach to Border Security

FRONTEX is developing a permanent regional border security concept, the European Patrols Network (EPN). It is based on the preceding studies MEDSEA and BORTEC, which explored the feasibility of the establishment of a Mediterranean Coastal Patrol Network (MEDSEA) and the technical feasibility of establishing a surveillance system covering the whole southern maritime border of the EU and the Mediterranean Sea. MEDSEA had already called for the implementation of a global approach to migration through the development of a permanent Coastal Patrol Network. It also has the advantage of offering Member States the possibility of coordinating their patrol schedules, pooling their civilian and military assets and exchanging strategic and tactical information in real time.[382] The integration of the EPN into the European Surveillance System (EUROSUR) is also foreseen. Both GMES and Galileo will contribute to EUROSUR.[383]

In a similar vein, the FP 7 project GLObal Border Environment (GLOBE) identifies the political and legal situation with regards to border security. EADS Defence & Security is developing an intelligent system for border

[380] Commission of the European Communities. SEC(2008)2337. op. cit.
[381] Mastracci, Federica. "European Maritime Security Services." MARISS NEWSLETTER # 1. 19 July 2006.

[382] Commission of the European Communities. COM (2006) 733 final. op. cit.
[383] Chapman, John. "The Questionmarks over Europe's Maritime Security." op. cit.; Commission of the European Communities. SEC(2007)1278. op. cit.

security which integrates cameras, radar, UAVs and satellites for border surveillance and security. Building on EADS DS-System IMARSEC (Integrated MARitime SECurity System), a system for safety and security to protect seaports, coastlines, territorial waters and the Economic Exclusive Zone (EEZ), the intelligent system protects sea and land borders and includes video surveillance, detection of intruders, secure communication, a database and computer network which integrates all the different facilities. It has already been used for building a "National Security Shield" in Qatar, particularly for its oil drilling stations and refineries.[384]

FRONTEX, together with EUSC, is working on a training programme related to GIS called FronBac in order to complete and enhance the concept of interoperability in the field of risk analysis. In 2007 the Member States decided to progress in developing and rolling-out analytical standards at FRONTEX and in the Member States[385]. Elaboration of structural analytical guidelines as well as an advanced programme of training have been identified as a first short-term objective. Through an overarching analytical effort of this kind, bringing together and training all relevant analysts from Member States and Frontex, the objective is to setup a European analytical community in the field of border security. FronBac contains a three year training plan aimed at setting up the analytical community and jointly reinforcing the analytical capacities of analysis units in the Member States and in FRONTEX.[386]

4.6. Preparedness for Emergency Management

The Community Action Programme in the field of Civil Protection (2000-2004) was established by a Council Decision of December 1999 (1999/847/EC), based on the Commission proposal of December 1998 (COM(1998) 768 final) with the intention to support and supplement Member States' efforts at national, regional and local levels for the protection of persons, property and, in so doing, the environment, in the event of natural and technological disasters. The aim was to facilitate co-operation, exchange of experience and mutual assistance between Member States in this field in the period 01/01/2000 to 31/12/2004.

The Feira European Council of June 2000 identified Civil Protection as one of the four priority areas within the civilian aspects of crisis management where the EU will undertaketo strengthen its capability. At the Göteborg European Council in June 2001, the Community Mechanism for the co-ordination of civil protection intervention in the event of emergencies was established on the basis of the Commission proposal, COM(2000)593 final, which supplemented the existing Community Action Programme in the field of Civil Protection (2000-2004). The Community mechanism will facilitate reinforced co-operation in civil protection assistance interventions (2001/792/EC, Euratom). It is intended to help ensure better protection, primarily of people but also of the environment and property, including cultural heritage, in the event of major emergencies, i. e. natural, technological, radiological or environmental accidents occurring inside or outside the Community, including accidental marine pollution. While not explicitly mentioning a terrorist attack as part of major emergencies, it explicitly refers to the use of new technologies, including the use of satellite technology.[387]

The Communication on a Community approach on the prevention of natural and man-made disasters aims to identify possible measures to be included in a community strategy for prevention of such disasters by building upon and linking already existing tools.[388] Proposed action at Community level focuses on areas where a common approach is more effective than separate national approaches, such as developing knowledge, linking actors and policies, and improving the performance of existing Community disaster prevention instruments. Amongst the concrete actions to be taken, the Commission suggested carrying out an assessment of current practices of hazard and risk mapping in the Member States. On this basis guidelines on hazard and risk mapping will be developed, building upon existing community instruments such as GMES. Additionally the EC called for supporting investments in research on the progress of interoperability of monitoring systems.

[384] EADS. "The Step Beyond – Sicherheit in einer veränderten Welt." Issue 1.2009. 50-1.
[385] FRONTEX. "General Report 2008." 30 July 2009. <http://www.eipa.eu/files/File/Migration/Frontex%20Annual%20Report%202008.pdf>.
[386] Ibid.
[387] The GMES Working Group on Security. op. cit.
[388] Commission of the European Communities. Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of the Regions. Community approach on the prevention of natural and man-made disasters. COM(2009)82 of 23 Feb 2009. Brussels: European Union.

The International Charter on Space and Major Disasters (hereinafter referred to as: The Charter)[389], initiated by ESA and CNES together with the Canadian Space Agency (CSA) following the UNISPACE III conference of July 1999 (operational since 1 November 2000),[390] provides for a requesting party to obtain easy and free of charge access to satellite data (Earth Observation resources), with top priority in satellite tasking. Representing a joint effort by global space agencies to put resources at the service of rescue authorities responding to major natural or man-made disasters, the civil protection authorities of the affected country (or the EU Civil Protection Unit) can invoke The Charter in case of major crises. The Charter's triggering is however limited to crises linked to natural or technological disasters.[391] Thus, in addition to technical limitations like insufficient revisit times and the limited coverage and a lack of services for data interpretation, The Charter cannot be triggered in the aftermath of a terrorist attack. Providing a good starting point, The Charter needs to be extended so as to include other forms of disasters, needing satellite imagery for emergency management. In the long run, The Charter could be replaced by a more user-driven system using a GMES infrastructure.[392]

In order to manage a disaster during crisis time, authorities have immediate information requirements such as:

- location and extent of affected zones
- human and material impact evaluation
- dimensioning rescue means
- deploying and establishing rescue means on the field

in which rapid mapping using satellite data can be a major response element. One can observe an increasing utilisation of satellite imagery for emergency response and crisis management by looking at the increased number of International Charter activations.

Users have highlighted the importance of having a first map assessing the impact of the crisis within less than 24 hours of the incident. Moreover, they require intra-daily revisits to provide updates every six hours. It has already been highlighted by the JRC supported by EUSC that Europe requires a European independent VHR data source specifically dedicated to security applications. Image products should contain both SAR and optical images as well as detailed maps and overview maps. In order to allow for detailed assessments, a resolution of less than one meter is necessary. Currently, accessibility to highest resolution imagery is not guaranteed for three reasons: (1) shutter control (e.g. Ikonos, Worldview, GlobalEye), (2) lower programming priorities on dual use missions (e.g. Cosmo-Skymed & Pleiades) and (3) military images cannot be distributed to users (Helios-2, SAR-Lupe). However rapid response does not only include EO data but also communication systems to rapidly transmit required data and ensure communication between emergency management agencies.

Similar to The Charter, the UN adopted Resolution 61/110 of 14 December 2006 on the UN Platform for Space-based Information for Disaster Management and Emergency Response (UN-SPIDER), with the objective to "[e]nsure that all countries and international and regional organisations have access to and develop the capacity to use all types of space-based information to support the full disaster management cycle" [393]. In comparison to many other initiatives, UN-SPIDER aims to ensure access to and use of such solutions during all phases of the disaster including in particular the risk reduction phase. Consequently, it focuses on "being a gateway to space information for disaster management support, by serving as a bridge to connect the disaster management and space communities and by being a facilitator of capacity-building and institutional strengthening". The UN-SPIDER staff monitors the development of additional value-added products and establishes contacts where needed.[394] Easy accessibility and dissemination of all types of space-based information to the end user is of particular importance. UN-SPIDER also identified awareness-raising and outreach activities as future activities.[395]

[389] Cf. Rathgeber, Wolfgang. "Space Policies, Issues and Trends 2008/2009." Report 18. Vienna: European Space Policy Institute. 103.
[390] "Charter activation brings space dimension to European Emergency Exercise." 12 Apr. 2005. European Space Agency 9 Feb. 2009 <www.esa.int/esaEO/SEM041W797E_environment_2.html>.
[391] International Charter Space and Major Disasters. op. cit.
[392] The GMES Working Group on Security. op. cit.
[393] United Nations Office for Outer Space Affairs. "United Nations Platform for Space-Based Informaiton for Disaster Management and Emergency Response – UN SPIDER." United Nations Office for Outer Space Affairs 26 Feb. 2009 <www.oosa.unvienna.org/oosa/eb/unspider/about_us.html>
[394] United Nations' Committee on the Peaceful Uses of Outer Space. Report on activities carried out in 2008 in the framework of the United Nations Platform for Space-based Information for Disaster Management and Emergency Response. A/Ac.105/929 of 14 Jan 2009. Vienna: United Nations.
[395] Ibid.

The Global Maritime Distress and Safety System (GMDSS) is a system intended to enable communication from a ship in distress.[396] Ships can alert authorities on shore as well as other ships in the vicinity in case of an emergency. Ships can also receive such alert messages, plus SAR information and navigational and weather warning messages. The COSPAS-SARSAT program, developed by Canada, France, the US and the former USSR, is an international satellite system intended to react to distress calls – from land, sea and air. COSPAS is operated by Russia, and SARSAT (Search And Rescue Satellite-Aided Tracking) is operated by Canada, France and the U.S., but they work as one system. One of its four parts are satellites, which function like ears in space (the others are emergency radio beacons, which call for help; ground stations, which receive the message; and control centres, which sound the alarm). The polar orbiting satellites (100 min period) are able to receive the signals from the beacons and relay them to ground stations. It may take some hours before a satellite passes over a beacon after it has been activated. If a ground station is in sight, a message received by a satellite is downlinked immediately, otherwise it is stored and downlinked later. The ground stations, in turn, process the signals to determine where the beacon is located within a radius of 2 km. The ground stations then relay this information to search and rescue authorities.

National Responses

None of the projects and satellites that are mentioned in the following is explicitly missioned to be used in the aftermath of a terrorist incident. However, satellites or projects dealing with emergency management regardless of whether they were primarily designed for a natural disaster should be applicable to disaster management for terrorism as well.

At the national level there are several existing or planned programmes including both communication systems such as Syracuse 3 (F), Skynet 5 (UK), Spainsat (E), GMILSATCOM (D) and Sicral (I), and earth observation systems such as Helios 2 and Pleiades (F), Cosmo- SkyMed (I), SAR-Lupe (D) and TerraSAR (D, UK), some of which are managed under Public-Private-Partnership (PPP) schemes (e.g. Skynet 5 (UK), Spainsat or XTAR-EUR).[397] The RapidEye constellation launched last year offers a maximum resolution of 6.5 meters and is applied to emergency management.

Many countries have specialised centres for satellite-based crisis information which can trigger the International Charter like, for example, the German Center for satellite-based Crisis Information (ZKI), offering technical expertise in data reception, data archiving and data management and applications competence in environment and geo-information as well as crisis information. The provided data can then be used by decision makers, situation centres, relief organisations and/or the public. ZKI uses data from IKONOS-2, TerraSAR-X, CartoSAT, Quickbird and others. Similarly, Austria conducted a project on the development of an Austrian Earth Observation Centre for Crisis Data (EO-KDZ / EO-CCD). Its main objective was the definition of requirements for such a centre and of the conceptual design of the centre and its components. Additionally it aimed at evaluating different earth observation satellites regarding their implementation facilities for disaster monitoring and crisis management and at setting-up a catalogue with satellite-specific data acquisition guidelines to be used in case of activation of the crisis data-centre. It also evaluated mobile mapping data regarding the implementation facilities for disaster monitoring and crisis management.

UK's TopSat satellite is missioned to provide satellite images with a maximum resolution of 2.8 meters in panchromatic modus free of charge to relief agencies responding to disasters anywhere in the world. Its pictures are good enough for individual houses or vehicles to be picked out. Though it is a technology demonstrator, its final objective is to provide imagery for the British national surveillance programme.

The Spanish SEOSAT will provide cloud-free land optical imagery for emergency management. Among its services it also offers thematic mapping. The Belgium research project HYSAR focussing on fusing polarimetric SAR and hyperspectral data to classify man-made objects in urban and industrial scenes has provided for useful insights for the acquiring of geospatial information of the area surrounding the disaster area.

France's project GALILEOCEAN, aiming at ensuring the optimisation of Galileo for use in terrestrial as well as maritime positioning, will improve search and rescue services. France is also conducting a project called MOBISIC as part of the competitiveness cluster

[396] European Commission/ Joint Research Centre, Ispra. op. cit.
[397] European Space Agency. Annual Analysis 2004. op. cit. 51.

SYSTEM@TIC, aimed at developing and testing a mobile plug and play security system using satellite navigation for localisation to be used indoors and outdoors. While it could be deployed during any social event (political meeting, soccer match etc.), it could also be used after a bomb attack close to an underground station.

Austria has conducted the GUSTAV project, developing a system for mobile documentation and online-management of catastrophic events based on satellite navigation applications. It mainly provides for spatial location by using GPS for event documentation (detection of the event, the damage) and for user information (fire brigades, police, civil services).

The Galileo Coordination Office of the Italian Prime Minister's Cabinet together with the Italian Space Agency (ASI) propose to integrate Earth Observation (Cosmo-SkyMed), Satellite Navigation (Galileo) and Satellite Communication (Athena-Fidus) in order to provide national and European Institutions with Public Regulated Services (PRS) for a rapid response to crises, disasters and emergencies in a project called SYNTHESIS. It particularly builds on the assumption that in case of damaged ground infrastructures space infrastructures might be the only alternative to guarantee continuous government action in a crisis situation protecting national interests and citizens. SYNTHESIS will provide the competent authorities real time space positioning/navigation, communication and Earth-Observation data, on the basis of EU security requirements for civil protection and civil defence purposes.

Italy additionally is looking at the development, integration and testing of satellite solutions for applications and communications services dedicated to security and emergency management through its EMERSAT project. The multinational Disaster Monitoring Constellation (DMC) consisting of six satellites is participating in the International Charter. The two British satellites provide data through the ESA project RESPOND.

European Responses

Whilst not part of EU policy, several Member States with an interest in strategic imagery have signed a joint document on the "Common Operational Requirements for a European Global Earth Observation System by Satellites". This document aims to explain why and how at European level, there could be an eventual autonomous European capacity in strategic imagery aimed at supporting all the information requirements necessary to undertake such an initiative.[398]

Several FP 6 and FP 7 projects tackle the issue of emergency response in the context of GMES. None of them explicitly mentions applicability in the aftermath of a terrorist attach. The main objectives of all these initiatives are to increase effectiveness of the humanitarian and security community through up-to-date and reliable geographic information, thereby covering all parts of the crisis cycle and delivering tailored solutions and services (e.g. assessment of events and impacts).[399] One can identify five categories of projects:

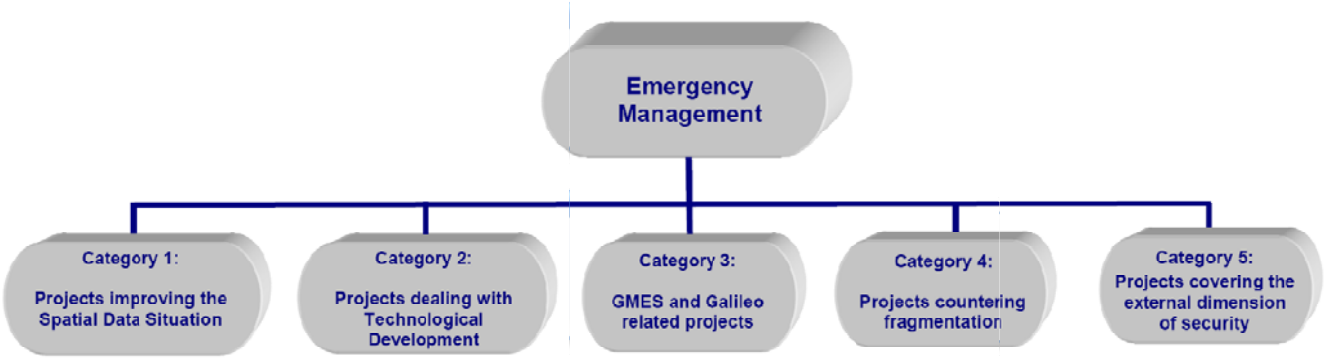


Figure 30: Categories of Projects for Emergency Management

[398] The GMES Working Group on Security. op. cit.
[399] Tobehn, Carsten. "OHB Systems complementary (very) high resolution satellites for emergency response." Presentation. ISU Annual Symposium. International Space University, Strasbourg, France. 18-20 Feb. 2008.

- projects improving the spatial data situation in the third phase
- projects dealing with technological development
- GMES related projects
- Projects countering fragmentation
- Project primarily tackling the external security dimension with implications for the internal one

Category 1: Projects improving the Spatial Data Situation

The FP 6 project OASIS (Open Advanced System for dISaster and emergency management) aiming at improving the spatial data situation in the response phase, can be seen as ORCHESTRA's counterpart for the last phase. In this regard OASIS focuses on the definition and development of an IT framework based on an open and flexible architecture using standards as a basis for a European Emergency Management system. OASIS will be linked with other EU projects focussing on the development of similar structures for other disaster and emergency cycle phases.

Category 2: Technological Development

ESA is currently examining the OHB System proposal to introduce "Very High Resolution Systems for Emergency Response and Security" focussing on highest resolutions of 0.5 metres over swath width, which can be provided by cost efficient SAR reflectors. In this context OHB Systems is currently looking at the development of a VHR SAR constellation as a stand-alone mission or in cooperation with Cosmo-Skymed and a VHR OPT Constellation stand-alone or with Pleiades both aimed at providing dedicated one metre resolution with fast response time. The FP 6 project GMOSS did research on the specific science and technology needed to provide rapid remote assessments of damage.

The FP 6 project PREVIEW is researching into crisis support services allowing for effective rescue operations through rapid mapping drawing on satellite observation in combination with other data. Through risk maps and early warning systems it aims at better anticipating short-term risks. Rapid mapping will support crisis management and rescue operations. While PREVIEW mentions its applicability to man-made disasters, its demonstrations have so far focussed on natural disasters like floods and tsunamis. ISFEREA is following a project on 3D urban structure information extraction from very high resolution SAR imagery used in emergency response, providing information on residential areas, city centres, industrial parks etc.. Moreover, ESA together with the World Health Organisation is conducting a project called SAFE on the use of satellites for epidemiology (i.e. health early warning). This could be used to counter epidemics in the aftermath of a terrorist incident.

Lockheed Martin SAR systems provide shore-based infrastructure including emergency communications, command and control, and rescue services called the MTM200™ system to save lives and property at risk in the maritime environment. This system collects and processes distress call information and becomes a platform for computer-aided search planning and modelling to quickly locate a lost or distressed vessel. It receives maritime distress and emergency response alerts, coordinates SAR personnel and assets (including aircraft, rescue boats, fire-fighting and de-watering equipment), communicates with mariners and provides command and control for operational missions in coastal and inland waterways.

JRC's ISFEREA is providing geospatial information and analysis in support of crisis management e.g. damage assessment and reconstruction monitoring. According to their experience, challenges persist in: large area coverage with high and very high resolution (optical) data of unpredictable data quality (multi-temporal, different sensors, processing), with regard to time constraints and in labour-intensive manual interpretation.[400]

ESA's GiANUS project would also considerably improve Europe's responsive capabilities. Primary research has shown that requirements are

- Telecommunications for initial actions in <24 hours
- Telecommunications for follow-on >> 100 Mbps
- EO constellation using improved sensors (SAR-MTI, hyperspectral, IR, MarSur,…) for reponsiveness in buildings
- Maritime Surveillance (AIS), advanced alert, interference mitigation and complementary with MUSIS.[401]

The GIANUS initiative aims to develop an integrated space infrastructure as part of a European security and crisis management

[400] Molch, Katrin. "Geospatial Data Analysis for Security Applications at the JRC – Challens and Opportunities for Information Mining." Presentation. Frascati, Italy. 4 March 2008. <http://earth.esa.int/rtd/Events/ESA-EUSC_2008/Presentation/Pr18_Molch.ppt>.
[401] Duhamel, Erwin. op. cit.

system, in coordination with other security-related European organisations. The general objective is to develop capabilities to support operations in case of crisis by providing them in a timely and affordable manner and in a coherent and holistic approach. In particular, it proposes an integrated solution that is capable of providing independent information as well as effective communications and navigation means. ESA is currently in the process of exchanging views with Member States and other European Organisations and potential users on the requirements for such a system.

EADS Astrium has developed an indoor satellite-based navigation system that helps search and rescue teams to understand the ground plan of any building and to track their own team members through positioning of sensors.[402] The FP 7 EULER aims at demonstrating how the benefits of Software Defined Radio (SDR) can be leveraged in order to enhance interoperability and fast deployment in case of a crisis which is jointly resolved. The outcomes will constitute one of the first European demonstrators of interoperability in a civil crisis situation based on SDR. ESA is currently examining the OHB System proposal to introduce "Very High Resolution Systems for Emergency Response and Security" focussing on highest resolutions of 0.5 metres over swath width, which can be provided by cost efficient SAR reflectors. In this context OHB Systems is currently looking at the development of a VHR SAR constellation as a stand-alone mission or in cooperation with Cosmo-Skymed and a VHR OPT Constellation stand-alone or with Pleiades, both aimed at providing dedicated one metre resolution with fast response time.[403]

Category 3: GMES and Galileo related Projects

One of the main European capabilities in the field of emergency response will be offered by the GMES' service element "Respond", as well as through its EU Emergency Response Core Service (ERCS). Respond provides for global mapping of crisis hotspots, an archive for detailed base and thematic mapping as well as rapid assessment maps. The EU Emergency Response Core Service (ERCS) is reinforcing the European capacity to respond to crises and emergencies. Focussing on the provision of rapid mapping and assessment services, it will provide support to civil protection and other relief agencies in the form of:

- Provision of geo-spatial database and information (in all forms and for a variety of sources including remote sensing) for regions of concern;
- Assessment of events and impacts;
- Access to monitoring tools for the duration of the crisis;
- Delivery of tailored solutions and services.[404]

Given the restriction of The Charter to natural or technological disasters,[405] and its technical limitations like insufficient revisit times and the limited coverage and lack of services for data interpretation, it could be replaced by a more user-driven system using a GMES infrastructure.[406] GMES thus also provides for a rapid mapping service that produces and delivers geographical information and maps covering disasters based on all Earth Observation sources and particularly the new satellites (e.g. ENVISAT, Spot 5). Providing worldwide coverage 24 hours a day all year long, it can provide the required imagery in less than twelve hours to civil protection services, decision makers, local authorities and/or the humanitarian aid community.[407] Up-to-date space-based maps provide for the event delimitation (e.g. extent maps), damage assessment (e.g. impact maps) and location needs (e.d. detailed maps focussing on severely affected regions). As part of the ERCS, the European Commission has been negotiating the SAFER project. A user-driven preoperational service has already been operating since 2008. SAFER can provide reference maps in less than six hours. It has already been activated several times this year for emergency management related to natural disasters.

The ESA funded GMES service element RESPOND aims to increase the efficiency and effectiveness of European and international humanitarian community action through appropriate and reliable application of geographic information to address all parts of the crisis cycle where geographic information is useful and to provide services in support of crisis preparedness and response operations

[402] EADS. "The Step Beyond – Sicherheit in einer veränderten Welt." Issue 1.2009. 54-57.
[403] Tobehn, Carsten. op. cit.

[404] "SAFER ERCS." RESPOND 30 July 2009 <http://www.respond-int.org/respondlive/public/html/what_is_respond/safer.html>.
[405] International Charter Space and Major Disasters. op. cit.
[406] The GMES Working Group on Security. op. cit.
[407] Tholey, Nadine. "Rapid Mapping: A GMES Service for Emergency Management." Presentation. ISU Annual Symposium. International Space University, Strasbourg, France. 18-20 Feb. 2008.

(e.g. mapping, alerting, early warning, field operations).

Similarly to RESPOND, the ESA funded GMES service element RISK-EOS aims at advancing rapid mapping. It offers a network of European service providers delivering geo-information services to support the management of floods and fires in particular, throughout all phases of the disaster cycle. While it could bring a huge advantage to crisis management in general it is limited to floods and fires and thus has very limited applicability to internal security. Building upon RESPOND, Risk-EOS and PREVIEW, which mainly focus on service delivery, Boss4GMES focuses rather on the future evolution of these services. It aims at developing an end-to-end information delivery lead time, ease of information access by users and to link emergency management with risk prevention. In this regard it will examine and build upon the existing service chains as provided by RISK-EOS and RESPOND as well as the existing EO data processing tools and the existing toolboxes and ongoing-developments for multi-satellite data access. It will also offer a rapid delivery mapping service (3-72 hours) as well as complementary and downstream support services for humanitarian aid, aiming to provide additional support throughout the whole crisis.

The Humanitarian Relief Cluster of the already frequently mentioned FP 6 project, LIMES, aims to improve and extend European capacities in the field of humanitarian relief. It provides added value through covering the whole crisis cycle in areas such as crisis preparedness, operational support of humanitarian relief during a crisis and reconstruction assistance after a crisis, by providing integration and fusion of satellite data as well as geo-information sources currently available from state of the satellite sensors and geo-information services along with development and incorporation of new technologies, techniques and sources particularly in the field of navigation and communication. For disaster preparedness and support of reconstruction, this means support of information services through population distribution and resource (food, water, infrastructure etc.) monitoring as well as contributing to an information system to support planning.[408] It foresees the rapid provision of updated geospatial information combined with the appropriate use of emergency satellite communication and navigation mobile systems. Galileo will advance the field of emergency response through location-based services and emergency calls. By integrating Galileo's system receivers in devices like mobile phones, customers can receive specific information on the surroundings (e.g. the nearest hospital).[409] Since the Public Regulated Services (PRS) and Galileo are designed for use by government bodies and thus their signal is characterised by robustness and continuity of service, it will be used to control civil protection and civil defence in a crisis such as natural disasters, terrorist attacks, military and humanitarian interventions.[410]

In particular, Galileo will contribute to the international search and rescue service in the maritime sector, enhancing the worldwide performance of the current COSPAR-SARSAT system. The positioning accuracy of today's system is insufficient (typically a few kilometres) and alerts are not always issued in real-time. The Galileo SAR (Search and Rescue) service will drastically reduce the time to alert, and the position of the distress beacon will be determined to within a few metres. It will be based on the Cospar-Sarsat satellite using special transponders, sending the emergency signal straight to land laboratories. People in distress will even receive proof that their emergency call has been received.[411]

Category 4: Projects Countering Fragmentation

Based on the concept of integration of geo-spatial data within a Geo-Information System (GIS), ISFEREA is and has been working on different dissemination mechanisms, thus covering the whole disaster cycle. ISFEREA is following an institutional action on crisis monitoring and response technologies (CRITECH), which builds upon the DMA and enhances the EU's crisis preparedness and response capability by means of fostering a better capacity for early warning, alerting, situational awareness, monitoring and impact assessment. Apart from offering the DMA, it is developing a crisis information management system including collaborative

[408] Telespazio. "LIMES Project Presentation." 28 Feb 2007. FP 6 LIMES 11 Mar 2009 <www.fp6-limes.eu/uploads/docs/LIMES-D1000.6-TPZ-2.0.pdf>.

[409] "Satellite Navigation Applications." op. cit.

[410] Roma, Alfredo. op. cit.

[411] Grippa, Andrea and Alessandra di Gangi. "New Technologies For Better Life: The Global Positioning System and the new European Project "GALILEO"." J. COMMODITY SCI. TECHNOL. QUALITY *47*.I-IV (2008): 209-227. <http://www.sci.unich.it/~jcs/2008/2008-02-11.pdf>. For further information on the Cospas-Sartsat concept cf: Cospas-Sarsat. "Concept." Cospas-Sarsat 30 July 2009 <http://www.cospas-sarsat.org/Description/concept.htm>.

web mapping tools and operational management tools as well as early warning and alerting models like the Global Disaster Alerting and Coordination Systems (GDACS).

GDACS provides near real-time alerts about natural disasters around the world. It led to the development of the spin-off application LiveMon which offers real-time monitoring and visualisation of disaster events. Additionally, CRITECH has been developing means to work and communicate more efficiently using classified electronic information. The FP 7 project GEOCREW also aims at improving the overall EU approach in emergency management by developing a Crisis Early Warning Situation Awareness Architecture Concept. In this regard, its main objective is to analyse the use of geo-spatial technology in the decision-making process of institutions dealing with internal crises and show best practices in their use. The overall aim will be the development of an architecture integrating different information sources as identified by user requirements.

Category 5: Projects covering the External Dimension of Security

Similarly to RISK-EOS, the FP 7 project G-MOSAIC provides for contingency planning, situation awareness, emergency mapping and damage assessment for conflicts, i.e. external relations only. It remains to be seen whether derived data and information can also be used for internal security. Moreover, ISFEREA was requested by DG ECHO[412] in 2000 to provide information on crisis areas around the world through a Digital Map Archive (DMA). So far it includes information of interest to the External Relations services only.

4.7. Comparative Matrix

The matrix enables a clear understanding of the ongoing activities and achievements but also shows the fragmentation, the intersection and the flaws and gaps of European projects related to the provision of internal security and space applications. On the basis of this analysis, the major objective of this chapter is to make a first step in the direction of putting forward a European concept for internal security.

[412] DG ECHO has as a mission to deal with external relations only.

Event Cycle		Projects and Services for the Provision of Internal Security involving Space Applications in Europe						
			Critical Mission Areas					
		Intelligence, Early Warning, and Risk Management	Critical Infrastructures	Transportation Security			Border Security	Emergency Management
				Land Transportation	Waterborne Transportation	Air Transportation		
Pre-Attack	Prediction / Prevention	– **CERISE** – **COSMO-SkyMed** – **TopSat** – **SPOT** – **EUSC GEOINT provision** – ***SVEA*** – ***PLEIADES-HR*** – ***HELIOS 2*** – ***SEOSAT*** – ***ORFEO*** – ***Pleiades*** – ***TanDEM-X*** – ***CLEMENTINE*** – ***ESSAIM*** – ***ELISA*** – ***SPIRALE*** – ***HELIOS 1B*** – CRITECH – GMOSS – LIMES – LOTUS – HAWKEY – PREVIEW – TANGO – BOC – MUSIS – GIANUS – Risk and vulnerability assessment of urban areas – *RISK-EOS*	– **ECI-GIS** – ***IMARSEC*** – ***GMES*** – EFFORTS – LIMES – PRESENSE – SECTRONIC – SOBCAH – VESPO – Space-Quest – MAHB – VASTS – VATDIS – Austrian Quantum – HYSAR – Greek SME Pipeline Project – SMARTRegions – T2L2 – Tesat Spacecom – *LOCOPROL*	– ***ERTMS*** – ***Galileo*** – *Bosch' initiative* – *SeCureSystem* – *ConTraffic* – GADEROS – ECO-RAIL – INTEGRAIL – TRACAR 1 / TRACAR 2 / TR 4026 – Logchain Translog Safety – LOCOLOC – VASTS – VATDIS – TOCHA – TransControl – MOBISIC – EMOGES – Italy's Safe Transport of Dangerous Goods – GaWaLoc	– **Bosch' initiative** – **SeCureSystem** – **IMARSEC** – **EADS' VTS** – **STIRES** – ***TanDEM-X*** – ***GMES*** – ***MASURE*** – *RIS* – *EMSA's VMS* – DECLIMS – EFFORTS – GMES/MCS – LIMES – MERSEA – ConTraffic – NAUPLIOS – VESPO – TRACAR 1 / TRACAR 2 / TR 4026 – GALILEOCEAN – Safety in Sea Traffic – SMARTRegions – Italy's Safe Transport of Dangerous Goods – SatHAV	- **Bosch' initiative** - ***Galileo*** - *SeCureSystem* - *ConTraffic* - NEWSKY - SAFEE - ARTES-10 Iris - *AATMS TR 1001* - *ANASTASIA* - *EOLIA TR 1021* - *ProATN TR 1042* - *SESAR* - HYSAR - *ASI-ENAV*	– **IMARSEC** – **STIRES** – ***GMES*** – ***TanDEM-X*** – LIMES – ConTraffic – GMOSS – EUROSUR – GMES – SOBCAH – VESPO – MODENA – GALILEOCEAN – SatHAV	- **UN-SPIDER** – **COSMO-SkyMed** – ***GMES*** – ***Pleiades*** - PREVIEW - GIANUS - CRITECH
	Detection	– **CERISE** – **TopSat** – **SAR-Lupe** – **COSMO-SkyMed** – **SPOT** – **EUSC GEOINT provision** – ***CLEMENTINE*** – ***ESSAIM*** – ***ELISA*** – ***SPIRALE*** – ***HELIOS 1B*** – ***SVEA*** – ***PLEIADES-HR*** – ***HELIOS 2*** – ***SEOSAT*** – ***ORFEO*** – ***TanDEM-X*** – ***Pleiades***	– **ECI-GIS** – ***GMES*** – ***IMARSEC*** – LIMES – PRESENSE – GMOSS – SECTRONIC – SOBCAH – G-MOSAIC – VESPO – T2L2 – Tesat Spacecom – SMARTRegions	– ***Bosch' initiative*** – ***SeCureSystem*** – ***GALILEO*** – GADEROS – GMES – ECO-RAIL – TRACAR 1 /TRACAR 2 /TR 4026 – Logchain Translog Safety – LOCOLOC – EMOGES – MOBISIC – TOCHA – TransControl – Italy's Vehicle Traffic Management – Italy's Safe Transport of Dangerous Goods – GaWaLoc	– **ENVISAT** – **TerraSAR-X** – **CosmoSky-Med** – **SPOT Commercialisation (SPOT 2, SPOT4, SPOT5, Formosat, Kompsat, Ikonos, EROS, Quickbird)** – **ERS-2** – **RADARSAT** – **IMARSEC** – **EADS' VTS** – **SeCureSystem** – **Bosch' initiative** – **STIRES** – ***MARUSE*** – ***MASURE*** – ***GMES*** – ***TanDEM-X*** – ***RIS***	- **Bosch' initiative** - ***SeCureSystem*** - ***GALILEO*** - *ConTraffic* - *ASI-ENAV*	- **ENVISAT** - **TerraSAR-X** - **SPOT Commercialisation (SPOT 2, SPOT4, SPOT5, Formosat, Kompsat, Ikonos, EROS, Quickbird)** - **CosmoSky-Med** - **ERS-2** - **RADARSAT** - **TerraSAR-X** - **IMARSEC** - **STIRES** - ***MARUSE*** - ***SEOSAT*** - ***TanDEM-X*** - ConTraffic - GMOSS - EUROSUR - G-MOSAIC	– **UN-SPIDER** – ***GMES*** – ***SEOSAT*** – ***Pleiades***

		– GMOSS – HAWKEY – LIMES – LOTUS – TANGO – BOC – MUSIS – GIANUS – ASARTECH – PolInSAR – *G-MOSAIC*		– *ConTraffic*	– ***EMSA's VMS*** – ***SAFER*** – SECTRONIC – LIMES – MARISS – MERSEA – VESPO – ConTraffic – Bosch' initiative – JRC Vessel Detection System – ERCS's SAFER – TRACAR 1 / TRACAR 2 / TR 4026 – SMARTRegions – Italy's Safe Transport of Dangerous Goods – GALILEOCEAN – SatHAV – *ConTraffic*		- MARISS - MERSEA - SOBCAH - VESPO - MODENA - GALILEOCEAN - *SatHAV*	
Attack	Alert	n.a.	– **ECI-GIS**	– ECO-RAIL – MOBISIC – Italy's Safe Transport of Dangerous Goods – TransControl	– **Bosch' initiative** – **SeCureSystem** – GALILEOCEAN – Italy's Safe Transport of Dangerous Goods – SatHAV	– **Bosch' initiative** – *ASI-ENAV*	– MODENA – GALILEOCEAN – SatHAV	– **UN-SPIDER** – **GDACS**
	Response	n.a.	– ***IMARSEC*** – ***GMES*** – GMOSS – SECTRONIC – SOBCAH – VESPO – VATDIS – Greek SME Pipeline Project – SMARTRegions	– ***GALILEO*** – GADEROS – INTEGRAIL – TRACAR 1 / TRACAR 2 / TR 4026 – Logchain Translog Safety – LOCOLOC – VATDIS – EMOGES – MOBISIC – Italy's Vehicle Traffic Management – Italy's Safe Transport of Dangerous Goods – Transcontrol – GaWaLoc	– **ERS-2** – **RADARSAT** – **MTM200™** – **IMARSEC** – **EADS' VTS** – **STIRES** – ***MARUSE*** – ***MASURE*** – ***SEOSAT*** – ***SAFER*** – ***RIS*** – ***EMSA's VMS*** – ***GMES*** – DECLIMS – LIMES – MARISS – MERSEA – NAUPLIOS – VESPO – TRACAR 1 / TRACAR 2 / TR 4026 – GALILEOCEAN – Italy's Safe Transport of Dangerous Goods – POB – SMARTRegions	– ***GALILEO*** – *ASI-ENAV*	– **ERS-2** – **RADARSAT** – **IMARSEC** – **STIRES** – ***MARUSE*** – ***GMES*** – ***OHB VHR-SAR*** – ***OHB VHR-OPT*** – EUROSUR – MARISS – MERSEA – SOBCAH – VESPO – MODENA – GALILEOCEAN	– **LiveMon** – **GDACS** – **SAFER** – **HELIOS 1B** – **TopSat** – **COSMO-SkyMed** – **TerraSAR-X** – **SPOT** – **RapidEye** – **UN-SPIDER** – **International Charter** – ***GMES*** – ***PLEIADES-HR*** – ***HELIOS 2*** – ***SEOSAT*** – ***TanDEM-X*** – ***OHB VHR System for Emergency Response*** – CRITECH – GMOSS – EULER – PREVIEW – RESPOND – GMES/ERCS – BOSS4GMES – 3D urban structure information extraction from very high resolution SAR imagery – ASARTECH – PolInSAR – HYSAR – EMERSAT – GALILEOCEAN – GUSTAV – PREVIEW – GIANUS – *RISK-EOS*

								– *G-MOSAIC* – *DMA*
Post-Attack	Relief / Impact Assessment	n.a.	- ***GMES*** - GMOSS - SOBCAH - VESPO - Greek SME Pipeline Project		– **International Charter** – ***GMES*** – ***EMSA's VMS*** – ***SAFER*** – LIMES – VESPO – GALILEOCEAN – *SatHAV*	– *ASI-ENAV*	– ***GMES*** – MODENA – GALILEOCEAN – SOBCAH – VESPO – *SatHAV*	– **UN-SPIDER** – **TopSat** – **RapidEye** – **TerraSAR-X** – **ENVISAT** – ***GMES*** – ***PLEIADES-HR*** – ***TanDEM-X*** – ***OHB VHR System for Emergency Response*** – CRITECH – LIMES – GMOSS – PREVIEW – RESPOND – GMES/ERCS – BOSS4GMES – 3D urban structure information extraction from very high resolution SAR imagery – HYSAR – EMERSAT – *G-MOSAIC* – *RISK-EOS*
Integrated Approach		– HUMBOLDT – ORCHESTRA – WIN – TIES	– **ECI-GIS** – **ERN-CIP** – SMARTRegions	– **ERTMS** – MAGNET A – MAGNET B – MULTITRACK – TRACAR 1 / TRACAR 2 / TR 4026	– **European Marine Observation and Data Network (EMODNET)** – **RIS** – **STIRES** – MAGNET A – MAGNET B – MERSEA – OPERAMAR – SeaDataNet – TRACAR 1 / TRACAR 2 / TR 4026 – TRACAR 1 / TRACAR 2 / TR 4026 – MyOcean – SMARTRegions – *DRAGONESS*	– MAGNET A – MAGNET B	– ***European Patrols Network*** – MERSEA – MyOcean – SeaDataNet	- **EO-KDZ** - ***GDACS*** - OASIS - ORCHESTRA - GEOCREW - SYNTHESIS

Figure 31: Comparative Matrix on Projects and Services for the Provision of Internal Security involving Space Applications in Europe

Matrix Key

- European R&D project, National R&D project
- **Implemented Service, operational Satellite, industry service**
- *European or national R&D projects with potential applicability, industry project with potential for contribution*
- ***technology demonstrators, envisaged satellites, implemented service with potential for contribution, industry project***

5. Developing a Concept for the Provision of Internal Security with the Support of Space Applications in Europe

The pillar structure leads to the fragmentation of the European approach to the provision of internal security by artificially dividing security, assigning elements of civil security to: Pillar I; military / external security to Pillar II; and Justice and Home Affairs (JHA) to Pillar III.

While most official EU documents have realised the urgency of tackling the matter, space as a distinct technology area as well as its added value to numerous operational areas are rarely mentioned.

The EU seems to have started to seriously look into the provision of internal security only recently with research programmes relating to this matter located in the Framework Programmes (FPs) (FP 5 to FP 7). These cover only the research and demonstration levels.[413] No full-fledged operational services to be derived from this have yet been put into practice.

While most official EU documents have realised the urgency of tackling the matter, space as a distinct technological area as well as its added value to numerous operational areas are mostly not mentioned. Projects included in the FPs mainly focus on Phase 1, the prevention phase. Those projects are difficult to assess with regards to the actors involved, the mission areas as well as the users and operators. Information is rather basic and does not provide insights to the actors involved, which would be necessary to identify their potential and make them part of a more comprehensive approach.

Often the target sector is limited to natural disasters like floods and fires, while the same applications would actually also be useful for terrorist incidents. Many projects look similar in nature and the relationship among them, if any, is not clear. With many projects having overlapping objectives, the added-value of the single projects is not transparent. While most projects claim to be user-driven, their involvement is still limited given the lack of understanding of the added value of space applications and rigid cultural or traditional working structures.

With regard to information dissemination, the websites of the single projects are fragmented, adding to the difficulties in assessing their value. As a consequence, hardly more than a simple introduction or factsheet presentation is given, with most FP5 and FP6 websites being no longer operational. Information can thus not be accessed and assessed. Evaluation of the projects for the purpose of clarifying the EU approach to internal security remains difficult if not impossible. Actors remain limited to their specific mission areas such as natural disasters or CFSP and cannot engage in emergency management following, for example, a terrorist incident.

5.1. Intelligence, Early Warning and Risk Management

At the national level there are several existing or planned dedicated missions to signal intelligence and early warning (e.g. CERISE, Clementine, Essaim, ELISA, SPIRALE, TopSat, Prisma, SVEA, SEOSAT) as well as high resolution EO satellites (e.g. SAR-Lupe, COSMO-Skymed, Pleiades, ORFEO) and projects on the improvement of SAR imagery techniques (e.g. ASARTECH, PolInSAR), but there is so far no dedicated European intelligence mission. MUSIS, GMES, BOC and Galileo will improve the current situation however they are so far not explicitly missioned to look for terrorist camps.

[413] For more information on the transition between demonstration and operation cf. Matthieu, Charlotte. "Space-Based Services in Europe – Addressing the Transition between Demonstration and Operation." Report 17. Vienna: European Space Policy Institute.

One of the main problems in geospatial information lies in the current data situation. Currently the EU has "islands of data" of different standards and quality. The current data situation in Europe often results in a lack of requireddata due to a lack of coordination across borders and between levels of government, and a lack of common standards resulting in incompatible information and information systems, fragmentation of information and redundancy. This situation is worsened by current data policy restrictions like pricing, copyright, access rights and licensing policy.

These "islands of data" need to be overcome. While the added value of geospatial information for military use as well as disaster management has been understood, this is not the case for disaster prevention related to counter terrorism and organised crime. Effective integration and assimilation of data and information from in situ and space-based observation is needed as governmental civil protection bodies and emergency services are becoming more and more dependent on integrated information systems to aid different phases of the disaster-cycle.[414] While the European Commission's initiative to develop an Infrastructure for Spatial Information in Europe (cf. INSPIRE) aims at solving the standardisation issues, other obstacles to satellite data exchange still persist. In this regard, information exchange in internal security, in particular, has to be improved.

While several projects of the Sixth Framework Programme aim at improving the current data situation (e.g. WIN, ORCHESTRA, OASIS), their added value does not become clear from publicly available sources. All three can be distinguished by virtue of covering different phases of the disaster cycle, yet the motivation for having different projects on data integration for different phases is not clear.

5.2. The three Critical Mission Areas

All mission areas share the need for clearly defined user requirements. As an overall requirement, the importance of interfaces between different components (e.g. SYNTHESIS, databases etc.) can be identified. Different satellite missions should be integrated into more all encompassing European missions (e.g. add AIS receiver to EO mission, integrate SATCOM, navigation and EO).

5.2.1. Critical Infrastructures

Many of the underlying more theoretical questions related to critical infrastructure remain unanswered. These include questions such as: what is vulnerable? What is critical? Which threats are meaningful? Which attacks are possible? Which level of protection is sensible? Which problems are national and which ones are cross-border in character? In addition, the related challenges to criteria, data, methods and tools are still unanswered and not clarified.[415]

The use of satellite applications for critical infrastructure protection is one of the most difficult-to-assess critical mission areas. Information is very limited and even though the application of space-based assets for e.g. protection of nuclear facilities seems obvious, there are almost no sources on the matter.

The JRC identifies two particular trends in the protection of critical infrastructures. First, energy systems are no longer only about energy but rather energy and information have become inextricably integrated. Thus electric power infrastructure consists of energy and information (E+I). Secondly, one can identify a shift towards systems-of-systems, i.e. increased inter-relationships within and across sectors.[416] This development is however not mirrored in current R&D projects. There is thus the need to re-evaluate existing projects so as to account for these new trends.

The study could not identify one single project that actually covers the whole disaster cycle for CIP. With the different phases being on a continuum, there is a need for coordination between different projects not only regarding different phases but also regarding connected critical infrastructure. In order to improve the overall European attempt there is thus the need to provide for a comprehensive approach to CIP. In order to adopt a fully integrated and sustainable system, one needs to know

- the system to be protected, its operation and its maintenance,
- the technologies used to protect it and to operated and maintain it,
- the vulnerabilities and the potential failures,

[414] European Space Agency/European Commission. op. cit. 23.

[415] Masera, Marcelo. op. cit.

[416] Masera, Marcelo. op. cit.

- existing counter measures and defences,
- the enemy,
- the potential threats,
- the attack mechanisms and their effects
- and finally,
- the space contribution.[417]

On the basis of this check-list one can build and maintain a comprehensive assessment of European critical infrastructures with catastrophic potential.[418]

While the promotion of Best Security Practices is a good starting point, international cooperation in the area of CIP should go beyond the mere exchange of best practices.[419]

With private firms being reluctant to provide extensive information on vulnerabilities that are needed for risk assessment due to fears that this information could become public and therefore adversely affect public or shareholder confidence, there is a need for Transparency and Confidence Building Measures in the industry. Thus access to this information (vulnerability assessment etc.) must be restricted to avoid terrorists from being able to download it and then use it to attack.[420] At the same time, cooperation on protecting critical infrastructure and information sharing should be exempt from antitrust laws in order to protect companies from unjustified lawsuits.[421]

Both governments and the EU should encourage indicator and warning systems as well as training on the protection of critical infrastructure, i.e. the use of space applications for that purpose. Also they should encourage the industry to seek greater cooperation with intelligence agencies as well as the law enforcement authorities.

As the analysis has shown, dependency on space assets is rising. Thus a space protection strategy is of vital importance. Any concept to internal security has to address this need as well. Thereby it should tackle the issue of space terrorism by providing for protection of any national/EU systems from intentional interferences but also unintentional ones from the space environment and collisions. It should further include ways of assuring access to space services.

5.2.1. Transportation Security

As indicated earlier, the sector of transportation security is the most scattered one in this analysis. There is an urgent need for an integrated approach to all sectors as to assure security of the supply chain in its entirety. In addition fragmentation also exists in the single sectors (land, sea, air). Thus, attempts have to be coordinated at the sectoral level (i.e. land, sea, air) and then at the overall system level (i.e. transport sector). The maritime sector is the only one showing attempts at a more integrated approach (e.g. JRC's institutional action MARSUR, MyOcean, OPERAMAR, EMODNET, SeaDataNet). No similar projects could be identified in the other transportation sectors.

There is a strong emphasis on waterborne transport security or maritime security as such. One example of this is the projects related to securing containers, which mainly tackle the maritime sector while containers are actually used in all transportation modes.

Incentives have to be created so as to achieve global coverage of satellite-based container integrity systems for all transportation modes. Satellite applications prove to be a promising alternative to physical verification, i.e. the actual opening of a container and manually verifiying its contents with the Bill Of Lading is a process that takes approximately eight hours and leads to significant delays, thereby introducing further uncertainties in the logistics chain, while X-ray scanners produce a fairly accurate image of the container's contents through scanning. Experience proves that systems ensuring container integrity are very effective in reducing the number of thefts. Yet, global coverage is far from being achieved.

Not all possibilities for sealing containers have been carefully studied since it has not received top priority due to the costs this would introduce to the chain. This barrier to improvement has to be resolved. Due to the high costs, only containers carrying high class loads are currently equipped and protected with these systems. Costs could be reduced by producing larger numbers. Although the public and some governmental bodies are willing to pay (a certain amount) for security improvements, the majority of security measures places costs on the shippers and carriers. Of course this is not insurmountable

[417] Masera, Marcelo. op. cit.
[418] European Security Research and Advisory Board. op. cit.
[419] Chapman, John. "Defending Europe's Vulnerable Infrastructures." Roundtable. Security and Defence Agenda, Brussels, Belgium. 2 Oct. 2006. <http://se1.isn.ch/serviceengine/FileContent?serviceID=ISN&fileid=9AE50350-A979-8E90-26D5-4654D3B53206&lng=en>.
[420] Wortzel, Larry M.. op. cit. 179.
[421] Ibid.

but does again raise a barrier for improvements.[422]

Efforts should also be made to incorporate advanced electronic container integrity systems in the container transport business. In this context container seals should be applied in a standard way, which should ensure the container's doors cannot be opened without damaging the seal.

Risk analysis needs to be improved. An expansion of criteria so as to be able to target more types of contraband, enhancement of algorithms, more information exchange within the logistics chain as well as between ports and more extensive use of computerised search tools would increase the effectiveness of risk analysis in container transportation.[423]

The EU has to develop a community framework for supply chain security instead of opting for a patchwork approach which divides the transport sector into different transportation modes.

Land Transportation

Of the transportation sectors the land sector is probably the most neglected. While recent documents aim at improving this situation, most applications deal with safety rather than security. Also, publicly available sources do not include passenger transportation particularly the urban mass transportation sector. Proposals complementing the existing Community transport security measures only cover freight transport and only touch upon passenger transport when looking at rail transportation. Mass transportation seems to be completely neglected in terms of added value through space applications.

Projects related to transportation security at roads on a national level can be divided into two categories: (1) dealing with pure traffic management and (2) dealing with the transport of dangerous goods. None of them explicitly tackles the possibility of terrorists interfering with road transportation. Rather, all of them aim at improving road transportation in general. Projects related to traffic management improve situational awareness on roads and thus have the potential to improve search and rescue. None of them however explicitly follow this objective.

Projects related to rail transportation security focus to a large degree on fleet management, track monitoring and train control. While primarily aimed at improving the detection and immediate response to accidents, some of these projects have the side effect of being able to identify whether derailing was accidental or due to malicious interference.

The public sector of mass transportation is probably the least transparent in this analysis. While satellite applications are used for fleet management and improved scheduling, no publicly available source could be identified proving their application for security of the mass transportation sector.

Generally speaking, the land transportation sector focuses on applications based on satellite navigation.

Waterborne Transportation

As indicated earlier, maritime security is currently high on the agenda and thus pushes the issue of waterborne transportation security up as well. One explanation for this development might be the increase of piracy attacks, a very topical issue for the media and the general public. Awareness among media and the public could, however, be further exploited so as to increase awareness of the added value of space applications in this context. Once established, this could be used as a building stone in further educating the media and public about the contribution of space applications to the provision of internal security as a whole.

Apart from the fragmentation and coordination problems of existing projects, Europe also faces some technological limitations, as already indicated in chapter 2. Many projects cover one particular phase of the supply chain only and thus risk neglecting important points that only become visible when looking at the whole supply chain. In this regard projects such as the Bosch initiative or SeCureSystem go in the right direction. Additionally, many projects cover one particular phase of the event cycle only and thus lack the overall perspective on the event. While the JRC for example claims to offer an integrated approach by covering all aspects of maritime security, it is unclear which role it plays in many of the FP projects it is involved in. There is thus a risk of duplication of efforts due to the lack of an overview of existing projects and actors involved.

Policies, strategies and projects aimed at improving waterborne transportation security seem to cover seas and oceans only. This is

[422] Ibid. 188-90.
[423] Voort, van de Maarten, Kevin A. O'Brien, Adnan Rahman and Lorenzo Valeri. op. cit. 192-3.

particularly striking given the high degree of fluvial transportation within the EU. In this way, waterborne transportation security as part of the more general maritime security policy becomes an area of external rather than internal security. Research projects should thus be extended to include applications for fluvial transportation as well.

Given the increase in projects and efforts in the field of maritime security, current approaches are very scattered, non-transparent and duplicative. This should be improved through enhanced coordination and communication between supranational instruments, national capabilities and regional needs. In this context it might be necessary to task one body with the supervision of all efforts. This would then report to an EU-level coordinating body. Building upon the Blue Paper and the accompanying Action Plan as first steps, a holistic EU Maritime Security Policy can be created.[424] IMO served and will serve as a market place for ideas.

While the EUSC is currently involved in the EU ATALANTA NAVFOR mission, its involvement in waterborne transportation security is limited to its links with ESDP and CFSP. Given EUSC's expertise, however, it could provide useful information in the transportation sector as well.

At the technological level, research has to be conducted into improving wake detection: moving ships show wakes in both radar and optical satellite imagery, which can provide information about the speed and direction of the ship. Their detection is still under-developed.

Different applications and users have different needs. Most scenarios require the combination of a wide swath (several hundred kms) with the capacity to detect also relatively small boats (tens of metres). In some cases even smaller vessels need to be detected.[425] Apart from detection problems, challenges such as the extension of the surveillance range and non-intrusive and continuous availability persist. Limits set on the volume of operations based on space assets are quickly reached under the current regime in Europe. The availability of space systems and their capacity to be replaced or augmented for operational purposes remains weak and hardly reactive. While

> some user needs, such as requests for continuous monitoring in real or near real time that is also able to track small-sized objects may not be feasible, especially in the short to medium term [and] some military requirements may likewise not be technically feasible[426]

there is still room for action. With current maritime missions not being originally envisaged for this task, there is a need for a dedicated maritime surveillance mission composed of a combination of both radar and optical satellites with requirements for future sensors varying according to the application. A full constellation is not necessarily needed - a group of radar and optical satellites would be enough.[427] Current capabilities are based on EO satellites that have already been operating for some time.

Given that radar (at medium resolution) is most suitable for ship detection but does not allow for ship classification, which in turn has to be derived from (very) high resolution optical imagery, a combination of radar (for day and night all-weather imagery) and optical (for large swath and high resolution) is needed for maritime surveillance. Current capabilities in maritime surveillance are based on Earth Observation satellites that have already been operating for some time and whose primary mission is not maritime surveillance. For this reason, they often cannot achieve "best delivery times" as revisiting is not sufficient. In a wide area, ship detection remains limited (< 8-10m (optical), < 30m (SAR)).[428] Thus, there is a need for a dedicated maritime surveillance system, which is better adapted to maritime surveillance in terms of surveillance, detection, classification and tracking.

In this context, Europe also needs to work on a European AIS-S system to contribute to the global aim ofproviding an AIS-S system with global coverage. The creation of a system capable of providing a global AIS-S solution is a major undertaking and represents a very large investment. Historically, such a system would only be the province of governmental space agencies. This is especially true for system designs that rely on limiting the coverage area of each satellite to solve the collision problem. Such systems must then deploy large numbers of satellites in order to provide a reasonable Vessel Refresh Rate. This is the fundamental trade-off in

[424] Chapman, John. op. cit
[425] A discussion on different scenarios and their requirements can be found in the related DECLIMS document: Greidanus, Harm. op. cit.
[426] Lindström, Gustav. "GMES: The Security Dimension." Seminar at EU Institute for Security Studies, Paris, 16 Mar 2007.
[427] Bressollette, Aurélie. op. cit.
[428] Bressollette, Aurélie. op. cit..

determining the Quality of Service for any AIS-S system.[429]

COMDEV has conducted extensive research into the needs and wants of potential users of a global AIS picture. Based on this research key needs are:

- Frequent updates from each ship – (Vessel Refresh Rate)
- High Capture Ratio – probability of seeing any ship during a pass should be as good or better than terrestrial systems.
- Data security and quality – data on historical ship tracks is both sensitive and valuable and should not be distributed without careful consideration.
- Access to historical ship tracks for environmental incident investigations and other applications.[430]

When defining how to effectively link AIS-S with the European space sector, ESA should take these user requirements into account. The main objective is to extend the scope of service trials to test the integration of LRIT and satellite AIS data streams beyond the current range of coastal AIS systems. The concept of satellite AIS is currently being tested in the USA with the intention of expanding AIS type reporting to open waters.[431]

With ports being on the one hand a critical infrastructure in itself and on the other hand handling tonnes of waterborne transported freight and passengers daily, an integrated port security system capable of providing accurate situational awareness and alerting security operators to required interventions is needed.[432]

Current attempts in the field of maritime surveillance are of a sectoral character covering environmental issues separately from security related matters. These approaches have to be integrated into one single European concept, taking into account sustainability and environmental issues at the same time as transportation and the creation of jobs. In this context, different technologies contributing to the same objective should also be integrated. Satellites are a major, but not the only, solution.

As indicated earlier, satellite applications for maritime surveillance cannot be used on their own but are rather understood as complementary to other data sources. This is a result of the rather limited information on vessel traffic provided by satellite images: data is available not continuously but only at one instant; within a limited swath; with limited resolution; and with limited detection probability and classification potential. In order to identify a potentially dangerous unknown ship from satellite SAR imagery which can be performed in near-real time, with detection and false alarm rates that are sufficient for some users and scenarios but need to be improved for others, resulting detections have to be fused with vessel traffic data from other sources (e.g. AIS, VMS, coastal radar, airborne patrol etc.), which is increasingly being done. Given the need for a comprehensive maritime picture, which can only be derived on the basis of a variety of data sources such as intelligence, conventional monitoring systems (coastal radars, AIS, VMS), aircraft patrolling and EO satellites (both optical and radar), there is a need for an integrated approach. Better shared resources between the Member States and the EU can be envisaged[433] to improve the situation. Furthermore, outreach activities would help to counter the lack of knowledge, misconceptions and the "community gap" which make space tools not fully accepted by users.[434]

Air Transportation

While the benefit of satellite applications for aviation security seems obvious, especially with the development of satellite-based container security systems for waterborne transportation which should theoretically be easily expanded to air transportation, publicly available sources give little insights on how space applications are used in this sector. Looking at the U.S., projects on audio-video cockpit monitoring, equipment based on irridium satellites could be identified. Similar projects in Europe could not be found.

International legislation and guidelines concerning aviation security do not explicitly mention satellite applications. As can be derived from the overview of the work of the international organisations, all of them lack sufficient enforcement powers and are therefore bound to only develop policies without being able to enforce them.

IATA is often considered] as the initiator among the international organisations in civil aviation. This is why some authors suggest that it should be used as a driving force for

[429] COM-Dev, ISU presentation / paper
[430] Cain, J.S. and E. Meger. op. cit.
[431] ibid
[432] ESRAB p.26
[433] Chapman, John. op. cit.
[434] Bressollette, Aurélie. op. cit..

practical implementation of effective security management.[435] Given however that the European Union has an even greater enforcement power, it could fulfil the same purpose even better. Given the primacy of European Union law over national law and the enforcement power of the European Court of Justice (ECJ), the European Union fills this gap and could therefore take the lead in the establishment of a global standard of aviation security, thereby acknowledgingthe added value of space applications.

Compared to ports, airports are not considered as critical infrastructures but rather form part of the transport sector. Thus different actors deal with this area, which might provide an explanation of the lower number of projects in this sector.

5.2.1. Border Security

Border security can be subdivided into land and sea border security, the latter achieving more attention in the context of the general publicity for maritime security as a whole. While from a technological viewpoint areas of application for land border security have been identified (e.g. remote sensing to detect hidden tunnels for illegal border crossing) practical applications are still limited and face difficulties in convincing users about the added value of space applications. Earth Observation technology is still of limited use given its limited applicability to on-spot detection. It can rather be used for planning only.

At a political level, there is still a need for increased cooperation with third countries. It is of utmost importance not only in the identification of persons and return of illegal migrants but operational cooperation is also needed to prevent illegal immigration and for the proper management of critical situations.[436] The basis of a successful policy is return strategies in particular. There is a need to adopt a comprehensive approach to migration with operational activities to fight illegal immigration, protect refugees and ensure uniform application of the Schengen acquis on external borders.[437]

5.3. Preparedness for Emergency Management

In looking at European projects covering emergency management, a strong emphasis on natural disasters like floods and tsunamis and forest fires can be identified. There are several projects covering rapid mapping and delivering different rapid mapping products (e.g. RISK-EOS, SERTIT). Some of them are related to the International Charter but their difference and added value is not obvious enough, leading to assumptions about potential duplication. While some of them seem to cover particular aspects such as rapid mapping for pandemics, it is still questionable why these cannot all be pooled into one database. Moreover, the delivery of rapid mapping products per se has to be improved. Faster delivery times are required (within 24 hours of the incident with updated maps every six hours).

As indicated earlier, EUSC's mission should be extended to include internal security threats. EUSC is currently missioned/mandated to support decision-making in ESDP and CFSP related issues only. This is particularly striking with EUSC having acquired some expertise already in rapid mapping and GeoInt and its access to military satellite data. Thus its analysis of satellite imagery, collateral data and related services, should be used to support decision-making in the provision of internal security as well.

Many projects currently focus on one area of satellite applications only. This should be improved through introducing a multi-purpose mission. Both GiANUS and GMES go in the right direction in this regard. Emergency management is not only about EO and rapid mapping but also about stable communication in the aftermath of an attack, or satellite navigation for search and rescue missions.

Compared to the U.S., Europe has the advantage of being able to use its own satellites for surveillance of its own territory without specific permission. In this context, Europe should take advantage of this situation and further improve its capabilities for emergency response.

[435] Rodney, Wallis. op. cit. 93.
[436] Commission of the European Communities. COM (2006) 733 final. op. cit.
[437] Commission of the European Communities. COM (2006) 733 final. op. cit.

Mission Area	Findings	Need for Action
Intelligence, Early Warning and Risk Management	- there is so far no dedicated European intelligence satellite mission - planned missions are not tasked to detect terrorist camps - lack of coordination across borders and between levels of government and a lack of common standards regarding data sharing resulting in incompatible information and information systems, fragmentation of information and redundancy: current data policy restrictions like pricing, copyright, access rights and licensing policy worsen the situation	- there is a need for a dedicated European intelligence mission pursuing the objective of providing internal security - need for a common data policy - effective integration and assimilation of data and information of in situ and space-based observation is needed
Critical Infrastructure Protection	- many of the underlying more theoretical questions related to critical infrastructure are still unanswered, like: What is vulnerable? What is critical? Which threats are meaningful? Which attacks are possible? Which level of protection is sensible? Which problems are national and which ones are cross-border in character? - related challenges to criteria, data, methods and tools are still not answered and clarified - while space-applications theoretically prove to be of high added-value for CIP, information on related projects and practical applications is very limited - energy systems are no longer only about energy but rather energy and information have become inextricably integrated. Thus the electric power infrastructure consists of energy and information (E+I). There is a shift towards systems-of-systems, i.e. increased inter-relationships within and across sectors - the study could not identify one single project that actually covers the whole disaster cycle for CIP	- clarify theoretical questions through formulating policy documents - increase information on existing projects for CIP - R&D projects for CIP have to mirror new trends in CI - with the different phases being on a continuum, there is a need for coordination between different projects not only regarding different phases but also regarding connected critical infrastructures - there is the need to provide for a comprehensive approach to CIP - while the promotion of Best Security Practices is a good starting point, international cooperation in the area of CIP should go beyond the mere exchange of best practices - both governments and the EU should encourage indicator and warning systems as well as training on the protection of critical infrastructure and particularly the use of space applications for that purpose - there is a need to encourage the industry to seek greater cooperation with intelligence agencies as well as law enforcement authorities
Transportation Security	- the transportation security sector is the most scattered sector under analysis. - fragmentation exists in the single sectors (land, sea, air) as well. - satellite applications prove to be a promising alternative to physical verification, i.e. the actual opening of a container and manually verifying its contents with the Bill Of Lading - experience proves that systems ensuring container integrity are very effective in reducing the number of thefts: global coverage is far from being achieved - due to the high costs, only containers carrying high class loads are currently equipped and protected with these systems. Costs could be reduced by producing larger numbers.	- urgent need for an integrated approach to all sectors to assure security of the supply chain in its entirety - attempts have to be coordinated at the sectoral level (i.e. land, sea, air) and then at the overall system level (i.e. transport sector) - extend container tracking systems to all transportation modes - Incentives have to be created so as to achieve global coverage of satellite-based container integrity systems for all transportation modes - risk analysis needs to be improved. An expansion of criteria as to be able to target more types of contraband, enhancement of algorithms, more information exchange within the logistics chain as well as between ports and more extensive use of computerised search tools would increase the effectiveness of risk analysis in container transportation. - the EU has to develop a community framework for supply chain security instead of opting for a patchwork approach which divides the transport sector into different transportation modes
Land Transportation	- among the transportation sectors the land sector is probably the most disregarded - most space applications deal with safety rather than security - publicly available sources also do not include passenger transportation particularly the urban mass transportation sector - proposals complement the focus on freight transport and only touch upon passenger transport when looking at rail transportation - projects related to transportation security at roads on a national level can be divided into two categories: (1) dealing with pure traffic management and (2) dealing with the transport of dangerous goods: None of them explicitly tackles the possibility of terrorists interfering with road transportation. Rather all of them aim at improving road transportation in general. - projects related to traffic management improve situational awareness on roads and thus have the potential to improve search and rescue. None of them however explicitly follow this objective. - projects related to rail transportation security focus to a large degree on fleet management, track monitoring and train control. While primarily aimed at improving detection and immediate response to accidents, some of these projects have the side-effect of being able to identify whether derailing was accidental or due to malicious interference. - while satellite applications are used for fleet management and improved scheduling in public mass transportation, no publicly available source could be identified proving their application for security of the mass transportation sector - in general the land transportation sector focuses on applications based on satellite navigation	- there is an urgent need for projects showing the added-value of space application for road and rail transportation and public mass transportation security
Waterborne Transportation	- highest sector (under analysis) on the agenda - maritime sector is the only one showing attempts at a more integrated approach (e.g. JRC's institutional action MARSUR, MyOcean, OPERAMAR, EMODNET, SeaDataNet). No similar projects could be identified in the other transportation sectors. - there is a strong emphasis on waterborne transport security or maritime security as such. One example of this would be the projects related to securing containers, which mainly tackle the maritime sector while containers are actually used in all transportation modes. - many projects cover one particular phase of the supply chain only and thus risk overlooking important points that only become visible when looking at the whole supply chain - many projects cover one particular phase of the event cycle only and thus lack overall perspective on the event - the role of institutional actors within certain projects is not clear - there is a risk of duplication of efforts due to the lack of an overview of existing projects and actors involved	- awareness among media and public could be further exploited to increase awareness of the added value of space applications in this context. Once established, this could be used as a building stone in further educating media and the public about the contribution of space applications to the provision of internal security as a whole - research projects should thus be extended so as to include applications for fluvial transportation as well - enhanced coordination and communication between supranational instruments, national capabilities and regional needs is needed to counter fragmentation - it might be necessary to task one body with the supervision of all efforts which will then report to a EU-level coordinating body. Build upon the Blue Paper and the accompanying Action Plan as first steps towards creating a holistic EU Maritime Security Policy - given EUSC's expertise, however, it could provide useful information in the transportation sector as well - at the technological level, research has to be conducted on improving wake detection - there is a need for a dedicated maritime surveillance mission composed of a combination of both radar and optical

	- policies, strategies and projects aimed at improving waterborne transportation security seem to cover seas and oceans only. This is particularly striking with the high degree of fluvial transportation within the EU. In this way waterborne transportation security as part of the more general maritime security policy becomes an area of external rather than internal security - while the EUSC is currently involved in the EU ATALANTA NAVFOR mission its involvement in waterborne transportation security is limited to its links with ESDP and CFSP. - moving ships show wakes in both radar and optical satellite imagery, which can provide information about speed and direction of the ship: their detection is still underdeveloped. - different applications and users have different needs. Most scenarios require the combination of a wide swath (several hundred kms) with the capacity to detect also relatively small boats (tens of metres). In some cases even smaller vessels need to be detected. - limits set on the volume of operations based on space assets are quickly reached under the current regime in Europe - the availability of space systems and their capacity to be replaced or augmented for operational purposes remains weak and hardly reactive. - current maritime missions are based on EO satellites that have already been operating for some time and were not originally envisaged for this task: for this reason, they often cannot achieve “best delivery times” as revisiting is not sufficient. In a wide area ship detection remains limited (< 8-10m (optical), < 30m (SAR)) - radar (at medium resolution) is most suitable for ship detection but does not allow for ship classification, which in turn has to be derived from (very) high resolution optical imagery - current attempts in the field of maritime surveillance are of a sectoral character covering environmental issues separately from security related matters	satellites with requirements for future sensors varying according to the application - Europe also needs to work on a European AIS-S system so as to contribute to the global aim of providing an AIS-S system with global coverage - with ports being on the one hand a critical infrastructure in themselves and on the other hand handling tonnes of waterborne transported freight and passengers daily, an integrated port security system capable of providing accurate situational awareness and alerting security operators to required interventions is needed - sectoral approaches to maritime surveillance have to be integrated into one single European concept, taking into account sustainability and environmental issues at the same time as transportation and the creation of jobs. In this context different technologies contributing to the same objective should also be integrated. Satellites are a major but not the only solution - outreach activities to counter the lack of knowledge, misconceptions and the “community gap” which make space tools not fully accepted by users are needed
Air Transportation	- while the benefit of satellite applications for aviation security seems obvious, especially with the development of satellite-based container security systems for waterborne transportation which should theoretically easily be expanded to air transportation, publicly available sources give little insights on how space applications are used in this sector - projects on audio-video cockpit monitoring equipment based on satellites could not be identified in Europe - international legislation and guidelines concerning aviation security do not mention satellite applications explicitly. As can be derived from the overview of the working of the international organisations all of them lack sufficient enforcement powers and are therefore bound to develop policies only without being able to enforce them. - compared to ports, airports are not considered critical infrastructures but rather form part of the transport sector: different actors deal with this area, which might provide an explanation as to the lower number of projects in this sector.	- increase enforcement power of international organisations in this sector - conduct further research on the use of satellite applications for aviation security - IATA is often seen as the initiator among the international organisations in civil aviation. This is why some authors suggest that it should be used as a driving force for practical implementation of effective security management. Given however that the European Union has an even greater enforcement power it could fulfil the same purpose even better. Given the primacy of European Union law over national law and the enforcement power of the European Court of Justice (ECJ) the European Union fills this gap and could therefore take the lead in the establishment of a global standard of aviation security, thereby indicating the added value of space applications.
Border Security	- border security can be subdivided into land and sea border security, the latter achieving more attention in the context of the general focus on maritime security as a whole - while from a technological view point areas of application for land border security have been identified (e.g. remote sensing to detect hidden tunnels for illegal border crossing) practical applications are still limited and face difficulties in convincing users about the added value of space applications. - Earth Observation technology is still of limited use given its limited applicability to on-spot detection. Rather, it can be used for planning only. - return strategies form the basis of successful border policies	- further research and development on the application of earth observation for on-spot detection is needed - at a political level, there is still a need for increased cooperation with third countries - it is of utmost importance not only in the identification of persons and return of illegal immigrations but also operational cooperation is needed to prevent illegal immigration and for the proper management of critical situations. - there is a need to adopt a comprehensive approach to migration with operational activities to fight illegal immigration, protect refugees and ensure a uniform application of the Schengen acquis on external borders
Emergency Management	- looking at European projects covering emergency management one can identify a strong emphasis on natural disasters like floods and tsunamis and forest fires - there are several projects covering rapid mapping and delivering different rapid mapping products (e.g. RISK-EOS, SERTIT). Some of them are related to the International Charter but their difference and added value is not obvious enough, leading to assumptions about potential duplication. - while some of them seem to cover particular aspects like rapid mapping for pandemics, it is still questionable why this cannot all be pooled into one database - moreover, the delivery of rapid mapping products per se has to be improved. Faster delivery times are required (within 24 hours of the incident with updated maps every six hours) - as indicated earlier, EUSC’s mission should be extended o include internal security threats. EUSC is currently missioned/mandated to support decision-making in ESDP and CFSP related issues only. This is particularly striking with EUSC having acquired some expertise already in rapid mapping and GeoInt and its access to military satellite data. Thus its analysis of satellite imagery, collateral data and related services should be used to support decision-making in the provision of internal security as well - many projects currently focus on one area of satellite applications only. - compared to the U.S., Europe has the advantage of being able to use its own satellites for surveillance of its own territory without specific permission. In this context, Europe should take advantage of this situation and further improve its capabilities for emergency response	- adopt a more comprehensive approach moving away from emergency management only for natural disasters towards inclusion of internal security threats - clarify differences and added value of projects related to rapid mapping - establish one reference database for emergency management in Europe - extend the mission of EUSC to provide support for decision-making in internal security - there is need to introduce multi-purpose missions. Both GiANUS and GMES go in the right direction in this regard. Emergency management is not only about EO and rapid mapping but also about stable communication in the aftermath of an attack, or satellite navigation for search and rescue missions.

Figure 32: Matrix summarising the findings and needs for action in the different mission areas

6. Conclusions and Recommendations

The urgent need to take action in the field of internal security is clear. Council Presidencies supported by both the European Commission and the European Parliament need to make use of the momentum created by the financial and economic crisis and promote space as a particularly useful contribution to the provision of internal security. They should not only provide for strategic guidelines in form of an Internal Security Strategy as a complementary tool to the European Security Strategy but should also advance the current state-of-the-art by drawing on the results of the FP projects and putting applications into practice. Many demonstrations have already provided for user feedback and requirements and have proven the added value of space applications for internal security.

While most of the official EU documents have realised the urgency of tackling the matter, space as a technological area as well as its added value is neglected and not directly mentioned in related documents. The Council Presidencies have to change this. With the need for more operational satellites being acknowledged, this needs to be more effectively presented and defended. Future projects covering space and internal security should move ahead and avoid focussing on the prevention phase as previous ones have done but rather take a holistic approach and counter fragmentation. Current projects are too similar in nature. Future projects have to be distinctly different, being characterised by a particularly and easily identifiable added value. Interaction among projects has to be increased so as to increase added value and to formulate an integrated approach.

In order for Europe to guarantee its "actorness", political support for internal security projects has to be increased to guarantee the move from demonstration to operation, particularly with regard to the GMES' Fast Track Services on Marine, Land Cover and Emergency Response. Gaining such support may be difficult given the fragmentation among stakeholders in the security domain. 2009 emerges as the year to take a decision. With many satellite missions needing decisions on extension or replacement, Europe has to take decisions in order to avoid national approaches leading to further fragmentation and duplication. There is an urgent need to present the case and take action in the areaof internal security. The community has to bring together the needs for internal security and apply for funds - the sooner the better. With the next Framework Programme starting in 2013, which means that the new financial perspective will be decided upon by the end of 2012, there is an urgent need to take action before 2012 so that internal security will be included in the next framework programme (FP8). Strategies have to be found to convince the Economic and Finance (ECOFIN) ministers about the investment needed. Thinking outside the box and introducing space as a solution in non-traditional fora will increase funding.

Given the changed security paradigm in the post-Cold Warera, Europe can no longer focus on traditional external security but has to move ahead and adopt a broader concept for the provision of security. Internal Security can no longer be treated in a fragmented and uncoordinated way. As an acto,r Europe needs to actively engage in the provision of internal security and the application of space-based applications in this context can no longer be neglected.

While primarily tackling the lack of strategic approach to the use of space applications for internal security, this study goes beyond the mere production of concrete recommendations in this context and provides a building block towards the development of a comprehensive joint European concept for internal security.

Recommendations

The recommendations can be grouped in three categories, thereby not only substantiating areas for need for action but also identifying addressees for immediate action and future steps.

6.1. Formulating a Common European Approach to Internal Security through an Internal Security Strategy

There is a strong need for further coordination (a) between EU and national approaches and activities, (b) within the different single mission areas and (c) in the overarching area of internal security at the Community level. A common European approach has to be adopted.

Guidance could be provided through the introduction of a European Internal Security Strategy, which should stress the importance of the use of technology - in particular space applications - for the provision of internal security, while at the same time providing for more coordination among projects and institutional actors. Such a comprehensive top-down policy-approach would be the most efficient way to ensure that space applications have the opportunity to receive the attention of decision-makers and ultimately lead to their implementation as tools in this area.

The current and upcoming EU Presidencies should actively promote this matter and finally move forward to establish an all-encompassing EU approach to internal security. The Commission as the "initiator" has to provide the policy background for an integrated approach in the single areas, but the future EU Presidencies should engage in discussions related to the introduction of an EU Internal Security Strategy. The European Parliament should aim at acting as a forum to bring together expertise and become involved through recommendations to the Council regarding the formulation of an Internal Security Strategy.

(a) Coordination between EU and national approaches and activities

The FORESEC study of 2008 has also pointed out that European security requires a comprehensive approach at both national and European level. Accordingly, while progress is undeniable, success is so far limited not least because fragmentation also stems from the number of distinct lenses through which security can be analysed within member states. Because of these different approaches there seems to be a lack of understanding of the linkages between technological and societal issues, between hard external threats and more domestic concerns, and there is a clear need to better link security research to the overall strategies and policies contributing to European security. With the terrorist threat being dynamic, the development of new technology on monitoring trends in terrorism, anticipating novel means of attack, distilling lessons learned from past incidents and continually updating best practices in form of research must be continued.[438] National security research and foresight activities are not adequately coordinated with European level research programmes thus resulting in gaps and overlap between activities.

There is a need for a common European approach. The EU needs to take the lead in providing internal security and getting its act together to prove its "actorness".[439] If Europe as a whole does not tackle the issue of internal security, or if individual member states choose national rather than common approaches, Europe's ability to act in the international arena may be diminished. A political dialogue needs to be developed in the EU Member States, particularly in those countries where the issue of internal security has so far received insufficient attention. The dialogues need not necessarily be based on risk perceptions (given the variations) but may focus on vulnerabilities instead.[440]

(b) Coordination within the different single mission areas

A comprehensive approach to security should not only cover all critical mission areas individually but also as a whole in order to account for a cascading effect. A security system focused on protecting modern society would seek to protect critical nodes of activity. It would integrate security considerations into the design and daily

[438] Hoffman, Frank. op. cit.145.

[439] The principle of "actorness" referrs to "the capacity to behave actively and deliberately in relation to others in the international system". Accordingly, an actor is an entity, which is: (1) delimited from others, and from its environment; (2) autonomous, in the sense of making its own laws and decisions, and (3) possesses certain structural prerequisites for action at the international level, such as legal personality, a set of diplomatic agents and the capability to conduct negotiations with third parties. Taking this basic definition a step further an actor is characterised as a "variable and multidimensional presence in international affairs". This analytical framework has first been introduced by Gunnar Sjöstedt. cf. Sjöstedt, Gunnar. The External Role of the European Community. Farnborough: Saxon House, 1977. David Allen and Michael Smith took Gunnar Sjöstedt's definition as a basic concept and came up with their own definition on actorness. cf. Allen, David and Michael Smith. "Western Europe's Presence in the Contemporary International Arena." Review of International Studies 16.1 (1990): 19-37.

[440] Föhrenbach, Gerd. op. cit. 55.

operations of such systems. It would identify potential vulnerabilities linked to the technological complexity of the modern world and seek to transform them into high reliability systems. It would seek to anticipate and prevent possible "cascading effects" of a breakdown or collapse of any particular node of activity. It would ensure that "connectiveness vulnerabilities" are not built into future systems.[441]

(c) Coordination in the overarching area of internal security at the Community level

The EU has to move beyond the mere interpretation of internal security as natural disasters and include organised crime and terrorism in threat scenarios to plan against as well. The current security scenario is dominated by a combination of asymmetric threats, suggesting the need for a comprehensive approach rather than the traditional separation of civil and military capabilities.

There is thus a need to move away from the separation of civil and military applications particularly in the field of technology and instead integrate both through the use of dual technologies. The European Defence Agency (EDA) has already taken some initial steps to integrate civil and defence capabilities. These approaches have to be acknowledged by the other institutional actors as well, leading to a re-definition of the EU approach. A European Internal Security Strategy should explicitly mention the use of dual-use technologies, naming space applications as a particularly valuable example. In this context it should provide for the integration of civil and military capabilities. The current security scenario is dominated by a combination of asymmetric threats, suggesting the need for a comprehensive approach rather than the traditional separation of civil and military capabilities.]

As identified by the Group of Personalities (GOP) in the field of security research, the technology base for defence, security and civil applications increasingly forms a continuum with challenges inside and outside of the EU often being similar. Across this continuum, applications in one area can often be transformed into applications in another area. Space is a prime example for such a development.[442] In this context, it seems current divisions in research programmes along the two key dimensions (internal versus external security and civilian versus military means) [443] have to be overcome.

Given its international standing the EU has a responsibility to play an active role in world affairs. By adopting a comprehensive approach linking the external and internal dimensions of security by combining the use of civil and military means, Europe can achieve full exploitation of potential synergies between defence, security and civil research. Preference should be given to multilateral institutions and agreements, rule of law and the treatment of root causes; effective multilateralism and preventive engagement. The EU should thrive by? agreeing on a distinctive European approach which can form the basis for increased coordination with the U.S. and other countries. As with its European Neighbourhood Policy, Europe should thrive through spreading its values to other countries in the context of the provision of internal security.

In this context, the missions of the institutional actors such as the European Union Satellite Centre (EUSC) have to be reviewed to establish a broader field of action taking into account their expertise. Given its international standing, the EU has a responsibility to play an active role in world affairs to take the lead in providing a comprehensive approach linking the external and internal dimensions of security combining the use of civil and military means so as to achieve full exploitation of potential synergies between defence, security and civil research.

The Internal Security Strategy should ensure that internal security is not provided at the cost of limiting civil liberties. Providing for internal security and ensuring civil liberties is not necessarily a zero-sum game. There is a variety of alternativesthat policymakers can consider. First, with respect to data retention, most individuals accept that some personal data may need to be collected to fight terrorism effectively. Nonetheless, how personal data is collected, used, and stored can vary. Some approaches protect civil liberties more than others. In many instances, citizens are concerned that data will be retained unnecessarily. To minimise such concerns, personal data should be retained only when warranted, for the shortest timeframe possible, and permanently destroyed when no longer needed. In other cases, concerns stem from

[441] Hamilton, Daniel S. op. cit. xxi.
[442] Group of Personalities in the field of Security Research. op. cit. 12.
[443] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. op. cit. 13-5.

the perception that databases will be increasingly cross-linked to facilitate the tracking of suspected terrorists. While combining databases can prove useful, it is important to resist the temptation to link databases that provide limited added value in the fight against terrorism. To limit societal concern, databases with information unlikely to yield results in the fight against terrorism should remain detached.[444]

6.2. Issues of Governance and Organisation

Areas of authority in the European institutions must be clarified and the provision of internal security established as a matter which cuts across institutions.While EU Member States have on many occasions confirmed that a more coherent and integrated approach to crisis management is needed, the EU's pillar structure artificially divides security into civil security (Pillar I), military / external security (Pillar II) and JHA (Pillar III). The ESS adopted by the European Council in December 2003, for example, explicitly indicated that more coherence is needed to improve effectiveness in the EU's crisis management.

The main challenge is to bring together the different instruments and capacities that have been created at different periods of time, within different institutional structures and with different rationales. The institutional set-up has to be reviewed taking into account the changing security context after the Cold-War and the new possible threat scenarios.

There are several stakeholders already active in the domain of, for example Earth Observation for security, including efforts by individual EU Member States, Commission services, and the EU Satellite Centre. Some products – such as imagery analysis in support of treaty verifications and border monitoring – fall within the scope of the GMES Security domain. In-situ data is typically managed by EU Member States, making it more challenging to provide data through a single or limited number of channels.[445] A more coherent approach is needed. Greater coherence requires complementarities in the use of EU instruments (civilian and military) and better co-ordination between them. With both the national and international dimensions being absolutely necessary to be able to counter the new threats, coordination of efforts at both the national and European level to counter fragmentation across the pillars and across the actors (Member States versus EU as well as EU internally: i.e. among the agencies) is needed. Such structural deficiencies at the institutional and political level hinder Europe in the exploitation of its scientific, technological and industrial strength. To overcome these Europe needs to improve the coherence of its efforts. This implies

- effective coordination between national and European research activities,
- systematic analysis of security-related capability needs from civil security to defence,
- full exploitation of synergies between defence, security and civil research,
- close cooperation between defence and internal security users
- specific legal conditions and funding instruments for security-related research at the European level,
- institutional arrangements that are both efficient and flexible enough to combine Member States' and Community efforts and to involve other interested partners.[446]

One single actor should thus pool resources and projects and provide the oversight needed. If such a body is not created, Europe risks delay and continued problematic approaches to internal security, comparable to those in the U.S., which is experiencing a variety of problems due to fragmented responsibilities of agencies. Such a body should coordinate European efforts, ensure post-9/11 measures are thoughtfully applied (i.e. halfway measures can bring about more attacks while too drastic counter-measures can easily be just as harmful as the attacks themselves), set European security criteria (i.e. for ports to comply with) and monitor the degree to which the security criteria are complied with so as to establish European standards[447].

Coordination should go beyond institutional coordination and should lead to cooperation among the industry owning and operating existing assets and among Member States. An immediately increased dialogue among the actors involved is needed. Currently talks exist only on an informal level and there is a

[444] Lindström, Gustav. op. cit. 130.
[445] Lindström, Gustav. "GMES: The Security Dimension." Seminar at EU Institute for Security Studies, Paris, 16 Mar 2007.
[446] Group of Personalities in the field of Security Research. op. cit. 9.
[447] Voort, van de Maarten, Kevin A. O'Brien, Adnan Rahman and Lorenzo Valeri. op. cit. 192-3.

strong need for agreement on formal arrangements. At the moment, each operator is for example limited to its own radar footprint. One can see that cooperation is also of technical interest. For the user neither the type (e.g. commercial, governmental) nor the country operating or owning the satellite are important - important is which satellite is passing at the right moment. Thus, international dialogue is needed. Constraints resulting from confidentiality should be limited. As the security dimension of GMES evolves, it will be important to ensure continued coherence across stakeholders. Cooperation between governments and the private sector plays a crucial role.

Commercial applications of space drive many of the systems developments. Thus many different users (commercial, internal security, defence) will be using the same satellites for similar types of applications. The provision of internal security with space applications can thus not be looked at in isolation. It should be part of a larger attempt to coordinate applications development.

With defence and internal security users using the exact same capabilities, close cooperation between these communities has to be fostered. Internal security users could also benefit from the experience of the defence community in making use of satellite applications for reconstruction and security efforts.

Efforts to introduce multi-functional missions have to be supported and fostered. In line with the principle of subsidiarity, an enhanced coordination and communication mechanism between supranational instruments, national capabilities and regional needs has to be created. Particularly in the maritime domain, efforts need to be pooled and coordinated by one single actor. The European Maritime Safety Agency (EMSA) could provide for this.

Given the technical limits of space applications such as the so far relatively low revisiting time, they need to be used as part of a wider system. This makes the integration of all technologies in use ever more important. There is also a need for an integrated approach addressing the overall challenge of prevention, mitigation and response.[448]

Sufficient funding is needed for successful provision of internal security. With European defence budgets lagging behind, commitment to space-based defence technology has remained rather low in an overall European context. By comparison, U.S. investment in this sector represents between eight and ten percent of total expenditure on defence equipment. Hence, the U.S. also accounts for more than ninety percent of global investment in defence-related space technology and Europe less than five percent. The bulk of European investment in space-based technology is accounted for by five countries that are actively planning or operating significant defence/dual-use programmes. France, the United Kingdom, Germany, Italy, Spain and Belgium account for over ninety-eight percent of the total European expenditure.[449]

The European Union and its Member States are part of a highly interdependent and complex world. Thus future needs of European security policy, strategy and research should not primarily be driven by internal developments related to the process of European integration.[450] A dialogue should take place both at the EU and the transatlantic level. Greater U.S.-European cooperation in the field of internal/homeland security is necessary in order to better guarantee security on both sides of the Atlantic. Neither the United States nor Europe is yet well organised to advance the provision of internal/homeland security. The different mechanisms set in place in Europe and the United States have each complicated transatlantic cooperation. Scattered efforts must now be incorporated into a systematic, high profile effort to “transform” homeland security in all of its many dimensions. Globalisation has given the effective provision of internal/homeland security an international dimension. Without systematic pan-European and transatlantic coordination, each side of the Atlantic is at greater risk of attack.[451]

Coordination and cooperation does not mean to adopt the other’s approach. It is rather about rapprochement, confidence building and understanding of each other’s policies, values and practical approaches. Thus, while

[448] Commission of the European Communities. Communication to the European Parliament and the Council on Reinforcing the Union's Disaster Response Capacity. COM (2008) 130 final of 5 Mar. 2008. Brussels: European Union.

[449] European Space Agency. Annual Analysis 2004. op. cit. 51.

[450] Antonenko Oksana, Bastian Giegerich Nigel Inkster, Raffaello Pantucci, Rahul Roy-Chaudhury. “FORESEC Deliverable D 2.3 Global Trends and Actors Shaping European Security.” FORESEC. 10. <http://www.foresec.eu/wp2_docs/GlobalTrendsAndActors.pdf>.

[451] Hamilton, Daniel S. op. cit. xv, xx.

the EU and the U.S., will each follow their own distinct approach to internal or homeland security, both should find a common ground basis so as to create a win-win situation for both. With Barack Obama as the new U.S. President, differences between American and European conceptions particularly in the fight against terrorism have narrowed substantially.[452] While complete transatlantic agreement is utopian, the EU and the U.S. should aim at presenting a unified message to the world both about their commitment and the fundamental values they espouse. In order to do so, the EU needs to internally assess its core principles in the provision of internal security and establish contacts so as to ensure that European opinions are conveyed to their U.S. counterparts. Given the narrowing of the EU and the U.S. approach to counter-terrorism, the Spanish Presidency could provide a good ground for advancing cooperation in this field with the objective of drafting a public declaration on fundamental principles for confronting terrorism, organised crime and illegal migration. Such a declaration should provide for a guiding strategy on cooperation, thereby underlining the principles and values both share in the fight against terrorism.

Cooperation between the EU and the U.S. in the provision of internal security and homeland security should however not be only about counter-terrorism. As indicated earlier, while the U.S. approach to homeland security is largely driven by its counter-terrorist agenda, European efforts to protect society have largely been derived from civilian emergency response communities working with domestic law enforcement agencies. Thus, cooperation has to go beyond mere cooperation in counter-terrorism and include other threats to internal and homeland security as well. The EU and the U.S. have to establish mechanisms with the objective of sharing best practices in the fight against organised crime and illegal migration. This should be of both a strategic and a practical nature ranging from data sharing to the sharing of expertise in technology development and particularly the use of space applications in this realm. In the past, EU-U.S. cooperation in the context of the Container Security Initiative (CSI) drew on expert groups. In areas with particular difficulties in coordination, expert groups could provide the bridge between the two positions and might be able to find common ground.

Changes in the U.S. Administration have led to increased cooperation between U.S. and European industry. While European industry was always open towards cooperation with their U.S. counterparts, cooperation has in the past been hampered by certain restraints from the U.S. side. Today, U.S. industry aims at benefitting from the complementarity and thus increasingly engages in cooperation with Europe. Europe on the other hand has become independent in terms of data and satellite imagery availability and no longer depends solely on the U.S. for information. Considering data policy, information flow and the way satellites are used, Europe is actually ahead. Technological cooperation in terms of user requirements, effective and well-equipped missions and data sharing should be fostered. The International Technology Alliance (ITA) in Network and Information Sciences, a research programme between the UK and the U.S., is a first step in this direction.

Increased coordination and cooperation is only possible if each sideshows a certain degree of openness towards the other side's approach. Particularly this entails that even though both have a completely different institutional set-up, mechanisms have to be found to arrange for cooperation and coordination. Within the EU, the provision of internal security is inherently distributed across different authorities and operators, with their respective roles, capabilities and resources. With transport policy falling under the first pillar, i.e. the community pillar, and border control falling mainly under the third pillar, i.e. the intergovernmental pillar, responsibilities as well as decision-making structures vary. There is no European "Minister for Homeland Security" available to the U.S. Secretary of Homeland Security as there is no DG for Internal Security as a counterpart to the U.S.' Department of Homeland Security. Thus, a possible request from the U.S. to have a single point of contact within Europe might be very difficult to accommodate at this stage. Therefore, it is of particular importance to clarify institutional responsibilities at EU level and to communicate those to U.S. counterparts as a basis for an increased dialogue even at agency level in order to have close cooperation even at lower levels for sharing best-practices.

Coordination and cooperation should also be sought among the international organisations. EU and ESA should engage in a dialogue with particularly the international organisations missioned to provide common security like NATO and OSCE.

[452] Dworkin, Anthony. op. cit. 2.

6.3. The Role of Space in a Comprehensive Internal Security Strategy of the EU

ESA and EC funded projects need to move beyond the demonstrations phase. The ESA and EC thus have to ensure that the actors involved properly deal with the transition from demonstration to operation. ESA and the European Commission also have to ensure that all critical mission areas in the provision of internal security are covered equally and that the agenda does not get "hijacked" by events like the piracy attacks off the coast of Somalia which moved maritime security up on the agenda while fluvial security and efforts in land border security were lowered. Existing projects have to be re-evaluated to include current trends like the increased integration of energy and information in the area of critical infrastructure.

Users are central to the development of new operational services but many potential users are unaware of the contribution of space-based services. They thus are often unable to demand space-based services. The space sector in turn has limited knowledge and understanding of the demand for space-based services and lacks the user requirements as a basis for new satellite missions.[453] There is an urgent need to improve communication with the public to achieve public understanding of the added value of space.

A systematic and large-scale approach is needed to further promote the use of space and to indentify, understand, federate and structure the demand for space-based services in Europe. Outreach projects can show the potential of space applications for internal security and bring users and the space community together to achieve mutual understanding and benefit. It is mostly a question of communication. There is a need to create a symposium with the general public. This could be achieved through the establishment of a User-Exchange mechanism. The development of dual use technology areas should be supported through the establishment of an exchange mechanism with users to identify their primary needs. This requires leadership from stakeholders and organisations.[454] GMES stakeholder consultation, which started with the 2005 Communication "From Concept to Reality", has indicated that progress towards the realisation of the political and operational goals of GMES is hindered by:

- The lack of a formal process to involve the users in the definition of the scope and architecture of the services despite the user-driven character of GMES and the establishment of expert groups (known as service implementation groups);
- The lack of a common approach and process consolidating the contributions of the various partners in the development of GMES (could result in a duplication of efforts in Europe).

A common approach is still missing among coexisting frameworks at EC, intergovernmental and national level which all have separate decision and financing mechanisms.[455]

There is a need for promotion of best practices for user involvement.

The adoption of marketable solutions and the introduction of a user-exchange mechanism should improve the current communication problem between the users and lead to an improved understanding of the added value of space applications for the provision of internal security through effective outreach activities. The industry should further get involved in outreach activities, moving beyond mere demonstrations so as to enlarge the group of addressees. Space-applications should be presented as innovative solutions in particular in times of financial and economic crisis as laid out in the Lead Market Initiative. At the same time, however, one should avoid creating unmatchable expectations. Space is an important instrument but not the only one and often has to be used in conjunction with other technologies. It is thus important to be realistic about the contribution of space assets to the provision of internal security.

Results from demonstrations and FP research should be considered seriously by ESA when planning and deciding on new satellite missions. User requirements conveyed to ESA should provide the building blocks for new satellite projects. Such projects should be outlined in such a way that they encompass as many opportunities as possible for applications related to internal security. This principle should be applied to single satellite missions as well as satellite constellations or the combined use of satellites with different

[453] Matthieu, Charlotte. op. cit. 6.
[454] Ibid.
[455] Commission of the European Communities. SEC(2008)2808. op. cit.

purposes. Increased cooperation among the different projects of the FP programme relating to space is of utmost importance in the context of formulating user requirements. On the other hand, users have to be educated as to demand for certain services.

Initiatives like GiANUS (Global Integrated Architecture for iNnovative Utilisation of space for Security) and programmes like the Integrated Applications Promotion (IAP), which was launched by the last ESA Council at Ministerial level, are means to provide frameworks for accomplishing interactions with users in order to establish user requirements as a basis for new satellite missions. These take a promising direction for establishing a focused dialogue between the users and ESA. Cooperation in this context has to be increased with users in the field of internal security demanding certain requirements and ESA engaging in this dialogue. The user-exchange mechanism proposed in this report would provide for formalisation of such a process.

Actors	Proposed Actions
Council and Council Presidencies	- actively promote “internal security” as a policy area - establish an all-encompassing EU approach to internal security - formulate an Internal Security Strategy, which - lays down the European Approach to the provision of internal security - provides overall guidance - highlights the added value of space applications - acknowledges EDA’s first steps towards integrating civil and military capabilities and re-defines the EU approach - explicitly mentions the use of dual-use technologies - provides for the integration of civil and military capabilities - ensures that internal security is not provided at the cost of limiting civil liberties - prove the EU’s “actorness” - review the mandates and missions of the EU agencies to account for new threat scenarios and in order to adopt a more comprehensive approach (e.g. EUSC) - create a new actor which - pools resources and projects - provides oversight - coordinates European efforts - sets European security criteria and the degree of compliance so as to establish European standards - improves coordination between national security research activities and European level research programmes - pools results and findings from projects (=user requirements) - establish a dialogue with the U.S. to increase transatlantic coordination - a dialogue should take place both at the EU and the transatlantic level - scattered efforts must now be incorporated into a systematic, high profile effort to “transform” homeland security in all of its many dimensions - make use of the narrowing of substantial differences between American and European conceptions due to the new U.S. Administration’s rhetoric - the EU and the U.S. should aim at presenting a unified message to the world about both their commitment and their fundamental values. In order to do so the EU needs to internally assess its core principles in the provision of internal security and establish contacts to ensure that European opinions are conveyed to U.S. counterparts - the Spanish Presidency could provide a good ground for advancing cooperation in this field with the objective of drafting a public declaration on fundamental principles for confronting terrorism, organised crime and illegal migration - cooperation and coordination has to go beyond mere cooperation in counter-terrorism and include other threats to internal and homeland security as well - the EU and the U.S. have to establish mechanisms with the objective of sharing best practices in the fight against organised crime and illegal migration. This should be of a strategic nature but also of practical value ranging from data sharing to the sharing of expertise in technology development and particularly the use of space applications in this area - increased coordination and cooperation is only possible if both sides provide a certain degree of openness towards the other side’s approach - even though both have a completely different institutional set-up, mechanisms have to be found to arrange for cooperation and coordination - it is of particular importance to clarify institutional responsibilities at EU level and to indicate those to U.S. counterparts as a basis for providing for an increased dialogue even at agency level in order to have close cooperation even at lower levels for sharing best-practices - expert groups for specific topics could provide for enhanced cooperation and coordination in particularly difficult fields - explore cooperation with other institutional actors in the field like NATO and OSCE - make an initiative to mandate EMSA to oversee all efforts in maritime surveillance - conduct outreach activities proving added value to (1) users, (2) the general public and (3) decision makers, leading to implementation as tools in the provision of internal security
Member States	- improve coordination between national security research activities and European level research programmes - develop a political dialogue focussing on vulnerabilities to increase attention for the importance of the provision of internal security - make national efforts and results and findings of studies publicly available as to show what capabilities and projects already conducted in order to decrease potential for duplication - share best practices in space applications for the provision of internal security - make data available so as to integrate national vessel monitoring systems (like the EADS VTS at the coast of Portugal) into a community vessel monitoring system
European Commission	- as the “initiator” provide the policy background for an integrated approach in the single areas - prove the EU’s “actorness” - related to the Framework Programmes: - establish a “security culture” providing a comprehensive framework in order to provide internal security in all event phases and covering all critical mission areas individually but also as a whole policy field accounting for cascading effects - encourage the active participation of the private sector owning and controlling most of the vulnerable networks - provide for more coordination among different projects (“top-down policy approach”) - ensure that the actors involved properly deal with the transition from demonstration to operation - encourage cooperation among the industry - foster multi-purpose missions - establish a user-exchange mechanism
European Parliament	- act as a forum to bring together expertise and get involved through recommendations to the Council regarding the formulation of an Internal Security Strategy
Industry	- show added value of space applications for the provision of internal security - make an effort to integrate new fields of users - present space-applications as innovative solutions particularly in times of financial and economic crisis - educate users so as to demand services

ESA	- encourage cooperation among the industry - foster multi-purpose missions - ensure that the actors involved in ESA funded projects properly deal with the transition from demonstration to operation - establish a user-exchange mechanism - consider results from demonstrations and FP research seriously when planning and deciding on new satellite missions - user requirements should provide building blocks for new satellite projects - educate users to demand services - GiANUS and IAP should form part of a greater effort to establish a dialogue between users
EMSA	- pool SafeSeaNet (SSN), CleanSeaNet (CSN) and other maritime surveillance actions into one integrated system addressing both environmental and security threats with one system, preferably a community vessel detection and monitoring system - integrate different data sources (in situ, coastal radar etc.) with satellite data in all systems
JRC	- extend the mission area of the Vessel Detection System (VDS) as to include internal security threats as well - integrated VDS with SSN and CSN and other community level maritime surveillance tools

Figure 33: Matrix summarising action items for the different actors

Annex

I. Funding and Personnel

ACTORS	PERSONNEL	BUDGET
European Space Agency (ESA)		
EMSA (SafeSeaNet, STIRES, Maritime Support Services 24/7, development of global requirements for a future upgrade of the application agree with Member States and the Commission, support to MS to develop and maintain an AIS Master Plan, incl. regional cooperation)	15 AD 6 AST 2 END	€ 4.968.369 (commitment appropriations) € 4.011.224 (payment appropriations)
EU LRIT DC	14 AD 8 AST 1 END 3 CA	€ 4.931.568 (commitment appropriations) € 6.921.205 (payment appropriations)
FRONTEX	n.a.	n.a.
EDA	n.a.	n.a.
DG JLS/ EUROSUR	n.a.	n.a.
DG MARE	n.a.	n.a.
DG ENTR (management of research framework programme)	40	€ 124.410.000 (commitment appropriations) € 121.835.500 (payment appropriations)
DG TREN	n.a.	n.a.
EC Maritime Affairs Task Force	n.a.	n.a.
DG ECHO	n.a.	n.a.
JRC	6-8	Ca € 1-1.5 million annually
Institute for the Protection and Security of the Citizen (IPSC)	n.a.	n.a.
European Laboratory for Structural Assessment (ELSA)	n.a.	n.a.
Western Sea Border Centre (WSBC)	n.a.	n.a.
Eastern Sea Border Centre (ESBC)	n.a.	n.a.
Civil Protection Units (CPU)	n.a.	n.a.
Traceability and Vulnerability Assessment (TRVA)	n.a.	n.a.
European Satellite Centre (EUSC)	100	Ca. € 9.5-12 million annually
SERTIT	n.a.	n.a.

European Union funding of common European space activities amounts to 5.25 billion Euros for 2007-2013.[456] The European Commission's 7th Framework Programme for Research and Technological Development (FP7) amounts to 50.5 billion Euros for 2007-2013[457] Space forms one of the areas of the block cooperation, with funding of 1.43 billion Euros over the seven years from 2007 to 2013. Some space activities are also funded through the security cluster, which also receives a total of 1.43 billion Euros over the seven years. Additionally, the transportation sector offers the possibility for funding of space applications for the provision of internal security in the transportation sector. This cluster receives a total of 4.16 billion Euros.

The European Parliament and the Council have allocated a budget of 3,4 billion Euros for 2007-2013 for the satellite navigation projects Galileo and EGNOS, including 400 million Euros from FP7.

[456] European Parliament Report on Space and Security, 10 June 2008, 2008/2030(INI).
[457] It features the four main blocks: cooperation, ideas, people and capacities; of which cooperation receives the largest share. Space forms one of the areas of the block cooperation, funded with 1.43 billion Euros over the seven years from 2007 to 2013. The bulk of the space funds (85% equalling 1,2 billion Euros) available in FP7 are earmarked for GMES activities.

Additionally, funding sources are given by funds of the Trans-European Transport Networks (TEN-T), part of Trans-European Networks (TEN), or the Entrepreneurship and Innovation Programme (EIP), which is part of the Competitiveness and Innovation Framework Programme (CIP) of the European Commission, amounting to a total amount of 3,4 billion Euros for European satellite navigation over a period of seven years.[458]

GMES

Within the 6th Research Framework Programme (FP 6), the EU has spent € 100 Million on GMES projects. ESA invested another € 100 Million in the GMES Service Elements projects. The EU Member States have earmarked a total of € 1.4 billion for funding the security theme and € 1.4 billion for funding the space theme over the duration of the 7th Research Framework Programme. Out of this the EU will make available € 550 Million for GMES service projects and procurement of data for these services. Additionally, € 650 million will be used to contribute to the development of the ESA Space component programme, which amounts to € 1236 million for its first segment. In addition to the funding of research projects through FP 7, the Commission is implementing the preparatory action (€ 3 Million in 2008) to prepare proposals with a view to the adoption of future actions.[459]

Funding Opportunities in the Justice, Freedom and Security Policy Areas for the Period 2007-2013

Three framework programmes have been established to provide coherent support to the area of freedom, security and justice under the financial perspectives 2007-2013. These will develop the three areas of freedom, security and justice in parallel and to the same degree of intensity, allowing for a balanced approach based on the principles of democracy, respect for fundamental rights and freedoms, and the rule of law. Each of the three key objectives is thus supported by a Framework Programme:

- *Security and Safeguarding Liberties* (€ 745 million)[460], consisting of two specific programmes (1) Prevention of and Fight against Crime (ISEC) and (2) Prevention, Preparedness and Consequence Management of Terrorism and other Security Related Risks (CIPS). Both are implemented via annual work programmes adopted by the Commission each year.
- *Fundamental Rights and Justice* (€ 542.9 million), consisting of five instruments: (1)Prevent and combat violence against children, young people and women and protect victims and groups at risk (Daphne III), (2) Drugs prevention and information, (3)Fundamental rights and citizenship, (4) Civil justice and (5) Criminal justice.
- *Solidarity and Management of Migration Flows* (€ 4020.37 million), consisting of four financial instruments (1) the External Borders Fund, (2) the Integration Fund, (3) the Return Fund, (4) the European Refugee Fund.

These also offer the possibility of funding space applications for the provision of internal security.

[458] Rathgeber, Wolfgang. “Space Policies, Issues and Trends 2008/2009.” Vienna: European Space Policy Institute. 62-63.
[459] Cf. Annex IV of the Commission of the European Communities. Commission Staff Working Document Accompanying the Communication from the Commission to the Council, the European Parliament, the European Economic and Social Committee and the Committee of Regions. Global Monitoring for Environment and Security (GMES) We Care for a Better Planet. SEC(2008)2808 of 12 Nov. 2008. Brussels: European Union.
[460] Amount for the period 2007-2013.

II. Projects and Satellites Related to the Provision of Internal Security in Europe

Project Name	JRC's 3-D Urban Structure information extraction from Very High Resolution SAR Imagery
Phase(s)	2: Information on urban structures is needed for immediate response 3: Information on urban structures is needed for reconstruction and impact assessment
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• 2006-2009
Current Status	• on-going
Country/Institution	• ISFEREA
Funding	• n.a
Partners	• n.a
Objective(s)	• The exploration of the information content of VHR SAR imagery of 3-D urban structures
In Practice	• The main scattering mechanisms of buildings in SAR, conducting scaled building model measurements in the European Microwave Signature Laboratory (EMSL), were investigated using a SAR simulator. • In VHR imagery, features from individual urban structures, like buildings, can be identified in their characteristic settings in urban settlement patterns (e.g. residential areas, city centres, industrial parks) • To assess the potential and the added value of VHR SAR for urban applications, a novel automatic 3D reconstruction method capable of determining the height of man-made structures from single detected SAR (power) imagery has been developed.
Demonstration(s)	• n.a
Users	• n.a
Added Value/Findings	• Novel automatic 3D reconstruction method capable of extracting man made structures from single detected SAR imagery
Collaboration(s)	• n.a
Follow-up	• n.a
Limitations	• n.a

Project Name	FP 4 AATMS TR 1001
Phase(s)	1: Air Traffic Management is useful for prevention
Critical Mission Area	Transportation Security/Air Transportation
Type	R&D
Project Duration	• 01.01.96 - 31.03.99
Current Status	• Completed
Country/Institution	• n.a.
Funding	• n.a.
Partners	• Daimler-Benz Aerospace NFS (DE) • Alenia Difesa (IT) • Dassault Electronique SA (FR) • Defence Research Agency DERA (UK) • GEC Marconi Avionics (UK) • Nationaal Lucht en Ruimtevaartlaboratorium (NLR) (NL) • National Avionics (IE) • OGMA (PT)
Objective(s)	• Increase navigation, communication and management capabilities for the improvement of aviation management and to tackle the expected growth in the aviation sector
In Practice	• The AATMS project has addressed user needs with regard to the onboard communication, navigation and surveillance/Air Traffic Management (CNS/ATM) equipment by building and validating an Aeronautical Telecommunications Network (ATN) compliant AATMS demonstrator covering a flight management system as an airborne end system that hosts the Automatic Dependent Surveillance (ADS), Air Traffic Control (ATC) and Airline Operations Control (AOC) application software and a communication management unit as the airborne router. • The AATMS demonstrator has implemented all the necessary CNS/ATM functions capable of meeting future ATM requirements taking into consideration the airline's point-of-view with regard to the objectives that must be applied when selecting an initial CNS/ATM architecture for implementation. • The AATMS modules were integrated in the flying test beds of DERA (BAC 1-11) and NLR (Cessna Citation) and validated in real-life scenarios. Making use of Eurocontrols PHARE ATN infrastructure and by simulating the ATC and AOC ground stations, the AATMS demonstrator could prove its operational and performance capabilities.
Demonstration(s)	• In the AATMS project the basis has been developed for an ATN compliant airborne architecture comprising an AFMS and an airborne router. The airborne components have been installed into two test aircraft and data communication has been established with

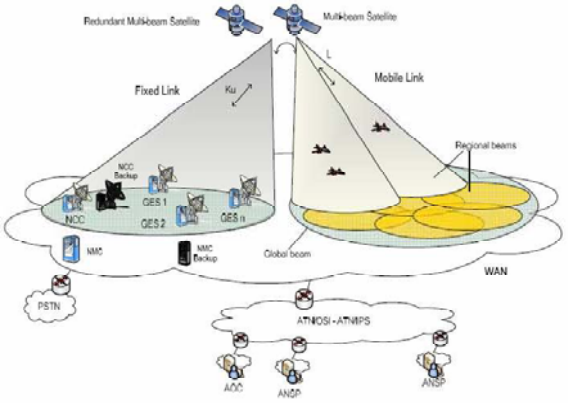
	ATC and AOC ground simulators making use of SATCOM and VHF data link systems. The trials have proved the selected concept.
Users	• Regional airplanes (could be used by all airplanes)
Added Value/Findings	• A better and faster running aviation management system • Due to the limited time available only a subset of the selected data link message set could be verified with some limitation given by the infrastructure and the data link system. • Limited to Air Traffic Management, thus of limited use for the provision of internal security. • Primary. objective to increase navigation, communication and management capabilities is not primarily aimed at improving the provision of internal security.
Collaboration(s)	• n.a.
Follow-up	• Follow-on activities will be proposed in the 5th Framework. These activities will make use of the developments made in AATMS and will in particular concentrate on a refinement of the data link applications towards pre-operational use.
Limitations	• Limited to air traffic management • Internal security not among the explicit objectives of this project

Project Name	**Projects related to the Establishment of a European AIS-S (Automatic Identification System picked up by Space)**
Phase(s)	1: Detection through non-cooperative AIS-S signals 2: Improved response through improved situation awareness based on AIS signals delivered
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security
Type	R&D
Project Duration	• n.a.
Current Status	• on-going
Country/Institution	• n.a.
Funding	• n.a.
Partners	• CNES/French Ministry of Defence • ESA • ThalesAlenia Space
Objective(s)	• Create a European AIS-S
In Practice	• CNES and the French Ministry of Defence started a study on maritime security based on space means in early 2008 with the objective of improving tracking of ships through space-based AIS. After some definition studies to be performed in bilateral cooperation, the programme could be propsed as a Euorpean service (possibly of GMES) • ESA is also exploring the feasibility of a system to pick up VHF signals emitted from ships' AIS from space: The European Commission through the JRC in close collaboration with the European Space Agency (ESA) is working to establish whether and how such a system could become operational, in which case it would considerably increase the added value of AIS data. Commission services are supposed to report on the feasibility project of picking up AIS signals from space by the end of 2009 • ThalesAlenia Space (TAS) is already working on an AIS-S system. TAS and its European partners are ready to deploy AIS-S, which will be capable of detecting an average of more than ninety percent of ships worldwide (including dense maritime areas) in less than two hours.
Demonstration(s)	• With current AIS reception being limited to the range of coastal antennas, trials have been conducted with the ESSAIM constellation to pick up the signals from space so as to provide a space-based AIS-
Users	• Stakeholders in waterborne transportation sector and border security
Added Value/Findings	• European component is a step towards establishing global coverage
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 6 ANASTASIA (Airborne New and Advanced Satellite Techniques and Technologies in a System Integrated Approach)**
Phase(s)	1: Air Traffic Management can have preventive results
Critical Mission Area	Transportation Security/ Air Transportation
Type	R&D
Project Duration	• April 2005-July 2009 (51 Months)
Current Status	• In progress
Country/Institution	• FP 6
Funding	• Total budget : € 19 628 427 • EU contribution: € 11 088 000
Partners	• Airbus • Thales Avionics (France) • DLR • Dassault
Objective(s)	• To provide on-board Communication, Navigation and Surveillance (CNS) solutions to cope

	with the foreseen doubling of air traffic by 2020. • The main goal of ANASTASIA is to pave the way for the introduction of new satellite-based technologies into aircraft operations in both navigation and communications.
In Practice	• To carry out research of future technology and system architectures for navigation, resulting in the development of a new generation of airborne GNSS receivers for all phases of flight. Such systems will offer accurate and safe global navigation while reducing avionics cost through the optimisation of the number and complexity of on board equipment. • On the communication side, work will concentrate on the design and implementation of a prototype SATCOM system that will meet the evolving European Air Traffic Management (ATM) requirements. • Research will also be conducted into higher bandwidth services, systems and airborne equipment to efficiently meet long term future aircraft communication requirements, including both ATM and passenger needs.
Demonstration(s)	• n.a.
Users	• Civilian Aircraft Operators • Stakeholders in the air transportation sector
Added Value/Findings	• The main outcome of ANASTASIA will be recommendations for future civil aircraft operation and a set of evaluated technologies and avionics architectures achievable from 2010 that will enable more autonomous, satellite-based aircraft operation. • When successful civilian aircraft operation becomes more autonomous enabling a great possible increase in civilian aviation that would otherwise be difficult to coordinate • Operates in all weather conditions • A beneficial consequence of this better management of airspace capacity will be reductions in air pollution, noise, and fuel consumption and increased safety
Collaboration(s)	• Airbus • DLR • Dassault
Follow up	• n.a.
Limitations	• Limited to traffic management and thus added value for provision of internal security is only a side-effect • Primary objective to provide on-board Communication, Navigation and Surveillance (CNS) solutions to cope with the foreseen doubling of air traffic by 2020; is not aimed at improving the provision of internal security

Project Name	**ESA's ARTES-10 - Iris** (Advanced Research in Telecommunications Systems))
Phase(s)	1: Prevention through secured communication systems
Critical Mission Area	Transportation Security/Air Transportation
Type	R&D
Project Duration	• n.a.
Current Status	• Development phase
Country/ Institution	• ESA
Funding	• Provides digital data links to airplane cockpits
Partners	• Canada
Objective(s)	• To develop a new Air-Ground Communication system for Air Traffic Management
In Practice	Communication System Design: • Communication protocols to be designed to operate with different types of satellites (GEO, LEO, HEO) so that interoperability between operators is possible The design should be such that ICAO acceptability criteria can be met (e.g. IPR) The spectrum used should be minimised Analysis and Definition of the Satellite System: • Dependability issues (esp. availability) are the main design driver Costs trade-off point towards a GEO solution, but Nordic countries request for high-latitude coverage led to consider complementary capacity from 3rd party satellites in Highly Elliptical or Low Earth Orbit. Antenna size and payload power are driven by ATS/AOC applications requirements

	 Figure: Concept of ANASTASIA[461]
Demonstration(s)	• n.a.
User	• Stakeholders in air transportation
Added Value/Findings	• n.a
Collaboration(s)	• Contributes to the SESAR (Single European Sky) Programme
Follow up	• n.a.
Limitations	• Not primarily aimed at countering malicious interferences by terrorists or criminals

Project Name	**Belgium's ASARTech** (Advanced SAtellite Remotes Sensing Techniques)
Phase(s)	1: Detection for Intelligence 2: Response
Critical Mission Area	Critical Infrastructures, Emergency Management
Type	R&D
Project Duration	• 15/12/2001 – 31/12/2004
Current Status	• Completed
Country/Institution	• Belgium
Funding	• Funded in the framework of the Belgian STEREO 1 (Support to the Exploitation and Research of Earth Observation data 1) programme of Belspo (Federal Science Policy Office)
Partners	• UCL (Telecommunications and Leledetection Laboratory) • UGENT (Image Processing and Interpretation Research Group) • CSL (Liège Space Centre) • ERM/KMS (Signal and Image Centre)
Objective(s)	• To develop and validate advanced techniques for the pre- and post-processing of SAR (Synthetic Aperture Radar) Spotlight remote sensing images
In Practice	• An analysis of the model sensitivity in regard to a set of input parameters was carried out, based on field data obtained by the Development of SAR Inversion Algorithms for Land Applications project, financed by ESA • The peculiarities and problems of Spotlight processing were identified (need for high numeric precision in calculations, interpolations, resampling), which will allow better command of this type of processing in the event of] future activities • An original and rapid filter, based on wavelets, was developed and validated to that end. A tool permitting edge detection was also developed
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• The peculiarities and problems of Spotlight processing were identified (need for high numeric precision in calculations, interpolations, resampling), which will allow better command of this type of processing on the in the event offuture activities • An original and rapid filter, based on wavelets, was developed and validated to that end. A tool permitting edge detection was also developed
Collaboration(s)	• An analysis of the model sensitivity in regard to a set of input parameters was carried out, based on field data obtained by the Development of SAR Inversion Algorithms for Land Applications project, financed by the European Space Agency. This work has continued under the PolInSAR project, also financed by the STEREO programme
Follow up	• n.a
Limitations	• n.a.

[461] Ongaro, Franco. "ESA Iris Programme Overview.". Presentation. NexSat – Eurocontrol. 24 Mar. 2009. 30 July 2009 <http://www.eurocontrol.int/nexsat/gallery/content/public/Steering%20Group/Meeting10/NexSat_Iris_Programme_overview_FO.pdf>.

Project Name	**PASR ASTRO+** (Advanced Space Technologies to suppoRt Security Operations)
Phase(s)	n.a.
Critical Mission Area	n.a.
Type	R&D
Project Duration	• 01/01/2005 – 31/03/2006, 15 months
Current Status	• Completed
Country/Institution	• PASR
Funding	• Total costs: € 2,95 million • EC contribution: € 2,20 million
Partners	• EADS Astrium Limited • EADS Astrium GmbH • Alcatel Space • Alenia Spazio • Telespazio • Centre National d'Etudes Spatiales • Deutsches Zentrum für Luft- und Raumfahrt • Alcatel ETCA • Ecole Royale Militaire de Belgique • Fondation pour la Recherche stratégique • Istituto Affari Internazionali • Indra Espacio • Landmäteriet Metria • Infoterra Limited • Nottingham Scientific Limited • Space Research Centre Polish Academy of Science • QinetiQ • Royal United Services Institute for Defence & Security Studies • SkySoft Portugal • European Union Satellite Centre • Infoterra GmbH
Objective(s)	• To demonstrate how space technologies and capabilities can enhance the security of Europe's citizens
In Practice	• Show the benefits of space technologies through the demonstration of precursor services for joint operations abroad • Answer to the implementation of a European security strategy by proposing new operational services available in the short term using space and by proposing a medium-term R&T action plan to exploit emerging mission concepts • Set up networking mechanisms between the space sector, research, users and stakeholders to create a multidisciplinary approach to address space services for security and to create a European framework for a "Space and Security Charter"
Demonstration(s)	• A security situation simulated in Poland in February 2006 demonstrated how ASTRO can support the requirements of operations on the ground by facilitating access to data from satellites, using high-performance equipment, and ensuring co-ordination between players to improve the overall efficiency of the crisis management system
Users	• Work with users of European security operations (peacekeeping forces, civil security forces, NGOs) in order to increase the value of space capabilities
Added Value/Findings	• Proposes, for user validation, new space services and infrastructures to support the mission concepts elaborated.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**BOC** (Besoin Opérationnel Commun)
Phase(s)	1: Prediction, detection
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Implemented service
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• n.a.
Funding	• n.a.
Partners	• Belgium, France, Germany, Italy and Spain in 2002 with Greece joining later on in 2003
Objective(s)	• To harmonise operational requirements and national Earth observation programmes of participating States • To identify requirements for building an independent European military satellite Earth observation system as a first step towards creating a European-level strategic imagery capability
In Practice	• Developed and implemented different multilateral agreements to achieve a common European Satellite Earth Observation System
Demonstration(s)	• n.a.
Users	• European satellite operators and users
Added Value/Findings	• Identification of common requirements for a Global European Earth Observation System of satellites
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

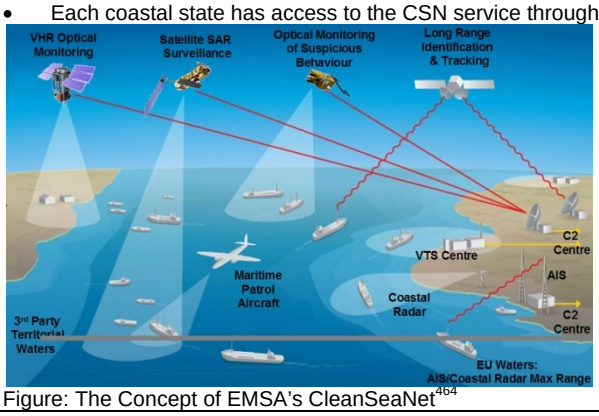
Project Name	**Bosch's transport supply chain security initiative**[462]
Phase(s)	1: Detection 2: Alert, response
Critical Mission Area	Transportation Security/Land Transportation
Type	Industry service
Project Duration	• n.a
Current Status	• On-going
Country/Institution	• Industry initiative
Funding	• Bosch
Partners	• n.a.
Objective(s)	• Worldwide container surveillance • Development of an international interventions network (with police and security authorities in all countries)
In Practice	• Data retrieved will be submitted to the network for taking preventive measures and emergency management • Route recommendations on the basis of risk assessment (data retrieved during past transports) for specific routes • 80 days autarkic plug and play container security box, which is tracked through GPS • Infrared sensor at the door • Both are connected to an alarm being transmitted to an alarm management centre, which checks the position of the container • Additional sensor in the container checking upon its load (warmth, movements etc.) • A similar concept can be used for tracking persons ("silent alarm option")
Demonstration(s)	• Positioning in Syria
Users	• Users involved in container transportation
Added Value/Findings	• International in character • Plug and play system can be added to all containers without large modifications being necessary • Can be used both on air and sea cargo (cost savings) • Potential for cost reductions in insurance payments
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• It remains to be seen whether this system can also be used for land transportation.

Project Name	**FP 6 BOSS4GMES** (Building Operational Sustainable Services for GMES)
Phase(s)	2: Response 3: Impact Assessment
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• 01/10/2006 – 31/03/2009, 30 months
Current Status	• In progress
Country/Institution	• FP 6
Funding	• Total costs: € 20.22 million • EU contribution: € 11.85 million
Partners	• 37 partners coordinated by Infoterra Ltd.
Objective(s)	• n.a.
In Practice	• To support the development of the three fast-track services and analyse the potential for synergies between current GMES services and will recommend the resulting optimal operational processes for and between those services, illustrating the resulting benefits through service operations of each of the EC specified "fast track" service topics • To define, study and test options for the most appropriate funding mechanisms, business models and organisational structures, mainly for the three fast-track services, and check how these findings can be applied to other GMES services • To build awareness of, and support for, the fast-track services and GMES as a whole amongst all its potential stakeholders in Europe and beyond. BOSS4GMES aims to leverage the knowledge and experience of many existing service providers and users, augmented by experts in the fields of financial and organisational analysis and in communications/lobbying to ensure that the project aims are met
Demonstration(s)	• n.a.
Users	• Stakeholders of the GMES' Fast Track Services
Added Value/Findings	• International in character • Respond, Risk-Eos and Preview projects focus mainly on service delivery. Boss4Gmes offers an opportunity to focus on the future evolution of these services. This will focus on

[462] Bosch Communication Center. "Initiative Transportkettensicherheit i. G." Presentation. Hamburg. 27 Sep. 2007. ISO 28000.de Transportkettensicherheit News. 30 July 2009 <http://www.iso28000.de/fileadmin/user_upload/iso28000.de/Download_area/070927_BoschMobileSecurity_Containersicherheit.pdf>.

	the end-to–end information deliver lead time, ease of information access by the users, and link with risks prevention
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Limited to GMES Fast Track Services

Satellite name	**CERISE** (Caractérisation de l´Environnement Radioélectrique par un Instrument Spatial Embarqué)
Phase(s)	1: Signal Intelligence for prevention
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Operational Satellite
Date of Launch	• 7 July 1995
Current Satellite Status	• Operational
Operating Country/Institution	• France, DGA/French Ministry of Defence
Capability	• n.a.
User	• n.a.
Projected lifetime	• Cerise collided with a piece of debris from an Ariane upper stage on 24 July 1996, taking damage at the stabilisation boom. After a software update, Cerise and its payload continued to work normally under full ground control. Operational status unclear.
Objective(s)	• Signal Intelligence
Service (in practice)	• To eavesdrop on weak HF communications • To carry out broadband radiometric measurements
Cooperation/data exchange	• n.a.

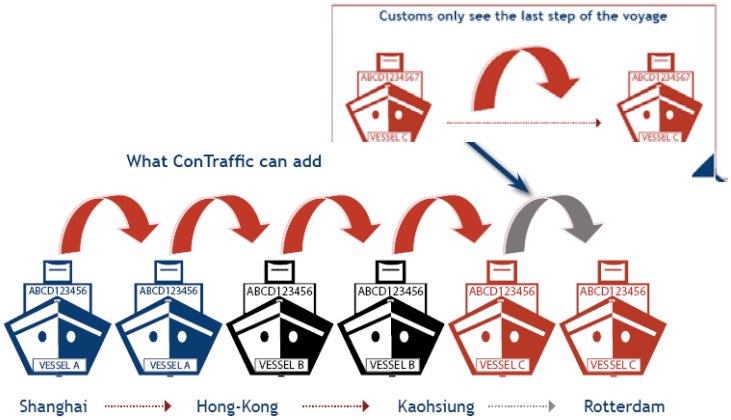
Project Name	**EMSA's CleanSeaNet (CSN)**
Phase(s)	n.a. (covers oil spill detection)
Critical Mission Area	Transportation Security/Maritime Transportation, Border Security, Emergency Management
Type	Implemented Service
Project Duration	• 04/2007 - ...
Current Status	• On-going
Country/Institution	• EMSA
Funding	• € 7.607.681 Commitment appropriations (EMSA) • € 7.951.681 Payment appropriations (EMSA)[463]
Partners	• JRC • EUSC • ESA (ENVISAT) • MDA (RADARSAT) • Eurimage (ASAR)
Objective(s)	• To detect potential discharges of oil from ships in near real time
In Practice	• Analysed satellite imagery available to the contact points within 30 minutes after satellite overpass • In the case of a detected oil slick, an alert message is delivered to the contact point • Each coastal state has access to the CSN service through the CSN browser  Figure: The Concept of EMSA's CleanSeaNet[464]

[463] European Maritime Safety Agency. "Work Programme 2009." 3 Feb. 2009 European Maritime Safety Agency 30. July 2009 <https://extranet.emsa.europa.eu/index.php?option=com_docman&task=cat_view&gid=108&Itemid=193>. 53.
[464] "THE TWINNING LIGHT PROJECT- Development of a procedure for the confirmation of CleanSeaNet satellite images and prefeasibility study of remote sensing and surveillance arrangements in Bulgarian territorial waters." Presentation. Sofia, Bulgari. 29 May 2009 <http://www.marad.bg/upload/docs/Presentation_Rytkonen__Lampela___development.ppt>.

Demonstration(s)	• n.a.
Users	• EU Coastal States
Added Value/Findings	• Detection at sea under most conditions – day and night and through cloud • The sensor can operate in a number of modes – from wide area and low detail to smaller area and higher detail
Collaboration(s)	• In the case of a major oil spill disaster in European seas and adjacent waters, EMSA, through the CSN service, is co-operating with the International Charter (Space and Major Disasters)
Follow-up	• EMSA has begun to prepare for the next generation of CSN to succeed the current round of contracts which end in December 2010. This 2nd generation CSN shall be a flexible and advanced system that will serve as a step forward providing Member States with enhanced information to respond to illegal oil pollution at sea and to support national prosecution activities • The new system will offer satellite radar products from ESA's ENVISAT ASAR, Sentinel-1, ERS-2, RADARSAT 1-2, and occasionally other SAR sensors. A central modular data management system will handle data dissemination, ingestion, archiving and cataloguing tasks as well as the web mapping browser for end users. Links between the new system and additional sources of marine and maritime data will be established
Limitations	• Limited to oil spill detection; currently no provision of additional services like detection of organised crime on illegal migration on sea • Could form a comprehensive EU Vessel Traffic Monitoring and Information System together with SNN

Satellite name	**Clementine**
Phase(s)	1: Intelligence
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Technology demonstrator
Date of Launch	• 3 December 1999
Current Satellite Status	• Operational
Operating Country/ Institution	• DGA / France
Capability	• n.a.
User	• DGA/French Ministry of Defence
Projected lifetime	• n.a.
Objective(s)	• Monitoring Earth's electro-magnetic environment (low frequency electronic signals)
Service (in practice)	• Electronic Intelligence/Signal Intelligence
Cooperation/data exchange	• n.a.

Project Name	**JRC's ConTraffic**
Phase(s)	1: Prevention (through deterrence), Detection 2: Alert
Critical Mission Area	Transportation Security/Waterborne Transportation
Type	R&D
Project Duration	• 2007 - ...
Current Status	• n.a.
Country/Institution	• JRC • OLAF
Funding	• JRC
Partners	• n.a.
Objective(s)	• To systematically gather and analyse data on global maritime container movements, and enable the identification of potentially suspicious consignments
In Practice	• To gather data from several dozen sources in parallel, allowing customs officers to retrieve movements of specific containers over the last 2-3 years • Con-Traffic can give customs additional information with respect to what is indicated in customs declarations, i.e. the intermediary transhipment ports and the origin of the goods
Demonstration(s)	• The system was tested very successfully for cases of false declaration of origin in order to circumvent anti-dumping duties and quotas or to smuggle prohibited or counterfeited goods
Users	• Users Involved in container transport
Added Value/Findings	• Makes entire supply chain more transparent • ConTraffic's experimental database is sufficiently populated to allow a statistically sound route-based risk analysis. Its 40% success rate in detection based on routes information alone can add great value to customs' own risk analysis systems • Through ConTraffic's risk analysis these abnormal events can be identified, thus allowing customs authorities to better target physical inspections

	• Improved risk analysis can improve targeting of physical inspections • ConTraffic can contribute to relevant EU and international regulatory activities (such as the Container Security and Proliferation Security initiatives) by providing new indicators based on global routes analysis for containerised cargo, that can support real-time control operations at ports and feed into a common set of minimum risk rules for container security in the EU  Figure: Added Value and Pracitcal Application of ConTraffic[465]
Collaboration(s)	• n.a.
Follow-up	• DG JRC is currently extending the ConTraffic system to address specific security related concerns including illicit trafficking of security-sensitive goods
Limitations	• Limited to waterborne transportation, should be extended to land and air transportation so as to cover intermodal transportation within the supply chain

Satellite family name	**Italy's COSMO-SkyMed** (Constellation of Small Satellites for Mediterranean Basin Observation - 4 satellites)
Phase(s)	1: Prevention for intelligence, detection for waterborne transportation security 3: Emergency Management
Critical Mission Area	Intelligence, Early Warning and Risk Management, Transportation Security/Waterborne Transportation, Emergency Management
Type	Operational satellite constellation
Date of Launch	• 8 June 2007 and 9 December 2007
Current Satellite Status	• Operational
Country/Institution	• ASI, Italian MoD / Italy
Capability	• Synthetic Aperture Radar • X-band • Spotlight mode: < 1 m (for an area of 10 km2) • HIMAGE/Strip map mode: 3-15 m (swath width 40 km) • WideRegion/ScanSAR mode: 30 m (swath width 100 km) • HugeRegion mode: 100 m (swath width 200 km)
User	• ASI, Italian MoD, civilian and military users
Projected lifetime	• 5 years
Objective(s)	• Provision of imagery
Service (in practice)	• Remote sensing for military (surveillance, intelligence, operations assessment) and civil (agriculture, forest, geology, environment, cartography, risk management) purposes
Cooperation/data exchange	• Duplication of ground segments for civilian and military users (architectural symmetry) • Designation of different classes of civilian and defence users, each characterised by a specific priority • Three modes with different response times: Routine Mode (72 h), Crisis Mode (36 h), Very-Urgent Mode (18 h) • Radar component of the dual-use ORFEO programme (with French optical Pleiades satellites) • Cooperation with Argentinean SAOCOM L-band SAR system for civil purposes (SIAGE) • Shall contribute to GMES • Imagery data is used by the European Union Satellite Centre • Formed a company to sell imagery gained by COSMO-SkyMed satellite constellation

[465] Joint Research Centre. "Maritime Surveillance ConTraffic: Monitoring Container Traffic." Joint Research Centre, 2008. <http://ipsc.jrc.ec.europa.eu/showdoc.php?doc=promotional_material/JRC42970_contraffic_leaflet.pdf&mime=application/pdf>.

Project Name	**JRC's Institutional Action CRITECH** (CRIsis monitoring and response TECHnologies)
Phase(s)	1: Early Warning Technology & Crisis Preparedness 2: Response Technology 3: Impact Assessmentren
Critical Mission Area	Intelligence, Early Warning and Risk Management, Emergency Management,
Type	R&D
Project Duration	• 2007 - ...
Current Status	• Ongoing
Country/Institution	• JRC
Funding	• n.a.
Partners	• UNDP, the United Nations Agency in charge of supporting field teams during their missions. • DG-SANCO • Dartmouth Flood Observatory
Objective(s)	• To enhance the EU's crisis preparedness and response capacity by means of fostering a better capacity for early warning, alerting, situational awareness and monitoring, and impact assessment
In Practice	• The development of web-based solutions for the collection, integration, analysis, visualisation and dissemination of crisis related information. • To expand expertise in numerical modelling, information technologies and GIS in view of providing novel solutions for crisis preparedness, early warning, monitoring the development of a crisis, and evaluating the consequences of crises such as those related to communicable diseases and natural disasters. Four main components: • Geo-visual and field gathering tools (to develop a field data collection system in order to support rescue or emergency teams that need to collect information in the field. The tool which will be developed will include light collection devices, based on PDA equipped with GPS and video camera, base-camp applications to collect the data gathered with the PDA devices and server applications to definitely store the data). • Crisis Information Management Systems (to develop and integrate additional modules / functionalities that will further support situation assessment, information exchange, field teams collaboration activities for crisis management operations). • Early Warning and Alerting Models (to develop procedures to have an automatic flood mapping of the affected areas. This procedure will allow always having automatic determination of the flooding extent and will allow estimation of the evolution of the phenomenon over time… • Secure IT systems.
Demonstration(s)	• n.a.
Users	• Emergency Responders • Early Warning to Local Authorities
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a
Limitations	• Definition of "crisis" too limited as it does not explicitly include terrorism.

Project Name	**FP 5 DECLIMS**
Phase(s)	All: Benchmarking of both optical and radar sensors
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security
Type	R&D
Project Duration	• 01/05/2003 – 30/04/2006, 36 months
Current Status	• Completed
Country/Institution	• n.a.
Funding	• FP 5
Partners	• 24
Objective(s)	• To provide a focus for research into the use of satellite imagery for maritime vessel detection, classification and identification • To allow researchers to develop a better understanding of the capabilities of such systems • To identify the advantages and drawbacks of different approaches • To strengthen the infrastructure capable of meeting the demands of users, and • To help drive the development of new sensors and platforms towards the operational needs of vessel monitoring • To investigate different techniques for vessel detection - SAR, optical imagery, wake detection, integration with GPS-based systems - on the basis that operational systems will integrate different methods
In Practice	• n.a.
Demonstration(s)	• In the 2005 benchmark on SAR vessel classification, 9 systems participated
Users	• Fisheries control (finding illegal fishing vessels); Defence (maritime situational awareness),

	pollution (oil spills), civil maritime security (smuggling, trafficking, piracy, terrorism), maritime traffic safety (compliance with traffic rules, collision avoidance)
Added Value/Findings	• The benchmark tests run in the DECLIMS project have made the operational state-of-the-art more clear, and have enabled developers to improve their systems and products • Ship detection from satellite SAR imagery is an operational application that can be performed in near-real time, with detection and false alarm rates that are sufficient for some users and scenarios but need to be improved for others. Its value for the end users is only attained when the resulting detections are fused with vessel traffic data from other sources, and this is increasingly being done. • Vessel classification from SAR data, however, is still difficult; automatic algorithms do not yet always provide reliable size estimates • Main users of space-based ship detection are fisheries control and defence authorities; civilian security use seems upcoming • Automatic ship detection software implemented for fast processing has quite good properties in terms of detection and false alarm rates, but still not as good as a human operator who is better at dealing with complex clutter situations • In combination with a good communication infrastructure, detected ship positions can be with the users within 30 minutes after acquisition • Many systems that call themselves operational have only recently installed the capacity for giving size estimates with their detections. Still, in many cases this estimate is rendered worthless on account of the azimuth smearing that the target, moving on the swells, is subject to • No operational system seems to be able to venture a guess as to the vessel type • The optimum way to deal with ENVISAT ASAR AP mode images is still under debate, and has not reached most operational systems yet • ENVISAT ASAR WSM receives some interest to detect larger (cargo) shipping, but its resolution makes it of very limited use for detection of (smaller) fishing vessels • With regard to future systems, maritime surveillance applications in most cases will favour swath width over polarimetric information. Full polarimetry is likely to be at best confined to coastal or port studies • The limited classification potential means that space borne SAR should primarily be used as a cueing sensor, implying the need for very fast turnaround and tasking capability
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Benchmarking was limited to 9 systems for testing and demonstration

Project Name	**JRC's DMA** (Digital Map Archive)
Phase(s)	2: Reference maps for emergency response
Critical Mission Area	Emergency Management
Type	Implemented service
Project Duration	• n.a.
Project Status	• n.a.
Country/Institution	• JRC
Funding	• JRC
Partners	• n.a.
Objective(s)	• To provide scientific support to EU policies, the JRC understands the technology involved through hands-on experience; the JRC can deal with information that is confidential to the Commission; and the JRC actively aims to integrate the effort of other organisations. • To provide aid and assistance to the European community by making imagery satellite data accecible to different user groups
In Practice	• The Digital Map Archive is developed and maintained by the CRITECH Action of the Unit for the Support of External Security of the Institute for the Protection and Security of the Citizen by the Joint Research Centre. • The digital map archive contains data from open sources, academic institutions and commercial vendors. The Digital Map Archive databases can be accessed by computer programs and websites using web service interfaces. DMA offers standard and non-standard interfaces to maps and mapping services, analysis services and more. Services developed are: • Google Interface • NASA WORLD Wind The DMA is also used as a decision support tool where several modelling and analysis systems have been developed or are in development: • Asgard: A model is run when the appropriate phenomenon is detected. Salient features of the event are passed to the model as parameter. The models comprise a series of sub-processes, including at least one geo-referenced analysis, and are computationally intensive. The models are long-running processes, and cannot be expected to provide 'real-time' responses. The models comprise an analysis phase that can trigger asynchronous messages. • GDACS is a web-based platform that combines existing web-based disaster information

	management systems. Its aims are to alert the international community in case of major sudden-onset disasters and to facilitate the coordination of international response during the relief phase of the disaster. GDACS is compromised of the following elements: • A set of standards to ensure interoperability between alert and response coordination systems • A set of working procedures to ensure predictability and reliability of information creation and exchange during crisis response • Stakeholders that commit to providing related information according to the standards and the working procedures
Demonstration(s)	• n.a.
Users	• National Governments • Academia • Scientific Community
Added Value/Findings	• By integrating different data sets a single database can be created for different users while at the same time enhancing security through these integrations
Collaboration(s)	• Europa Technologies Ltd. (Global Discovery data) • NASA Earth Science Enterprice (Landsat data) • Google (Google Interface) • GDACS • Oak Ridge National Labatory • Touratech (Russian Military Map)
Follow-up	• n.a
Limitations	• n.a

Satellite Constellation	**DMC** (Disaster Monitoring Constellation - 6 satellites)
Phase(s)	1 & 3: Emergency Management
Critical Mission Area	Emergency Management
Type	Operational Satellite Constellation
Date of Launch	• Algeria: 2002 • China: 2005 • Nigeria: 2003 • Turkey: n.a. • UK: 2003 and 2008 • Spain: 2008
Current Constellation Status	• Turkey's satellite is the only one not operational
Operating Country/Institution	• Algeria, China, Nigeria, Turkey, UK, Spain • The satellites are all designed and built at Surrey Satellite Technology Ltd. (SSTL) • Each satellite is independently owned and controlled by a separate nation (2 for UK), but all satellites have been equally spaced around a sun-synchronous orbit to provide daily imaging capability
Capability	• Algeria: 32 m (multispectral) • China: 32 m (multispectral) / 4 m (panchromatic) • Nigeria: 32 m (multispectral) • Turkey: completed • UK: 22 m and 32 m (multispectral) • Spain: 22 m (multispectral)
User	• Emergency Responders • Algeria, China, Nigeria, UK, Spain
Projected lifetime	• n.a.
Objective(s)	• Disaster Monitoring
Service (in practice)	• Disaster Monitoring
Cooperation/data exchange	• DMC Members agree to provide 5% of capacity free for daily imaging of disaster areas • The DMC Consortium has agreed to consider participation in the International Charter for Space in Major Disasters • UK-DMC also provides data through an ESA project called RESPOND • DMC Members are interested in encouraging the use of DMC data for scientific and commercial applications

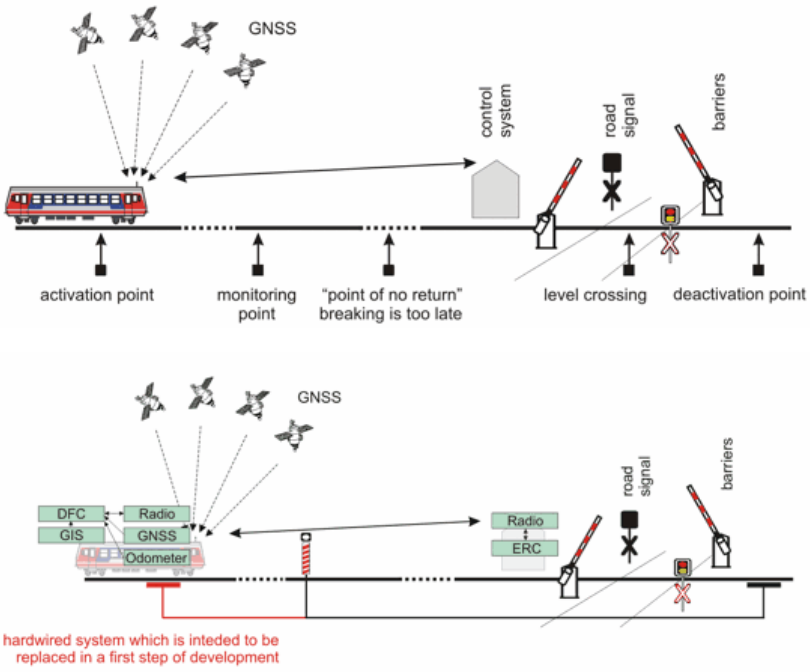
Project Name	**FP 6 DRAGONESS** (DRAGON in Support to the Development and Integration of a Chinese Marine Monitoring for Environment and Security System)
Phase(s)	n.a. (added value for global approach)
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Secuirty
Type	R&D
Project Duration	• 01/09/2007 – 31/08/2010, 36 months
Current Status	• In progress

Country/Institution	• FP 6
Funding	• Total costs: € 500.542 • EU contribution: € 500.542
Partners	• 13, coordinated by Nansen Environmental and Remote Sensing Centre (Norway)
Objective(s)	• To consider harmonisation of methods and approaches in the frame of international programs such as GOOS, GEO and GMES
In Practice	• To assess current Chinese and European services and information products arising from integrated use of networks of remote-sensing, in-situ observations, models and data assimilation methods against the GEOSS requirements • To identify service/data gaps and barriers, such as for instance restrictive data dissemination and availability and re-use policies • To develop activities to disseminate and implement products and services derived or customised from existing development and operational activities • To study and identify the potential for existing and foreseen European GMES services (both funded through EU FP6 and ESA) to be transferred to P.R. of China and provide the building blocks for the EU contribution to Chinese marine monitoring for environment and security, and hence to GEOSS • To stimulate exchange and initiation of new partnership in Earth Observation science and technology in support of global environmental monitoring by bringing together scientists from Europe and China • To make an inventory of Chinese and European capacities in marine monitoring for environment and security
Demonstration(s)	• n.a.
Users	• GMES services, GEOSS, China
Added Value/Findings / Findings	• Exchange of knowledge between China and the EU
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Limited to China, not internal security BUT could be useful to improve border securiy (coastal zones) and waterborne transportation security

Project Name	**JRC ECI-GIS (CEI-GIS)** (Geographic Information System of the European Critical Energy Infrastructure)
Phase(s)	1: Detection/Prevention through modelling of of security scenarios 2: Alert 3: Lessons learned for reconstruction
Critical Mission Area	Critical Infrastructures
Type	R&D, some preliminary application
Project Duration	• 2008/2009
Project Status	• n.a.
Country/Institution	• JRC • IPSC • Institute for Energy
Funding	• n.a.
Partners	• DG TREN
Objective(s)	The purpose of the CEI-GIS project is to create an interactive Geographic Information System, for supporting: • The identification and representation of the CEI over the European territory; • The modelling and simulation of security relevant behaviours of the CEI • The storage and the visualisation of the appropriate information, mainly focusing on security scenarios; • Further risk analysis elaborations and models.
In Practice	• The system will be built upon a database storing all the relevant information related to the CEI, including social, administrative and organisational data. The aim is to support the modelling and simulation of security scenarios, establishing connections with the map representation for the correct visualisation of both the infrastructure as a whole and the particular critical components. The CEI-GIS is being built using ARC-GIS tools. The web applications of CEI-GIS are developed according to the following procedure: • Start point: SDE geo data bases, • Creation using Microsoft SQL Server • Population using ArcCatalog. The map services are created in the ArcGIS Server, based on the available databases; the web applications are prepared using the ArcGIS Server Manager, with all the required functionalities (editing, query attributes, geo processing…).
Demonstration(s)	Making use of: • GISCO Database

	• Teleatlas Multinet • Platts • The Union for the Co-ordination of Transmission of Electricity (UCTE) • EUROSTAT
Users	• Stakeholders in energy sector
Added Value/Findings	• The raw data and the information resulting from the analyses are automatically stored in a central database. Those information sets are of different kind: administrative, technical, and geographical security-related. • By creating specific links associated to each element in the map, each user has the possibilityof directly connecting to the database for accessing all the information related to the element. • Qualified users have also the ability to modify the database, by using the specific ArcGIS Server editing tools. • With the ArcGIS Server tools, users can do complex data geo-processing, run models, create elaborations showing the behaviour of the infrastructure, and or visualise specific scenarios.
Collaboration(s)	• MOSAIC • GISCO Database • Teleatlas Multinet • Platts • The Union for the Co-ordination of Transmission of Electricity (UCTE) • EUROSTAT
Follow-up	• n.a.
Limitations	• n.a.

Project Name	ESA's ECO-RAIL
Phase(s)	1: Prevention, Detection 2: Monitoring rail crossings could provide basis for alert
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• September 2001-June 2003
Current Status	• Completed
Country/Institution	• ESA
Funding	• ESA
Partners	• TECHNICATOME, France • ALCATEL, Austria • STERN & HAFFERL, Austria • TELECONSULT, Austria • ST MICROELECTRONICS, Italy • SYSTRA, France • ICROELECTRONICS
Objective(s)	• The ECO-RAIL project was aimed at designing and testing a positioning system which is based on satellite navigation in a safety critical railway application. In order to show the benefit of the innovation, the project focused on the specific application of the control of automatic level crossings (ALX): Based on an on-board navigation unit employing a multi-sensor technique, it is possible to activate an ALX via a radio link. Consequently, there is reduced need for cost-intensive track-side facilities, especially for positioning (e.g. track circuits, beacons) and for signal transmission (cables).
In Practice	• The EGNOS Controlled Railway Equipment (ECO-RAIL) project was one of several GNSS-1 Rail User Navigation Equipment projects dealing with the application of satellite navigation io the railway domain to demonstrate the feasibility and benefits of GNSS. The ECO-RAIL application is based on the following components: • On-board unit (OBU) for determining the train's position on the track: The navigation unit is composed of three navigation sub-systems: • GNSS-1 receiver (GPS positioning with EGNOS augmentation) • odometer (measurement of travelled distance) • GIS (cartographic information) • The data of these components is processed in a data fusion computer (DFC) that extracts the best combined solution by means of sophisticated filtering techniques (Kalman filtering). • Communication system: Conventional radio equipment establishes the communication link between on-board and ground equipment. • Electronic Recording Computer (ERC): This part of the equipment is in charge of recording radio messages coming from the on-board equipment as well as the actions of the existing hardwired system of the ALX. • To examine the quality of localisation, the activation of a level crossing was simulated and compared with the data derived from conventional equipment. In addition, a time-optimised activation based on the current speed of the train was tested. Therefore, it was demonstrated that the satellite navigation is not only an equivalent alternative to

	conventional track-side equipment but it has to be considered as a more efficient solution.  Figure: Practical Working of ECO-RAIL[466]
Demonstration(s)	• After the specification of requirements and technical delineation, the ECO-RAIL project comprised the design and testing of the on-board equipment as well as a demonstration on a pilot track in the Austrian province Upper-Austria. The demonstration aimed at the evaluation of system performance, regarding the efficiency of the developed localisation compared to the existing hardwired localisation system.
Users	• Stakeholders in the railway transportation sector
Added Value/Findings / Findings	• Cost reduction for investment and operation (reduce costs by saving the expensive wiring) • Performance improvement with an safety level equivalent to existing systems • Save environmental resources • Increase the commercial competitiveness of railway operators • The capability to integrate additional functions based on satellite navigation data
Collaboration(s)	• European Train Control and Rail Traffic Management System (ETCS/ERTMS)
Follow-up	• n.a.
Limitations	• Demonstrations were limited to GNSS • Management not per se intended to improve internal security • Limited to train traffic

Satellite Constellation	**ELISA** (ELectronic Intelligence SAtellites - 4 satellites)
Phase(s)	1: Intelligence
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Envisaged technology demonstrator
Date of Launch	• 2010
Current Satellite Status	• Envisaged
Country/Institution	• DGA / France
Capability	• Elisa will replace Essaim with enlarged radio spectrum, improved sensitivity and localisation accuracy
User	• DGA/French Ministry of Defence
Projected lifetime	• 3 years
Objective(s)	• Identification of radar emitters worldwide
Service (in practice)	• Electronic intelligence
Cooperation /data exchange	• The French government is reportedly advocating the development of a European signal intelligence capability based on its technology demonstrators.

[466] Thevenot, V. "ECORAIL: GNSS For Safe Railway Road Cross Applications." Presentation. Noordwijk, The Netherlands. 17 Janary 2006. 29 July 2009 <http://esamultimedia.esa.int/docs/NavigationProjects/ECORAIL_FP_.pdf>.

Project Name	Italy's EMERSAT
Phase(s)	2: Response 3: Impact Assessment
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• Italy/ ASI
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Development, integration and testing of satellite solutions for applications and communications services dedicated to security and emergency management • Give the institutional agencies' operators devoted to emergency security and management all the technology for receiving and transmitting the information needed for the most effective and safest management of emergency interventions • The enhancement of the satellite's role considering essential problems such as: interoperability between different networks, strength and "survival", security of communications and access, back up and complementarity of solutions, enhancement of all synergetic applications that optimizes the use of satellite technology
In Practice	• Development of a multiservice applications architecture that foresees providing integrated low-band (LB) communications (fixed, mobile or relocatable) of satellite navigation and georeferenced localization and high-definition remote sensing • To consider interoperability between different networks, security of communications and access, back up and complementarity of solutions, enhancement of all synergetic applications that optimise the use of satellite technology
Demonstration(s)	• n.a.
Users	• Emergency management forces and responders
Added Value/Findings	• Development, integration and testing of satellite solutions for applications and communications services dedicated to security and emergency management
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• n.a.

Project Name	**Austria's EMOGES** (Evaluation of Positioning Technologies for Monitoring Dangerous Goods Transports on the Austrian high-ranking Road Network)
Phase(s)	1: Prevention and Detection through monitoring 2: Improved Search and Resque through improved monitoring and positioning
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• 02/03/2003 – 30/09/2004
Current Status	• Completed
Country/Institution	• Austria
Funding	• Funded in the framework of the Austrian Space Programme
Partners	• ARC Österreichisches Forschungs- und Prüfzentrum Arsenal GmbH (coordinator) • c.c.com Andersen & Moser GmbH • ARC Seibersdorf research Ges.m.b.H. • Abteilung Höhere Geodäsie, TU Wien • Forschungsgemeinschaft Mobilität Ges.m.b.H. • Via donau
Objective(s)	• Since the functionality of conventional and future satellite navigation systems cannot be guaranteed on all sections of the Austrian high-ranking road network (e.g. in tunnel shadings) it is necessary to use complementary technologies for these areas to provide a seamless monitoring system of dangerous goods on the whole road infrastructure. • Consistent monitoring of dangerous goods transports on all sections of the road network is the only method to obtain sufficient data quantity to guarantee the safety of motorists and all abutters on those critical road sections • Evaluation of positioning technologies for monitoring dangerous goods transport on the Austrian high-ranking road network
In Practice	• For that purpose a functional model of an on-board-unit (laptop) including all relevant software solutions as well as a virtual tunnel centre to track the hazardous goods transport during its tunnel passage was implemented. • A system architecture was created and implemented that carries out automatically the handshake-procedure at the crossover point between out-door-positioning (using GPS) and in-door-positioning and the whole communication with the tunnel control centre.
Demonstration(s)	• n.a.
Users	• Civil authorities, transportation companies
Added Value/Findings	• The evaluation of the test run of the system showed not only a high grade of usability in hazardous goods monitoring systems but also that the accuracy of the positioning is highly

	dependent on the quality of the used acceleration sensor. Finally, concerning the usability of video image processing the feasibility study showed that this technology is merely appropriate for the use in hazardous goods monitoring systems in tunnels. • Integration of satellite navigation with complementary technologies for areas where satellite navigation systems cannot be guaranteed (e.g. mountains)
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• Limited to monitoring of dangerous goods transport on high-ranking road network • Has the potential to be extended so as to cover road monitoring in general • Should become a part of a layer attempt to improve the land transportation sector and to secure the whole transport supply chain

Project Name	**FP 4 EOLIA TR 1021**
Phase(s)	1: Communications, Navigation and Surveillance 2: Air Traffic Management and Control
Critical Mission Area	Transportation Security/Air Transportation
Type	R&D
Project Duration	• n.a.
Current Status	• Completed
Country/Institution	• Aerospatiale Matra Airbus
Funding	• EC DGXIII • EUROCONTROL
Partners	• Sofreavia/STNA (FR) • Airsys ATM (FR), • NATS (UK) • NLR (NL) • Thomson/CSF Airsys • Airbus
Objective(s)	• To develop software to demonstrate and evaluate Communications, Navigation and Surveillance, Air Traffic Management (CNS/ATM) for Air Traffic Control (ATC) data link services for eventual operational use in European airspace.
In Practice	• For the first time, both ground and airborne systems were developed according to the industrial rules. In particular, the airborne software was installed in a commercial avionics computer, the Aerospatiale Matra Airbus Air Traffic Services Unit (ATSU) recently used for FANSA operations in the South Pacific on A340 aircraft
Demonstration(s)	• EOLIA services were also demonstrated at partner sites: • NLR used its NARSIM facility to assess its ground HMI, • Sofreavia assessed HMI design and datalink procedures on CHARME, its internal ProATN compatible infrastructure • NATS performed local verification of a subset of services. • The airborne data link services were tested using the Aerospatiale Matra - Airbus A340 simulator, which fully represent the behaviour of an actual A340 aircraft, and were then installed on a Cessna Citation in conditions that are more representative of a real operational environment. EOLIA offered a series of successful demonstrations at the Le Bourget Air show in Paris in June 1999. The demonstrations during the show were a very good opportunity to show the results to the European and worldwide aeronautics community.
Users	• Air Traffic Control Centre • Air Transportation Sector
Added Value/Findings	• Reduced ATC and aircrew workloads • Reduced airline costs • Increased safety and reliability. • The results of the evaluations shown that benefits can be expected when these services are used in an operational context.
Collaboration(s)	• n.a.
Follow-up	• n.a
Limitations	• Objectives mainly concerned with improving air traffic management • It is not primarily concerned with internal security threats • Should become part of a bigger attempt to improve the provision of internal security in the sector of air transportation

Satellite Constellation	**ESA's ERS-2** (European Remote Sensing satellite)
Phase(s)	1: Imagery for detection 2: Improved response through improved situational awareness
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security
Type	Operational Satellite Constellation
Date of Launch	• 1995

Current Satellite Status	• ERS-2 Operational
Country/Institution	• European Space Agency (ESA).
Capability	• Comprehensive payload including an imaging Synthetic Aperture Radar (SAR), a radar altimeter and other powerful instruments to measure ocean surface temperature and winds at sea. • RA: The Radar Altimeter is a Ku-band (13.8 GHz) nadir-pointing active microwave sensor designed to measure the time return echoes from ocean and ice surfaces. • ATSR: The Aling Track Scanning Radiometer consists of an Infra-Red Radiometer (IRR) and a Microwave Sounder (MWS). On board ERS-1 the IRR is a four-channel infra-red radiometer used for measuring sea surface temperatures, whereas on board ERS-2 the IRR is equipped with additional visible channels for vegetation monitoring. • GOME: Global Ozone Monitoring Experiment, a nadir-scanning ultraviolet and visible spectrometer for global monitoring of atmospheric Ozone, was launched on-board ERS-2 in April 1995. Since summer 1996, ESA has been delivering to users three-dayGOME global observations of total ozonem nitrogen dioxide and related cloud information. • MWR: The main objective of the microwave radiometer (MWR) is the measurement of the integrated atmospheric water vapour column and cloud liquid water content, as correction terms for the radar altimeter signal. • SAR: Synthetic Aperture Radar wave mode provides two-dimensional spectra of ocean surface waves. In image mode the SAR provides high resolution two-dimensional images with a spatial resolution of 26 m in range (across track) and between 6 and 30 m azimuth (along track). Image data is acquired for a maximum duration of approximately ten minutes per orbit. • ES: The purpose of the Wind Scatterometer is to obtain information on wind speed and direction at the sea surface for incorporation into models, global statistics and climatological datasets. It operates by recording the change in radar reflectivity of the sea due to the perturbation of small ripples by the wind close to the surface. • PRARE: The Precise Range And Range-Rate Equipment PRARE is a compact space-borne, two-way, two-frequency microwave satellite tracking system
User	• Transportation sector • Maritime services • First Responders/Organisations
Projected lifetime	• n.a.
Objective(s)	• To test and demonstrate the feasibility of using EO satellite data in different maritime policy areas
Service (in practice)	• Apart from supporting development in environmental applications the ERS mission also advanced coastal bathymetry mapping and vessel detection • ERS-2 provides coastal surveillance capabilities
Cooperation/data exchange	• SPOTimage

Project Name	**ERTMS** (European Rail Traffic Management System)
Phase(s)	1: Train control, signalling and traffic management can have a preventive effect Integrated approach: introduction of European standard in control, signalling and traffic management
Critical Mission Area	Transportation Security / Land Transportation
Type	Implemented service
Project Duration	• n.a.
Current Status	• On-going
Country/Institution	• n.a.
Funding	• n.a.
Partners	• Alstom Transport, Ansaldo STS, Bombardier Transportation, Invensys Rail Group, Siemens Mobility and Thales
Objective(s)	• To harmonise the current national train control and command systems in Europe so as to increase European railway's competitiveness
In Practice	ERTMS has two basic components: • ETCS, the European Train Control System, is an automatic train protection system (ATP) to replace the existing national ATP-systems; • GSM-R, a radio system for providing voice and data communication between the track and the train, based on standard GSM using frequencies specifically reserved for rail application with certain specific and advanced functions.
Demonstration(s)	• n.a.
Users	• stakeholders in the railway transportation sector
Added Value/Findings	• ERTMS is becoming the European standard for train control, signalling and traffic management. • ERTMS brings significant advantages in terms of maintenance costs savings, safety, reliability, punctuality, and traffic capacity • Improved positioning will advance the transportation security of dangeours goods
Collaboration(s)	• European Union

	• Railway stakeholders • GSM-R industry
Follow-up	• n.a.
Limitations	• Primarily intended to introduce a standard for train control, signalling and traffic management • Not primarily aimed at improving the provision of internal security (side-effect)

Satellite Constellation	**ESSAIM (4 satellites)**
Phase(s)	1: Prevention, detection
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Technology demonstrator
Date of Launch	• 18 December 2004
Current Satellite Status	• Operational
Country/Institution	• DGA / France
Capability	• Analysis of the electromagnetic environment on the ground in a number of frequency bands used exclusively for military communication
User	• DGA/French Ministry of Defence
Projected lifetime	• 3 years
Objective(s)	• The Essaim micro-satellite constellation is designed to demonstrate the feasibility of listening to terrestrial electromagnetic signals in order to determine the density of strategically interesting communications in different parts of the world.
Service (in practice)	• Electronic Intelligence / Communication Intelligence • Risk Management
Cooperation/data exchange	• The development of a future fully operational communications intelligence capability is only regarded possible within the framework of a European programme

Project Name	**European Commission's EMODNET** (European Marine Observation and Data Network)
Phase(s)	n.a., (data harmonisation, integrated approach)
Critical Mission Area	Transportation Security/Waterborne Transportation
Type	Implemented service
Project Duration	• 2009-2013
Current Status	• On-going
Country/Institution	• European Commission (DG MARE, EC Maritime Task Force)
Funding	• n.a.
Partners	• n.a.
Objective(s)	• To collect data once and use it many times • To develop standards across disciplines as well as within them • To process and validate data at different levels. Structures are already developing at national level but infrastructure at sea-basin and European level is needed • To provide sustainable financing at an EU level so as to extract maximum value from the efforts of individual Member States • To build on existing efforts where data communities have already organised themselves • To develop a decision-making process for priorities that is user-driven • To accompany data with statements on ownership, accuracy and precision • To recognise that marine data is a public good and discourage cost-recovery pricing from public bodies
In Practice	• The "proof of concept" of EMODNET is being tested through preparatory actions. Portals for a number of maritime basins are being set up for hydrographic, geological, biological and chemical data as well as functional habitat maps. These portals will provide access to marine data of a standard format and known quality and identify gaps in coverage.
Demonstration(s)	• n.a.
Users	• Maritime services
Added Value/Findings	• The projects will identify the main challenges in moving from an ur-EMODNET to an operational EMODNET. • Facilitation of GMES Marine Core Service
Collaboration(s)	• Complementary service to GMES/Kopernikus, EMBRC, ESFRI and EURO-ARGO
Follow up	• n.a.
Limitations	• n.a.

Satellite name	**ESA's ENVISAT** (ENVIronment SATellite)
Phase(s)	1: Imagery for detection 3: Rapid mapping
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security, Emergency Management

Type	Operational Satellite
Date of Launch	• March 2002
Current Satellite Status	• Operational
Country/Institution	• ESA
Capability	• Michelson Interferometer for Passive Atmospheric Sounding (MIPAS); • Global Ozone Monitoring by Occultation of Stars (GOMOS); • Scanning Imaging Absorption Spectrometer for Atmospheric Cartography (SCIAMACHY); • Medium Resolution Imaging Spectrometer (MERIS); • Advanced Along Track Scanning Radiometer (AATSR); • Advanced Synthetic Aperture Radar (ASAR); • Radar Altimeter 2 (RA-2); • Microwave Radiometer (MWR); • Doppler Orbitography and Radio positioning Integrated by Satellite (DORIS); • Laser Retro-Reflector (LRR)
User	• Emergency Management Services • First Responders
Projected lifetime	• 10 years
Objective(s)	• The main objective of the Envisat programme is to endow Europe with an enhanced capability for remote sensing observation of Earth from space, with the aim of further increasing the capacity of participating states to take part in the studying and monitoring of the Earth and its environment.
Service (in practice)	• To provide for continuity of the observations started with the ERS satellites, including those obtained from radar-based observations • To enhance the ERS mission, notably the ocean and ice mission • To extend the range of parameters observed to meet the need of increasing knowledge of the factors determining the environment • To make a significant contribution to environmental studies, notably in the area of atmospheric chemistry and ocean studies (including marine biology) • To allow more effective monitoring and management of the Earth's resources To better understand solid Earth processes
Cooperation/data exchange	• Images commercialised by SpotImage • Contribution to GMES

Project Name	**FP 7 EULER** (European Software Defined Radio for Wireless in Joint Security Operations)
Phase(s)	3: Emergency Management
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• 01/03/2009 – 29/02/2012, 36 months
Current Status	• In Progress
Country/Institution	• FP 7
Funding	• EU contribution: € 8.72 million • Total costs: € 15.47 million
Partners	• 18, coordinated by Thales Communications SA
Objective(s)	• To demonstrate how the benefits of Software Defined Radio (SDR) can be leveraged in order to enhance interoperability and fast deployment in case of crises that need to be jointly resolved
In Practice	• Three main components will be combined: • A reference high-data-rate radio technique • A communication system architecture allowing integration of heterogeneous radio standards • And SDR as a key enabler for this
Demonstration(s)	• The outcomes will constitute one of the first European demonstrators of interoperability in a civil crisis situation based on SDR
Users	• Various security organisations
Added Value/Findings	• First European demonstrator of interoperatability • End user driven
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Crisis does not necessarily include emergency management after terrorist attack

Project Name	**Belgium's EUREKA's "Logchain Translog Safety"**
Phase(s)	1: Prevention and detection through monitoring of dangerous goods transports 2: Fast response mechanism
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D

Project Duration	• 1/10/01 – 1/10/03, 24 months
Current Status	• Completed
Country/Institution	• Belgium (Coordinated by Krakow University Of Technology/Rail Vehicle Institute, Poland)
Funding	• € 900.000 by Eureka
Partners	• Ingenieurgesellschaft Für Angewandte Technologie Mbh, Germany • Industrial Institute for Automation And Measurements (PIEP), Poland • Science and Technical Railway Centre, Poland
Objective(s)	• Development of a system that monitors cargo flow and guarantees safety in transcontinental corridors especially transportation of dangerous materials and standard cargo units for rail transportation
In Practice	• The status of railway vehicles is gathered by sensors, which then transfer the data to the transceiver and via satellite to the control centre. • When an accident occurs, e.g. a derailment, the sensors automatically alert the control centre. • The position of the train is given so that a report can be sent immediately to the civil protection services in the area, reporting the accident and the nature of the potential hazard
Demonstration(s)	• The system was successfully demonstrated to railway authorities, and the Polish Railway Authority has already agreed to adopt it
Users	• Transport sector
Added Value/Findings	• Efficient tracking and monitoring of cargo • Global coverage allows the system to operate anywhere on Earth • No need for terrestrial GSM for operation
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• Limited to tracking of dangerous materials, standard cargo units and rail transport • it has so far only been tested between Europe and Asia

Project Name	**FP 7 FORESEC**
Phase(s)	n.a. (analysing "security" drivers, trends and scenarios)
Critical Mission Area	n.a.
Type	R&D
Project Duration	• 01/02/08 - 31/12/2009, 22 months
Current Status	• In progress
Country/Institution	• FP 7
Funding	• Total Cost : € 942,202 • EU Contribution : € 942,202
Partners	• Crisis Management Initiative (Finland) • Austrian Research Centers System Research (Austria) • International Institute for Strategic Studies (U.K.) • Swedish Defence Research Agency (Sweden) • The Centre for Liberal Strategies (Bulgaria) • Joint Research Centre (Italy)
Objective(s)	• Tying together the multiple threads of existing work on the future of European security • Enhancing the common understanding of the complex global and societal nature of European security • Enhancing the shared vision and facilitate the emergence of a coherent and holistic approach to current and future threats and challenges for European security, through a participatory foresight process • Identifying the security responses in which there is a particular added value and shared interest to work at the European level • Building a pan-European network around the European security foresight processes
In Practice	• Establish the new norm for the field of forward looking assessments of European security issues, • Create a systematic, participatory, medium-to-long-term vision-building process, • Analyse the political contexts, trends and drivers for European security issues, • Initiate a public debate on security research and its implications, • Identify potential future threats and technological opportunities, • Identify and examine ethical issues presented by security technologies, • Formulate and evaluate holistic scenarios on particular threats, linking human rights, cultural and ethical aspects with the security technologies utilized to confront them, • Prioritise what issues future efforts should concentrate upon, • Recommend a new holistic concept of European security that is inclusive of state, societal and human strands, • Link this work to other European and national security foresight projects and other security research projects, • Make recommendations on further action in the context of European Security Research. • The participatory process will involve representatives from European institutions, member states governments and administrations, NGOs and think tanks, public users, the private

	sector including SMEs as well as research institutes and universities The foresight process will include: • Scanning, with a view to exploiting relevant elements, previous pertinent work, • Participatory methods: an e-platform for exchange and interaction among stakeholders and experts in between face-to-face meetings, Delphi studies and focus groups where expert and non-expert opinions on a specific issue are collected and analysed, • Scenario analysis: the development of descriptions of possible future crisis situations in order to anticipate and prepare for potential future scenarios, • Technology assessment: the analysis of technological opportunities on the horizon with a view to analysing their impact for use in policy-making contexts. • A series of reports: global and country reports, Delphi report, trend-assessment reports, drivers and threats, scenario descriptions and analysis, technological opportunities and a fi nal summary report, • An interactive website, • Vision-building and dissemination events, • Increased public interaction and involvement as a stakeholder in the process.
Demonstration(s)	• n.a.
Users	• Security Community
Added Value/Findings	• To understand the future threats and challenges in order to be able to not only envisage the future but also prepare a well-grounded set of responses to it • Through a participatory foresight process, enhance the shared vision and facilitate the emergence of a coherent and holistic approach to current and future threats and challenges for European security. • Deepen the dialogue with society on security issues and nurture broad pan-European participation by including partners from both non-governmental organisations and new Member States
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FRONTEX' and EUSC's FronBac**
Phase(s)	n.a. (training programme for GIS analysis)
Critical Mission Area	Border security
Type	Training Programme
Project Duration	• On-going
Current Status	• In progress
Country/Institution	• FRONTEX, EUSC
Funding	• n.a.
Partners	• n.a.
Objective(s)	• To bring together and train all relevant analysts from Member States and Frontex for setting up a European analytical community in the field of border security • To promote the joint training and work of all analysts
In Practice	• A three years training plan aimed at setting up the analytical community and jointly reinforcing the analytical capacities of analysis units in the Member States and in Frontex
Demonstration(s)	• n.a.
Users	• all relevant analysts
Added Value/Findings	• Enhances better and more efficient border security throughout Europe and fosters cooperation • FronBac is open to staff of other units in FRONTEX and, based on ad-hoc decisions, to representatives from Third Countries border guard authorities
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 5 GADEROS** (GAlileo DEmonstrator for Railway Operation System)
Phase(s)	1: Prevention 2: Response: improved search and rescue through reliable positioning
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• 28.12.2001 -27.6.2004, 30 months
Current Status	• Completed
Country/Institution	• FP 5
Funding	• n.a.
Partners	• INECO (Spain) (project lead)

	• European Rail Research Institute (Netherlands) • Railway Safety (UK), Nottingham Scientific • Limited (UK) • THALES Navigation, (France) • TIFSA (Spain), GMV Sistemas (Spain) • SENER (Spain) • AENA (Spain)
Objective(s)	• To demonstrate the use of GNSS Integrity and Safety of Life service characteristics for defining a satellite-based system to perform train location for safe railway applications • Definition of the Users Requirements focused on low traffic lines to be satisfied by the GNSS application, including as input the User Requirements from other related railway projects, in order to define a common ground for this type of application. These requirements will be discussed in a workshop with the other projects and stakeholders. • Development of principle, conception and architectures for the GNSS component in the ERTMS/ETCS architecture, to reach the Safety Integrity Level requested for the application of the train location function in railway operation systems • Development / adaptation of standard test and evaluation procedures for GNSS based trials for railway safety applications. • Adaptation of simulation tools, based on existing UIC ETCS simulator level 2 and 3, to provide compatibility and interoperability of train location by satellite with the ETCS Kernel. • Development of a reference site for railway safety-related GNSS applications. • Definition of the functional specification to apply to the future prototype equipment of train location by satellite • Realisation / adaptation of test support tools (including Digital Route Map, on-board Database, set of qualification criteria and their dynamic update, corresponding software modules and applications within an on-board computing mainframe -portable professional PC for demonstration) • Realisation of a functional module for the GNSS component based on existing UIC ETCS simulator software modules, within an advanced PC computing system. • Establishing test bed, test specification and test procedures to demonstrate conformity of GNSS locators with ERTMS/ETCS concept and architecture.
In Practice	• The project pursues the integration of the so called GNSS subsystem within ERTMS/ETCS by means of the definition of the functional and interface specification of the navigation module to be used as the reference specification for further developments outside the project. These specifications will be firstly verified using the ERTMS/ETCS simulator platform, followed by a real railway environment in the local, low density traffic railway line 310 (Aranjuez – Valencia), in a stretch defined by Aranjuez – Cuenca – Buñol, and within the operational and regulatory framework of RENFE (Spanish railways). Besides the test specification, the project will design, test tools and the test procedures. Finally, it will carry out functional evaluation tests on at least one industrial prototype from several possible providers, as the first step towards the certification of a GNSS subsystem within ERTMS/ETCS. The project will take advantage of the APOLO prototype for fixing the test bed.
Demonstration(s)	• n.a.
Users	• Stakeholders in the railway transportation sector
Added Value/Findings	• The system will offer another technological approach for the train location function, mainly for conventional and low density traffic lines. • The demonstration on a Low Density Traffic Line, through tests on a number of prototypes developed by other projects outside GADEROS, will provide real-life implementation of train location based on a GNSS receiver with integrity and augmentation
Collaboration(s)	• Galileo • ERTMS • EFCS
Follow-up	• n.a.
Limitations	• Emphasis on safety in rail transportation rather than security

Project Name	**France's GALILEOCEAN**
Phase(s)	1: prevention and detection for waterborne transportation security and sea border security 2: Improved immediate response based on improved maritime positioning
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security
Type	R&D
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• France
Funding	• Funded in the framework of the French competitiveness cluster Brittany Sea
Partners	• GET/ENSTB (Brittany advanced telecommunications institute) • ENSIETA (National advanced institute for weapons industry and technology) • IRENAV, (Naval academy research institute) • Thales Aeronautics Division • BOOST Technologies

	• Antenessa • MARTEC SERPE IESM
Objective(s)	• Ensuring the optimisation of Galileo for use in terrestrial as well as maritime positioning
In Practice	• The project involves the full range of maritime activities: navigational aids, coastal surveillance, maritime safety and search and rescue, exploitation of marine resources, pollution monitoring, laying/repairing undersea cables, marine science and engineering relating to oceanography and the environment – meteorology, cartography, currents, winds, the fishing industry and maritime trade • Creation of a prototype receiver using a simulation platform • Critical analysis of existing solutions based on GPS • To study and create a multi-constellation (GPS and Galileo) prototype infrastructure: simulation platform with marine environment signal modelling and service and user platforms – terminals, antennae, applications, communications networks and information management systems
Demonstration(s)	• Validation of technologies in real environments
Users	• Stakeholdes in the mission areas mentioned above
Added Value/Findings	• Optimisation of Galileo for the use in terrestrial as well as maritime positioning
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• n.a.

Project Name	**Austria's GaWaLoc** (Galileo Wagon Location)
Phase(s)	1: Prevention and detection through constant tracking 2: Improved response through reliable positioning
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• n.a.
Project Status	• n.a.
Country/Institution	• Austria
Funding	• Austrian and Bavarian state
Partners	• Era-Star • Thales Rail Signalling Solutions (Coordinator) • Fraunhofer Institute • Kayser-Threde • SELSYS • TriaGnoSys
Objective(s)	• An independent tracking system without being reliant on any infrastructure operators' information • Accurate localisation • Increasing the efficient use of the existing infrastructure and rolling-stock, in the same way that the Single European Sky initiative is developing more efficient use of European airspace
In Practice	• Is examining both the commercial and technical aspects of implementing a wagon tracking system using Galileo. • Provides real-time monitoring of wagon movements and identifies any deviations from the schedule or location. • Provides logistic support and surveillance in European ocuntries; outside Europe, support will depend on a suitable communication infrastructure, such as GSM. • Sensors detect unauthorised door openings, changes in load weight, excessive shocks and load temperature monitoring where appropriate. • Enhances information about the status and condition of each vehicle, which will drive higher productivity through a faster reloading rate; real-time monitoring to help reduce train operation costs and allow the introduction of condition-based rather than periodic preventative control. • Automatic provision of information on the vehicle order, brake system status, train integrity checks and more precise calculations of the gross train weight, allows operating staff to increase efficiency by replacing various manual tasks with automatic ones during train assembly GaWaLoc uses a next generation GALILEO transponder which makes use of two key attributes: • Additional modules can be added to a GSM transmitter and extend coverage cost effectively with satellite services. • The low-energy transponder harvests power intelligently, meaning its maintenance free operating period will outperform existing solutions on the market.
Demonstration(s)	• n.a.
Users	• European rail freight industry
Added Value/Findings	• Helps to improve productivity, and therefore reduce costs, through more effective operational and maintenance management. • The information is independent of the train-operating companies, giving increased control to freight owners
Collaboration(s)	• GALILEO
Follow-up	• n.a.
Limitations	• Limited to Rail Transportation not specifically focused on the provision of internal security

Project Name	**JRC's GDACS** (Global Disaster Alert and Coordination System)
Phase(s)	2: Alert, immediate response Integrated approach
Critical Mission Area	Emergency Management
Type	R&D, Implemented Service
Project Duration	• n.a.
Current Status	• On-going
Country/Institution	• JRC
Funding	• n.a.
Partners	• DG ECHO • United Nations Office for the Coordination of Humanitarian Affairs • ReliefWeb • Virtual OSOCC • JRC • GDACS Portal and Asgard • Digital Map Archive • European Media Monitor • UNOSAT • AlertNet • HEWSWeb
Objective(s)	• A set of standards to ensure interoperability between existing alert and response coordination systems • A set of working procedures to ensure predictability and reliability of information creation and exchange during crisis response • Stakeholders that commit to provide related information according to the standards and the working procedures
In Practice	• GDACS is a web-based platform that combines existing web-based disaster information management systems with the aim of alerting the international community in case of major sudden-onset disasters and facilitating the coordination of international response during the relief phase of the disaster
Demonstration(s)	• n.a.
Users	• Disaster managers in donor countries, response organisations and disaster-prone countries, related research community
Added Value/Findings	GDACS services: • Initial disaster alert and impact/loss estimation immediately after the occurrence of a disaster • Alert of the international disaster response community • Coordination of relief activities • UNOSAT provides customised satellite images and maps that can be requested on-line by registered users of the Virtual OSOCC or directly from UNOSAT • Integration of related information management systems (ReliefWeb report and map repository, News feeds (e.g. AlertNet, EMM, Yahoo), Disaster early warning sites (HEWSWeb))
Collaboration(s)	• n.a.
Follow-up	• Led to Spin-off application LiveMon
Limitations	• No indication that terrorist attacks are considered a disaster • Limited to natural disasters • Should be extended to cover all types of disasters

Project Name	**FP 7 GEOCREW** (Crisis Early Warning Situation Awareness Architecture Concept)
Phase(s)	n.a. (countering fragmentation)
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• 01/01/2005, 13 months
Current Status	• Completed
Country/Institution	• FP 7 • Actors involved: ISFEREA: Isferea was responsible for (1) interacting with decision makers of COM department and EU Member States crisis rooms (2) providing material on good examples of best practice. Isferea consolidated the understanding of the use of geo-spatial data based on interviews with RELEX, DG DEV, ECH, and ENV as well as with staff that iweredispatched to crisis areas.
Funding	• Total Cost : € 695,573 • EU Contribution : € 532,900
Partners	• BAE Limited • Asemantics S. R. L. • Deutsches Zentrum für Luft- und Raumfahrt e. V • Eurosense S. R. O. • GMV S. A • Joint Research Centre European Commission • Open Geospatial Consortium Europe • Planetek Italia • Stichting National Lucht-en Ruimtevaartlaboratorium
Objective(s)	• Aimed at analysing the use of geo-spatial technology in the decision-making process of

	institutions dealing with international crises • To show best practice in the use of geo-spatial technologies in the decision-making process in practice
In Practice	• The study will comprise an overall architecture for integrating different information sources identified through user requirement soft security-related services, the Geodata study concentrates on a more detailed technical approach on the specific utilisation of geospatial data for improving situation awareness
Demonstration(s)	• n.a.
Users	• Early warning and first responders, decision makers
Added Value/Findings	• Improved overall European approach through countering fragmentation
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Added value depends on the definition of "crisis"

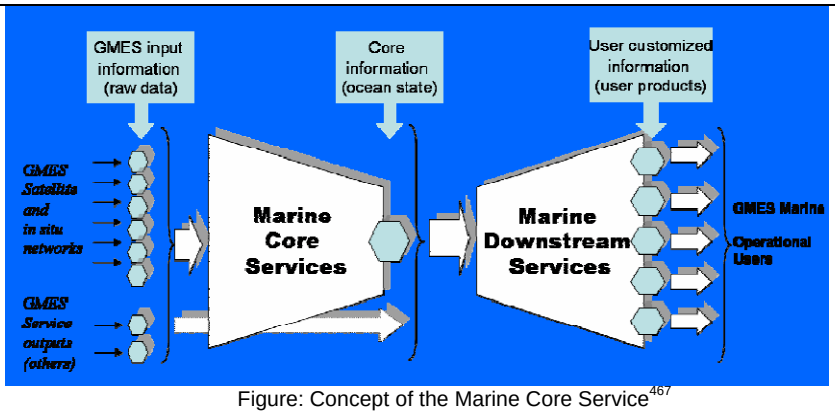
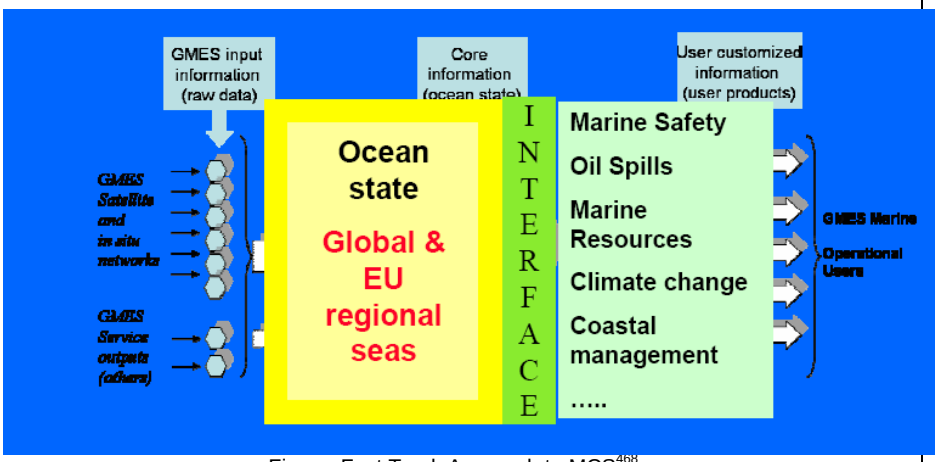
Project Name	**ESA's GIANUS** (Global Integrated Network for iNnovative Utilisation of space for Security)
Phase(s)	1: Intelligence, preparedness for emergency management 2: Improves Europe's response capabilities
Critical Mission Area	Intelligence, Early Warning and Risk Management, Emergency Management
Type	R&D
Project Duration	• n.a.
Current Status	• Preparatory phase
Country/Institution	• ESA
Funding	• ESA.
Partners	• European Commission
Objective(s)	• Improve preparedness of Europe for a variety of threats • To develop an integrated space infrastructure as part of a European security and crisis management system in coordination with other security-related European organisaitons • To develop capabilities to support operations in case of crisis by providing data in a timely and affordable manner and in a coherent and holistic approach
In Practice	• Providing very high resolution in a fast response time (less than 8 hours) for internal and external security purposes • Integration of SATCOM, satellite navigation and EO for crisis response • Telecommunications for initial actions in <24 hours • Telecommunications for follow-on >>100Mbps • EO constellation using improved sensors (SAR-MTIm hyperspectral, IR, MarSur,…) for responsiveness even in buildings • Maritime Surveillance (AIS), advanced alers, interference mitigation
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• Integrated Approach
Collaboration(s)	• Comeplements MUSIS
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 7 GLOBE** (GLOBal Environment)	
Phase(s)	n.a. (s. limitations)	
Critical Mission Area	Border Security	
Type	R&D	
Project Duration	• 01/07/2008 – 1/07/2009; 12 Months	
Current Status	• completed	
Country/Institution	• n.a.	
Funding	• European Union	
Partners	• Telvent Interactiva S.A. • Amper Sistemas S.A. • GMV Aerospace and Defence S.A. • Fundación Robotiker • Instituto Nacional de Técnica Aerospacial • Altran Technologies • SETTCE	• Econet Polska • Eurosense Belfotop N.V. • Skysoft Portugal • CES vision • PRIO • Empresa de Servicos e Desenvolvimento de Software • Cogenc Systems GMBH
Objective(s)	• Providing a comprehensive framework in which an integrated border management system must be developed	
In Practice	• Provide a comprehensive Roadmap including political and legal situation on border security • Provide full coordination between institutions • Providing all relevant EU and national institutions with intelligence required to improve decisions on issues such as illegal immigration and movements of illegal goods.	

Demonstration(s)	• n.a.
Users	• National and international authorities
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• It remained unclear whether satellite data is included.

Project Name	**GMES Land Information Service/ Land Monitoring Core Service (LMCS)**
Phase(s)	n.a. (limited to environmental purposes)
Critical Mission Area	n.a.
Type	GMES service element
Project Duration	• n.a.
Current Status	• on-going
Country/Institution	• ESA
Funding	• GSE project (GMES Service Element) funded by the European space Agency (ESA) within the European GMES initiative
Partners	• n.a.
Objective(s)	• To operationally provide sound, reliable and affordable geo-information products throughout Europe and over the globe
In Practice	• The GSE Land Consortium has mapped an area of almost 300.000 sqkm and delivered the resulting services. • All validated products are now accessible through the GSE Land geodata server • More services: European Urban Atlas, Impervious Areas and Sealing Levels; Inland Water Quality and Contamination; Irrigation / Agricultural Water Consumption
Demonstration(s)	• n.a.
Users	• All GSE Land services are particularly designed to support public authorities at various levels of responsibility in their reporting obligations and improved management of natural resources.
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Limited to natural resources. Could be extended as to include Critical Infrastructures.

Project Name	**GMES Marine Core Service (MCS)**
Phase(s)	n.a. (systematic reference information on the state of the ocean over global ocean & European regional seas)
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security, Emergency Management
Type	n.a.
Project Duration	• n.a.
Current Status	• Development phase
Country/Institution	• ESA
Funding	• ESA funded GMES Core service
Partners	• n.a.
Objective(s)	• Systematic reference information on the state of the ocean over global ocean & European regional seas
In Practice	• To establish an integrated European capacity for: • On sea state and dynamics (e.g. 3D currents, temperature, salinity) and primary ecosystem (surface phytoplankton and primary production) characteristics; and at global scale (all world ocean) with daily updates on 5-10 km horizontal grids and over European regional seas (Baltic, Mediterranean, North Sea, North East Atlantic Basin), with daily updates at customised space resolution (1-5 km horizontal grids). • Thematic Assembly Centres will be responsible for specific pre-processing of different parameters to be used as observation time-series and input to the global and regional Monitoring and Forecasting Centres. Regional models are foreseen for the Baltic, Mediterranean, Black Seas, Arctic Ocean, Northwest Shelf and the North Atlantic.
Demonstration(s)	• Technical and strategic help to the Spanish Navy (September 2007): Different actors had to find the best way to coordinate their activities. The aim was to support the navy in the detection and evaluation of anomalies in maritime traffic in the Straits of Gibraltar. The actors had to provide AIS correlation with optical/SAR imagery, analysis of the optical/SAR imagery, and a final product based on the fusion of data coming from different sources of information.

	 Figure: Concept of the Marine Core Service[467]  Figure: Fast Track Approach to MCS[468] The MCS should deliver • Systematic reference information on the state of the ocean/seas • Observational and model data • Real time predictions: from day to week • Re-analyses: time series of information • Ocean scenario simulations (with climate modeling centers) • At the resolution required by intermediate users & service providers • Of known quality and accuracy • For the global ocean & European regional seas, with downscaling capacity based on specific needs
Users	• n.a.
Added Value/Findings	• n.a.
Collaboration(s)	• Part of GMES
Follow up	• n.a.
Limitations	• Currently its main objective is to provide “systematic reference information on the state of the ocean over global ocean & European regional seas”. It should be extended as to include maritime surveillance for security purposes.

Project Name	**GMES Emergency Response Core Service (ERCS)**
Phase(s)	2: Rapid mapping 3: Reconstruction
Critical Mission Area	Emergency Management
Type	n.a.
Project Duration	• n.a.
Current Status	• Development phase
Country/Institution	• ESA
Funding	• ESA funded GMES Core service
Partners	• n.a.
Objective(s)	• To support at global level all stages of the intervention cycle from early warning to crisis

[467] Minster, J.F. “Report – Workshop on the Marine Core Service.“ Presentation. Marine Core Service Workshop. Brussels, Belgium. 27-8 October 2005. <http://www.gmes.info/pages-principales/library/implementation-groups/marine-core-service-mcs>.
[468] Ibid.

	management and recovery in response to natural, environmental emergencies (e.g. droughts, floods, storms, earthquakes, tsunami, volcano eruptions, landslides fires, etc).
In Practice	• To provide rapid mapping services (reference and damage assessment maps) to gather the necessary information to know the incoming anomalous events and the following critical effects, the effective damages and to forecast and design the restoration of normal life and ordinary conditions • The applications covered by the ERCS include the following: Active seismic zone and volcano observatory, Regional landslide risk monitoring, Rapid Flood mapping & damage assessment, Regional Forest Fire Area Mapping & status, Urban subsidence risk assessment
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• Seems to involve natural disasters and does not consider terrorist attacks as disasters needing rapid mapping.

Project Name	**FP 6 GMOSS** (Global MOnitoring for Security and Stability)
Phase(s)	1: Prevention/detection through effective treaty monitoring and monitoring of energy facilities 2: Improved response through monitoring 3: Damage assessment
Critical Mission Area	Intelligence, Early Warning and Risk Management, Critical Infrastructures, Border Security, Emergency Management
Type	R&D
Project Duration	• 01/03/2004 – 29/02/2008, 48 months
Current Status	• Completed
Country/Institution	• FP 6
Funding	• Total costs: € 6.0 million • EU contribution: € 6.0 million
Partners	• 22, coordinated by DLR
Objective(s)	• Primarily, to integrate the European community concerned with the use of EO for civil security applications through joint research and training activities, sharing resources and infrastructure, and exchange of expertise and staff • Additionally, an analysis of the potential of remote sensing data for threat analysis and early warning, monitoring of critical Infrastructures and damage assessment, monitoring of borders and the migration of peoples, as well as benchmarking of recent data analysis and visualisation concepts and techniques
In Practice	• To research specific science and technology needed to provide effective monitoring of international treaties protecting against proliferation of WMD, better estimates of static and dynamic populations on a global scale, better monitoring of infrastructure and borders, rapid remote assessments of damage • To develop methodologies such as rapid mapping, specific feature extraction, image automatic interrelation, geographic information system modelling for division support
Demonstration(s)	• 4 test cases relevant for European security: Iraq, Iran, Kashmir, Zimbabwe • the Review Meeting in The Hague • the third Summer School on 'Early Warning and Monitoring of Agreements' in Madrid • a Seminar on 'Environment and Conflict' in Bonn
Users	• Academic research, service providers, user organisations, Commission services
Added Value/Findings	• Concrete proposals for new applications for remote sensing data analysis involves, among other matters, the use of: night lights data for monitoring the impact of disasters and the movements of refugees, low-spatial high-temporal resolution sensors for monitoring conflict-related explosions and/or fires, automatic satellite image matching techniques for precise damage assessment of buildings, radar and optical sensors for automatic detection of moving targets • It contributes to the awareness of the security dimension of GMES • It Integrates the existing EO for the civil security community
Collaboration(s)	• Contributes to the awareness of the security dimension of GMES • RESPOND • LIMES • RISK-EOS • PREVIEW • BOSS4GMES • SAFER • G-MOSAIC • TANGO
Follow-up	• As a Network of Excellence, GMOSS acted as a think tank for related projects and consequently aims to share its knowledge with the remote sensing community, especially with partners of ongoing and future GMES projects
Limitations	• n.a.

Project Name	**FP 7 G-MOSAIC** (GMES Services for Management of Operations, Situation Awareness and Intelligence for regional Crises)
Phase(s)	1: prevention through monitoring of treaties for non proliferation, monitoring of illegal activities, routes and borders, improved emergency preparedness through contingency planning 2: situation awareness; emergency mapping (3: damage assessment (BUT: explicitly only for conflicts))
Critical Mission Area	Intelligence, Early Warning and Risk Management, Border Security, Emergency Management
Type	R&D
Project Duration	• 01/01/2009 – 31/12/2011, 36 months
Current Status	• In progress
Country/Institution	• FP 7
Funding	• Total costs: € 15.3 Million • EU contribution: € 9.3 million
Partners	• 36 partners, coordinated by Telespazio
Objective(s)	• Within the context of the GMES Initiative, to identify and develop products, methodologies and pilot services (intelligence, early warning, crisis management…) for the provision of geo-spatial information in support to EU external relations policies and to contribute to define and demonstrate sustainability of GMES global security perspective • The activities will be devoted: to investigate the operational exploitation of GMES services as assets supporting security related activities in external regional crises situations; to contribute to identify the core and downstream geo-spatial intelligence services in the framework of current status of the GMES initiative; to disseminate knowledge on the potential impact of GMES' on the security related user community; to contribute to build a political consensus on GMES services for security; to promote the construction of a European inter-pillar capability for the monitoring services and infrastructures; to assess a sustainable provision and funding model for GMES security services; to provide recommendations for next sensors for security services
In Practice	• Pilot services will be related to: • Situation awareness on major threat warning factors (illegal activities, critical assets monitoring, WMD proliferation control, crisis indicator assessment, extended routes surveillance) • Intelligence applications • Crisis management operations
Demonstration(s)	• n.a.
Users	• DG-RELEX (and DGs Development, ECHO, Environment, etc.) • Council entities (EU Military Staff, Situation Centre) • National institutions (Ministries of Foreign Affaires, intelligence centres)
Added Value/Findings	• It provides services in all three phases of the disaster and emergency cycle and this way provides an all-encompassing approach
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Aimed at external security (primary objective and main users); is a "Stability Instrument" • Only one out of three pilot services is applicable to internal security as well • It provides services in all three phases of the disaster and emergency cycle instead of focussing on one phase only and specialising there • It does not cover the alert phase sufficiently – after having identified the threat how does G-MOSAIC propose to alert?

Project Name	**Austria's GUSTAV**
Phase(s)	2 & 3: Emergency Management
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• 01/05/2005 – 30/04/2006
Current Status	• Completed
Country/Institution	• Austria
Funding	• Funded by the Austrian Space Programme
Partners	• PRISMA solutions EDV-Dienstleistungen G.m.b.H. (coordinator) • ARC Österreichisches Forschungs- und Prüfzentrum Arsenal GmbH • Salzburg Research Forschungsgesellschaft m.b.H. • GeoExpert Research & Planning GmbH • HiTec – Vereinigung High Tech Marketing • Peter Stix, Ingenieurkonsulent für Vermessungswesen • die.wildbach und lawinenverbauung Sektion WIEN, NÖ und BGLD • BM für Land- u. Forstwirtschaft, Umwelt und Wasserwirtschaft Sektion Forst, Referat 4b • Abteilung Feuerwehr und Zivilschutz
Objective(s)	• To develop a system for mobile documentation and online-management of catastrophic events
In Practice	• GUSTAV focuses on information about events and their consequential actions. Through the use of GPS, accuracy and reliability of given information, especially in spatial dimensions, should be guaranteed.

	• The following proceedings of catastrophic management are supported by GUSTAV: • Registration on-the-spot: event declaration, damage declaration • Local support current information: about the event and about attributes • Event handling, subsequent invoices respectively local information queries as a basis for: analysis and reconstruction, commission of damage] The following technical components are used by GUSTAV: • A server to provide and store data, to validate event based notices, to allocate information in a cartographical environment and to integrate different web services • Client components via portable terminal, cartography – user profiles, use of existing expert – knowledge • Communication via GPS as well as checking and verification of alternative communication tools in case of natural catastrophically events • Spatial location through the use of GPS for event documentation (detection of the event, detection of damage, measuring points) and user information (fire brigades, police, civil service) • Legal restrictions • Cost effectiveness and potential of realisation • Alternative means of communication • Adoption and external information sources
Demonstration(s)	• n.a.
Users	• Emergency responders • Local authorities
Added Value/Findings	• Better information for event analysis
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• Added value depends on definition of "catastrophic event"

Project Name	**FP 6 HAWKEYE** (Thermal Infra Red Hyperspectral sensing Assistance to Clandestine Weapon Surveillance under WorKing Conditions Linking fixed Airborne or space borne Systems)
Phase(s)	1: Land monitoring
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	R&D
Project Duration	• 01/08/2005 – 31/07/2008, 36 months
Current Status	• Completed
Country/Institution	• FP 6
Funding	• Total costs: € 1.81 million • EU contribution: € 0.98 million
Partners	• 8, coordinated by ATIS
Objective(s)	• To evaluate the benefits of Thermal Infra Red Hyper Spectral Imaging (TIHSI) technologies, as remote sensing units integrated into surveillance systems for the detection of clandestine weapon development or activities putting at risk civil security
In Practice	• n.a.
Demonstration(s)	• The airborne prototype will be tested in real life conditions giving input data that will allow the consortium to validate the algorithms and give the necessary information to extrapolate the performances of the detector. On the basis of the experimental results through field and airborne experiments, a preliminary evaluation of a space borne solution will be made
Users	• International community for treaty monitoring, intelligence community,
Added Value/Findings	Main achievments: • Identification of potential applications of the developed technology for non-proliferation enforcement and homeland security/internal security • List of scenarios and detectable products established and discussed with external experts through the advisory board • Development of methods for validating and extrapolating the results including the validation of a simulator • Performance assessment of existing and new algorithms • Construction of a spectral data base • Improvement of the software used in signal analysis • Specifications of the prototype for airborne experiments taking into account the performance requirements needed compared to the present ground based sensor • Manufacture of an airborne device with improved hardware performance for the sensitivity and to adapt the device to specific constraints related to flight • Preparing the prototype for test in real-life conditions • Preparing flight tests
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Satellite System	France's Helios 1 (2 satellites)
Phase(s)	1: Intelligence
Critical Mission Area	Intelligence, Earrly Warning and Risk Management
Type	Technology demonstrator
Date of Launch	• Helios 1A: 7 July 1995
Current Satellite Status	• Operational
Country/Institution	• DGA (runs the programme), CNES (space segment responsibility) / France
Capability	• Helios 1A is no longer operable • Helios 1B: optical, visible spectrum
User	• French, Italian and Spanish armed forces
Projected lifetime	• Helios 1A: 5 years
Objective(s)	• Imagery
Service (in practice)	• Signal intelligence
Cooperation/data exchange	• France concluded agreements with Italy and Spain • Cost distribution: France: 79%, Italy 14%, Spain: 7% • Ground segments for the acquisition and processing of data in France, Italy, and Spain ensure data access for all partners; onboard encryption restricts the access to data to one or more of the partners • Imagery data is used by the European Union Satellite Centre

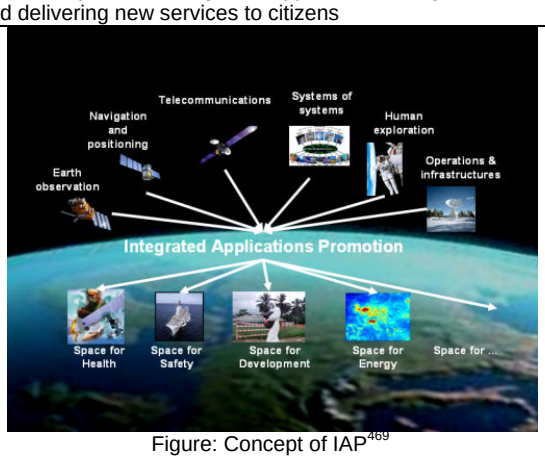
Satellite System	France's Helios 2 (2 satellites)
Phase(s)	1: Intelligence
Critical Mission Area	Intelligence; Early Warning and Risk Management
Type	Operational Satellite
Date of Launch	• Helios 2A: 18 December 2004 • Helios 2B: launch planned in 2009
Current Satellite Status	• Helios 2A operational • Helios 2B to be launched in 2009
Country/Institution	• DGA (runs the programme), CNES (space segment responsibility) / France
Capability	• Helios 2A: two optical instruments operating in visible and near-infrared spectra (high resolution camera and wide-field camera) • Helios 2B: two optical instruments operating in visible and near-infrared spectra (high resolution camera and wide-field camera)
User	• French, Belgian, Spanish, Italian, Greek and German armed forces
Projected lifetime	• Helios 2A: 5 years • Helios 2B: 5 years
Objective(s)	• Imagery
Service (in practice)	• Geospatial Intelligence
Cooperation/data exchange	• France cooperates with Belgium and Spain (since 2001), with Italy (since 2001), and with Greece (since 2007) • Cost distribution: France: 90%, Belgium, Spain, Italy and Greece: 2.5% each • In exchange for SAR-Lupe data Germany receives data from the Helios 2 system based on an agreement signed in 2002. The beginning of data exchange is envisaged for 2009. The request for satellite imagery is transmitted to the partner who integrates it into the planning process of its satellite system. According to the request the data is directly provided to the partner for analysis and archiving. • Data access is ensured by the installation of ground segments in each cooperating country, the ground segments are capable of acquiring and processing Helios (1+2) images and can program the Helios satellites • Imagery data is used by the European Union Satellite Centre

Project Name	**FP 6 HUMBOLDT** (Framework for Data Harmonisation and Service Integration)
Phase(s)	n.a./ aims at the implementation of a European Spatial Data Infrastructure (ESDI)
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	R&D
Project Duration	• 01/10/2006 – 30/09/2010, 48 months
Current Status	• In progress
Country/Institution	• FP 6
Funding	• EU contribution: € 7,93 million • Total costs: € 13,52 million
Partners	• 28, coordinated by Fraunhofer Gesellschaft zur Förderung angewandten Forschung e.V.
Objective(s)	• To manage and advance the implementation process of a European Spatial Data Infrastructure (ESDI) by enabling organisations to document, publish and harmonise their spatial information
In Practice	• To provide tool sets of software module plus connecting technology that can be used to build a Spatial Data Infrastructure (SDI), where different organisations can share their

	spatial information in an effective and sustainable way without having to develop all parts of their SDI applications from scratch
Demonstration(s)	• An essential element of the Humboldt project is the development of scenarios in which the different Humboldt components are applied and tested under realistic conditions. The Midterm scenario call was launched in May 2008
Users	• Broad user group, including all fields using spatial data
Added Value/Findings	• Besides facilitating the harmonisation of the spatial data, HUMBOLDT endeavours to reduce harmonisation costs by automating the necessary processes as far as possible; harmonisation is understood as a process of transforming the available data into usable data • Fosters a broader use of geo-data by overcoming the hurdles between the different domains and supporting the reuse of data without further processing • The applications scenario include border security
Collaboration(s)	• INSPIRE & GMES - The tools and processes created would demonstrate the feasibility and advantages of an ESDI as planned by INSPIRE, meeting the goals of GMES • FP6 BOSS4GMES project • FP6 CASCADOSS project
Follow-up	• n.a.
Limitations	• No explicit reference to counter-terrorism or organised crime

Project Name	**Belgium's HYSAR** (Man-made Object Classification Using Fused Polarimetric SAR and Hyperspectral Imagery Data)
Phase(s)	1: Intelligence 1: Airport security 3: emergency management: provides for geospatial information on the surrounding area
Critical Mission Area	Critical Infrastructures, Transportation Security / Air Transportation
Type	R&D
Project Duration	• 01/06/2004 – 31/05/2005
Current Status	• Completed
Country/Institution	• Belgium
Funding	• Funded in the framework of the Belgian STEREO 1 (Support to the Exploitation and Research of Earth Observation data 1) programme of Belspo (Federal Science Policy Office)
Partners	• ERM/KMS (Signal and Image Centre) • VITO (Remote sensing and Earth Observation Processes) • DLR (German Aerospace Centre)
Objective(s)	• Because of the complexity of urban systems, their spatial and spectral characteristics become more complicated. Affected by human and natural activities, the urban surface is very heterogeneous and makes the classification of man-made objects in those scenes very difficult using a single sensor with limited spectral range. The goal of this research is to fuse polarimetric SAR and hyperspectral data to classify man-made objects in urban and industrial scenes.
In Practice	• In May 2004 E-SAR and HyMap data were acquired. The detected images included various man-made materials combining residences, industrial buildings, nuclear power plants, airport and roads in rural-urban and industrial scenes. The E-SAR images are high spatial resolution scene (1.5 m) multi channel (X, L bands), full polarimetric (L-HH, VV, VH, HV) and interferometry (single path) data. The HyMap images are 4 meters spatial resolution data collected using 126 spectral bands. • Various (semi-)automatic detection, different classification and feature extraction methods were applied to the E-SAR and the hyperspectral data. In particular the following experiments were carried out: - Road detection using spectral information from hyperspectral data, - Object detectors based on logistic regression for application to buildings and roads, - Classification using logistic regression for application to buildings and roads, - Discrimination of roof types, - Classification of roof types, - Nuclear power structure, - Fences and roads using the MTMF (Mixture Tuned Match Filtering) technique , - Detection of dominant point scatterers in SAR images, - Classification of man-made structures using SAR data. • Feature level fusion techniques were applied for extraction of man-made structures combining SAR and PolSAR data and combining SAR and Hyperspectral data.
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• It has been found that data fusion is useful for road detection. However, fusion of E-SAR and hyper spectral was not sufficient for building detection. Moreover, it appears that point scatterers detected in SAR images are strongly correlated with man-made structures signals • It is not possible to detect individual houses in an urban environment using SAR, mainly due to the imaging geometry. A classifier based on polarimetric decomposition methods is

	useful for detection of large roads in SAR. However, a specific detector for linear feature is more effective for smaller roads.
Collaboration(s)	• Belgium • Germany
Follow up	• n.a.
Limitations	• Only potentially linked to internal security

Project Name	**ESA's IAP** (Integrated Applications Promotion)
Phase(s)	n.a. (integrated approach)
Critical Mission Area	n.a.
Type	Outreach activity
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• ESA
Funding	• ESA
Partners	• a.a.
Objective(s)	• To promote the development of Integrated Applications using a combination of space capabilities and delivering new services to citizens
In Practice	 Figure: Concept of IAP[469] • An "Integrated Application" for a particular user group is a user-driven service delivering, on a routine basis, high quality and innovative information products, generated through the combined utilisation of space and non-space technologies. Routine services require sustainability which necessitates users/partners able to take over responsibility for delivering the service beyond the promotion phase. The aim of the IAP is to identify users/partners as early as possible in the overall process • For Europe to obtain a maximum social and economic return from its space assets, providers of space-based services therefore must come to view the market consistently in terms of thematic uses rather than the major space technologies of Earth observation, satellite navigation and satellite telecommunications. Such satellite systems must be combined seamlessly and used optimally in conjunction with terrestrial systems if they are to deliver their potential for benefiting society. In the first instance, the following themes have been identified. • Space for Safety • Space for Health • Space for Development • Space for Energy • These themes presently have a different degree of maturity and other themes might be added during the course of the programme. • The process includes the following objectives • Promote awareness of space capability for new user communities and establish new partnership models. • Develop a systematic user-driven approach and new partnership for the development of new services and information systems (incubator of new services). • Promote new applications based on the integration of data generated by several systems, space based and non-space based. • Organise open platforms to analyse cross links effects such as the impacts of climate changes on health.
Demonstration(s)	• n.a.
Users	• n.a.

[469] "Integrated Application Promotion (IAP) Programme Information Day." 15 Jan. 2008. European Space Agency 29 July 2009 <http://www.congrex.nl/08c10/>.

Added Value/Findings	• Integrated approach • Outreach activitiy
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**EADS IMARSEC** (Integrated MARitime SECurity System)
Phase(s)	1: Prevention and detection 2: Response
Critical Mission Area	Critical Infrastructures, Transportation Security/Waterborne Transportation, Border Security
Type	Industry service
Project Duration	• n.a.
Current Status	• Implemented service
Country/Institution	• EADS
Funding	• EADS
Partners	• n.a.
Objective(s)	• Develop a system for safety and security to protect seaports, coastlines, territorial waters and the Economic Exclusive Zone (EEZ)
In Practice	• The intelligent System protects sea and land borders and includes video surveillance, detection of intruders, securi communication, a database and computer network which integrates all the different facilities
Demonstration(s)	• Has been used for building a “National Security Shield” in Qatar for its oil drilling stations and refineries • Is already implemented in the Tangier Med port
Users	• Coast guards, port authorities etc.
Added Value/Findings	• Intelligent system combining different technologies
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 6 INTEGRAIL**
Phase(s)	1: Monitoring 2: Improved search and rescue through reliable train positioning
Critical Mission Area	Transportation Security / Land Transportation
Type	R&D
Project Duration	• 1.1.2005 - 31.12.2008
Current Status	• Completed
Country/Institution	• FP 6
Funding	• Total costs: € 20 Million • EU contribution € 11 Million
Partners	• 39 Partners, project lead: UNION OF EUROPEAN RAILWAY INDUSTRIES, Belgium
Objective(s)	• Reliable train positioning using GNSS and EGNOS • To integrate the major railway sub-systems and deliver a higher level of coordination and cooperation between the key railway processes • To achieve higher levels of performance of the railway system in terms of capacity, average speed and punctuality, safety and the optimised use of resources • Development of vehicle and vessel concepts for both passengers and freight, characterised by interoperability and inter-connectivity, for cross-operation between different transport routes and networks supported by advanced mechatronics, on-board electronics, information and communication systems
In Practice	• InteGRail will enable information to be shared within the Railway World to enable optimisation of decision making for improved performance. • It will identify the information that is needed to be shared (SP2) • It will ensure that the right information can be obtained (SP3A) • It will identify ways of using the information more effectively for maintenance optimisation (SP3B) • It will identify ways of combining and using information effectively for management requirements (SP3C) • It will ensure that the information can be transmitted effectively to the decision makers (SP3D) • It will demonstrate that creating the right information and sharing it will enable performance improvement (SP4) • InteGRail will NOT replace existing systems. It will be used in conjunction with existing systems
Demonstration(s)	• n.a.
Users	• All railway stakeholders
Added Value/Findings	• Increased reliability and performance of the rail system through intelligent vehicles and

	infrastructure systems and their maintenance
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Primary purpose is improved fleet management and traffic management • Not primarily aimed at providing or improving the provision of Internal Security

Satellite name	**JASON-2**
Phase(s)	n.a.
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security
Type	Envisaged satellite
Date of Launch	• n.a.
Current Satellite Status	• Envisaged
Country/Institution	• EUMETSAT, NOAA, CNES and NASA
Capability	• n.a.
User	• n.a.
Projected lifetime	• n.a.
Objective(s)	• n.a.
Service (in practice)	• CNES is together with NASA working on a radar ocean surveillance successor for TOPEX-Poseidon and Jason-1. Jason-2 is a four party Programme of
Cooperation/data exchange	• n.a.
Demonstration(s)	• n.a.
Users	• Different security organisations
Added Value/Findings	• Detection at sea under most conditions – day and night and through cloud • The sensor can operate in a number of modes – from wide area and low detail to smaller area and higher detail
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Monitoring tool primarily aimed at fisheries enforcement, needs to be extended to other sectors (e.g. Waterborne Transportation, Sea Border Security etc.)

Project Name	**JRC's Vessel Detection System (VDS)**
Phase(s)	n.a.
Critical Mission Area	n.a.
Type	R&D
Project Duration	• Since 12/2002
Current Status	• On-going
Country/Institution	• JRC
Funding	• JRC
Partners	• n.a.
Objective(s)	• To determine the number of vessels and their position in a given area • To cross-check the positions of the vessels detected by Vessel Detection System (VDS) with position reports from Vessel Monitoring System (VMS) • To signal the possible presence of vessels from which no position reports have been received through VMS
In Practice	• Computer Tool

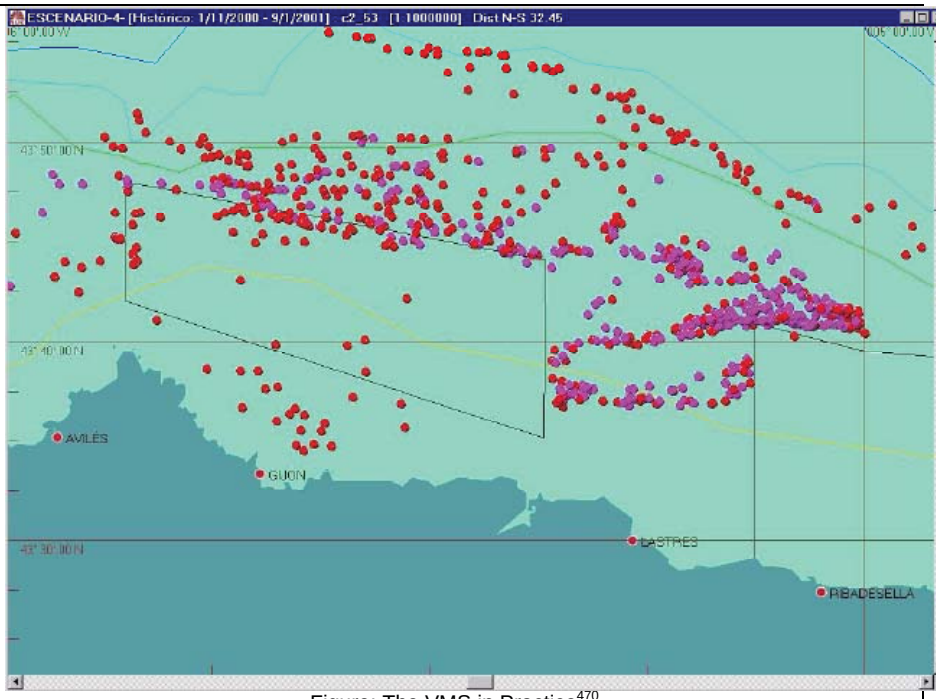
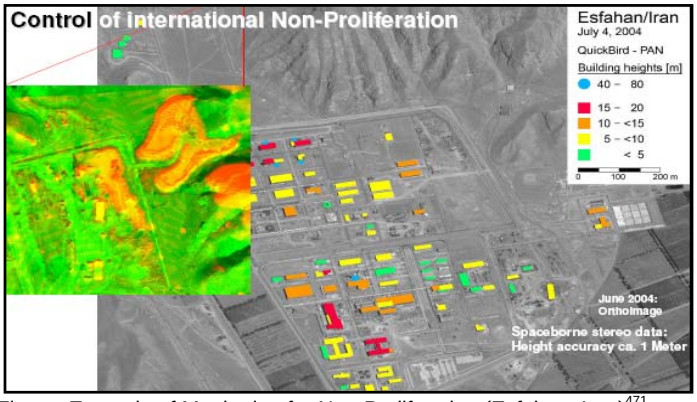


Figure: The VMS in Practice[470]

Project Name	FP 6 LIMES
Phase(s)	1: Treaty monitoring, land monitoring, critical infrastructures monitoring 2. Improved response through improved situational awareness based on surveillance 3: Post crisis impact assessment
Critical Mission Area	Intelligence, Early Warning and Risk Management,Critical infrastructures, Transportation Security/ Waterborne Transportation, Border Security, Emergency Management
Type	R&D
Project Duration	• 01/12/2006 – 31/05/2010, 42 months
Current Status	• In progress
Country/Institution	• FP 6
Funding	• Total costs: € 21.25 million • EU contribution: € 11.98 million
Partners	• 46, coordinated by Telespazio S.p.A.
Objective(s)	To develop prototype information services to support security management at EU and global level in the following areas of interest: • Organisation and distribution of humanitarian aid & reconstruction • Surveillance of the EU borders (land and sea) • Surveillance and protection of maritime transport for sensitive cargo • Protection against emerging security threats (e.g. terrorism, illegal trafficking, and proliferation of WMD)
In Practice	• Development of new algorisms for change detection • Testing of change detection techniques Services are clustered in three groups: 1. protection against emerging security threats 2. Area surveillance outside the EU including non-EU coasts and sensitive spots (e.g. land borders) 3. CI monitoring: (pipelines, nuclear power plants and ports) • demonstration along sea lanes and corridors of special interest • surveillance of the EU borders: 4. Coastal and open-water surveillance 5. surveillance of maritime border and over the exclusive economic zone of the EU • The core of the service is the integration of EO data with non EO systems date like IAS Organisation and distribution of humanitarian aid & reconstruction: • This will be achieved through integration and fusion of satellite data as well as geo information sources currently available from state-of-the-art satellite sensors and geoinformation services along with development and incorporation of new technologies,

[470] Delincé, Jaques. "The Vessel Detection System." Presentation. 14 June 2007. 29 July 2009 <http://ec.europa.eu/research/press/2007/maritime-briefing/pdf/11-presentation-jacques-delince-vds_en.pdf>.

	techniques and sources in particular in the field of navigation and communication. • For Disaster Preparedness and Support of Reconstruction after a crisis, the information services will support population distribution and resource (food, water, infrastructure, etc) monitoring, as well as the contribution to an information system to support planning. • For Operational Support of Humanitarian Relief, the fast provision of updated geospatial information and set up and validation of emergency satellite communication and navigation mobile systems are foreseen. • Incorporation, processing, integration and fusion of data and information currently available from state-of- the art surveillance systems and services, along with extension of these systems and services through development and incorporation of new technologies, techniques and sources, based on satellite assets. The services are based on the capacity of Very High Resolution satellites, used in conjunction with medium to high resolution data and aerial imagery, to enable critical 4D spatial analysis of updated reference data with the aim to assess risks, improve security and enhance preparedness  Figure: Example of Monitoring for Non-Proliferation (Esfahan, Iran)[471]
Demonstration(s)	Developments are in progress and demonstrations of pre-operational services are foreseen in clusters: Early Warning: a NPT Monitoring Area Maritime Surveillance:  Figure: Maritime Surveillance[472] Cargo Surveillance: • demonstration along sea lanes and corridors of special interest • Test users: shipping companies

[471] Cannizzaro, Giovanni. "The LIMES and GMOSAIC EC Integrated Projects to support Security Management in EU." Presentation. ISU Annual Symposium. International Space University, Strasbourg, France. 18-20 Feb. 2008.
[472] Ibid.

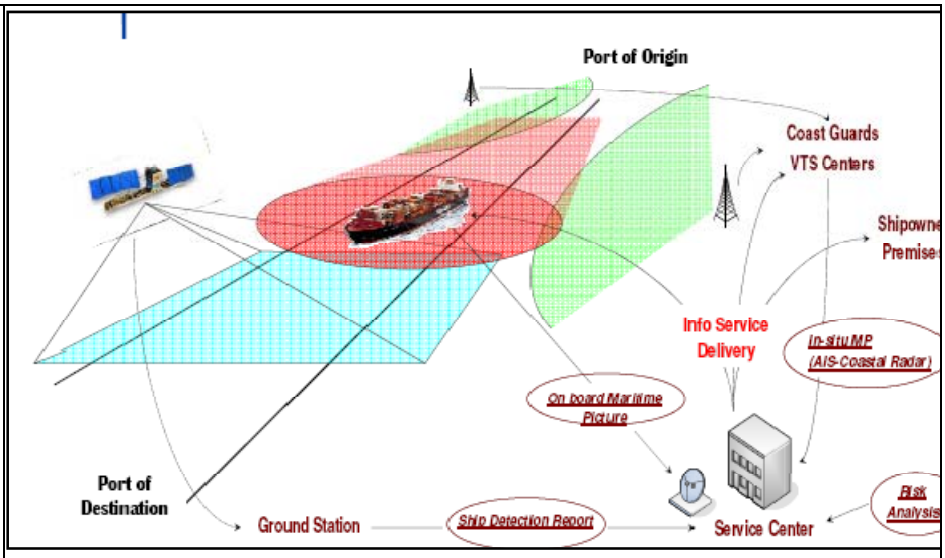


Figure: Structure of the provided Service[473]

- A set of services is being developed in Spain for gasification plants and pipelines in collaboration with ENAGAS, future developments include port security; services are also supporting event planning (Zaragoza expo)
- Surveillance Outside Europe - Caribbean Sea (Support to French Navy)
- Routine surveillance of routes in the Atlantic Ocean
- Systematic wide swath SAR data acquired and made available to EUSC
- (N.B. included for the sake of completeness only partial relevance to internal security)
- Demonstrations at several European and non-European ports, i.a. demonstration of terrorist attack

Infrastructure surveillance:

- Defines CI based on ECIP
- World EXPO 2008: Monitoring of side-planning for contingency planning and the location of exits, security measures
- Spain (Liquated Natural Gas (LNG) treatment premises, storage facilities and pipelines at Escombreras, around Madrid and in Asturias) and UK for Infrastructure Surveillance, a G8 Summit for Event Planning and (3) humanitarian relief and reconstruction support
- Analysis of Pipeline accident in Moscow 2008 (mainly archive research)
- Monitoring of non-cooperating Nuclear power plant in China

Border:

- test areas are Eastern EU land borders (with Frontex)
- sea borders: Nautilus 2008 Operation coordinated by Frontex
- Period: July 17 – Aug 06, 2008
- Two SAR satellites, acquisitions in the morning and/or in the evening
- Different observing modes, defined on a day to day basis, 1 to 3 acquisitions per day
- 2 hours between data acquisition and ship detection report

Humanitarian Relief Cluster:

- Test areas are Yemen and Nigeria and Tsunami affected areas in Indonesia for Preparedness and Reconstruction support. For Operational Support of Humanitarian Relief, the geographical areas will be selected during the first phase of the project based on presence/absence of crisis situations. Among the main users are EU and nternational Agencies, Civil Protection, NGOs.

Users	• Land Border Police, Civil Protection, EU (e.g. FRONTEX) and International Agencies, civil protection authorities and local authorities responsible for the security of citizens; energy distributing companies • Cost guards, customs and navy, planning and execution of patrolling missions • Civil protection agencies, NGOS active in the humanitarian aid sector, the Red Cross family and UN agencies
Added Value/Findings	• The increase of prevention capabilities and improvement of the response time • Maritime security involves locating and identifying threats and/or illegal activities, in a 'background' of empty ocean expanse or more crowded coastal maritime traffic; today, coastal radar is used for this task, alongside airborne and ship-based surveillance, but these and the other existing tools are inadequate to establish anything near 'watertight' security → LIMES aims to bring in satellite sensors that supplement existing monitoring

[473] Ibid.

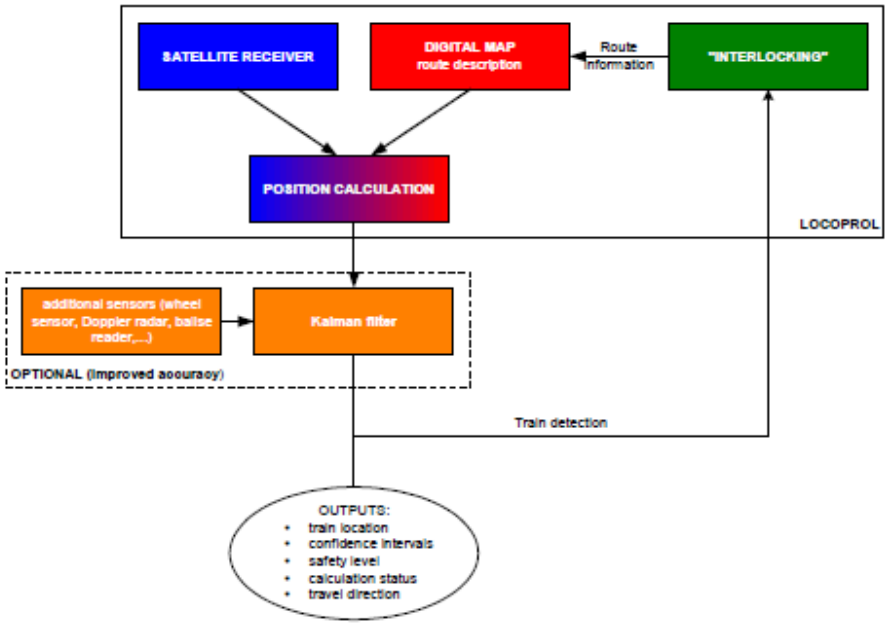
	and surveillance tools • Improving the ability of EU authorities (such as Coast Guard, Customs, FMC) to react faster in cases of emergency, suspicious/illegal activity or environmental threat. • Improving dissemination of data to interested parties through improved information sharing. • Improving capability of authorities to provide evidence in cases of fraud or illegal activity. • Aiding conflict resolution through improved information sharing in the open sea. • Providing a sound statistical basis on which to test and justify legislation through repeated and reliable information acquisition and management.[474] User feedback: • Satellite imagery useful to follow maritime routes and to plan reconnaissance missions • Developments are in progress and demonstrations of pre-operational services are foreseen in the following three clusters: maritime surveillance, land and infrastructure surveillance, humanitarian relief and reconstruction support • GMES products are very promising tools, as they are well suited for large oceanic areas with little ship traffic. They also allow for the sort of low-profile surveillance that such operations require. Finally, performance is likely to get even better as: • GMES imagery becomes integrated with data from other sources, • Processing techniques further progress to allow detection of ships under ten metres, while processing time is brought to under half an hour, • More optical and radar satellites become available, allowing the obtaining of at least a daily coverage of the area." • Difficulties in convincing border brigades at Eastern European borders about the added-value of satellite imagery for land border surveillance given their traditional army education and working structures • Earth observation satellite imagery is too limited, thus it proves to be very good for certain elements of analysis (proof of evidence to restructure teams, resources, technical capabilities) but does not provide for immediate action based on on-spot detection • Lack of thermal infra-red data at high-special resolution; asks ESA to equip sentinel satellites with this as otherwise critical infrastructures are linked to thermal releases and there is a range of the spectrum where there is no information for the detection of normal and abnormal behaviour • Problem in delivery mechanisms of satellite imagery: sometimes requested coordinates not shown in delivered satellite imagery → delays due to waiting time for the next image (revisit time) • The service combines local geospatial information and very high resolution satellite imagery. Outcomes provide updated infrastructure and terrain analysis to benefit the security of premises, population, land uses and economic activities • 2D modelling of CI, land uses at risk under urban burst, infrastructures at risk due to land displacements • Surveillance and protection of maritime transport for sensitive cargo • Coastal and open-water surveillance • Satellite SAR data combined with localisation data (coastal radar, aircraft information) and vessel identification systems (e.g. AIS) by means of the analysis of the wake detected-provides information on their velocity and direction • Maritime cluster: (user feedback) - revisit time insufficient - detection of vessels below 30m is critical - costs need to be better assessed
Collaboration(s)	• LIMES claims to be building on the work already achieved through other relevant FP and other projects, and ensuring a combination of expertise across the EU in the different domains in order to develop an integrated solution to maritime monitoring • GMES • FRONTEX • ENAGAS
Follow up	• Nauplios
Limitations	n.a.

Project Name	**ESA funded LOCOLOC**
Phase(s)	1: Prevention through location system 2: Improved response through improved locating
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• 01/03/2002 – 01/09/2004
Current Status	• Completed
Country/Institution	• ESA
Funding	• ESA

[474] "Organisation: wp 2000 – Maritime Surveillance." 11 Mar 2009. LIMES Project Website 30 July 2009 <http://www.fp6-limes.eu/index.php?page=custom&page_id=33>.

Partners	• n.a.
Objective(s)	• To develop an innovative, cost-effective satellite-based train location system based on GNSS as a primary component of a train protection, control and command system
In Practice	• to set up and validate new cost effective satellite-based vital train length, speed and acceleration measurement algorithms
Demonstration(s)	• A successful demonstration of the system was carried out in Gembloux, Belgium in April 2004.
Users	• Stakeholders in railway transportation
Added Value/Findings	• new algorithms are necessary to supplement satellite based train localisation algorithms to achieve an integrated train navigation system that could in future fruitfully fullfil both train protection, control and command as well as fleet management and end user service requirements
Collaboration(s)	• Complements LOCOPROL
Follow-up	• n.a.
Limitations	• Deals with train location for safety rather than security • No explicit mentioning of applicability for counter-terrorism or organised crime

Project Name	**FP 5 LOCOPROL**
Phase(s)	1: Prevention, detection through rail traffic management
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• 1/08/2001 – 31/07/2004
Current Status	• Completed
Country/Institution	• ALTSOM, Belgium
Funding	• CORDIS through a cost sharing contract • EC DG INFSO
Partners	• Alstom Belgium SA (Alstom BSI) (B), • INRETS (F), • Honeywell Regelsysteme Gmbh (D), • Alstom Transport Spa (Alstom STL) (I), • TRASYS SA (B), • BPV (D), • SEPTENTRIO NV (B), • RFF (F), • NMBS/SNCB (B), • Northern Jiaotong University (CN), • ERTICO (B), • CFTA(F).
Objective(s)	• To offer the same level of safety as on high-density lines and to enhance the efficiency of these lines in order for rail transport to become more attractive. • To define, develop, test and validate an innovative cost-effective satellite based (GNSS) vital fail-safe train location system as the core of a train protection, control and command system • To define a new multi-technology location system based on satellite positioning combined with fail-safe on-board track mapping and interlocking • To study its application to ERTMS/ETCS • To study and prove its short term applicability in Low Density Traffic Lines • To study and prove its applicability in order to increase track side workers protection
In Practice	• The system will enhance and extend the ERTMS/ETCS system, currently covering high density lines, to low density lines The new LOCOPROL location system is based on simultaneous use of information supplied primarily by the following three sub-systems: • Fail-safe track mapping (giving two equations to define a line segment), • Fail-safe interlocking functions (linking the line segments in one complete line), • Satellite positioning (locating the train on the complete line). • Unlike classical methods, LOCOPROL calculates the position using the untreated satellite signals, namely, the distance between train and satellite rather than the coordinates of the train after internal computation. This information is processed to achieve fail-safe satellite positioning.

	 Figure: The Functioning of LOCOPROL[475]
Demonstration(s)	• n.a.
Users	• Civilian sector
Added Value/Findings	• The system will enhance and extend the ERTMS/ETCS system, currently essentially covering high density lines, to low density lines • Will reduce costs significantly • Creates opportunities for European industry to conquer a large part of the large world market for this type of systems applied to Low Density Traffic Lines • Has the potential to have a direct positive impact on the day-to-day life of European commuters, and facilitate the creation of seamless intermodal chains in freight transport
Collaboration(s)	• EC DG INFSO • CORDIS
Follow-up	• n.a.
Limitations	• Primarily aimed at improving safety and functioning of rail traffic • No direct link to Internal Security

Project Name	**FP 7 LOTUS (LOcalisation of Threat substances in Urban Society)**
Phase(s)	1: Prevention of terrorist attack involving chemical substances through detection
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	R&D
Project Duration	• 01/01/2009 – 31/12/2011, 36 months
Current Status	• In progress
Country/Institution	• FP 7
Funding	• Total costs: € 4,30 million • EU contribution: € 3,19 million
Partners	• 10, coordinated by the Swedish Defence Research Agency (FOI)
Objective(s)	• To develop a new anti-terrorism tool for law enforcement agencies in the form of an integrated surveillance system for continuous chemical background monitoring with fixed site and/or mobile detectors in order to identify “chemical hotspots”, such as bomb or drug factories
In Practice	• Integrated Surveillance System
Demonstration(s)	• n.a.
Users	• Security organisations of different nature
Added Value/Findings	• A system like LOTUS would allow law enforcement agencies to become proactive, to act during a phase where there is low threat to citizens. The same system could also be used for combating organised crime by detecting e.g. drug production • By using existing global infrastructures for positioning (GPS) and networking (GSM, GPRS or 3G) the LOTUS system can be used more or less anywhere in the world at a relatively small cost for supporting installations and extra personnel

[475] Mertens, Pierre and Antonin Stark. “LOCOPROL – Project Presentation.” 2002. 8. 30 July 2009 <http://www.ertico.com/download/locoprol%20documents/presenta.pdf>.

Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Limited to chemical substances

Project Name	**JRC's LiveMon**
Phase(s)	2: Real-time monitoring and visualisation of disaster
Critical Mission Area	Emergency Management
Type	Implemented service
Project Duration	• n.a.
Current Status	• Implemented
Country/Institution	• JRC
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Real time Monitoring and Visualisation of Disaster Events
In Practice	• Real time Monitoring and Visualisation of Disaster Events
Demonstration(s)	• n.a.
Users	• First responders
Added Value/Findings	• n.a.
Collaboration(s)	• Is a spin-off of GDACS
Follow-up	• n.a.
Limitations	• Definition of "disaster" remains unclear

Project Name	**FP 4 MAGNET A TR 1034**
Phase(s)	1: Performing assessment of ENSS 1 involving all types of civil transportation
Critical Mission Area	Land Transportation, Transportation Security
Type	R&D
Project Duration	• 01/01/96 – 31/12/98, 36 months
Current Status	• Completed
Country/Institution	• FP 4
Funding	• n.a.
Partners	• Thomson-CSF Detexis, France (project lead) • Alcatel Space Industries (FR) • Daimler-Benz Aerospace NFS ()DE • Racal Telecommunications Ltd (UK)
Objective(s)	• To develop GNSS 1 user segment prototypes and to assess their performance in terms of availability, accuracy, integrity and continuity of service through field trials involving civil, land, maritime and aviation users. • The achievement of a multimodal core module common to all user segments
In Practice	• The project started with a first phase intended to update user requirements and the qualification and operational constraints that would apply to GNSS 1 user devices, leading to the definition of the validation plan. • The second phase of the project consisted of the development of GNSS 1 user segments and the set-up of a laboratory test-bed designed to simulate GNSS-1 like signals. This test-bed was used to verify the behaviour of the user equipment. The third phase of the project aimed at the evaluation of the performance of GNSS 1 in field trials. Since at the time of the trials the EGNOS infrastructure was limited to the ranging capability, the project team proposed to upgrade the SKYFIX network with real time wide area differential and integrity monitoring algorithms. The experiments performed with the users then constituted a good validation of the user segments. Civil aviation experiments were conducted on a Dasa Piper Navajo to demonstrate en-route to CAT I performances. In addition, railway network management and maritime navigation in coastal waters experiments were performed.
Demonstration(s)	• n.a.
Users	• Civil Aviation Services • Civil Maritime Services • Civil Rail Services
Added Value/Findings	• Demonstrations showed that the accuracy was better than the 10 m level which was targeted. In fact, with adequate filtering, the accuracy was in the range of 2.5 m, when using the WAD corrections. Experimentation shows the feasibility of satisfying new user applications who are not satisfied by GPS service, by using EGNOS SIS (WAD and GIC). This experimentation will help to strengthen European industry and will facilitate the development of new user markets • Increased availability and accuracy of information • Can be used by various sectors and is thus not limited to one sector only • The GLONASS capability is included, but the exploitation of the signal has faced lack of short term market opportunities
Collaboration(s)	• n.a.
Follow-up	• A complete family of products has been developed by DSNP, based on the MAGNET

	architecture and also used by Thomson-CSF Detexis. The exploitation of the aviation and the rail applications is more in the medium term • MAGNET B
Limitations	• Primary focus is to improve the Land transportation Sector and is not directly linked to internal security

Project Name	**FP 4 MAGNET B TR 1035**
Phase(s)	1: Performing assessment of ENSS 1 involving all types of civil transportation
Critical Mission Area	Land Transportation, Transportation Security
Type	R&D
Project Duration	• 01/01/96 – 30/06/99, 41 months
Current Status	• Completed
Country/Institution	• FP 4
Funding	• n.a.
Partners	• Thomson CSF Detexis, France • Alcatel Space Industries (FR) • Daimler-Benz Aerospace NFS (DE)
Objective(s)	• To augment the MAGNET A project performances for Local Area Applications (docking for maritime, CAT II/III landings for Aviation) through the design and development of user segment prototypes and to assess their performance in terms of availability, accuracy, integrity and continuity of service through field trials involving civil, land, maritime and aviation users
In Practice	• The first phase intended to update the user requirements and the qualification and operational constraints that would apply to GNSS 1 user devices, leading to the definition of the validation plan. • The second phase of the project consisted of the development of GNSS 1 user segments and their associated Local Augmentations (GLONASS differential augmentations, odometer coupling, map matching, kinematic techniquesÉ). • The third phase of the project aimed at the evaluation of the performance of GNSS1, enhanced locally, in field trials. Since at the time of the trials the EGNOS infrastructure was limited to the ranging capability, the project team proposed to upgrade the SKYFIX network with real time wide area differential and integrity monitoring algorithms. • Civil aviation experiments were conducted on a NLR Fairchild Metro to demonstrate CAT II/III landing performances. In addition, railway network management and maritime navigation in narrow channels and docking operations experiments were performed.
Demonstration(s)	• The field trials demonstrated the performances of these receivers and of the core module. Accuracy was better than the 1 m level which was targeted. In fact, with the kinematic mode experimented in the marine and the aviation a segment, accuracy was close to the cm level. • Experimentation showed the feasibility of satisfying new user applications, not satisfied by GPS service, by using EGNOS SIS (WAD and GIC) and improving locally these performances through local augmentation techniques • Performing GLONASS differential corrections were possible as well as transferring the corrections through the GSM data-link (Rail applications).
Users	• Aviation sector • Maritime sector • Railway sector
Added Value/Findings	• The MAGNET technical developments and experimentation contributed to strengthening the competitive position of European industry in manufacturing satellite navigation receivers and test bed, also setting the scene for the stimulation of new markets for various users and applications. Can be applied to multiple users
Collaboration(s)	• Followed up on MAGNET A
Follow-up	• A complete family of products based on the MAGNET architecture has been developed by DSNP. This line is now promoted by Thomson-CSF Detexis (which results from the merge between Dassault Electronique and Thomson-CSF RCM
Limitations	• On the Civil Aviation area, NFS who built the Air user segment was unfortunately closed by its mother company Daimler in late 1998, and the prototype will no longer be exploited • For non-safety applications in the Railway area, the MAGNET A receiver is too expensive and can not replace low cost GPS receiver yet. An improved MAGNET B equipment can be used for safety Railway applications in the future. • The main concern from Railway customers is the integrity performance of EGNOS locally completed. Under these circumstances, exploitation for the aviation and rail applications would remain in the medium term. On the other hand, short-term Exploitation Plans for medium and high performance receivers dedicated to land and marine users are (very?) promising • Technological benchmarking not primarily aimed at improving the provision of internal security in the transportation sector

Project Name	**JRC's MAHB** (Major Accident Hazards Bureau)
Phase(s)	1: Prevention through risk management and lessons learnt analysis of past incidents
Critical Mission Area	Critical Infrastructures
Type	Implemented service
Project Duration	• n.a.
Current Status	• On-going
Country/Institution	• JRC
Funding	• JRC
Partners	• n.a.
Objective(s)	• To contribute to consistent and efficient control of industrial hazards through the development of tools and methods for risk assessment and management
In Practice	• Monitoring of accidents and incidents through an accident reporting system (accident data base) • Analyses of accident and incident cause • Drawing lessons for competent authorities • Mainstreaming lessons into risk management process
Demonstration(s)	• n.a.
Users	• Stakeholders in the Critical Infrastructures sector
Added Value/Findings	• Improved risk management process
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Definition of "incident" is unclear; it remains open whether terrorist interference is included

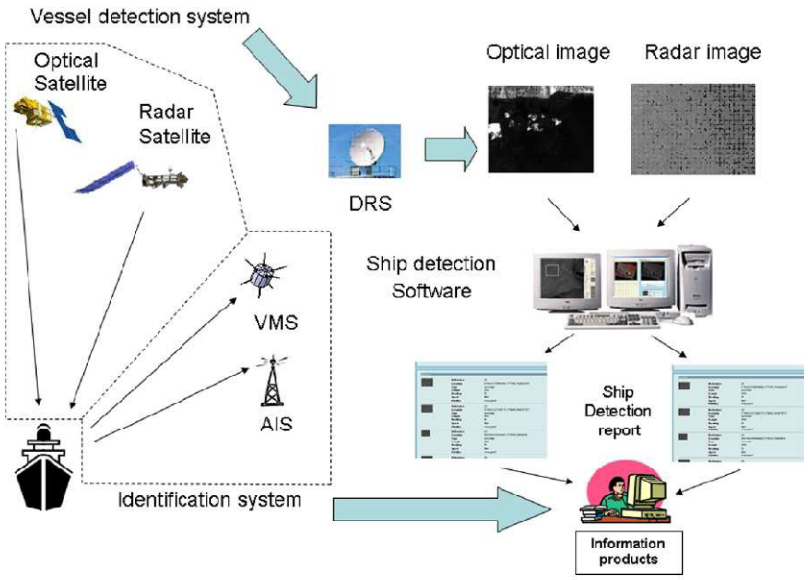
Project Name	**FP 6 MarNIS** (MARitime Navigation and Information Services)[476]
Phase(s)	2: Response
Critical Mission Area	n.a. (integrated approach)
Type	Implemented service
Project Duration	• 01/11/2004 – 30/10/2008, 36 months
Current Status	• completed
Country/Institution	• FP 6
Funding	• European Union • European Commission (co-funding)
Partners	• Dutch Ministry of Transport, Public Works and Water Management • ABP Marine environmental research Ltd. • Belgian Ministry of the Flemish Community Administration Waterways and Maritime Affairs • Bureau Veritas • Cetemar SL Centro Estudios Técnico-Maritimos Sociedad • Centre for the development of Transport and Logistics in Europe • D'Appolonia • Det Norske Veritas • ECORYS Nederland B.V. Transport division • European Harbour Masters' Committee • European Maritime Pilots' Association • Hochschule Wismar, University of Technology, Business and Design • Institut Francais de Navigation • Institut National de Recherche sur les Transports & leur Sécurité • Instituto Superior Tecnico • Kongsberg Seatex AS/Norcontrol IT • Koninklijke Scheepsagentuur Dirkzwager • LogIT Systems a.s. • Lyngsoe Marine A/S • Maritime Research Institute Netherlands • Maritime and Coastguard Agency • Nautical Enterprise Centre Ltd • Norwegian Maritime Directorate • Norwegian Coastal Administration (NCD) • Norwegian Meteorological Institute • Marine Forecasting Centre • Oceangraphic Company of Norway • Portel Servicios Telematicos S.A. • QinetiQ • Port of Rotterdam • Sociedad de Salvamento y Seguridad Maritima • Scheepvaart en Transport College b.v. • Sequoyah NV • Sistemi e Telematica • SevenCs AG & Co. KG • Swedish Maritime Adminstration • SSPA Sweden AB • Telespazio S.p.A. • Thetis S.p.A. • Organisation for Applied Scientific Research • University of Liege • University of Strathclyde • World Maritime University • Christian Michelsen Research AS • Laboratório Nacional de Engenharia Civil • TRESCO Navigation Systems • Transas • OMC Europe

[476] "MarNIS - Maritime Navigation and Information Services." Maritime Navigation Information Service 30 July 2009 <http://www.marnis.org/home.asp>.

Objective(s)	• Improvement of safety and the protection of the environment • Improvement of security • Improvement of the economic aspects of sea transport • Improvement of efficiency and reliabilty
In Practice	• The development of a safety structure in European waters through the use of Vessel Traffic Management (VTM) in the littoral seas continuously monitoring high risk vessels along the European coast using AIS. • The development of practical solutions as to how VTM can contribute to monitoring of vessels and tracking of cargoes from consignor to consignee (down to pallet and package size) within its jurisdictional boundaries • Enhancement of traffic management using, among others, the GALILEO interface • Developing a co-ordinated and harmonised system for the collection and presentation of dangerous goods related information to all relevant stakeholders in a port environment
Demonstration(s)	• 23 to 25 September 2008 in Genoa presenting the main findings; emphasising the interaction and interoperability of the concepts of Maritime Information Management (MIM) and Maritime Operational Services (MOS) as proposed in MarNIS • 15 to 16 October 2008 in Lisbon highlighting innovations for more efficient port entry
Users	• European maritime sector
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• Concept will be realised between 2012 and 2015
Limitations	• n.a.

Project Name	**ESA Funded MARISS** (GMES Maritime Security Service Element)
Phase(s)	1: Anomaly detection 2: Improved response due to improved situational awareness
Critical Mission Area	Transportation Security, Waterborne Transportation, Border Security
Type	R&D
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• ESA
Funding	• ESA funded GMES service element
Partners	• 17 coordinated by Telespazio
Objective(s)	• Intends to integrate and correlate different types of data as well as to further improve the information currently provided and finally achieve the consolidation of operational services
In Practice	• Vessel Detection System, VTS, AIS with EO • Service portfolio is devoted to enhance Maritime Border Surveillance capabilities and it is based on the integration of coastal radar information, Vessel Detection Systems, Vessel Traffic Management Systems (VTS) and Automatic identification Systems (AIS), with EO products such as ship detection reports, critical feature detection • Vessel Detection in Open Sea Areas: This service results from the integration of AIS or vessel monitoring systems (VMS) ship reports, with EO Synthetic Aperture Radar (SAR) Ship detection products, and size estimation products. The final results are delivered to the user terminal. • Vessel tracking at Sea Borders: This service produces results from the correlation of VTS radars with SAR Ship Detection products. Specifically, vessels tracked by VTS radars while approaching the coastline can be tracked back as much as 3 hours through integration with EO SAR Ship Detection products acquired in the preceding 3 hours; vessels tracked by VTS radars while leaving territorial waters can be tracked forward as much as 3 hours through integration with EO SAR Ship Detection products. The correlation is performed at User premises where EO service chain is integrated with VTS Data Base; results (extended tracks) are displayed at user terminal. • Anomalies Detection in Specific Critical Areas: This services focus on the detection of specific features in already identified coastal critical areas comparing the data provided from different sources or at different times. Such critical features could be, for example: mustering points for people, fuelling areas, refugee camps, etc. The analysis of this information can reveal anomalies and trigger specific surveillance activities in specific sensitive areas. The final product will be accessible through secure internet connections, together with ship detection reports acquired off shore. MARISSapproach:[477]

[477] "A solution to Maritime Security" Nov. 2006 MARISS Newsletter #1 30 July 2009 http://www.gmes-mariss.com/sections/publications/documents/MARISS-NEWSLETTER-19072006.pdf Page 4



- Border surveillance / anomalies detection: The MARitime Security Service (MARISS) project has been working with Coast Guards, Border Guards, Navies and other national and European organisations to begin the process of qualifying an enhanced monitoring capability that integrates satellite based vessel detection with conventional systems listed above

- Decision Making Support: Decision making support services result from the combination of EO ship detection outputs with ancillary data to provide intelligence support information, through a centralised management of data distribution/request. The final product will be delivered integrated in a GIS.
- Near-Real time Monitoring Services: These services provide tactical support by enabling the creation of an enhanced Recognised Maritime Picture within a maximum of 60 minutes of satellite imagery being acquired:
- By integrating satellite radar based vessel detection with AIS/VMS/VTS based vessel identification data, rapid detection and location of non-cooperative vessels can be obtained for a particular area.
- This is performed by correlating the positions of vessels' reporting using cooperative systems (AIS, VMS) with the vessels detected on the SAR imagery so that all known targets can be identified. Any remaining vessels can then be considered as potentially suspicious and can be investigated using airborne survey or surface patrol craft.
- This can include backward or forward tracing of vessel movements to better understand vessel movements and detect suspicious behaviour.
- These services extend the coverage of traditional surveillance systems and optimise the deployment of airborne and surface patrol craft.
- Strategic analysis service: Offline analysis can improve the available intelligence and support identification of priority areas for suspicious behaviour (e.g. offloading) or trafficking routes or the tracking of suspect vessels. In addition offline analysis can support the detection of infrastructure and changes related to illegal trafficking such as boat building facilities, new mustering and embarkation facilities, storage facilities or the build up of people in sensitive areas close to the coast or to national borders. These provide data complementary to established intelligence sources by integrating imagery from both radar and optical satellites
- Tactical information service: MARISS Tactical Information Services are based on the integration of satellite SAR based vessel detection with more conventional vessel identification systems. Examples of some of the different integrated information sets are shown below.

Demonstration(s)

- Today, MARISS services have been tested on a pre-operational basis in the following areas:
- South Spanish Coast
- North Atlantic Channel
- Western Atlantic
- Southern Baltic Sea
- Sicily Channel
- Aegean Sea
- North African Coastal area
- Canary Islands
- Portugal continental coast (and Gorringe fishing area)
- Azores area
- Libyan coastline

Users

- Core user group: Italian coast guard, Ministero dell'Interno, Italy; DAMGM, France; Hellenic Coast Guard, Greece; Puertos del Estado, Spain; Guardia Civil, Spain; HM Revenue & Customs, UK; Irish Naval Service, Ireland; Swedish Coast Guard

Added

- Although current service performance levels are strongly constrained by present

Value/Findings	availability of satellite assets, this situation is evolving radically. Over the next 2-3 years, more than 10 new radar satellites will be launched by European or Canadian operators. This year alone, 4 high resolution radar satellites (2 civil, 2 military) were launched and more are planned for later in the year. This will ensure significant improvement in update times and the MARISS project is intended to enable Coast Guards, Border Guards, Navies and other relevant organisations to reap the benefits from these new surveillance and vessel detection capabilities • Hard to detect small boats (user feedback from demonstration over Canary Channel) • Integrates optical and adar satellite imagery with VMS and AISi into one information product
Collaboration(s)	• LIMES (strongest technical cooperation); TANGO, national financed initiatives in maritime surveillance and security, MARCOAST, CleanSeaNet • In particular it is hoped that LIMES activity in the maritime security domain can improve the performance characterisation of the different satellite radar systems and their various operating modes with respect to the detection and identification of types of vessel and the impact of different environmental and viewing conditions. In addition, LIMES activity in developing new algorithms can provide an improved basis for integrating the different data streams which can then be trialled on a more operational basis within any future follow-on activity to MARISS. • GMES
Follow up	• n.a.
Limitations	• n.a.

Project Name	**MARUSE** (MARitime USEr community)
Phase(s)	1: Prevention 2: Improved Search and Rescue
Critical Mission Area	Transportation Security / Waterborne Transportation, Border Security
Type	Industry Project by Kongsberg
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• n.a.
Funding	• n.a.
Partners	• Kongsberg Seatex • Datamat • EADS Astrium • European Commision Joint Research Centre • European Satellite Service Provider • France Développement Conseil • Septentrio Satellite Navigation • Skysoft Portugal • Sogei • Telematica • Terma • Thales Research & Technology • Trinity House Lighthouse Service • Ursa Minor Space & Navigation • via donau
Objective(s)	• Introducing GALILEO and EGNOS (European Geostationary Overlay Service) in the maritime domain
In Practice	• supporting a seamless, efficient and harmonised end-to-end service • support intermodal transport, short sea shipping and cargo tracking • increase the transport efficiency • allow revenue collection • provide a logistic company interface • satisfy safety needs • add security and authentication • provide maritime user terminals • provide maritime local elements • provide local and regional control centres for local elements • provide use of Galileo GSTB-V2 signal • provide use of Galileo pseudolites • improved navigation performance in terms of accuracy, integrity, continuity and availability of the service Key technical development: • Maritime Galileo pseudolites • Galileo/GNSS receiver prototype capable of tracking GSTB-V2 signal and Galileo Pseudolites • Integrated Navigation Test Bed • Use of communication links (VHF, IALA, AIS, WLAN, GPRS/UMTS)
Demonstration(s)	• 9 May 2007 in Trondheim demonstrating high accuracy maritime applications using available GNSS, AIS, EGNOS and software simulations to raise awareness of the potential of the future European satellite navigation system Galileo in the maritime sector. • 10 October 2007 in the United Kingdom revealing how GALILEO can provide continuous

	performance in the transition from coastal navigation, through port approach and port navigation • 21 June 2007 in Djerdap, Serbia demonstrating how EGNOS and Galileo signals are applicable to be used for supportive vehicle guidance on critical river sections, such as locks
Users	• Maritime sector
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**JRC's Institutional Action MASURE** (MAritime SURveillancE)
Phase(s)	1: Prevention and detection 2: Response through provision of completed picture of vessel traffic Integrated approach
Critical Mission Area	Transportation Security / Waterborne Transportation, Border Security
Type	R&D
Project Duration	• n.a.
Current Status	• On-going
Country/Institution	• JRC
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Fusion of information from all sources to create a more effective Reference Maritime Picture • Pollution control • Fisheries control • Maritime border control • Maritime and port security • Security & Defense • Enforcement and compliance monitoring • Identification of risks and threads (environmental, security, and economic)
In Practice	• Pollution and vessel surveillance supported by EGEMP (European Group of Experts on satellite Monitoring of sea-based Pollution) • Long-term monitoring of oil pollution by mapping spills and identifying hotspots • Using new tools such as the Automatic oil spill detection algorithm for satellite SAR and the GIS layers for environmental and shipping data • Maritime Domain Awareness • Working in collaboration with EU stakeholders
Demonstration(s)	• n.a.
Users	• EMSA (European Maritime Safety Agency) • FRONTEX (European Agency for the Managemen of Operational Cooperation at the External Borders) • CFCA (Community Fisheries Control Agency) • EUSC (European Union Satellite Centre)
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 6 MERSEA** (Marine EnviRonment and Security for the European Area**)**
Phase(s)	1/ 2/ 3: Ocean Monitoring System
Critical Mission Area	Transportation Security / Waterborne Transportation
Type	R&D
Project Duration	• n.a.
Current Status	• Completed
Country/Institution	• FP 6
Funding	• n.a.
Partners	• 39, coordinator: IFREMER
Objective(s)	• Develop an ocean monitoring system in an operational network that systematically acquires data (EO from satellites, in-situ from ocean observing networks and surface forcing fields from numerical weather prediction agencies).
In Practice	• These data from diverse sources are combined and merged with numerical ocean circulation models to produce best estimates of the actual state of the oceans (now casts)

	and forecasts. • That information is then disseminated to enable various users to develop specific applications. The project itself has identified specific applications on marine safety concern improved wave forecasts, offshore operations, ship routing and oil spill drift[478] • Remote sensing from satellites • In situ data form ships and buoys • Atmospheric forcing field
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• Potential of tackling all three phases • Integrates data from different sources
Collaboration(s)	• GMES • GOOS • GODAE • also contributes to the implementation of the Global Ocean Observing System (GOOS) and federated European contribution of the Global Ocean Data Assimilation Experiment (GODAE)
Follow up	• n.a.
Limitations	• n.a.

Project Name	**France's MOBISIC** (Mobile Security System for Critical Infrastructures)	
Phase(s)	1 & 2: Detection, alert 3: Emergency management	
Critical Mission Area	Transportation Security / Land Transportation, Emergency Management	
Type	R&D	
Project Duration	• 10/2007 – 09/2010, 36 months	
Current Status	• In progress	
Country/Institution	• France	
Funding	• Funded in the framework of the French competitiveness cluster SYSTEM@TIC • 9.4 million Euros (3,25 from the government)	
Partners	• Thales (coordinator) • Alcatel-Lucent • Bertin • CEA • Evitech	• Gemalto • INRIA • Sagem • Sinovia • Sodern
Objective(s)	• Development of mobile security systems linked to critical infrastructure and emergency management	
In Practice	• Development of a fitted system • Development of fitted sensors („Plug & Play", geo-localisation...) • Development of a mobile unit with cooperative visualisation means • Demonstration	
Demonstration(s)	• n.a.	
Users	• Civil and military authorities	
Added Value/Findings	• Up to demonstration, collaborative project	
Collaboration(s)	• n.a.	
Follow up	• n.a.	
Limitations	• Limited use of satellite data • Limited to positioning	

Project Name	**France's MODENA** (Modélisation de l'Observation à Distance de l'ENvironnement mAritime)
Phase(s)	1: Detection through surveillance 2: Improved search and rescue
Critical Mission Area	Border Security
Type	R&D
Project Duration	• n.a.
Current Status	• In progress
Country/Institution	• France
Funding	• Funded in the framework of the French competitiveness cluster Brittany Sea
Partners	• Research centres: Ecole nationale supérieure des télécommunications de Bretagne (GET/ENST Bretagne – Brittany National Advanced Telecommunications Institute) in Brest, project director: TAMCIC Lab (télédétection aérienne et spatiale en environnement marin et sous-marin/Remote Sensing of Marine and Undersea Environment) and LEST Lab (modélisation de la propagation en télécommunications/Propagation Modelling in Telecommunications), Institut d'électronique et de télécommunication (IETR – Electronics and Telecommunications Institute), Rennes – SAPHIR team specialising in radar imaging,

[478] MERSEA. "Project Summary." 26 Feb 2009 <www.mersea.eu.org/ProjectOverview/Summary.html>.

	Institut de recherche en électrotechnique et électronique de Nantes Atlantique (IREENA – Nantes Atlantique Electro-technology and Electronics Research Institute) – Radar team specialising in the field of remote sensing particularly at sea, ENSIETA (National Advanced Weapons Industry and Technology Institute), Brest – Team studying "Extraction et Exploitation de l'Information en Environnements Incertains" (Extraction and Exploitation of Information in Uncertain Environments), Ecole nationale d'ingénieurs de Brest (ENIB – Brest National Engineering College) is involved through the Centre européen de réalité virtuelle (CERV – European Centre for Virtual Reality) which it has created within its establishment, IFREMER, Brest (French Marine Research and Development Institute) – Department of Physical and Spatial Oceanography • Companies: MAREE, Lorient – Consultancy specialising in the development of new marine geophysics and undersea acoustics tools, BOOST Technologies, Plouzané – Consultancy services specialising in remote surveillance of the marine and coastal environment and, in particular, in high resolution, radar satellite imaging, Antenessa, Plouzané – One of the leading European companies for the design and manufacture of radiant systems and components, Thales Systèmes Aéroportés, Brest – European leader and global player in aeronautics and avionics, TSA is developing surveillance systems for civil and military purposes, Artal Technologies, Labèges (31, Haute Garonne) – Development of computerised supervision systems, command and control systems and simulators for use in aeronautics, defence and space industries.
Objective(s)	• MODENA is a marine environment modelling platform for simulating maritime surveillance systems
In Practice	• Numerous methods are used for ocean surveillance – airborne, satellite, ship, coastal, etc. – and various technologies are employed. These call on a range of active and passive sensors varying widely from optronic sensors to radar, sonar, etc. with the result that data collected then has to be compiled • A great deal is at stake involving the successful acquisition of a share in rapidly expanding markets relating to the field of homeland safety and security – anti-pollution measures, fishing ground surveillance, combating illegal immigration and all types of trafficking, sea rescue operations, etc. – as well as relating to monitoring of coastal environment and sea conditions, weather/climate forecasting, flood zone surveillance, etc • MODENA is an R&D tool which will make it possible, at a very early stage of a product design process, to take account of the huge diversity of parameters in the monitored environment and the constraints imposed on inter-data dialogue, thereby facilitating development of new and more intelligent sensors and surveillance systems.
Demonstration(s)	• n.a.
Users	• Stakeholders in maritime sector
Added Value/Findings	• It is anticipated that advances in this area will enable the various technologies and surveillance methods to "speak to one another" within the context of a "systems' system" thereby moving from an accumulation of heterogeneous information to a synthesized vision of situations under observation. Current advances also focus on problems involved in taking account of the specifics of particularly complex fields of observation, such as with moving objects and substances adrift on or in a moving environment
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• n.a.

Project Name	**Lockhead Martin's MTM200™ system**
Phase(s)	2: Quick search and rescue
Critical Mission Area	Transportation Security / Waterborne Transportations
Type	R&D
Project Duration	• n.a.
Current Status	• on-going
Country/Institution	• n.a.
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Saving lives and property at risk in the maritime environment
In Practice	• Providing shore-based infrastructure including emergency communications and by collecting and processing distress call information • Quickly locating lost or distressed vessels • Providing command and control with intelligence required for rescue missions in coastal and inland waterways
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

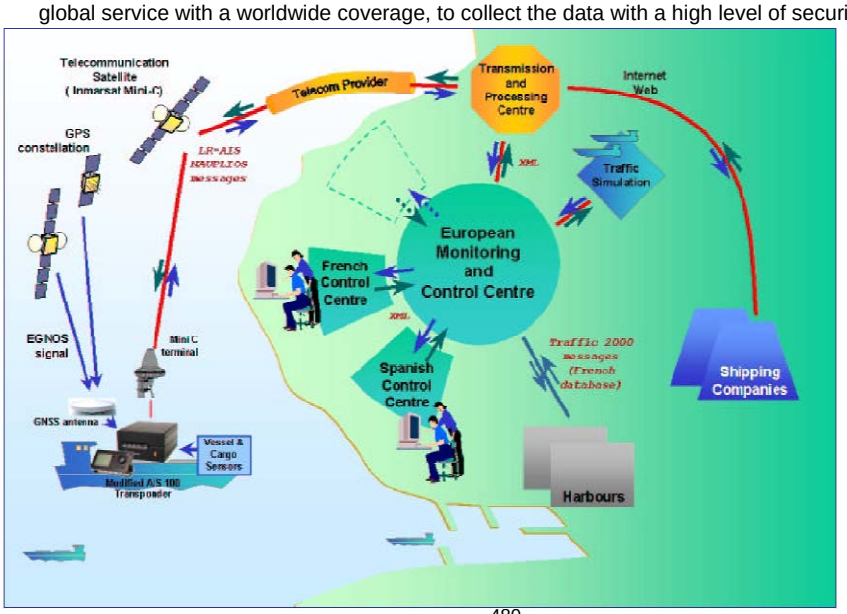
Project Name	FP 4 MULTITRACK TR1039
Phase(s)	n.a. addressed problem of intermodal transportation
Critical Mission Area	Transportations Security
Type	R&D
Project Duration	• 01/01/96 – 31/05/99, 41 months
Current Status	• Completed
Country/Institution	• Strategy Analytics (UK)
Funding	• n.a.
Partners	• Atos -ODS (FR) • Atos-ODS (ES) • Autoridad Portuaria de Santander (ES) • Eutelsat (FR) • FI Systems (FR) • Instituto de Automatica Industrial (ES) • Intermec International (FR) • Ronda Grupo Consultor (ES) • SEMAT (ES)
Objective(s)	• To develop and market a European service that enables different players to monitor, track and locate a shipment in the most transparent way possible throughout the whole logistic chain, whatever the means of transport used (road, rail and sea). • To address the problem and improve management of intermodal logistics chains. It was to provide an accurate time-stamp of the passage of the merchandise at different points along the chain. • To meet different end users' needs in terms of multi-level access classes, confidentiality and level of access to information for each class, and ease of use
In Practice	• Monitoring, Tracking and Tracing of Goods • Use of GIS data
Demonstration(s)	• The validation results have clearly demonstrated the ability of the system to track a whole consignment of Honda cars from Bristol (UK) to Port of Santander (SP) and onto mainland Spain via road and rail • 275 Intermec tags have been fitted to 200 Walon transport vehicles and 75 Semat wagon trains, and 8 EutelTRACS' terminals fitted to 6 Walon trucks and 2 UECC boats. This installation will be used to track shipment of Honda cars manufactured in the UK and exported to Spain
Users	• Entire logistic chain
Added Value/Findings	• Integration of different intermodal transportation modes for improving security of the entire supply chain. • To improve the usability of the information, tracking data are presented using a powerful Geographic Information System. • In terms of tracking technology, the initial implementation of Multitrack uses Intermec International vehicle and train tags and EutelTRACS satellite tracking terminals. The system is designed to interface with sensors that transmit cargo status data (e.g. temperature, shocks, and leaks).
Collaboration(s)	• n.a.
Follow-up	• In the future, Multitrack will be integrated with other key tracking technologies that have not yet been integrated in the Multitrack System. The most notable ones are GSM cellular system, newer satellite services based on Low/Mid Earth Orbit (LEO and MEO), and third-generation mobile technology that will be available in the near future
Limitations	• Could be further improved by adding an alert function and cooperation with emergency responders

Project Name	**MUSIS** (Multinational Space-based Imaging System)
Phase(s)	1: Prevention and detection
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Satellite system
Project Duration	• First satellite envisaged for early 2010
Current Status	• Definition phase
Country/Institution	• n.a.
Funding	• n.a.
Partners	• French Ministry of Defence • CNES • Austria • Belgium • Sweden • EDA involvement
Objective(s)	• European intelligence capability
In Practice	• Three satellites to be launched on orbits with two different altitudes • Overall design is derived from the Pleiades system (high resolution and great manoeuvrabilitiy)
Demonstration(s)	• n.a.
Users	• n.a.

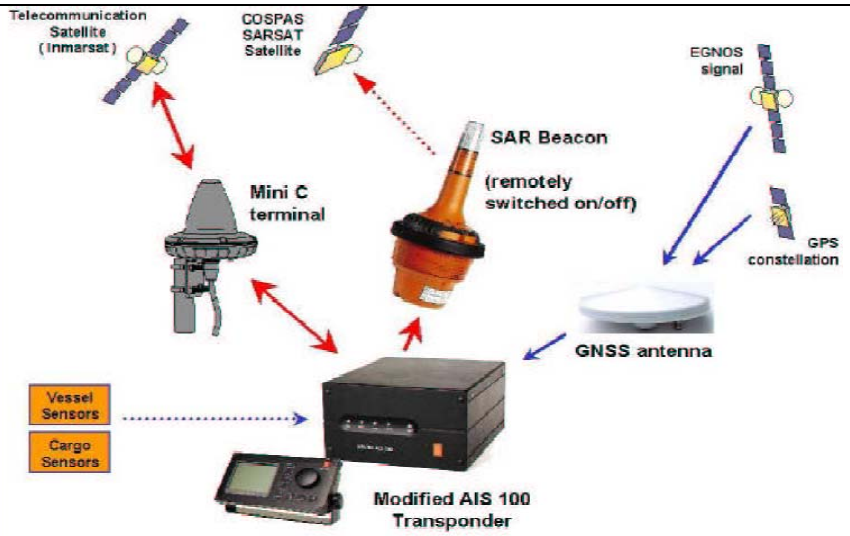
Added Value/Findings	• European intelligence capability
Collaboration(s)	• Complements GMES
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 7 MyOcean**
Phase(s)	n.a. (integrated approach)
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security
Type	R&D
Project Duration	• 2009 – 2012, 39 months
Current Status	• On-going
Country/Institution	• FP 7
Funding	• FP 7
Partners	• ESA • Fisheries and Ocean Canada • TechWorks Marine • University of Plymouth • Plymouth Marine Laboratory • UK Met Office • European Centre for Medium Range Weather Forecast • The british National Environment Research Council • HR Wallingfort • The Norwegian Meterological Institute • Norsk institutt for vannforskning • Nansen Center • Brockmann Consult • Bundesamt für Seeschifffahrt und Hydrographie • Romanian Marine Research Institute
Objective(s)	• Providing best possible information available on oceans • Integrating Member States' capacities and procedures to avoid resource duplications and define European procedures and organisations as well as a European architecture
In Practice	• To set-up an integrated pan-European capacity for ocean monitoring and forecasting • Working on the combination of space and in-situ observation and their assimilation in numerical models
Demonstration(s)	• n.a.
Users	• Marine safety • Marine and coastal environment
Added Value/Findings	• Integration of Member States' capacities and procedures to avoid resource duplications and define European procedures, organisations as well as a European architecture
Collaboration(s)	• A first step towards the implementation of the GMES' Marine Core Service
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 5 NAUPLIOS (Navigation and Perilous Goods Input and Output System)**
Phase(s)	1: Prevention through monitoring and traffic management 2: Improved Search and Rescue due to improved positioning
Critical Mission Area	Transportation Security / Waterborne Transportation
Type	R&D
Project Duration	Feb. 2002 – April 2004
Current Status	Completed
Country/Institution	FP 5
Funding	n.a.
Partners	• CNES, France • DAMGM, France • INAVE, Spain • BlueFinger, United Kingdom • CLS, France • KONGSBERG SEATEX, Norway • CETEMAR, Spain • ECROYS, The Netherlands
Objective(s)	• To heighten maritime surveillance mainly for pollution but also: • Monitoring (advanced traffic monitoring for better identification of ships and cargoes in Europe's coastal waters) • Improve safety aspects incl. mitigation actions and improved traffic management, as well as stronger crisis management and maritime SAR • To provide means for averting acts of piracy and terrorism, and protecting the security of ships, crews and destination ports (e.g. if a bomb is placed in a shipping container) • To demonstrate the added value of the Galileo positioning and SAR services for maritime transportation of goods and hazardous materials; • To evaluate the measures that will have to be prepared at European level to take full advantage of Galileo; • To enhance safe navigation based on earlier identification of ships coming into coastal

	zones; • To contribute, through a Mission Control Centre simulating the main functions, to the setting-up of a European Maritime Safety agency.[479] • To demonstrate new maritime surveillance and safety services, based on GNSS and satellite communication systems • To enhance maritime transport surveillance, with vessels identification from open sea up to coastal zone, ship data and dangerous goods data for every vessel in surveillance zone • To support search and rescue with remote SAR beacon activation and transmission of message towards beacon, improvement in localisation • To foster information sharing with harbours and commercial companies
In Practice	• Provides navigation information to ships (location, heading). In order to demonstrate the added value of Galileo, the EGNOS signal broadcast by the EGNOS System Test Bed (ESTB) will complement GPS signals to add integrity navigation data; • Telecom, to exchange navigation data between the ship and the control centre; • SAR, to represent the Galileo baseline configuration and to experiment and validate the SAR/Galileo operational concept; • Distress forward link, to relay the distress messages from the ships to the rescue centres via Cospar-Sarsat satellites; • Return (or feedback) link, for SAR acknowledgement and coordination message (to inform ships in the distress area). The added value of Galileo is this return link. The return message will be transmitted via the telecommand link and does not necessitate any modification of the ESTB
Demonstration(s)	• A six months demonstration involving 6 vessels and hundreds of simulated ones. The types used were a high-speed ferry, a tanker, a fishing vessel, a container carrier, a standard ferry and a French patrol vessel. The Vessels: • Elisa-B from Naviera PINILLOS shipping company is a pure container carrier. Every week, she crosses the Gibraltar strait, from Barcelona and Valencia (Sp) on Mediterranean Sea to Canaries Islands (Sp) in Atlantic Ocean and performs the return voyage the week after. • Chassiron from PETROMARINE is a chemical / oil tanker, built in 2000. She operates mainly off the French Atlantic coast with occasional voyages in the Channel. • Condor Express from Condor Ferries is a high speed craft offering fast ferry passenger and freight service between the UK, the Channel Islands and the French port of St. Malo. • The fishing vessel VILLO from JEAN NICOT organisation operates mainly off the French and Irish Atlantic coasts and is able to cool its own catch. • IRIS from DAMGM (French Maritime Affairs) is a patrol vessel (46m) based in Lorient (Fr). Capable of operating on all the coasts of continental France, it operates off the Atlantic coast during the demonstration. • Pride of Bilbao, a ferry from P&O Nedlloyd, crosses the Channel several times each day (Portsmouth - Le Havre) and does at least every week the Portsmouth -Bilbao voyage. • NAUPLIOS demonstrates that using a satellite communication system coupled together with an existing standardised system for identification like AIS, should provide an efficient service for both maritime safety and security application. The satellite is able to perform a global service with a worldwide coverage, to collect the data with a high level of security.  Figure: Example of the working Structure of Nauplios[480]

[479] Levy, David. "NAUPLIOS Presentation Sheet." 17 Mar 2009 <http://nauplios.cnes.fr/NAUPLIOS-PresentationSheet.pdf>.
[480] Source: Maréchal, Jean. "Final Report Nauplios Galileo Pilot Project." July 2004. CNES 31 31 July 2009 <http://nauplios.cnes.fr/Nauplios-Final-Report-ExecutiveSummary-V1.0-pdf.zip>.

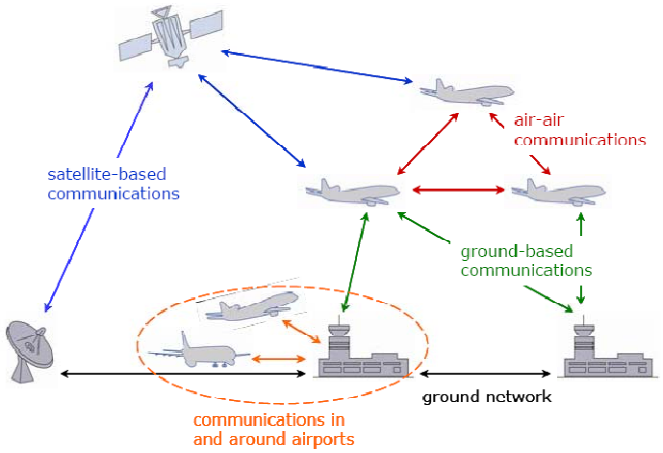
	 Figure: Different Components of the Nauplios Architecture[481]
Users	• n.a.
Added Value/Findings	• Adapt Galileo services to the needs
Collaboration(s)	• Galileo • SafeSeaNet
Follow up	• n.a
Limitations	• Mainly covering safety aspect, should thus be extended to include internal security threats

Project Name	**FP 6 OASIS** (Open Advanced System for dISaster & emergency management)
Phase(s)	n.a. (data harmonisation)
Critical Mission Area	Emergency Management
Type	R&D and Demonstration
Project Duration	• 01/09/2004 – 31/12/2008, 48 months
Current Status	• Completed
Country/Institution	• FP 6
Funding	• Total costs: € 19.54 million • EU contribution: €10.5 million
Partners	• 14 coordinated by EADS
Objective(s)	• The objective of OASIS is to define and develop an Information Technology (IT) framework based on an open and flexible architecture and using standards, existing or proposed by OASIS, that will be the basis of a European disaster and emergency management system
In Practice	• To define a standard instrument, the Tactical Situation Object (TSO), which carries the description of the situation (event, resources engaged and tasks in progress) between systems
Demonstration(s)	• The TSO has been tested with several applications developed by OASIS partners during trials throughout Europe
Users	• Police • Civil protection • Fire fighters • NGO such as the Red Cross, Médecins Sans Frontière, etc. • Ambulance services • Liaison officers of military organisations.
Added Value/Findings	• The TSO is defined as a cornerstone achievement of the project • Improved emergency management through the definition and development of an IT framework and standardisation for the development of a European emergency management
Collaboration(s)	• OASIS is limited to the response phase and will thus at a later stage be linked with other EU ICT projects deal with other phases of the disaster and emergency cycle • INSPIRE and GMES European initiatives • FP6 ORCHESTRA project • FP6 WIN project • FP6 PREVIEW project • Ongoing related projects developed under ESA contracts
Follow-up	• The next versions of the TSO have been discussed and set up in the CEN Workshop (the 'Information System for Disaster and Emergency Management' workshop)
Limitations	• ORCHESTRA WIN and PREVIEW should all be integrated in one approach/concept to achieve data harmonisation

[481] ibid

OHB Very High Resolution Systems for Emergency Response and Security	
Phase(s)	2: Imagery for direct response 3: Imagery for impact assessment
Critical Mission Area	Emergency Management
Type	Industry project
Project Duration	• Starting date: TBC • implementation date: 2015
Current Status	• ESA investigates Architecture & Precursor Mission • Full System under EC/Agency/GMES governance – TBD
Country/Institution	• OHB
Funding	• ESA • EC
Partners	• OHB Lead • Partners for optical and SAR payloads
Objective(s)	• Focused only on highest resolution of 0,5m over swath width can be provided by cost efficient reflector SAR
In Practice	• No new technologies are required • Space Segment needs to be added to achieve autonomy • Ground Segment need to be upgraded for timeliness and high data volumes. • Data Processing chains and data distribution need to be improved for near-real-time operations • European Governance for Security/Emergency Response needs to be established • Consisting of two constellations: • 1. VHR-SAR constellation: Simple Reflector SAR provides: Highest Resolution & excellent SAR image quality for fast response day & night imaging: 0,5 m spatial resolution for 10km x 10km images Real-time commanding and data downlink to local stations and/or via GEO data relay satellite Secondary Security Payloads: IR Imager for detecting thermal signatures and AIS to track and monitor global ship traffic. Constellation currently under ESA study Satellite characteristics Mass ca. 800 kg Downlink 600-800 Mbps Direct Downlink to local user Image Characteristics: Pan: 0.5m / MS: 2m Swath 15 km Off-nadir pointing: ± 45 Degrees Image Mosaics: up to 60 x 100 km • 2. VHR-OPT constellation: Provide dedicated cost efficient 1m resolution with fast response time stand-alone or with Pleiades Mission Characteristics: 2 Satellite Constellation Altitude 614km, SSO Launcher: VEGA dual launch (ROCKOT) Timeliness (95% @ 40 deg lat.) System response time < 24 h Image Characteristics PAN: 1m / MS: 4m Swath 12km Off-nadir pointing: ±45 deg Image mosaics: up to 48 x 100 km Satellite characteristics Mass ca. 690 kg Downlink 320 Mbps
Demonstration(s)	• Field demonstrations with existing systems (i.a. Cosmo-Skymed, Helios, SAR-Lupe, …) • First European satellite as precursor mission may be from ESA
Users	• EC Agencies for Security/Emergency Response (EUSC, JRC, …) • National Organisations for Humanitarian Aid / Emergency Response
Added value/Findings	• Commercial (US) VHR data under shutter control • Military VHR data not usable outside classified environment • Dual use mission availablility unclear • No coordination / Limited exchange between Mil / Security / Civil systems • High resolution optical and SAR systems requires high cost systems • Today optical systems are preferred for easy interpretation • Only SAR systems can guarantee fast access to remote areas • Fast Response time requires constellations of satellites • Focused only on highest resolution of 0,5m over swath width can be provided by cost efficient reflector SAR • Autonomy / independent and fast access to VHR data for EC relevant tasks • Global revisit <6 h (with CS) - 12 h (stand-alone) • System response time <11-17 h
Collaboration(s)	• EU level: EUSC, JRC as users • National level: Cosmo-Skymed, Pleiades, TerraSAR, … Space and Ground Infrastructures. • ESA projects: GMES Sentinel-1 & 2 for medium resolution – overview maps, VHR precursor mission
Limitations	• n.a.

Project Name	FP 6 NEWSKY
Phase(s)	1: Prevention of interception of communication through introduction of redundant communication system
Critical Mission Area	Transportation Security / Air Transportation
Type	R&D
Project Duration	• 26/02/2007 – 26/08/2009, 30 months
Current Status	• In progress
Country/Institution	• FP 6
Funding	• Total Costs €3.600.000

	• EU Contribution €2.100.000
Partners	• German Aerospace Center (DLR) • Thales Alenia Space-France • QinetiQ Ltd-United Kingdom • University of Salzburg-Austria • Frequentis GmbH-Germany • TriaGnoSysy GmbH-Germany • Deutsche Flugsicherung GmbH-Germany
Objective(s)	• The aim of NEWSKY is not the development of new link technologies; rather, NEWSKY aims to develop an innovative networking concept to integrate different existing and emerging link technologies into a single, global network for a secure, seamless and robustly redundant communication system, which is scalable to cope with future long-term increasing demands. High benefits are expected from the increased information availability and information sharing possibilities that the NEWSKY network will enable.
In Practice	• The NEWSKY project aims at integrating the different systems for aeronautical communication – ground-based, satellite-based, airport systems, and systems based on ad-hoc networks between aircrafts – into a global heterogeneous communications network • Realising a communication system which is completely transparent to the users. • This global, heterogeneous network is intended to transport both safety related ATC/ATM and AOC information and non-safety related AAC and APC information.
Demonstration(s)	 Integrated IP Networking for Air-Ground Communications[482]
Users	• ATM service provider, airspace user, airport operator, passenger, and military user
Added Value/Findings / Findings	• The NEWSKY concept enables real air-ground integration and the information sharing concepts of System Wide Information Management (SWIM) and Collaborative Decision Making(CDM) could be made available to the aircraft. • Compared to a single communication link, an integrated network inherently achieves increased redundancy which in turn increases availability and reliability of the overall communications system. • The coverage of the overall system is increased since different communication systems with different application areas are combined. • NEWSKY integrates several communication links which are specially adapted to different environments, e.g. IEEE 802.16 (WiMAX) for airport communication and satellite-based communication for remote or oceanic areas. This enables globally optimised network performance. • The integrated NEWSKY approach ensures inter-operability between different communication links and, thus, provides a seamless aeronautical communication system which is fully transparent to the end users. • The different communication links within NEWSKY cover the whole available Aeronautical Mobile (Route) Services (AM(R)S) and Aeronautical Mobile Satellite (Route) Services (AMS(R)S) spectrum. As global integrated network NEWSKY realises the most flexible and efficient use of the available aeronautical spectrum allocations • A single integrated airborne avionics system covers the whole variety of communication links and can be used for different application classes (ATS, AOC, AAC, and APC). This is a business case enabler making the introduction of new communication technologies more attractive to airlines. • The NEWSKY network architecture is a modular concept which is easily reconfigurable and extendable. New communication link technologies can be easily included and phased-out technologies can be withdrawn. • The modular NEWSKY network architecture enables easier introduction of future communication links. Note that deployment of new communication links within an existing

[482] Schreckenbach, Frank. "The NEWSKY Project: Networking the Sky for Aeronautical Communications." Presentation. NEWSKY. 28 Aug 2009 <http://www.newsky-fp6.eu/NEWSKY_Presentation.pdf>.

	architecture is a crucial problem nowadays. • Cost savings, high reliability and an optimal alignment with the evolution of communication and security technologies by using Commercial-Off-The-Shelf (COTS) components (e.g. based on IPv6) wherever possible. Besides its benefits the NEWSKY approach also includes several challenges which have to be addressed and solved before implementation of the NEWSKY concept is possible. First of all, an adaptation layer as a common interface towards the different link technologies has to be designed to ensure that the transmission over different communication links is completely transparent to the end users. This entails the design of appropriate: • Routing algorithms, capable of coping with the fast changes in the network topology • system-level resource management strategies, to make sure that the "best" available link is selected at any point in time and that the available network resources are globally used in an optimised way, bearing in mind that different applications may have very different QoS requirements • Seamless handover techniques, to ensure that the aforementioned QoS requirements can be maintained during the handover from one communications link technology to another; this is a relevant technological challenge, since different link technologies might offer different data transmission capacities and different latencies.
Collaboration(s)	• ICAO, Eurocontrol, FAA, SESAR
Follow up	• n.a.
Limitations	• While aiming at integrating SATCOM in a secure seamless and robustly redundant communication system this is not primarily done in the context of improving the provision of internal security.

Project Name	**FP 7 OPERAMAR** (InterOPERAble Approach to the European Union MARitime Safety & Security Management)
Phase(s)	n.a.
Critical Mission Area	Transportation Security / Waterborne Transportations
Type	R&D
Project Duration	• 01/03/2008 – 31/05/2008, 15 months
Current Status	• Completed
Country/Institution	• FP 7
Funding	• EU contribution € 669132.00 (total estimated cost: € 669134.00)
Partners	• 9 coordinated by Thales Underwater Systems SAS, France
Objective(s)	• Meant to provide the foundations for pan-European Maritime Security Awareness by addressing the insufficient interoperability of European and national assets with a view to generating unified data models for seamless exchange and contributing to address the discrepancies of the behavioural, organisational and cultural issues. • fuse available data into a common and comprehensive picture to be shared as appropriate among the organisations of participating Member States→the achievement is hampered by the fragmentation affecting the Maritime Security domain in the EU, i.e.: • At national level various missions dealing with Maritime Security are the responsibility of a wide range of sectoral organisations • The EU still lacks a coordinating framework for Maritime Security which could maximise synergy with Maritime Safety initiatives Suggest to the EC recommendations in terms of future research programmes, projects and new standards: • Step 1: from the analysis of the present situation the stakeholder's network will identify the key interoperability challenges that will produce significant improvements in the operational performances. • Step 2: OPERAMAR will translate these interoperability requirements into guidelines for technical requirements, common architectures and systems specifications • The strategic roadmap developed by OPERAMAR will describe the evolution of an interoperable approach to the EU maritime security management rorm the multiple perspective or organisations, institutions, legislation and regulations
In Practice	• Define interoperability models • Develop common interoperability requirement and translate these into technical requirements • Study the consequences of such requirements and recommend a relevant strategic research roadmap
Demonstration(s)	• The effectiveness of the methodological results will be tested in three scenarios: Mediterranean, Black Sea and Atlantic Ocean (Canary Island)
Users	• n.a.
Added Value/Findings	• Is countering the current fragmented approach to maritime security • It will identify priority areas for additional security research to facilitate development at Regional and European levels. The roadmap will contribute to future FP7 and other European security linked activities taking into account the work of the ESRIF
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP6 ORCHESTRA** (Open Architecture and Spatial Data Infrastructure for Risk Management)
Phase(s)	n.a. (establishment of standards for data sharing)
Critical Mission Area	Intelligence, Early Warning, Risk Management, Emergency Management
Type	R&D
Project Duration	• 01/09/2004 – 29/02/2008, 42 months
Current Status	• Completed
County/Institution	• FP 7
Funding	• Total costs: € 13,75 million • EU contribution: € 8,20 million
Partners	• 14, coordinated by ATOS
Objective(s)	• ORCHESTRA is designing and implementing the specifications for a service oriented spatial data infrastructure for improved interoperability among risk management authorities in Europe
In Practice	• To deliver technical results such as the Reference Model-ORCHESTRA Architecture (RM-OA) plus developed services and applications • To bring together and consolidate the risk management community
Demonstration(s)	• Development and implementation of applications for four specific scenarios: flash floods and forest fires in North East Spain; the effects of earthquakes / floods on road traffic on the French / Italian border; the impact of ship traffic on pollution in the German Bight and forest fires and floods on a pan-European level. The ORCHESTRA architecture is also being applied for the tsunami early warning system in the Indian Ocean
Users	• System-users: providers of data and application services that are used for risk management • End-users: civil protection agencies and private companies involved in risk management
Added Value/Findings	• Intelligent data handling of both spatial and non-spatial data assuring the interchange of information among different actors at different levels (from local to national to global) • On-demand access for end users in the office, the field, an airplane or in a cavern • ORCHESTRA will integrate the results and recommendations of previous and current European and national projects and initiatives
Collaboration(s)	• Input for INSPIRE and GMES • Collaboration with nearly twenty on-going projects and initiatives, incl. projects not funded by the EC and projects outside Europe
Follow-up	• Future plans include extending the architecture and risk technology to other areas, developing field applications to deal with events such as earthquakes and man-made risks
Limitations	• All projects on data harmonisation should eventually be brought together in order to provide an approach to data sharing in all phases and critical mission areas

Satellite name	**ORFEO** (Optical and Radar Federated Earth Observation)
Phase(s)	1: High resolution imagery for prevention and detection
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Envisaged satellites (dual system consisting of PLEIADES and COSMO-SkyMed)
Date of Launch	• COSMO-SkyMed: 2007-2009 • Pleiades: envisaged for 2010-2011
Current Satellite Status	• Envisaged / partially operational
Country/Institution	• ASI, Italian MoD, CNES / Italy, France
Capability	• Pleiades: 0,7 m (panchromatic) • COSMO-SkyMed: < 1 m (spotlight mode for an area of 10 km2)
User	• Military (French and Italian armed forces) and civil. According to their contribution users Austria, Belgium, Spain and Sweden will have access to Pleiades Imagery
Projected lifetime	• 5 years
Objective(s)	• The Optical and Radar Federated Earth Observation (ORFEO) programme aims at establishing an Earth observation system with optical and radar capabilities
Service (in practice)	• The Earth observation capabilities will be provided by the French optical Pleiades and the Italian radar COSMO-SkyMed satellites. The agreement enshrines the dual-use character of the programme and the definition of principles for adequate resource sharing, imagery ownership and diffusion. A steering group composed of French and Italian representatives is managing the agreement. Defence institutions will have direct access to the satellite data on dedicated sites, whereas non-defence users will have to go through a civilian operator to have their request processed and distributed. The ground facilities consist of several user centres located in France, Sweden, Spain, and Italy. These centres have the capacity to undertake task orders including image acquisition as well as product generation and dissemination. Furthermore, the dual-use centres ground facilities will perform command and control activities considering the priorities given by the defence users. France and Italy each carry the costs for their respective system • Concentrate on three-dimensional aspects, detection of changes, texture analysis, shape recognition, and the complementary nature of optical and radar images. • Analysis of HR X band radar data • From pixel to object: models and analysis of static or dynamic images • From object to scene: models and analysis of static or dynamic images – indexing

	• 3D models in an urban environment using optical images: modelling, representation and visualisation • Study of movements: deformation, displacements, evolution (subsidence, glaciers, etc.) • HR radar imaging in urban areas: radargrammetry, interferometry and coupling with optical imaging Services concerned with: • Sea and Shore • Risks and Humanitarian Aid • Cartography and Land Management • Forests • Agriculture
Cooperation/data exchange	• Pleiades: besides the ORFEO programme Austria (1%), Belgium (up to 4%), Spain (up to 3%), and Sweden (up to 3%) also contribute financially to Pleiades • COSMO-SkyMed: duplication of ground segments for civilian and military users (architectural symmetry) • Designation of different classes of civilian and defence users each characterised by a specific priority • Three modes with different response times: Routine Mode (72 h), Crisis Mode (36 h), Very-Urgent Mode (18 h) • Cooperation with Argentinean SAOCOM L-band SAR system for civil purposes (SIAGE) • Shall contribute to GMES • Imagery data is used by the European Union Satellite Centre

Satellite family name	**Pleiades-HR (2 satellites)**
Phase(s)	1: Emergency Management
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Envisaged satellite
Date of launch	• Envisaged in 2010 (1st satellite) and 2011 (2nd satellite)
Current Satellite Status	• Development phase
Country/Institution	• CNES / France
Capability	• Optical, visible and near infrared • Panchromatic and multispectral • Maximum resolution: 0.7 m (panchromatic)
User	• Military (French and Italian armed forces) and civil and, according to their contribution, users Austria, Belgium, Spain, and Sweden will also have access to Pleiades imagery
Projected lifetime	• 5 years
Objective(s)	• Optical imagery
Service (in practice)	• Intelligence • Geospatial Intelligence • Humanitarian aid • Cartography • Geology and geophysics • Hydrology, agriculture and forestry
Cooperation/data exchange	• Optical component of the dual-use ORFEO programme (with Italian SAR COSMO-SkyMed satellites) • Besides the ORFEO programme Austria (1%), Belgium (up to 4%), Spain (up to 3%), and Sweden (up to 3%) also contribute financially to Pleiades • Shall contribute to GMES • Defence requirements are given priority • Might form part of the envisaged OHB Very High Resolution Systems for Emergency Response and Security

Project Name	**UK's POB** (Person OverBoard)
Phase(s)	2: Immediate response
Critical Mission Area	Transportation Security/Waterborne Transportation
Type	Implemented service
Project Duration	• n.a.
Current Status	• Implemented service
Country/Institution	• United Kingdom
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Improving search and rescue at Sea
In Practice	• A system monitoring the position of people over board in real-time through GNSS
Demonstration(s)	• n.a.
Users	• Stakeholders in maritime sector, search and rescue authorities

Added Value/Findings	• Improved search and rescue at sea
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**Belgium's PolInSAR** (Polarimetric SAR Interferometry)
Phase(s)	n.a.
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	R&D
Project Duration	• 15/05/2004 – 31/03/2007, 34 months
Current Status	• Completed
Country/Institution	• Belgium
Funding	• Funded in the framework of the Belgian STEREO 1 (Support to the Exploitation and Research of Earth Observation data 1) programme of Belspo (Federal Science Policy Office)
Partners	• CSL (Space Centre Liège) • ERM/KMS (Signal and Image Centre) • UCL (Telecommunications and Teledetection Laboratory)
Objective(s)	• To develop know-how in Polarimetric SAR Interferometry (PolInSAR); polarimetric techniques are incorporated in SAR interferometry applications to provide a combined sensitivity to the vertical distribution of scattering mechanisms. This makes it possible to investigate the three dimensional structure of vegetation cover using a polarimetric SAR
In Practice	• A new processor named PolInSAR was developed, implemented and validated at CSL, taking the ESA's POLSARPRO processor as reference (and often helping to improve it). An improved version of the UCL polarimetric radiative transfer model, POLSCAT, and its corresponding code, was developed in order to integrate the interferometric coherence information
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• The project was able to demonstrate that the merged characteristics of different SAR frequencies are complementary and appropriate for land use classification; the PolInSAR characteristics are complementary to the PolSAR information and essential for producing accurate classification of different land use types such as man-made constructions, water bodies, forest, crops and bare soil; the optical data provide complementary information for the SAR data, but which is not necessary for producing a precise land use classification
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Benchmarking, advancing the field of provision of internal security only as a side effect

Project Name	**FP 5 PRESENSE** (Pipeline Remote Sensing for Safety and the Environment)
Phase(s)	1: Detection
Critical Mission Area	Critical Infrastructures
Type	R&D
Project Duration	• 01/01/2002 – 30/09/2004, 33 months
Current Status	• Completed
Country/Institution	• FP 5
Funding	• Total cost: € 4 million • EU contribution: n.a.
Partners	• 17, coordinated by Advantica
Objective(s)	• To develop and integrate the elements of a pipeline management system to improve safety, reduce survey costs and improve efficiency
In Practice	• Assessment of operators requirements • Review of available sensing technologies • Selection of most promising solutions • Preparation of tests sites to collect remote sensing data with documented ground activities • Flight campaigns to assess new sensors technologies and combinations of sensor technologies • Automated change detection, feature recognition, classification and extraction • Development of a concept computing system architecture to manage the processing chain, provide the communications infrastructure, and deliver prioritised hazard warnings to the pipeline operator's desktop • Detection of third party interference e.g. earth moving vehicles, pipe-laying • Detection of land movement e.g. ground subsidence, land slip • Detection of gas leakage through vegetation stress
Demonstration(s)	• A new approach to the management of pipeline monitoring, based upon remote satellite surveillance, and providing pipeline operators with a computer based information system capable of alerting them to potential pipeline damage under all weather and lighting conditions has been demonstrated
Users	• Gas industry
Added Value/Findings	• Developments in sensing technology and use of satellites • Increasing probability of detection of 3rd party interference

	• Increasing standards for pipeline surveillance • Environmental benefits • Cost savings
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 6 PREVIEW** (PREVention, Information and Early Warning)
Phase(s)	1: Prevention, anticipation, early warning, risk maps 2: Crisis management, asset mapping, rapid mapping, damage assessment 3: Post-crisis and discovery
Critical Mission Area	Ingelligence, Early Warning and Risk Management, Emergency Management
Type	R&D
Project Duration	• 01/04/2005 – 31/12/2008, 45 months
Current Status	• Completed
Country/Institution	• FP 6
Funding	• Total costs: € 22.9 million • EU contribution: € 14.33 million
Partners	• 58 partners, coordinated by Infoterra France
Objective(s)	• To develop new geo-information services for atmospheric, geophysical and man-made risk management on a European level in order to better prevent, anticipate and manage different types of disasters
In Practice	• To develop thematic services to address floods, forest fires, windstorms, earthquakes and volcanoes, landslides, man-made risks (dispersion of airbone hazardous substances) • Additional transverse services are developed for asset mapping, rapid mapping, and damage intensity assessment
Demonstration(s)	• Civil protection in the Euro-Mediterranean region: MED2007 exercise • Tsunami simulation model
Users	• European civil protection units at all levels
Added Value/Findings	• The services benefit from the most recent research methodologies and scientific models, and are developed through extensive integration of space data, meteorological data, in-situ measurements, other data sources • All crisis phases covered • User-driven approach • On-site test campaigns, which improve integration of user feedbacks
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• It mentions man-made disasters but it seems to be focussed on floods and natural disasters like tsunamis

Satellite name	**Italy's PRISMA** (PRecursore IperSpettrale of the application mission / Hyperspectral Precursor of the Application Mission**)**
Phase(s)	n.a.
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Technology demonstrator
Date of launch	• Foreseen by the end of 2010
Current Satellite Status	• Envisaged satellite
Country/Institution	• ASI / Italy
Capability	• Daily imaging capability Satellite • Hyperspectral (10 m resolution) / Panchromatic Payload (2,5 m resolution) • Data Handling Payload and Transmission (PDHT) • Chemical-physical composition recognition
User	• Civil, ASI, scientific community
Projected lifetime	• 5 years
Objective(s)	• Monitoring Earth's electro-magnetic environment (low frequency electronic signals)
Service (in practice)	• Mapping of land cover and agricultural landscapes • Signal Intelligence • Pollution monitoring • Quality of inland waters • Coastal zones and Mediterranean Sea • Soil moisture • Carbon cycle monitoring
Cooperation/data exchange	• PRISMA takes advantage of previous investments made by ASI in the field of "small missions" (for example, AGILE), hyperspectral payloads (for example, Hypseo, JHM), satellite platforms (MITA/PRIMA) and receiving and remote sensing data processing centres (COSMO-SkyMed and CNM) that have given the opportunity to national industries to acquire competences and develop/consolidate vitally important products

Project Name	**FP 4 ProATN TR 1042** (Prototype Aeronautical Telecommunication Network)
Phase(s)	1: Air Traffic and Transport Management
Critical Mission Area	Transportation Security / Air Transport
Type	R&D
Project Duration	• 01/02/96 – 31/12/99, 46 months
Current Status	• Completed
Country/Institution	• Sofreavia, France
Funding	• n.a.
Partners	• Aerospatiale (FR) • Airsys ATM (FR) • DFS (DFS) • NATS (UK) • SITA (FR)
Objective(s)	• To deploy a pre-industrial, pre-operational European prototype of the Aeronautical Telecommunication Network (ATN) of which the key elements are certifiable on an end-to-end basis • To make possible the exchange all types of data securely (control order, weather messages, flight parameters, position reports, passenger communications, airlines applications, etc.) according to standardised message formats and communication protocols using satellite and terrestrial communication systems • To demonstrate the cost-effectiveness of ATN technology when employed in a pre-operational aeronautical environment
In Practice	• The ProATN prototype had to conform to the ICAO SARPs and Guidance Material for the ATN. To validate the ATN and the ICAO CNS/ATM concept, it was deployed in a real-time pre-operational aeronautical environment. • Both ground and airborne systems were developed according to industrial rules. In particular, the airborne software has been installed in a commercial avionics computer, the Aerospatiale Air Traffic Services Unit (ATSU). • Each of seven ground sites have been equipped with one ground router and one End System, connected via an Ethernet/LAN, and developed on Unix Work Stations. All of them are connected via various public and private national and international X.25 networks. • ProTAN makes use of mobile terrestrial communication applications as well as satellite communication applications and has therefore two airborne platforms equipped with one ATSU computer, one SATCOM Data Unit and one VHF Data Radio. • Two air / ground sub-networks are available in ProATN infrastructure, the satellite (AMSS Data 3) having been available earlier.
Demonstration(s)	• Validation of the infrastructure was the key activity for 1999, including the "Le Bourget" demonstration. Validation of the network management on an end-to-end basis followed during the first quarter of 2000. End-to-end testing of VDL Mode 2 was scheduled for the summer of 2000 • The VHF (VDL Mode 2) sub-network is deployed in Paris, Toulouse and Amsterdam
Users	• International aeronautical community
Added Value/Findings	• The validation of this infrastructure provides a unique opportunity for the validation of the ATN concept as a whole. • The availability of a representative ATN infrastructure is a key advantage for Europe.
Collaboration(s)	• n.a.
Follow-up	• The results will be now be re-used in other projects such as the EUROCONTROL PETAL II trial and in the programme of the Fifth R&D Framework. The need to maintain the ProATN in the future has already been expressed in the scope of other CNS/ATM projects.
Limitations	• Mainly about Air Traffic Management, no direct link with internal security

Project Name	**Austrian QUANTUM** (Quantum Entanglement in Space Experiments)
Phase(s)	1: Prevention through secured communication
Critical Mission Area	Critical Infrastructure
Type	R&D
Project Duration	• 01/03/2006 – 30/06/2007, 15 months
Current Status	• Completed
Country/Institution	• Austria
Funding	• Funded in the framework of the Austrian Space Programme
Partners	• Institut for Quantum Optics and Quantum Information of the Austrian Scientific Academy
Objective(s)	• To place a source of entangled photons on a LEO (low earth orbit) platform such as the international Space Station (ISS)
In Practice	• n.a.
Demonstration(s)	• n.a.
Users	• Satellite designers, operators and owners
Added Value/Findings	• The objective could not be fulfilled. Discussions continue to take place at ESA
Collaboration(s)	• n.a.
Follow up	• Space-Quest is following up on this
Limitations	• n.a.

Satellite family name	Germany's RapidEye 1 to 5
Phase(s)	2: Response 3: Relief
Critical Mission Area	Emergency Management
Type	Operational Satellites (Satellite family)
Date of Launch	• 29 August 2008
Current Satellite Status	• Operational
Country/Institution	• RapidEye AG, Brandenburg Germany
Capability	• Optical imagery • Multispectral
User	• Consumers of RapidEye AG
Projected lifetime	• 7 years
Objective(s)	• To provide geospatial information: • Emergency and security • Environment • Energy & infrastructure • Spatial solutions • Forestry • Agriculture
Service (in practice)	• RapidEye AG operates a commercial satellite constellation to cover a broad range of derived products
Cooperation/data exchange	• Potential contribution to GMES

Project Name	ESA Funded GMES Service Element RESPOND
Phase(s)	2: Response 3: Relief / Impact Assessment
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• ESA
Funding	• ESA funded GMES Service Element
Partners	• Partners: 20 partners led by Infoterra
Objective(s)	• Objectives: five staged programme • Guaranteed access to global mapping for crisis hotspots • Access to an archive fo detailed base mapping and thematic mapping • Access to rapid assessment maps for major crisis • Aims to increase the efficiency and effectiveness of the European and international humanitarian community through appropriate and reliable application of geographic information to address all parts of the crisis cycle where geographic information is useful and to provide service in support of crisis preparedness and response operations (mapping, alerting, early warning, field operations
In Practice	• In practice: • Basic mapping (digital, paper, EU, non- EO) • Crisis and damage mapping • Situation mapping (Map and GIS) • Refugee / IDP support mapping • Thematic maps for prevention / reconstruction (planning purpose) • Health maps • Environmental impact / assessment maps • Alert services • Communication / reporting resources (ad-hoc mapping) • Infield mapping (ad-hoc mapping)
Demonstration(s)	• n.a.
Users	• Emergency management community
Added Value/Findings	• Rapid assessment map in major crisis
Collaboration(s)	• ESA funded GMES Service Element
Follow-up	• Also active in the development of the OCHA/ECHO GDACS through UNOSAT (which is the United Nations Institute for Training and Research (UNITAR) Operational Satellite Applications Programme, implemented in co-operation with the European Organization for Nuclear Research (CERN). The programme, created in 2000 provides satellite solutions to relief and development organisations and comprises UN fieldworkers, satellite imagery experts, geographers, geologists, development experts, database programmers and internet communication specialists. Their stated mission is "to deliver integrated satellite-based solutions for human security, peace and socio-economic development, in keeping with the mandate given to UNITAR by the UN General Assembly since 1963".), the JRC and Reuters AlertNet
Limitations	• Unclear whether terrorist incidents fall within RESPOND''s scope

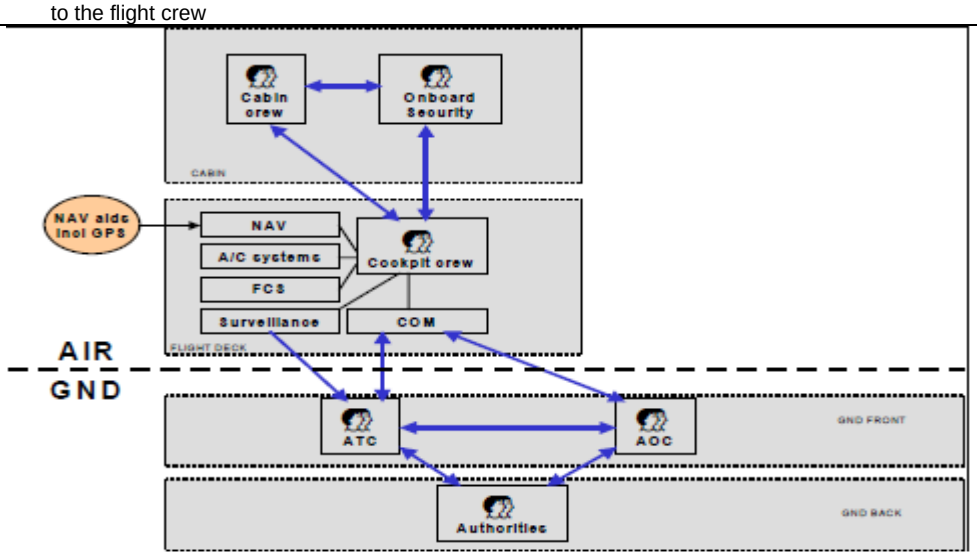
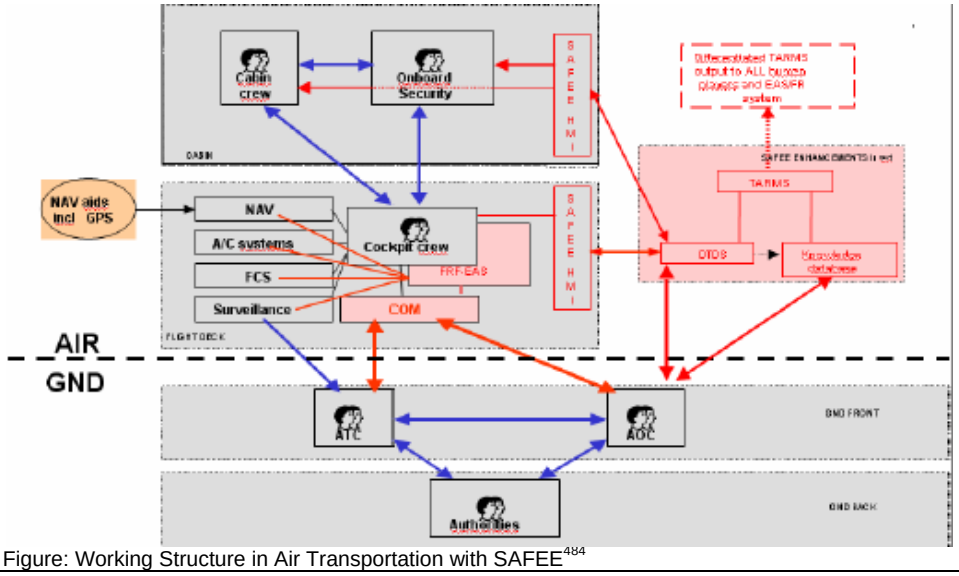
Project Name	**RIS** (River Information System)
Phase(s)	1: Traffic management, use of satellite navigation for information on ship movement 2: Improved response through improved situational awareness - Harmonisation of RIS on inland waterways in the community
Critical Mission Area	Transportation Security / Waterborne Transportation
Type	Implemented Initiative
Project Duration	• On-going
Project Status	• On-going
Country/Institution	• European Union
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Transforming inland waterway transport into a transparent, reliable, flexible and easy to access transport mode
In Practice	• Provides traffic information and traffic management services through AIS and satellite navigation for high accuracy of data on ship movements allowing for improved tracking
Demonstration(s)	• n.a.
Users	• Inland water way transportation sector
Added Value/Findings	• Harmonisation of RIS on inland waterways in the Community
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Limited to inland waterways. • Concerned with traffic management rather than security.

Project Name	**ESA's RISK-EOS** (GMES Service Element)
Phase(s)	1: Intelligence, Early Warning and Risk Management 2: Emergency Response, rapid mapping
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• Since 2003
Project Status	• In progress
Country/Institution	• ESA
Funding	• ESA funded GMES service element
Partners	• Infoterra France • ADISA – Associação para o Desenvolvimento do Instituto Superior de Agronomia • ARPA Piemonte • Critical Software • CRP-GL • DLR • DDSC • DDSC • GEOMER • GEOMER • Infoterra GmbH Germany • INOV-Inesc Inovação • Ingeniería y Servicios Aeroespaciales, S.A., INSA • LATUV • Luxspace S.à.r.l • METEO France International • METRIA • NOA/ISARS • SERTIT • TELESPAZIO
Objective(s)	• Rapid mapping; specifically designed for floods and fires
In Practice	• Is a network of European service providers delivering geo-information services to support the managment of flood, fire and other risk throughout all phases: Prevention, Early Warning, Crisis, Post-Crisis • Combines use of satellite EO data with exogenous data and modelling techniques User driven approach Services: • Assets mapping • Flood risk analysis • Flash flood early warning • Rapid mapping • Burn scars mapping • Automatic burn scars mapping
Demonstration(s)	• n.a.
Users	• Risk management actors at European, National and Regional levels, • Civil Protection, Early Warning Operators • Catchment authorities • Land planning authorities, Territorial authorities • Environment agencies, Forestry agencies • Local administrations like Municipalities
Added Value/Findings	• n.a.
Collaboration(s)	• RESPOND
Follow-up	• n.a.
Limitations	• Limited to floods and fires

Project Name	Germany's Risk and Vulnerability Assessment of Urban Areas
Phase(s)	1: Risk and vulnerability assessment
Critical Mission Area	Critical Infrastructures
Type	R&D
Project Duration	• Research and applications field of DLR's Applied Remote Sensing Cluster
Current Status	• On-going
Country/Institution	• Germany / DLR
Funding	• Germany / DLR
Partners	• n.a.
Objective(s)	• Risk and vulnerability assessment of urban areas
In Practice	• Enables spatial analysis of the pattern of vulnerability of the potentially affected system. The built-up density distribution reflects the quantity of potentially affected structures. The identification of open spaces, their location and dimension provides information on potentially safe and sheltered areas. In addition the main street network extracted from the land-cover classification supports the assessment of accessibility. • Orographic information or distance to expected sources of natural hazards allows a first assessment of a spatial pattern of exposure. The utilization of a digital elevation model enables the integration of orographic surface information when assessing high risk zones like flood- or tsunami-prone areas based on height information, or landslide prone slopes derived from the terrain's steepness.
Demonstration(s)	• n.a.
Users	• Decision makers, emergency planners, risk assessment
Added Value/Findings	• Improved risk management
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Limited to urban areas

Project Name	RUNE
Phase(s)	n.a.
Critical Mission Area	Transportation Security / Land Transportation
Type	R&D
Project Duration	• n.a.
Current Status	• Completed
Country/Institution	• n.a.
Funding	• n.a.
Partners	• n.a.
Objective(s)	• An integrated solution to improve the train driver's situational awareness
In Practice	• Integrating EGNOS/GPS with other on-board positioning sensors, and signalling and speed restriction information from a central control centre significantly improving safety as a result of improved situational awareness
Demonstration(s)	• n.a.
Users	• European railway companies
Added Value/Findings	• n.a.
Collaboration(s)	• EGNOS
Follow-up	• n.a.
Limitations	• Railway only

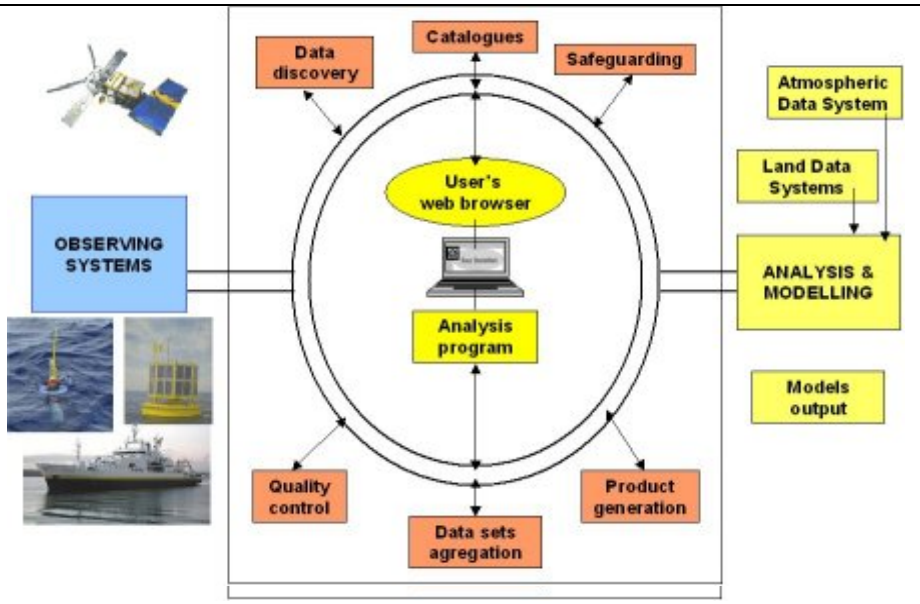
Project Name	**SAFEE** (Security of Aircraft in the Future European Environment)	
Phase(s)	1: Prevention, secured communication	
Critical Mission Area	Transportation Security / Air Transportation	
Type	R&D	
Project Duration	• 1 Feb. 2004 – 30 April 2008, 51 months	
Project Status	• Completed	
Country/Institution	• n.a.	
Funding	• European Community	
Partners	• SAGEM • AIRBUS FR • AIRBUS DE • BAE SYTEMS • THALES Avionics • NLR • Gs-3 • SELEX • SITA • EADS	• ONERA • SKYSOFT • Siemens • RCF France • SODIELEC • Cenciarini • IEEA • Environics • Miriad • ECORYS

	• EADS • ISDEFE • Galileo • BAM • HAI • AIRTEL • TUM • University of Reading • CEDITI • ENERTEC • SIA
Objective(s)	• To construct advanced aircraft security systems to prevent onboard threats with the implementation of onboard threat detection systems and the provision of reliable threat information to the flight crew
In Practice	 Figure: Working Structure in Air Transportation without SAFEE[483]  Figure: Working Structure in Air Transportation with SAFEE[484]
Demonstration(s)	• n.a.
Users	• Airlines
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Airport / airplane security

Project Name	**FP 6 SeaDataNet**
Phase(s)	n.a. (integrated approach)
Critical Mission Area	Transportation Security / Waterborne Transportation, Border Security
Type	Integrated Infrastructure
Project Duration	• n.a.

[483] Gaultier, Daniel. "SAFEE (Security of Aircraft in the Future European Environment) Final Report." SAFEE, 2008. 9.
[484] Ibid. 10.

Project Status	• In progress
Country/Institution	• FP 6
Funding	• FP 6
Partners	• IFREMER • MARIS BV • NERC • B. SEESCHIFFARHT UND HYDROGRAPHIE • SWEDISH METEOROLOGICAL & HYDROLOGICAL I. • Instituto Español de Oceanografía • HCMR • IN-OGS • RIHMI • UNESCO/IOC • ENEA • INGV • Middle East Technical University • Collecte Localisation Satellites (CLS) • ALFRED WEGENER INSTITUT (AWI) • UNIVERSITE DE LIEGE • RESEARCH COUNCIL FOR OCEANOGRAPHY • National Environmental Research Institute (NERI) • CIEM/ICES • EUROPEAN COMMISSION • MARINE INSTITUTE • INSTITUTO HIDROGRAFICO DE PORTUGAL • NODC • Management Unit of the North Sea Mathematical Model • Flanders Marine Institute (VLIZ) • MARINE RESEARCH INSTITUTE • FINNISH INSTITUT OF MARINE RESEARCH • INST.OF METEOROLOGY & WATER ECONOMY • Tallinn Technical University • Latvian Institute of Aquatic Ecology • Center of Marine Research • Shirshov Institute of Oceanology • National Academy of Science of Ukraine • Institute of Oceanology, Bulgarian Academy of Sciences • National Ins. Marine Research & Development "Grigore Antipa" • TBILISI STATE UNIVERSITY • INST. NATIONAL DE RECHERCHE HALIEUTIQUE • INSTITUTE OF OCEANOGRAPHY AND FISHERIES • Polytechnic University of Tirana • NATIONAL INSTITUTE OF BIOLOGY • University of Malta • University of Cyprus • ISRAEL OCEANOGRAPHIC & LIMNOLOGICAL RES. • National Center for Marine Sciences (NCSR/NCMS) • DEPARTMENT OF PHYSICAL OCEANOGRAPHY • CONSIGLIO NAZIONALE DELLE RICERCHE • ISMAL • ENVIRONNEMENT, AMÉNAGEMENT ET ECOSYSTÈMES LITTORAUX • INSTITUT NATIONAL DES SCIENCES ET TECHNOLOGIES DE LA MER
Objective(s)	• Construct a standardized system for managing the large and diverse data sets collected by the oceanographic fleets and the new automatic observation systems • Networking and enhancing current infrastructures (national oceanographic data centres and satellite data centres of 35 countries)

In Practice	 Figure: Functionality of the SeaDataNet Infrastructure[485] • In the countries bordering the European seas of the partnership, more than 600 scientific data collecting laboratories from governmental organizations and private industry
Demonstration(s)	• n.a.
Users	• Maritime community
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 7 SAFER** (Services and Applications For Emergency Response)
Phase(s)	1: Maritime Surveillance 2: Improved Search and Rescue 3: Rapid Mapping
Critical Mission Area	Transportation Security / Waterborne Transportation, Emergency Management
Type	R&D
Project Duration	• 30 January 2009 – 30 January 2012
Project Status	• In progress
Country/Institution	• FP 7
Funding	• GMES Emergency Response Core Service. • 40 Million (27 Million is funded by the FP 7)
Partners	• 54 Partners (coordinated by Infoterra, France)
Objective(s)	• To produce over 500 background maps • To support GMES and its preparation for its operational implementation of the fast track service on emergency response. Every year in the world, fires, floods, earthquakes and volcanic eruptions, landslides and other humanitarian crises claim the lives of thousands of citizens • Rapid mapping capacity in response to disastrous events for the improvement of response time.
In Practice	• SAFER will deliver a comprehensive set of geoinformation products, based on satellite images Two main products: • Rapid damage assessment (phases 2 and 3) and reference maps (1), which SAFER prepares in advance for the most exposed areas: mountain valleys, dry lands or Mediterranean forests, neighbourhood of a volcano, dense urban coastal zones. • This is the largest initiative launched at European level in the framework of GMES services. SAFER ERCS: three core services • Service on emergency response (ERCS): rapid mapping • Service on land monitoring (LMCS) • Service on marine (MCS)
Demonstration(s)	• SPOT International Conference June 2009, Toulouse, France • 33rd International Symposium on Remote Sensing of Environment, May 4-8 2009. Stresa, Italy.

[485] "SeaDataNet Context and Objectives" SeaDataNet 30 July 2009 <http://www.seadatanet.org/overview/context_and_objectives>.

	• European Geoscience Assembly 2009, Vienna, Austria, April 19-24 2009 • Integration Environmental Knowledge in Europe, March 25-27, 2009
Users	• Emergency responce services • European civil protection authorities • International UN agencies
Added Value/Findings	• Rapid damage assessment and reference maps
Collaboration(s)	• RISK-EOS • TERRAFIRMA • RESPOND • PREVIEW • BOSS$GMES • LINKER
Follow-up	• n.a.
Limitations	• Catastrophes and natural disasters

Project Name	**EMSA's SafeSeaNet (SSN)**
Phase(s)	n.a.
Critical Mission Area	Transportation Security / Waterborne Transportation
Type	Implemented service
Project Duration	• On-going
Project Status	• Implemented
Country/Institution	• EMSA
Funding	• EMSA
Partners	• 20 EU coastal states
Objective(s)	• The development of a comprehensive EU vessel traffic monitoring and information system
In Practice	• Implementation of a central hub for all communication between data requesters and data providers • Development of a Community vessel traffic monitoring and information system according to Directive 2002/59/EC
Demonstration(s)	• n.a.
Users	• Coastal security authorities
Added Value/Findings	• A Vessel Traffic Monitoring system covering all member states of the European Union as well as Iceland and Norway
Collaboration(s)	• EMSA will cooperate closely with national administrations on several issues, such as the implementation of plans for places of refuge and the development of means for long range identification and tracking of ships. It will monitor the set-up of shore-based traffic monitoring and information infrastructure by member states and will facilitate cooperation relating to interconnectivity (until the end of 2008)
Follow-up	• EMSA will additionally examine the possibility of integrating other applications and functionalities into SafeSeaNet (e.g. waste notification messages
Limitations	• Does not include satellite data; will examine the possibility

Project Name	**Italy's Safe Transport of Dangerous Goods**
Phase(s)	1: Detection, prevention through deterrence? 2: Alert through monitoring/ in case of changes of the route 3: Support of management of emergencies
Critical Mission Area	Transportation Security / Land Transportation / Waterborne Transportation
Type	R&D
Project Duration	• 2000 - ...
Current Status	• a.a.
Country/Institution	• Italy/ASI
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Promoting experimental satellite navigation applications and services for the transport of dangerous goods also including intermodal transport and management of transport emergencies.
In Practice	• Transport planning • Monitoring: Control of vehicle positioning (also along tunnels) • Remote control of vehicle conditions and their loads • Detection of driving behaviour (in order to signal to the driver and to the traffic control station any inadequate driving behaviour due to negligence, carelessness, tiredness or incapacity) and adoption of relevant corrective measures • Control of the speed of the vehicles to be adjusted in compliance with road network structures, traffic and weather conditions and spacing between incompatible transports • Real time remote monitoring of dangerous commodity critical parameters • Automatic alarms in case of change of the route • Continuous data link between vehicles and control stations

	• Voice communication between vehicles and Service centres • Coverage of the Service in the area concerned (local, regional...) Interface with Public Authorities (Emergency Planning Department, Police Forces, Firefighters, Hospitals) • Automatic Identification of any Operators.
Demonstration(s)	• includes a detailed definition of applications and technological products and the setting of operational validation agreed upon with relevant institutions and users
Users	• Stakeholders in transport of dangerous goods
Added Value/Findings	• Tackling entire supply chain: focuses on the transportation of dangerous goods along roads and inland waters, also tackling the intermodal connections with railways stations and ports • Integration of one system for land and inland waterborne transportation
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**Italy's Safety in Sea Traffic**
Phase(s)	1: Efficient implementation of sea roads, reducing need for interventions at sea for rescuing pleasure boaters
Critical Mission Area	Transportation Security / Waterborne Transportation
Type	R&D
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• Italy/ASI
Funding	• n.a.
Partners	• ASI
Objective(s)	• Promoting, in cooperation with corresponding Authorities, the development and testing of pre-operative applications and services, based on the employment of satellite navigation infrastructures in sea traffic, to contribute to a greater efficiency of safety-at-sea applications, such as search and rescue, management of navigation and sea traffic support services • Reducing the need of interventions at sea for rescuing pleasure boaters through preventive actions, targeting as far as possible the search operations, supporting the SAR activity by means of satellite navigation instruments provided by GALILEO • Aimed at developing prototype applications and services for a greater safety in pleasure boating, thus supporting the activity of Harbour Offices in search and rescue (SAR) operations • Aimed at developing and testing satellite navigation services and systems, capable of providing solutions for the prevention of risks and accidents and guaranteeing ready and adequate rescue in case of emergency
In Practice	• The completion of the research for innovative technologies and the design and development of prototype apparatuses and systems, as well as technical and operative validations and public demonstrations, on a reduced geographic scale, • Develops and tests innovative advanced satellite navigation technologies supporting efficient implementation of Sea Roads • Reduces the need for interventions at sea for rescuing pleasure boaters through preventive actions • Targeting as far as possible the search operations • Supporting the SAR activity by means of satellite navigation instruments provided by GALILEO and by the Project application product as tested • Addresses the approaching, manoeuvring and berthing of Sea Road vessels and designing the infrastructure required for those operations; does not neglect the requirements of other vessels operating in the Port thus guaranteeing the general efficiency of this transport modality
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• Contributes to a greater efficiency of safety-at-sea applications, such as search and rescue, management of navigation and sea traffic support services
Collaboration(s)	• GALILEO
Follow-up	• n.a.
Limitations	• Emphasis on safety rather than security

Satellite family name	**Germany's SAR-Lupe (5 satellites)**
Phase(s)	1: Intelligence 3: Emergency Management
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Operational Satellite Family
Date of Launch	• SAR-Lupe 1: 19 December 2006 • SAR-Lupe 2: 2 July 2007 • SAR-Lupe 3: 1 November 2007 • SAR-Lupe 4: 27 March 2008 • SAR-Lupe 5: 22 July 2008
Current Satellite	• Operational

Status	
Country/Institution	• Bundeswehr (German armed forces): User ground segment • OHB System AG: Satellite ground segment / Germany
Capability	• SAR-Lupe 1, 2, 3, 4 and 5: Synthetic Aperture Radar, X-band • Maximum resolution: 0,5 m
User	• German and French armed forces
Projected lifetime	• SAR-Lupe 1, 2, 3, 4 and 5: 10 years / operational
Objective(s)	• Imagery
Service (in practice)	• Signal Intelligence • Imagery for Emergency Management
Cooperation/data exchange	• Cooperation agreement between Germany and France on the exchange of SAR-Lupe and Helios 2 data was signed in 2002. The beginning of data exchange is envisaged for 2009. The request for satellite imagery is transmitted to the partner who integrates it into the planning process of its satellite system. According to the request, the data is directly provided to the partner for analysis and archiving. • Imagery data is used by the European Union Satellite Centre

Project Name	**Norway's SatHav**
Phase(s)	1: Marine Mapping, monitoring
Critical Mission Area	Transportation Security / Waterborne Transportation
Type	R&D
Project Duration	• 01/01/2003 – 31/12/2006, 48 months
Current Status	• Completed
Country/Institution	• Norway
Funding	• Norwegian Space Centre
Partners	• Principal Norwegian space actors
Objective(s)	• Develop the use of satellite data within ocean monitoring and management • Operationalization of satellite based products and services for the marine and meteorological community in Norway
In Practice	• The use of SAR data and information extracted from SAR data (wind, fronts, waves, current, ..) will be developed and operationalized
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• n.a.
Collaboration(s)	• SAR, Radasat-II
Follow up	• n.a.
Limitations	• Norway

Project Name	**FP 7 SECTRONIC** (Security System for Maritime Infrastructures, Ports and Coastal Zones)
Phase(s)	1: Detection 2: Improved situational awareness for response
Critical Mission Area	Critical Infrastructures
Type	R&D
Project Duration	• 01/02/2008 – 31/01/2011, 36 months
Current Status	• In progress
Country/Institution	• FP 7
Funding	• Total costs: € 7.08 million • EU contribution: € 4.5 million
Partners	• 11 coordinated by Marine & Remote Sensing Solutions LTD
Objective(s)	• To address observation and protection of critical maritime infrastructures: passenger and goods transport, energy supply and port infrastructures • To prevent them being damaged, destroyed or disrupted by deliberate acts of terrorism, natural disasters, negligence, accidents or computer hacking, criminal activity and malicious behaviour
In Practice	• The project aims to develop an integrated security system that: • Accurately observes, characterises and tracks any object of significance, 360 degrees around an infrastructure, 24h a day in all weather conditions • Communicates security information of significance to the infrastructure authorities and to selected authorised third parties of importance for the overall security situation in real time • Aggregates, reports and displays any security-related information of significance in an intuitively understandable way. Reliably raises alarms in identified situations • Enables response procedures and actions to be undertaken in situations that require effective use of protective measures • Demonstrates system effectiveness in real maritime infrastructures
Demonstration(s)	• n.a.
Users	• Major market player in each of the three infrastructures: passenger transport, energy production, energy transport, commercial ports and combined military/commercial ports
Added Value/Findings	• 24hrs surveillance makes detection and tracking during a crisis easier

Collaboration(s)	• Collaboration with LIMES remains unclear
Follow-up	• n.a.
Limitations	• n.a.

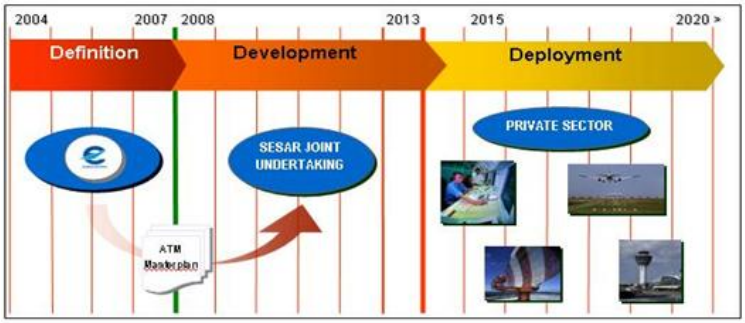
Project Name	**EADS Astrium's SeCureSystem**
Phase(s)	1: Detection 2: Alerting
Critical Mission Area	Transportation Security / Waterborne Transportation
Type	Industry project
Project Duration	• n.a.
Current Status	• Industry service
Country/Institution	• EADS Astrium
Funding	• EADS Astrium
Partners	• EADS Astrium (coordination), EUROGAT, Hellmann Worldwide Logistics, MSCGATE
Objective(s)	• To improve the security of container transport
In Practice	• Combining local RFID technology and global satellite communications through IRIDIUM and cost-effective GSM connections • Real-time data transfer via communications satellites enables world-wide availability of all data and documents on their entire transfer route. Route or cargo changes can be transmitted in real time to the destination ports and the container intended for import can pass through the Green Lane. Time and thus cost consuming customs procedures will be reduced and subsequent transports can be optimised • SeCureSystem creates electronic shipping documents, which provide information such as content details, arrival data and state of container integrity. Then the system transmits these data in real time to the control centre to allow targeted and selective forwarding of such data to the process partners. • In addition to basic technologies such as RFID and GPS, Smart Containers are of particular security relevance. Smart Containers are capable of sensing changes and are connected with the control centre through bidirectional communication pathways. • Ground-independent and worldwide position determination of containers by GPS satellites. • Permanent real-time data transfer via communications satellites enables world-wide availability of all data and documents. • Electronic creation and signing of loading and customs documents by authorised persons. • Goods-condition monitoring and coding of all data • Highly secure coding technologies
Demonstration(s)	• In spring 2007, first tests were carried out in transatlantic container service, confirming all the expectations placed on SeCureSystem. The Monitoring Unit is currently being developed further at Astrium's facilities in Bremen.
Users	• Container Transort sector • Customs
Added Value/Findings	• Ground-independent and worldwide position determination of containers by GPS satellites not only enhances container security, it also provides scope for optimising capacity utilisation and the flow of information between all those involved. • Seamless container monitoring, reduction of time spent on administrative processing at destination ports
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Limited to container transport at sea

Satellite family name	**Sentinel (6 satellites)**
Phase(s)	1: Detection 2: Response
Critical Mission Area	Border Security
Type	Operational Satellite Family
Date of launch	• Sentinel 1: envisaged in 2011 • Sentinel 2 & 3: envisaged in 2012
Current Satellite Status	• Operational
Country/Institution	• ESA / EU
Capability	• Sentinel 1: SAR, C-band, 5 m resolution • Sentinel 2: 13 optical channels, visible – near-infrared – shortwave infrared – spectra, 10 m resolution • Sentinel 3: CryoSat-derived microwave radiometer orbit determination device MERIS-like instrument (deployed at ENVISAT) for sea/land temperature observation
User	• Institutional and private
Projected lifetime	• Sentinel 1,2 &3: 7 years
Objective(s)	• High resolution imagery
Service (in practice)	• Eventually, the following services shall be provided by GMES:

	• Mapping (topography, road maps, land-use, harvest, forestry monitoring, mineral and water resources) • Support for emergency management in case of natural hazards; real-time and secure information for civil protection and security • Forecasting for marine zones, air quality, and crop yields
Cooperation/data exchange	• Earth observation data provided for GMES, an initiative aimed at implementing information services dealing with environment and security

Project Name	**PASR SENTRE**
Phase(s)	n.a.
Critical Mission Area	n.a.
Type	R&D
Project Duration	• 14 Months
Current Status	• Completed
Country/Institution	• ASD Belgium
Funding	• €1.056,327 (EU Contribution €792.245)
Partners	• European Association of Aerospace Industries-Europe • Totalförsvarets forskningsinstitut-Sweden • Nederlandse Organisatie voor toegepast-natuurwetenschappelijk onderzoek-The Netherlands • Industrieanlagen-Betriebsgesellschaft mbH-Germany • QinetiQ-United Kingdom • Joint Research Centre – Institute for Protection and Security of the Citizen-Europe • Istituto Affari Internazionale-Italy • Foundation for Strategic Research-France • Austrian Research Centers-Austria • Délégation Générale de l'Armement (Centre d'Etude du Bouchet)-France • Fraunhofer-Gesellschaft-Germany • VTT Technical Research Centre of Finland-Finland • EADS Astrium-United Kingdom • Finmeccanica-Italy • Dassaul-Francet • Sagem-France • Rheinmetall-Germany • EADS--France • Thales Avionique-France • Herstal Group-Belgium • Saab Ericsson Space-Sweden • BAE Systems-United Kingdom-mail : frederic.perlant@asd-europe.org
Objective(s)	• Develop a strategic research plan for European security by establishing and consulting with a network of users and technology experts at national and European levels, in direct link with the EC Advisory Board on Security (ESRAB) .The study, performed by a number of European organisations with relevant expertise, has provided the EU with comprehensive input for planning its programme for security research (ESRP). • Establish a European network to identify user needs at European, national and multi-lateral levels
In Practice	• At national and multi-lateral level: • Top down and bottom up (bilateral exchange) • Identify European/common capability needs and research priorities for comprehensive security missions • Use the structures provided by the SMIG network to engage with a large industrial and research community (Formed in 2004, the Security Mission Industry Groups (SMIGs) serve as expertise panels for the members of ASD, and others by agreement, to exchange views on potential R&T projects to be pursued with the European Commission under the umbrella of the EC Preparatory Action on Security Research) - Provide the EC and MS with a broadly supported view of the key technology needs in Europe; • Define the priorities for filling the gaps • Propose a programme for achievement of EU PASR • Suggest a strategy for EUROPE in the field of comprehensive security • Identify key research issues for FP7 ESRP, to start in 2007
Demonstration(s)	• n.a.
Users	• Security Community
Added Value/Findings	• Development of a strategic plan • Identification of gaps and needs
Collaboration(s)	• n.a.
Follow-up	• PASR STACCATO • FP 7 STRAW
Limitations	• n.a.

Satellite name	**Spain's SEOSAT** (Spanish Earth Observation SATellite)
Phase(s)	1: Intelligence 1: Border security 1 & 3: Emergency management
Critical Mission Area	Intelligence, Emergency Management, Border Security, Early Warning and Risk Management
Type	Envisaged satellite scheduled
Date of Launch	• In 2010
Current Satellite Status	• Envisaged
Country/Institution	• Centre for the Development of Industrial Technology (CDTI) / Spain
Capability	• Panchromatic, 2,5 m resolution • Multispectral: 10 m resolution
User	• Spanish government authorities for territory surveillance, commercial users
Projected lifetime / operational status	• 7 years
Objective(s)	• Cloud free land optical imagery
Service (in practice)	• Intelligence • Cartography • Thematic mapping • Agriculture and forestry mapping • Environmental monitoring • Emergency management • Border Control
Cooperation/data exchange	• Development will be managed by ESA • Will contribute to GMES

Project Name	**SESAR**
Phase(s)	1: Prevention
Critical Mission Area	Transportation Security / Air Transportation
Type	R&D
Project Duration	• 2004 – 2020, 192 months
Current Status	• In progress  Figure: The SESAR Project[486]
Country/Institution	• n.a. (A consortium of 30 companies)
Funding	• Total costs: € 60,000,000 • Co-funded bei the European Commission and Eurocontrol
Partners	• n.a.
Objective(s)	• Modernising existing European air traffic control infrastructure • Developing the new generation air traffic management system capable of ensuring the safety and fluidity of air transport worldwide over the next 30 years
In Practice	• Delivering an ATM master plan defining the content, the development and deployment plans of the next generation of ATM systems. • Production of the required new generation of technological systems and components as defined supported by the creation of a joint undertaking, based on the GALILEO model, which will federate public and private funds • Build the new infrastructure at a wide scale both in Europe and in partner countries without further public funding
Demonstration(s)	• n.a.
Users	• Airport security, Airlines
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.

[486]"What is the SESAR project." 31 March 2009. European Commission Transport 30 July 2009 <http://ec.europa.eu /transport/air/ sesar/sesar_en.htm>.

Limitations	 Figure: SESAR only covers Air Traffic Management[487]

Project Name	**SmartRegions** (Space and Security Knowledge for Maritime Regional Development)
Phase(s)	1: Prevention / Detection 2: Response Integrated approach
Critical Mission Area	Transportation Security / Waterborne Transportation, Critical Infrastructures
Type	R&D
Project Duration	• 2007
Current Status	• Completed
Country/Institution	• Regional project of the two Hanseatic cities Ventspils (Latvia) and Bremen (Germany)
Funding	• European Union
Partners	• OHB
Objective(s)	• Container Security • Environment monitoring • Ship emission monitoring
In Practice	• Implemention of emission-based ship tolls through emission monitoring and processing of routing data • Enhance benefits gained from environmental monitoring programs by using hyperspectral information for more detailed and reliable classification results
Demonstration(s)	• n.a.
Users	• Hanseatic maritime community
Added Value/Findings	• Space-based technologies can contribute to user needs in Ventspils and Bremen • Networking can grant significant benefits
Collaboration(s)	• GMES • Galileo
Follow-up	• n.a.
Limitations	• Allthough other regions are welcome within the project it was originally designed for the Hanseatic regions only.

Project Name	**Greece's SME**
Phase(s)	1: Prevention through surveillance 2: Improved response given continuous monitoring and availability of imagery before and after event 3: Impact assessment through imagery
Critical Mission Area	Critical Infrastructures
Type	Industry service
Project Duration	• n.a.
Current Status	• on-going
Country/Institution	• Greece/ Industry
Funding	• n.a.
Partners	• n.a.
Objective(s)	• Pipeline security
In Practice	• An innovative pipeline surveillance system using remote sensing and EO techniques • Involves image operation in different spectral bands (visible, infrared, thermal), lineament analysis and geostructural investigations as to assure safety of pipelines
Demonstration(s)	• n.a.
Users	• Stakeholders in CI and particularly using pipelines

[487] Ludwig, K.-P. and J. Federhen. "Satellites in Air Traffic Management ." Presentation. Sächsisches GIS-Forum. 25 Feb. 2008. <http://www.gdi-sachsen.de/skripte/gis09/GIS-Forum%20Pres_SatellitesInATM_25022009.pdf>.

Added Value/Findings	• Innovative concept for pipeline surveillance
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**FP 7 SOBCAH**
Phase(s)	1: Surveillance for Ports as Critical Infrastructures 2: Improved response through improved situational awareness 3: Improved response
Critical Mission Area	Border Security, Critical Infrastructures
Type	R&D
Project Duration	• 18 Months
Current Status	• Completed
Country/Institution	• FP 7
Funding	• € 3,007,109 • EU contribution € 2,010,600
Partners	• SELEX Sensors and Airborne Systems Ltd. • HELLENIC AEROSPACE INDUSTRY S.A. • PATRIA Advanced Solutions Oy • THALES RESEARCH & TECHNOLOGY LIMITED • THALES UNDERWATER SYSTEMS SAS • The Netherlands Org. for Applied Scientific Research • Fraunhofer-Gesellschaft zur Förderung der angewandten Forschung e.V. • OBR Centrum Techniki Morskiej • SELEX Sistemi Integrati • INDRA SISTEMAS S.A. • RHEINMETALL Defence Electronics GmbH • Sistemi e Telematica S.p.A. • Software e Tecnologias de Informacao S.A. • DELOITTE • Autorità Portuale di Genova
Objective(s)	• Identify the main threats relevant to ”green” and “blue” borders, • Elaborate the most suitable architectural solutions based on the most advanced existing sensors and network technologies, • Execute a proper modelling of the established solution, • Carry out the technology validation of the selected solution, first in the laboratory and then in the port of Genoa (Italy), • Elaborate a consistent Road Map. • SOBCAH will maximise their effectiveness through innovative application of fusion and high levels of integration for the protection of border coastlines and harbours through surveillance
In Practice	• The SOBCAH demonstration will provide an efficient, real-time, user-friendly, highly automated surveillance system to be deployed in one of the larger European Ports, i.e. an area exposed to possible terrorist attacks, covering all kinds of border security threat.s • Preparatory Action for Security Research (PASR 2005) • Current security systems are effective in isolation (i.e. customs, goods tracking, VTS, personnel identification) but are poorly integrated: SOBCAH will maximise their effectiveness through innovative application of fusion and high levels of integration. SOBCAH aims to overcome complex systems engineering issues including: • Integration of incumbent systems with new technologies, • Real time situational awareness, • Data fusion, • Multiple users with multiple views, • Multi-level security with segregated access to controlled • Information, • Net centric and interoperability.
Demonstration(s)	• n.a.
Users	• Port authorities
Added Value/Findings	• n.a.
Collaboration(s)	• Uses CCTV, Radar,… as part of the Preparatory Action in the field of Security Research activites
Follow-up	• n.a.
Limitations	• Port security

Project Name	Space-QUEST
Phase(s)	1: Quantum Communication as secure communications
Critical Mission Area	Critical Infrastructure
Type	R&D
Project Duration	• 2002 – n.a.d
Current Status	• In progress
Country/Institution	• n.a.
Funding	• ESA • Austrian Space Programme
Partners	• ESA • A European research consortium led by Vienna University
Objective(s)	• To demonstrate unconditional secure global distribution of cryptographic keys from space, based on novel quantum communication techniques (e.g. Quantum Key Distribution), and fundamental quantum physics principles beyond the capabilities of earth-bound laboratories by utilising the added value of the space environment
In Practice	• Internal security related space applications: • Secure access to a satellite • Secure communications between gateways / ground stations • Secure satellite-to-satellite communication
Demonstration(s)	• n.a.
Users	• Satellite Operators
Added Value/Findings	• With the success of the inter-island link experiment, European teams are a step ahead of the U.S. and Asia, and in cooperation with ESA, they are determined to keep this lead
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Long term project. In 2015 the first ever space-to-ground demonstration of innovative quantum communication schemes could take place

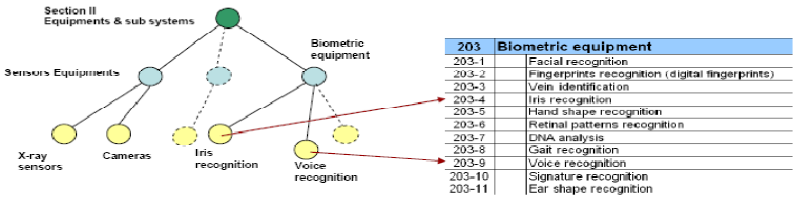
Satellite family name	**SPIRALE** (Système Préparatoire Infra-Rouge pour l´Alerte - 2 satellites)
Phase(s)	1: early warning
Critical Mission Area	Early warning
Type	Technology demonstrator
Date of launch	• 12 February 2009
Current Sat. Status	• n.a.
Country/Institution	• DGA / France
Capability	• Optical and infrared cameras
User	• DGA/French Ministry of Defence
Date of launch	• 12 February 2009
Projected lifetime	• n.a.
Objective(s)	• Detection of a missile's launch during the boost phase through assessment of the camera's resolution and sensitivity and improvement of image analysis
Service (in practice)	• Detection of missile launch
Cooperation/data exchange	• n.a.

Project Name	SPOTimage and Spot Infoterra Group Constellation for Maritime Surveillance
Phase(s)	1: Prevention, Detection, Surveillance, Intelligence 2: Emergency Response
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security
Type	Satellite constellation
Project Duration	• n.a.
Current Status	• Operational
Country/Institution	• SPOTImage, Spot Infoterra Group
Funding	• Industry
Partners	• French Navy
Objective(s)	• Combat maritime piracy
In Practice	• Monitoring in a short loop as part of its close cooperation with the French Navy QuickBird 2 (18 Oct. 2001) • 0,6 m for panchromic images • 2,5 m for multispectral images Korea's KompSat-2 (28 July 2006) • Imagery in black and white (Pan) at a resolution of 1 m • Imagery in colour (MS) across 4 bands in the visible (red, green, blue) and near-Infrared at a resolution of 4 m

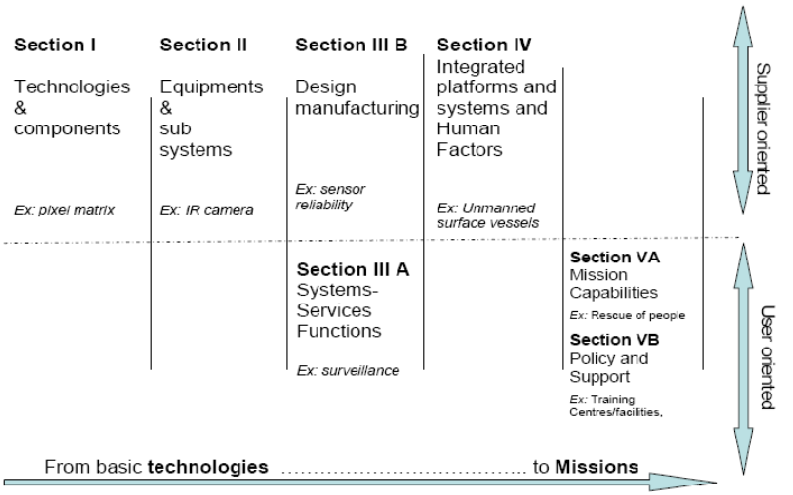
	• Footprint of 15 km x 15 km Ikonos (24 September 1999) • 1 m for chromatic optical images • 4 m for multispectral images Taiwan's FormoSat-2 (21 May 2004) • Image Processing System (IPS) • X-band antenna • High resolution of 2 meter panchromatic data and 8 meter multispectral satellite image data Israel's EROS (EROS A: 5 December 2000, EROS B: 25 April 2006, EROS C: envisaged in 2009) • EROS A: CCD camera, maximum resolution 1,8 m (black & white) • EROS B: new telemetry communications and control systems, news global positioning system and new battery • EROS C: maximum resolution 0,7 m Canada's RADARSAT (RadarSat-1: 4 November 1995, RadarSat-2: 14 December 2007) • RadarSat 1 & 2: Synthetic Aperture Radar, C-band • RadarSat-2 has left- and right-looking capability in contrary to Radarsat 1 • Maximum resolution: 7,6 m (RadarSat-1) and 3 m (RadarSat-2)
Demonstration(s)	• n.a.
Users	• Military and commercial users
Added Value/Findings	• Unrivalled constellations of optical and radar satellites
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Satellite family name	**France's SPOT** (Satellites Pour l´Observation de la Terre - 5 satellites)
Phase(s)	1: intelligence
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Operational satellite
Date of Launch	• Spot 2: 22 January 1990 • Spot 4: 24 March 1998 • Spot 5: 4 May 2002
Current Satellite Status	• Spot 2: Operational • Spot 4: Operational • Spot 5: Operational
Country/Institution	• CNES, Spot Image / France
Capability	• Spot 1 and 3 are no longer operable • Spot 2 o High Resolution Visible (HRV) o Spectra: panchromatic, green, red, near infra-red • Spot 4 o High Resolution Visible Infra-Red (HRVIR) o Spectra: panchromatic, green, red, near infra-red, short-wave infra-red o Vegetation: wide angle earth observation radiometer (resolution: 1 km, swath width: 250 km) o Maximum resolution: 10 m • Spot 5 o High Geometric Resolution (HGR) o Spectra: panchromatic, green, red, near infra-red, short-wave infra-red o Vegetation: wide-angle earth observation radiometer (maximum resolution: 1 km, swath width 250 km) o Maximum resolution: 2,5 m (panchromatic)
User	• Civil and military
Projected lifetime	• Spot 2: 10 years / Operational • Spot 4: 5 years / Operational • Spot 5: 5 years / Operational
Objective(s)	• Optical imagery
Service (in practice)	• Intelligence • Geospatial Intelligence • Environmental monitoring • Agriculture and forestry mapping • Geology • Hydrology • Oceanography • Cartography • Urban and rural planning
Cooperation/data	• Imagery data are distributed via Spot Image on a commercial basis (Spot Image shareholders:

exchange	EADS-Astrium Group 81%, Telespazio 7,7%, SSC 6,7%, IGN 2,7%, Banks 1%, Belgian government 0,6%) • Within the scope of the Vegetation instruments aboard Spot 4 and 5, Belgium, Sweden, Italy and the European Commission contributed to the Spot programme. Imagery data is used by the European Union Satellite Centre

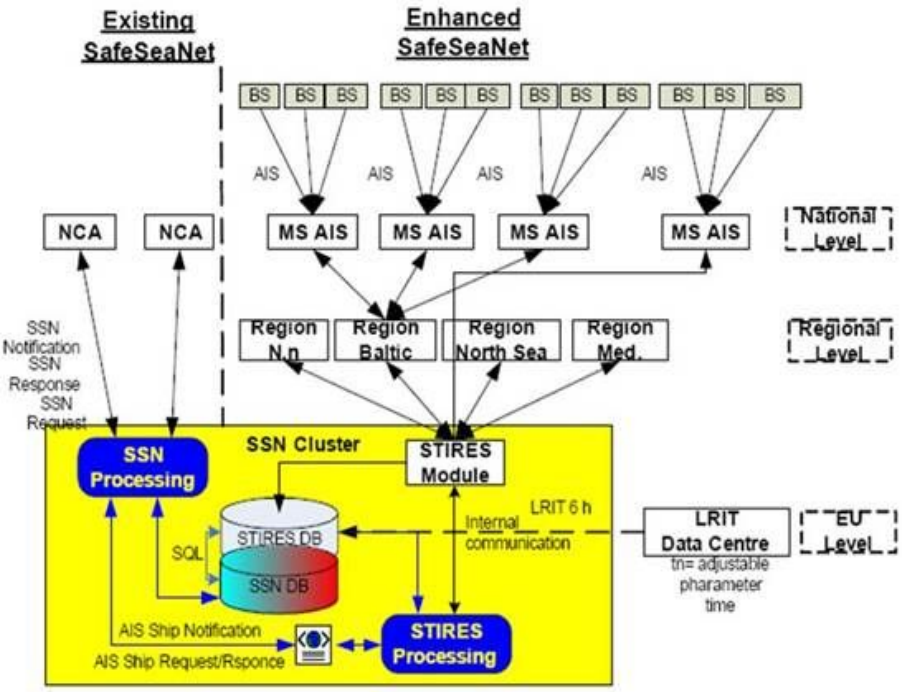
Project Name	**PASR STACCATO**
Phase(s)	n.a.
Critical Mission Area	n.a.
Type	R&D
Project Duration	• 16 months
Current Status	• completed
Country/Institution	• PASR
Funding	• PASR
Partners	• AeroSpace and Defence Industries Association of Europe – ASD- Europe • Dassault Aviation – DASSAV-France • Diehl BGT Defence GmbH & Co. KG – DIEHL-Germany • European Aeronautic Defence and Space Company – EADS-France • Astrium SAS – ASTR-France • Electricité de France – EDF-France • Finmeccanica – Società per Azioni – FNM-Italy • Indra Sistemas S.A. – INDRA-Spain • Sagem Défense Sécurité – SAGEM-France • Thales Avionics SA – THAv • Österreichisches Forschungs- und Prüfzentrum Arsenal Ges.m.b.H – Arsenal-Austria • Commissariat à l'Energie Atomique – CEA-France • Swedish Defence Research Agency – FOI-Sweden • Industrieanlagenbetriebsgesellschaft – IABG-Germany • Netherlands Organisation for Applied Scientific Research – TNO-The Netherlands • Technical Research Centre of Finland – VTT-Finland • Fondation pour la Recherche Stratégique (FRS)-France • Istituto Affari Internazionali – IAI-Italy • European Biometric Forum – EBF-Ireland • European Association for Bio Industries – EUROPABIO-Belgium • Commission of the European Communities, Directorate General JRC, EC, DG JRC-European Commission
Objective(s)	• Proposing methods and solutions for the creation of a security market and a structured supply chain in Europe. In line with ESRAB (European Security Research Advisory Board) recommendations, it goes beyond research needs and gap analysis already undertaken through efforts supported by PASR, by identifying implementation measures.
In Practice	The STACCATO taxonomy is structured along 3 levels. Upper level or Sections Middle level or Subsections ("blue-lines", in the text) Lower level (specific items) *Example:*  Figure: STACCATO's Taxonomy[488]

[488] STACCATO. "PASR Security Research Deliverable D 1.2.2: STACCATO Final Taxonomy." 15 Jan. 2007. STACCATO 30 July 2009 <http://www.asd-europe.org/Objects/2/Files/STACCATO%20final%20taxonomy.pdf>.

	 Figure: STACCATO's Taxonomy Structure[489]
Demonstration(s)	• n.a.
Users	• AeroSpace and Defence Industries Association of Europe (ASD)
Added Value/Findings	• Identification of implementation measures
Collaboration(s)	• n.a.
Follow-up	• Builds upon SENTRE
Limitations	• n.a.

Project Name	**EMSA's STIRES** (SafeSeaNet Tracking Information Relay and Exchange System)
Phase(s)	1: Prevention and Detection of malicious behaviour 2: Improved response given the awareness of the maritime picture Integrated approach
Critical Mission Area	Transportation Security/Waterborne Transportation, Border Security, Emergency Management
Type	R&D/ eventually service
Project Duration	• 2008 – n.a.
Current Status	• On-going
Country/Institution	• EMSA
Funding	• n.a.
Partners	• SAAB
Objective(s)	• Research on the introduction of a SafeSeaNet Tracking Information Relay and Exchange System (STIRES)

[489] Ibid.

In Practice	 Figure: SafeSeaNet Architecture[490] • EMSA has together with SAAB conducted a study on SafeSeaNet Tracking Information Relay and Exchange System (STIRES) aimed at facilitating relaying and exchanging information between the EU Member States, Norway and Iceland. It relies on the existing AIS systems of the EU Member States and/or regional AIS servers and the existing SSN core system, the EU LRIT Data Centre and the EMSA Geodatabase system
Demonstration(s)	• n.a.
Users	• Coastal security authorities
Added Value/Findings	• n.a.
Collaboration(s)	• SafeSeaNet
Follow-up	• n.a.
Limitations	• Currently no use of space data • Given that EMSA is thinking about including satellite data into SafeSeaNet, those might eventually also be included in STIRES

Project Name	**FP 7 STRAW** (Security Technology Active Watch)
Phase(s)	n.a.
Critical Mission Area	n.a.
Type	R&D
Project Duration	• 01/10/2008 – 31/03/2010, 18 months
Current Status	• In progress
Country/Institution	• FP 7
Funding	• Total costs: € 1,3 million • EU contribution: € 1 million
Partners	• Atos Origin SAE (Spain) • AeroSpace and Defense Industries (Belgium) • Thales Services SAS (France) • Stiftelsen SINTEF (Norway) • Fraunhofer IFF (Germany) • InstitutoNacional de Técnica Aeroespacial (Spain) • Elsag Datamat S.p.A (Italy) • Asociación de Empresas de Electrónica, Tecnologías de la Información y Telecomunicaciones de España (Spain) • Fondazione Rosselli (Italy) • European Organisation for Security (Belgium)
Objective(s)	• To provide a European service of technology watch on security technologies

[490] "Overview of the anticipated architecture." 3 Oct. 2008 EMSA 30 July 2009
< https://extranet.emsa.europa.eu/index.php?option=com_content&task=view&id=95&Itemid=152>

In Practice	• STRAW Website to increase awareness of industry and defence research related to European security • STRAW Network to enhance cooperation between Member States and Associated Countries as well as EU organisations in the industry and defence research community • STRAW panel to identify opportunities for and constraints on developing and strengthening markets related to European security and even provide input on futures security themed programs and their implementation
Demonstration(s)	• n.a.
Users	• Security community, security providers (industry, local authorities)
Added Value/Findings	• Ministries of Interior belong to the Advisory Board
Collaboration(s)	• n.a.
Follow-up	• Is building on SENTRE and STACCATO
Limitations	• n.a.

Satellite name	**Sweden's SVEA**
Phase(s)	1: intelligence, Early Warning and Risk Management
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	Envisaged Satellite
Date of Launch	• Expected spring 2010
Current Satellite Status	• Operations Expected in 2010
Country/Institution	• Swedish Space Corporation / Sweden
Capability	• Spatial resolution: 1.5 m (black/white), 3 m (colour) • Scene size: 10 km x 10 km • Observation capacity: ~300 scenes/day
User	• Swedish armed forces, actors related to civil crisis management
Projected lifetime	• n.a.
Objective(s)	• It is designed to provide detailed, time-critical and autonomous geographical data from any region on the Earth. The Svea mission concept comprises the complete chain of events from observation scheduling, image acquisition, data down-link, data pre-processing and subsequently, product delivery to the users within minutes from data down-link. The system provides full security in every step, to assure complete integrity of sensitive information.
Service (in practice)	• Signal Intelligence
Cooperation/data exchange	• The satellite platform design takes advantage of on-going technology development for the PRISMA satellite mission and the space camera will be selected to meet the desired user requirements for performance.

Project Name	**SYNTHESIS**
Phase(s)	n.a. (aims at integrating SATCOM satellite navigation and EO for emergency management)
Critical Mission Area	Emergency Management
Type	R&D
Project Duration	• n.a. (long-term)
Current Status	• On-Going
Country/Institution	• Italy
Funding	• Galileo Coordination Office • Italian Space Agency (ASI)
Partners	• n.a.
Objective(s)	• Provide National and European Institutions with Public Regulated Services (PRS) for a rapid response to crises, disasters and emergencies.
In Practice	• Provide combined Public Regulated Services (Navigation AND Communication AND Earth-Observation) to accredited European institutional users, based on certified security standard & procedures. • Place side-by-side Earth Observation, Positioning-Navigation, and Communication Systems
Demonstration(s)	• n.a.
Users	• Civil protection authorities, emergency responders
Added Value/Findings / Findings	• Integrated approach
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• n.a.

Project Name	**France's T2L2** (Transfert de Temps par Lien Laser)
Phase(s)	1: Prevention
Critical Mission Area	Critical Infrastructure
Type	R&D
Project Duration	• 2005 - ...

Current Status	• In progress
Country/Institution	• CNES, France
Funding	• CNES is prime contractor
Partners	• CNES • Observatoire de la Côte d´Azur
Objective(s)	• Use of optical signals in data exchange
In Practice	• Using optical signals to compare atomic clocks for GPS and TWSTFT • Validation of T2L2 system performance, • Time/frequency metrology (establishing time scales, calibrating other comparison systems, etc) • Fundamental physics • Characterising the Doris onboard ultra-stable oscillator carried by Jason 2
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• It is essential to have a way of comparing the performance of the various atomic clocks installed around the globe, not only to evaluate their metrological precision but also to establish international time scales and contribute to experiments in fundamental physics. Until now, such comparison has mainly been based on the manipulation of radiofrequency signals (GPS and Two-Way Satellite Time and Frequency Transfer - TWSTFT). The T2L2 (Time Transfer by Laser Link) mission will break new ground with the use of optical signals. • The researchers behind T2L2 hope to increase the accuracy of these measurements by one or two orders of magnitude in comparison with systems such as GPS, which use microwaves. • There are many immediate applications for atomic clock synchronisation, including banking operations or geolocation systems.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Satellite Name	**Germany's TanDEM-X**
Phase(s)	1: Intelligence, Risk Management & Early Warning 3: Emergency Response
Critical Mission Area	Early Warning and Risk Management, Intelligence, Transportation Security / Land Transportation / Waterborne Transportation, Border Security and Emergency Management
Type	Envisaged satellite
Date of Launch	• September 2009
Current Satellite Status	• Under development
Country/Institution	• German Aerospace Center (DLR) • EADS Astrium GmbH
Capability	• Tracking, Occultation and Ranging (TOR) provided by GFZ. TOR consists of a two-frequency CHAMP type GPS receiver and a CHAMP Laser Retro-Reflector (LRR). • Capable of generating a high-accuracy global Digital Elevation Model (DEM). • bi-static applications (e.g., polar metric SAR interferometry) • along-track interferometry (e.g., for measurement of ocean currents • Laser communication • Active phased array X-band SAR
User	• n/a
Projected lifetime	• 5 years with a 3 year overlap with TSX
Objective(s)	• To create a high-accuracy global Digital Elevation Model (DEM) • Active phased array X-band SAR • Tracking, Occultation, and Ranging (TOR) package • GPS receiver • Retro reflector array • Laser Communication Term • HRTI-3 (High Resolution Terrain Information) specifications (12 m posting, 2 m relative height accuracy inal (LCT)
Service in Practice	• Geospatial Intelligence • New SAR technology test • Research • Agriculture • Forestry • Crisis management • Urban planning • Cartography • Geology • Hydrology • Oceanography • Coastal zones monitoring
Cooperation/Data exchange	• Infoterra GmbH • Microwaves and Radar Institute - HR • German Remote Sensing Data Center - DFD • Remote Sensing technology Institute - IMF • German Space Operation Center - GSOC

Project Name	**FP 6 TANGO** (Advanced Networks for GMES Operations)
Phase(s)	1: Detection and alert
Critical Mission Area	n.a. (early warning for all GMES services)
Type	R&D
Project Duration	• 36 months (starting data 01/11/2006)
Current Status	• In progress
Country/Institution	• FP 6
Funding	• EU contribution: € 4.995.500 (estimated total cost: € 8.883.300)
Partners	• 24 partners, coordinator EADS Astrium
Objective(s)	• Identifying the needs for satellite telecommunication in GMES • To develop, integrate, demonstrate and promote new satellite telecom services dedicated to GMES requirements • Aims at providing operational telecommunication solutions to immediate GMES services needs, and at preparing the definition of optimised satellite telecom infrastructures to expand future GMES services • Improvement of the reactivity and freshness of the data through faster scene and in-situ data collection → improved data collection: includes Data Relay infrastructure for EO satellites to enable faster data scene acquisition and enhanced data distribution. Inclusion of UACs and HAPS in the global relay infrastructure enables quasi real time and continuous access to dedicated zones for monitoring or surveillance. Improvement of data collection systems to meet the growing needs of GMES (higher transmission rates and volume of data, sensor command) are also considered within CLS offered and adapted solutions based on Argos satellite system • Speed up the transfer of data expected as these prototype services become operational and allow higher volumes to be processed • Data distribution: improve the transfer of data form data providers to GMES service providers and enable efficient dissemination of GMES products wherever it is needed and down to the end-users. This includes adaptation of broadband standard solutions such as DVB/ DVB-RCS solutions or SATMODE solutions to meet the needs of data dissemination in regions poorly served by terrestrial networks, in oceans, or after natural disasters. Transportability, compactness and, in some cases, mobility are key criteria for some GMES applications, with the needs of product dissemination in the field and interactive mapping. Operational solutions as Inmarsat BGAN will also be considered
In Practice	• Explore synergies between EO and SATCOM • adaptation from Worldspace (satellite radio broadcast) satellite networks solutions with satellite return link will permit autonomous – independent from terrestrial network – early warning associated with crisis management • Adaptation from SATMODE solutions for alert implementation also foreseen • To explore the synergy between EO and SATCOM: take advantage of the ARTEMIS inter-satellite communications link to provide real-time EO info from anywhere in the planet • To facilitate the access of GMES community to telecommunication solutions, in particular through the development of the TANGO “common Telecommunication Service Platform”, which offers a privileged interface to GMES service providers to get, at optimised cost, fast access to the telecommunications solutions
Demonstration(s)	• n.a.
Users	• GMES (marine services, emergency response incl. risk & crisis management and humanitarian aid); civil security or humanitarian aid forces networks
Added Value/Findings	• Has complemented the collection of telecommunication requirements through contacts with major GMES projects • Systematic approach for assessing the needs has been established, relying on the def of a clear and agreed terminology and TANGO reference architecture
Collaboration(s)	• GMES
Follow up	• n.a.
Limitations	• Limited to GMES services, while exploration of synergies between EU and SATCOM can actually help all early warning systems

Satellite name	**Germany’s TerraSAR-X** (twin asset TanDEM-X foreseen)
Phase(s)	1: Intelligence, Detection 2: Alerting 3: Emergency management
Critical Mission Area	Intelligence, transportation, border security and emergency management
Type	Operational Satellite
Date of Launch	• 15 June 2007
Current Satellite Status	• Operational
Country/Institution	• DLR, Infoterra GmbH (100% owned subsidiary of EADS Astrium) / Germany
Capability	• Synthetic Aperture Radar • X-band • Maximum resolution: 1m, 3 m or 16 m per pixel (depending on the image size) • TerraSAR-X also offers along track interferometry which can be used for automatically detecting vehicles • TerraSAR-X has also been tested for the use in AIS

User	• Scientific and commercial users, government authorities including armed forces	
Projected lifetime	• 5 years	
Objective(s)	• Imagery	
Service (in practice)	• Geospatial intelligence • Agriculture • Forestry • Crisis management • Urban planning	• Cartography • Geology • Hydrology • Oceanography • Coastal zones monitoring
Cooperation/data exchange	• TerraSAR-X is realised as a public private partnership; EADS Astrium/Infoterra contributed to the development costs in exchange for the exclusive commercial utilisation rights, depending on the revenue EADS Astrium/Infoterra are also contributing to the operational costs • Infoterra offers data at different levels of refinement: Basic Image Products, Enhanced Image Products, Geoinformation Products • Infoterra arranges distribution either through direct access of the customer to the data (Direct Access Services) or by further distribution by regional partners (Direct Access Partner) • The scientific use of TerraSAR-X data is coordinated via DLR's TerraSAR-X Science Service System planned formation flying with upcoming twin asset TanDEM-X • Shall contribute to GMES • Imagery data is used by the European Union Satellite Centre	

Project Name	**Germany's Tesat Spacecom** (Laser Crosslinks)
Phase(s)	1: Prevention and detection
Critical Mission Area	Critical Infrastructures
Type	Implemented
Project Duration	• n.a.
Current Status	• In Progress
Country/Institution	• Germany
Funding	• n.a.
Partners	• Tesat Spacecom
Objective(s)	• Secured data transmission
In Practice	• Enhancing data transmission through use of laser technology
Demonstration(s)	• Transmission of 5.625 Gpbs across 142 km between La Palma and Tenerife in 2005
Users	• Communication industry, military
Added Value/Findings / Findings	• n.a.
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• n.a.

Project Name	**TIES**
Phase(s)	n.a. (integrated approach)
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	R&D
Project Duration	• May 2006 – September 2008, 28 months
Current Status	• Completed
Country/Institution	• European Defence Agency
Funding	• European Defence Agency
Partners	• European Satellite Centre
Objective(s)	• Compensate for the lack of interoperability between European satellite imagery systems and ground systems
In Practice	• Creating a capacity for receiving imagery data in different formats provided by different satellite systems and fusing it into one usable and workable intelligence product
Demonstration(s)	• European Union Satellite Center, Torrejon, Spain 2008 proving the basic functionality of an EU Imagery Architecture
Users	• Intelligence, surveillance, target acquisition and reconnaissance community
Added Value/Findings / Findings	• n.a.
Collaboration(s)	• n.a.
Follow up	• n.a.
Limitations	• n.a.

Project Name	**France's TOCHA** (Continuous and Operational Tracking with High Autonomous)
Phase(s)	1: Prediction 1: Protection
Critical Mission Area	Transportation Security / Land Transportation
Type	Future Implemented Service
Project Duration	• 02/2008 – 07/2010, 30 months
Current Status	• In progress
Country/Institution	• France
Funding	• Funded in the framework of the French competitiveness cluster SYSTEM@TIC • € 1.96 million (€ 1.10 million from ANR)
Partners	• Deveryware (coordinator) • CEA • Photospace • IETR
Objective(s)	• The project aims at developing new object, person or vehicle trackers. In detail, TOCHA should enable monitoring in real time a small autonomous tracker over several years, with maximal isotropy, operational even covered, at short and long distances, with accuracy, and with the ability to send information about its energy level.
In Practice	• Combining of different localisation techniques (GPS, A-GPS, GSM...) • Advanced algorithms for the most appropriate choice of technique • Location of tracking unit through a web service
Demonstration(s)	• n.a.
Users	• Security community
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	•

Satellite name	**TopSat**
Phase(s)	1: Intelligence 1 &3: Emergency management
Critical Mission Area	Intelligence, Early Warning and Risk Management, Emergency Management
Type	Technology demonstrator
Date of Launch	• 27 October 2005
Current Satellite Status	• Operational
Country/Institution	• British National Space Centre (BNSC), QinetiQ Ltd. / UK
Capability	• Optical • Maximum resolution: 2.8 m (panchromatic)(good enough to pick out houses or vehicles)
User	• British armed forces
Projected lifetime	• 1 year
Objective(s)	• Imagery for British national surveillance programme
Service (in practice)	• Intelligence, disaster management, agricultural and environmental monitoring • Mapping and land-use monitoring • Includes special lightweight portable ground stations • Images are provided free of charge to relief agencies responding to disasters anywhere on Earth
Cooperation/data exchange	• Data can be received and programmed by fixed and mobile ground stations • Infoterra is leading the exploitation activities for TopSat • Imagery data is used by the European Union Satellite Centre • Imagery operates within the International Charter: Space and Major Disasters • TopSat platform is already being developed for future projects such as RapidEye

Project Name	**FP 4 TRACAR 1 & 2 TR 1059 & TR 4026**
Phase(s)	1: Traffic monitoring and tracking
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• 01/01/96 – 31/12/99
Current Status	• Completed
Country/Institution	• Cable & Wireless (GB)
Funding	• n.a.
Partners	• Blue Water Shipping (DK) • TSS (DK)
Objective(s)	• To encourage users of transport to improve the management of their intermodal cargo movement by automating aspects of tracking, tracing and monitoring from end-to-end of the supply chain • To reduce manual monitoring and replace it with an inexpensive, automatic system of control
In Practice	• Part I of the project involved the transport of frozen cargo by road and rail. Part II concerned

	a road and sea demonstrator for both dry and frozen cargo even when that cargo is stowed below deck on a ship • The system that has been developed can position a cargo to a high degree of accuracy, when necessary, by using the tag and reader technology developed in the project.
Demonstration(s)	• n.a.
Users	• Transportation sector • Originators of transport orders and the • Final recipients of the cargo
Added Value/Findings	• Using the combination of tag, reader and short range radio technology with GPS positioning, TRACAR can provide end-to-end positioning along a transport corridor by so that no blind spots exist, even when a train is travelling at high speed. • Through providing a means of collecting planning data that is meaningful for purposes of future planning. • Costs remain relatively high and this could still be a determining factor for a purchaser as the value of the cargo plays a part in determining the economic benefit • The main obstacle to users choosing T&T systems is the uncertainty about which technology will succeed.
Collaboration(s)	• n.a.
Follow-up	• The technology has been used for Bus and Emergency Vehicle Priority Systems and for light rail.
Limitations	• n.a.

Project Name	**France's TransControl**
Phase(s)	1: Prevention and detection 2: Response
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• 2007 - 2009
Current Status	• In Progress
Country/Institution	• France
Funding	• Funded in the framework of the French competitiveness cluster Aerospace-Valley
Partners	• M3 Systems (coordinator) • 8 other industrial partners (Thales Alenia Space, ACTIA...) • Centre d'études techniques du Sud-Ouest • TeSA laboratory • Ecole nationale supérieure des officiers sapeurs-pompiers
Objective(s)	• Innovative monitoring and inspection application for the transportation of hazardous materials through EGNOS, GALILEO
In Practice	• n.a.
Demonstration(s)	• n.a.
Users	• Civil authorities
Added Value/Findings	• Innovative monitoring and inspection application for the transportation of hazardous materials through EGNOS, GALILEO
Collaboration(s)	• EGNOS; Galileo
Follow-up	• n.a.
Limitations	• Limited to regional Land transportation

Project Name	**JRC's VASTS** (Vulnerability Assessment and Surface Transport Security)
Phase(s)	1: Prevention
Critical Mission Area	Transportation Security / Land Transportation, Critical Infrastructures
Type	R&D
Project Duration	• 2004 – 2005
Current Status	• Completed
Country/Institution	• JRC
Funding	• JRC
Partners	• n.a.
Objective(s)	• Provide an adequate and balanced level of security over European territory • Protect Critical Infrastructures (EPCIP) and other European key assets against the threat posed by terrorism and other malicious intentional acts
In Practice	• Creation of the Critical Infrastructure Warning Information System (CIWIN) to facilitate exchange of information on shared threats and vulnerabilities and appropriate counter-measures and strategies • Address, in a single comprehensive security management model, the existing and emerging threats to critical infrastructures, their vulnerability and criticality, and the defense layers and other cost-effective protective measures that can be implemented
Demonstration(s)	• A report presenting the preliminary results of the review and comparison of methods for threat, risk and vulnerability assessment of Critical Infrastructures, and recommendations for

	improving their protection, taking into account, primarily, the threat (01/11/2006) • A study report presenting the progress of a case-study on the security needs of stakeholders of a trans-European transport corridor, with user requirements, technical specifications and IT prototype demonstrators. (01/08/2006) • A study of the threat of terrorist actions to regional and local (including subways) passenger transport and of the existing security management models currently adopted to protect against - and respond to - such actions. (01/10/2006) • A report presenting the current technical requirements for the monitoring and surveillance of public transport infrastructure and a critical review of existing and emerging technologies that support these requirements. (01/07/2006) • A technical report summarizing the work carried out in 2006 for DG TREN for the functional and security design of a second generation of Digital Tachograph equipment. (31/12/2006) • A document describing the high-level functional architecture of a Universal On-board Platform for road vehicles that would meet basic requirements requested primarily in security, safety and logistics. (01/12/2006) • A report presenting an analysis of the threat associated with the transport of dangerous substance, scenario analysis and possible recommendation of counter-measures. (01/10/2006) • A document describing the basic requirements and possible architecture for on-board equipment to allow real-time monitoring and telemetry of dangerous substances while being transported, and timely management of alarms and emergency response. (01/09/2006)
Users	• European Commission • Swiss Federal Institute of Technology (ETH) • Swedish Rescue Service Agency (SRSA) • International Risk Governance Council (IRGC)
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• Currently use of space applications unclear.

Project Name	**FP 7 VATDIS** (Vulnerability Assessment in Transport and Distribution System)
Phase(s)	1: Prevention 2: GIS (Geographical Information System) for response 3: GIS (Geographical Information System) for reconstruction Integrated approach_ contribution to definition of European Critical Infrastructures
Critical Mission Area	Transportation Security / Land Transportation, Critical Infrastructures
Type	R&D
Project Duration	• n.a.
Current Status	• Completed
Country/Institution	• FP 7
Funding	• FP 7
Partners	• n.a.
Objective(s)	• Conduct research during the 7th Framework Programme on the issue of vulnerability of transport and distribution systems against man-made risks, natural hazards and intentional acts.
In Practice	• Harmonization and improvement of risk analysis, vulnerability assessment and protection methods of critical infrastructures • Identification, study and harmonization of best practices and supporting methods and tools for scenario analysis. emergency preparedness and response to accidents or intentional acts related to the transport of dangerous substances • Comparison of technological architecture and study of functional specifications for on-board electronic platforms supporting safety, security and logistic requirements • Comparison and development of open service-oriented software architecture for integrated territorial vulnerability assessment and risk management
Demonstration(s)	• A report on methods and tools for vulnerability analysis, impact assessment and protection of critical infrastructures (31/12/2008) • A report on the vulnerability of a trans-national gas distribution system and on the impact of its disruption on depending infrastructures (31/12/2008) • A report and a published paper on the application of vulnerability assessment tools (e.g. attack trees) on a hypothetical critical infrastructure (31/12/2008) • A report on possible scenarios of terrorist acts associated with the transport of dangerous goods (31/12/2008) • A report on functional and security requirements of on-board units, with an emphasis on the design of the new generation of Digital Tachograph (31/12/2008)
Users	• European Commission • EUROMOT • UIC/UNIFE
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**JRC's Institutional Action VESPO** (VEssel Surveillance and POrt security project)
Phase(s)	1: Prevention and detection 2: Reponse Integrated Approach
Critical Mission Area	Critical Infrastructures, Transportation Security/Waterborne Transportation, Border Security
Type	R&D
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• JRC
Funding	• JRC
Partners	• n.a.
Objective(s)	• Examine scientific and technical issues related to maritime safety and security and monitoring of container traffic, specifically with respect to maritime surveillance • Address maritime surveillance through an integrated approach
In Practice	• Look into the desirability and feasibility of an integrated plan for EU-wide maritime surveillance, taking into account the many actors and their remits • Selectively detect anomalous movements based on route information by further developing the ConTraffic System providing collection and storage of data about container movements from (web) open sources as well as a set of tools for risk assessment analysis and monitoring
Demonstration(s)	• n.a.
Users	• EMSA • FRONTEX • CFCA
Added Value/Findings	• n.a.
Collaboration(s)	• Naturally with other JRC projects such as MASURE. • Builds on ConTraffic
Follow-up	• n.a.
Limitations	• n.a.

Project Name	**Italy's Vehicle Traffic Management (VTM)**
Phase(s)	1: Detection 2: Response
Critical Mission Area	Transportation Security/Land Transportation
Type	R&D
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• Italy
Funding	• n.a.
Partners	• ASI
Objective(s)	• The project aims at developing prototypes and at analysing and testing applications for the payment of tolls and the control of road traffic and of gates to urban centres (virtual gates), traffic monitoring and control, first aid and assistance in case of accident.
In Practice	• n.a.
Demonstration(s)	• n.a.
Users	• Stakeholders in land transportation sector
Added Value/Findings	• The innovative employment of satellite navigation potentialities that can generate huge social advantages and savings in public direct and indirect costs • The possibility of cutting costs incurred and pollution caused by traffic jams as a result of shorter journey times and, thanks to the exploitation of existing road infrastructures, thus reducing the need for building new roads; the possibility of quickly providing help and assistance in case of accident, thus increasing the number of lives saved and of adapting the speed of the vehicle to limits prescribed along the route as well as to the different weather and traffic conditions, are only some examples of the number of undisputed benefits provided by satellite navigation to road transport.
Collaboration(s)	• GALILEO • By investing in this field, national industry and technological research centres might acquire and consolidate a transnational leadership by producing patents, products and services which might turn into European standards
Follow-up	• n.a.
Limitations	• n.a.

Project Name	EADS's Vessel Traffic Service (VTS)
Phase(s)	1: Prevention / detection through tracking 2: Improved response through reliable positioning
Critical Mission Area	Transportation Security / Waterborne Transportation, Border Security
Type	Industry service
Project Duration	• n.a.
Current Status	• n.a.
Country/Institution	• EADS
Funding	• EADS
Partners	• n.a.
Objective(s)	• Provide active monitoring and navigational advice for vessels in particularly confined and busy waterways
In Practice	Two systems encompassing a wide range of techniques and capabilities aimed at preventing vessel collisions, rammings, and groundings in the harbor, harbor approach and inland waterway phase of navigation: • Surveilled systems consist of one or more land-based sensors (i.e. radar, AIS and closed circuit television sites), which output their signals to a central location where operators monitor and manage vessel traffic movement • Non-surveilled systems consist of one or more reporting points at which ships are required to report their identity, course, speed, and other data to the monitoring authority
Demonstration(s)	• n.a.
Users	• n.a.
Added Value/Findings	• n.a.
Collaboration(s)	• n.a.
Follow-up	• n.a.
Limitations	• n.a.

Project Name	FP 6 WIN (Wide Information Network for Risk Management)
Phase(s)	n.a. (data harmonisation)
Critical Mission Area	Intelligence, Early Warning and Risk Management
Type	R&D
Project Duration	• 01/09/2004 – 31/12/2007, 40 months
Current Status	• Completed
Country/Institution	• FP 6
Funding	• Total costs: € 8.08 million • EU contribution: € 4.4 million
Partners	• 16 coordinated by Thales Alenia Space France
Objective(s)	• To integrate all existing reference results or initiatives to contribute to the design, the development, and the validation of what could be referred to as a "European Risk Management information infrastructure"
In Practice	• To define a data/information model valid for several thematic issues of risk management • To optimise the info-structure in terms of use of state-of-the-art information technologies (web, grid), and high capability to inter-operate data, services, and risk management actors • To cover business and organisational aspects through the sub-network and charters concepts
Demonstration(s)	• n.a.
Users	• All actors involved in risk management
Added Value/Findings	• Integration of existing results and initiatives as to achieve a more coherent approach avoiding unnecessary duplications • The developed information models, the service oriented architecture with its set of generic services and workflow management facility, constitute an efficient and open framework
Collaboration(s)	• The "European Risk Management information infrastructure" will be a major element of the future overall European Spatial Data Infrastructure (ESDI) • FP6 ORCHESTRA project • Marine and Coastal Environment Information Services implementation project • GMES
Follow-up	• n.a.
Limitations	• Collaborations are unclear • Information on findings and follow-up are lacking

List of Acronyms

A	
AATMS	Airborne Air Traffic Management System
ACARE	Advisory Council for Aeronautics Research in Europe
ADAS	Advanced Driver Assistance System
AEO	Authorized Economic Operator
AIS	Automatic Identification System
AIS-S	Automatic Identification System- Satellite
AIS-SART	Automatic Identification System Search And Rescue Transmitter
ALOS	Advanced Land Observing Satellite
AMOR Radar	Advanced Meteor Orbit Radar
ANASTASIA	Airborne New and Advanced Satellite Techniques & Technologies in A System Integrated Approach
ARES	Automated Search and Rescue System
ASAR	Advanced Synthetic Aperture Radar
ASAT	Anti-SATellite
ASI	Agenzia Spaziale Italiana (Italian Space Agency)
ASI-ENAV Joint Programme	Italian National Satellite Navigation Programme for Civil Aviation
ASTRO+	Advanced Space Technologies to support secuRity Operations
ATM	Air Traffic Management
AUSIAS	Automatic Transport Telematics (ATT) in Urban Sites with Integration and Standardisation
AVSEC	Aviation Security Regulatory Committee
B	
BOC	Besoin Opérationnel Commun (Common Requirements for a Global European Earth Observation System by Satellites)
BOSS4GMES	Building Operational Sustainable Services for GMES, project
BTC	Baku-Tbilisi-Ceyhan
C	
CALDOVREP	Bilateral Reporting System for Dover Strait
C-TPAT	Customs-Trade Partnership Against Terrorism
CCD	Charged-Coupled Device
CDI	Common Data Index
CDP	Capability Development Plan, European Defence Agency
CEI	Critical European Infrastructure
CEI-GIS	Critical European Infrastructure Geographic Information System project
CERISE	Caractérisation de l´Environnement Radioélectrique par un Instrument Spatial Embarqué, satellite
CFCA	Community Fisheries Control Agency
CFSP	Common Foreign and Security Policy
CHORIST	Integrating Communications for enHanced envirOnmental RISk management and citizens safeTy, project
CI	Critical Infrastructures
CIA	Central Intelligence Agency, U.S.
CIC	Civil Intelligence Cell within the Joint Situation Center
CIP	Critical Infrastructures Protection
CIWIN	Critical Infrastructure Warning Information Network
CNES	Centre National d´Études Spatiales (National Centre for Space Studies), France
COMINT	COMmunication INTelligence
COP	Common Operating Picture
COREPER II	COmité des REPrésentants PERmanents (Committee of Permanent Representatives
COSMO-SkyMed	Constellation of Small Satellites for Mediterranean Basin Observation
COSPAS	COsmicheskaya Sistema Poiska Avariynyh Sudov (Space System for

	the Search of Vessels in Distress)
COTER	Working party on COunter TERrorism (Second pillar Common Foreign and Security Policy)
CPU	Civil Protection Unit, DG Environment – European Commission
CRITECH	Institutional action on CRIsis Monitoring and response TECHnologies, JRC
CSA	Canadian Space Agency
CSB	Container Security Box
CSBCI	Centre for Satellite Based Crisis Information
CSI	Container Security Initiative
CSN	CleanSeaNet
CSTDMA	Carrier Sense Time Division Multiple Access
CT	Counter Terrorism Cell within the organisation Situation Centre
D	
DECLIMS	DEtection and CLassIfication of MArine Traffic from Space
DG	Directorate General, European Commission
DG ECHO	Directorate General for Humanitarian Aid, European Commission
DG ENTR	Directorate General for Enterprise and Industry, European Commission
DG JLS	Directorate General for Justice, Freedom and Security, European Commission
DG MARE	Directorate General for Maritime Affairs and Fisheries, European Commission
DG RELEX	Directorate General for External Relations, European Commission
DG TREN	Directorate General for TRansport and ENergy, European Commission
DHS	Department of Homeland Security, U.S.
DIGSTER GO&MAP	DIGital Satellite Based TERrain Mapping – User Requirements
DLR	Deutsches Zentrum für Luft- und Raumfahrt (German Centre for Aeronautics and Space), Germany
DMA	Digital Map Archive
DMC	Disaster Monitoring Constellation, satellite family
DOC	Document
E	
EC	European Commission
ECAC	European Civil Aviation Conference
ECI	European Critical Infrastructure
ECJ	European Court of Justice
ECOFIN	Economic and Finances
ECO-RAIL	European Geostationary Overlay Service (EGNOS) COntrolled RAILway Equipment
EDA	European Defence Agency
EDRS	European Data Relay Satellite
EDTIB	European Defence Technological and Industrial Base
EEZ	Economics Exclusive Zone
EFFORTS	EFFective operations in pORTS, project
EGM	EuroGlobal Map
EGNOS	European Geostationary Navigation Overlay Service
ELINT	ELectronic INTelligence
ELISA	ELectronic Intelligence Satellites
EOLIA	European pre-Operational DataLInk Applications
ELSA	European Laboratory for Structural Assessment, Joint Research Centre
EMERSAT	Real Time EMERgency Management via SATellite
EMODNET	European Marine Observation and Data NETwork
EMOGES	Evaluation of Positioning Technologies for Monitoring Dangerous Goods Transports on the Austrian high-ranking Road Network, project
EMSA	European Maritime Safety Agency
ENAGAS	Empresa NAcional del GAS (National Gas Compagny), Spain
ENP	European Neighbourhood Policy
ENVISAT	ENVironmental SATellite

EO	Earth Observation
EO-KDZ (EO-CCD)	ErdbeObachtungsbasiertes Regionales KrisenDatenZentrum (Earth Observation Based Regional Centre for Crisis Data)
EP	European Parliament
EPCIP	European Programme for Critical Infrastructure Protection
EPN	European Patrols Network
ERCS	Emergency Response Core Service, GMES
ERN-CIP	EU Reference Network for Critical Infrastructure Protection
EROS	Earth Resources Observation Satellites
ERTMS	European Rail Traffic Management System
ERS satellites	European Remote Sensing satellites
ERSP	European Research Security Programme
ESA	European Space Agency
E-SAR	Experimental Synthetic Aperture Radar
ESBC	Eastern Sea Borders Centre
ESDI	European Spatial Data Infrastructure
ESDP	European Security and Defence Policy
ESPI	European Space Policy Institute
ESRIF	European Security Research and Innovation Forum
ESRAB	Security Research and the European Security Research Advisory Board
ESS	European Security Strategy
ETA	Euskadi Ta Askatasuna (Basque Homeland and Freedom)
ETCS	European Train Control System
ETML	European Train Management Layer
EU	European Union
EU LRIT DC	EU Long Range Identification and Tracking of Ships (LRIT) Data Centre
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
EURATOM	EURopean ATOMic Energy Community
EUROJUST	European Union's Judicial Cooperation Unit
EUROPOL	EUROpean POLice office
EUROSUR	EUROpean External Border SURveillance System
EUSC	European Union Satellite Centre
F	
FORESEC	FOREsight SECurity, project
FP	Framework Programme for Research and Technological Development
FP4, FP5, FP6, FP7	4th, 5th, 6th, 7th Framework Programme
FPs	Framework Programmes
FRONTEX	FRONTières EXtérieures (External Borders), European Agency for the Management of Operational Cooperation at the External Borders of EU's Member States
G	
GADEROS	GAlileo DEmonstrator for Railway Operation System
GaWaLoc	GAlileo WAgon LOCating System, project
GDACS	Global Disaster Alerting and Coordination System, JRC
GEO-CREW	GEOdata and CRisis Early Warning Situation Awareness
GEOINT	GEOspatial INTelligence
GEOSS	Global Earth Observation System of Systems
GIANUS	Global Integrated Architecture for iNnovative Utilisation of space for Security
GIS	Geographic Information System
GISCO	Geographic Information System for the European Commission
GLOBE	GLObal Border Environment
GLONASS	GLObal NAvigation Satellite System, Russia
GMDSS	Global Maritime Distress Safety System
GMES	Global Monitoring for Environment and Security
GMILSATCOM	Global MILitary SATellite COMmunications
G-MOSAIC	GMES services for Management of Operations, Situation Awareness and Intelligence for regional Crises

GMOSS	Global MOnitoring for Security and Stability, project
GMTI	Ground Moving Target Indication
GNSS	Global Navigation Satellite System
GOFREP	Golf Of Finland REPorting Systems (bilateral cooperation)
GoP	Group of Personalities
GPS	Global Positioning System, U.S.
GSM-R	Global System for Mobile Communications – Rail(way)
H	
HAWKEYE	Thermal Infra Red Hyperspectral sensing Assistance to Clandestine Weapon Surveillance under WorKing Conditions Linking fixed Airborne or space borne Systems, project
HAZMAT	HAZardous MATerial reporting
HF	High Frequency
HPRDS	Human Portable Radiation Detection System
HUMBOLDT	Framework for Data Harmonisation and Service Integration, project
HYSAR	HYperspectral Synthetic Aperture Radar
I	
IAEA	International Atomic Energy Agency
IAP	Integrated Applications Promotion
IATA	International Air Traffic Association
ICAO	International Civil Aviation Organisation
ICT	Information and Communication Technologies
IEC	International Electrotechnical Commission
IED	Improvised Explosive Device
IMARSEC	Integrated MARitime SECurity System
IMINT	IMage INTelligence
IMO	International Maritime Organization
INFRA	Innovative and Novel First Responders Applications, project
INSPIRE	INfrastructure for Spatial InfoRmation in Europe, project
INTEGRAIL	INTElligent inteGration of RAILway systems, project
IPSC	Institute for the Protection and Security of Citizen, JRC
IR	Infra-Red
IRA	Irish Republican Army
ISAR	Infrared Synthetic Aperture Radar
ISFEREA	Institutional Action on Information Support for Effective and Rapid External Action, JRC
ISPS	International Ship and Port facility Security code
IT	Information Technologies
ITA	International Technology Alliance
ITS	Intelligent Transport System
J	
JAA	Joint Aviation Authorities
JHA	Justice and Home Affairs
JRC	Joint Research Centre, European Commision
L	
LEO	Low Earth Orbit
LIMES	Land and Sea Monitoring for Environment and Security, project
LiveMon	Live Monitoring project for disaster management
LMCS	Land Monitoring Core Service, GMES
LOCOLOC	LOw-COst LOCcalisation, ESA project
LOCOPROL	LOw COst satellite based train location system for signalling and train PROtection for low density Lines
LOTUS	LOcalisation of Threat substances in Urban Society, project
LRIT	Long Range Identification and Tracking
M	
MAGNET	Multi-modal Approach for GNSS-1 in European Transport, project
MAHB	Major Accident Hazards Bureau

MANPAD	MAN Portable Air Defence System
MASURE	Institutional Action on Maritime Surveillance, JRC
MARISS	GMES Maritime Security Services, project
MarNIS	MARitime Navigation and Information Services
MARSUR	MARitime SURveillance, project
MARUSE	MARitime USEr community
MCS	Marine Core Service, GMES
MEDSEA	MEDiterranean SEA, project
MERSEA	Marine EnviRonment and Security for the European area, project
MilsatCom	Military Satellite Communication system
MM/MS	Multi-Mission/Multi-Sensor
MOBISIC	Système de Sécurisation MOBIle des Infrastructures Critiques (Mobile Security System for Critical Infrastructures), project
MODENA	Modélisation de l´Observation à Distance de l´ENvironnement mAritime, project
MS	Multi Spectral
MSS	Maritime Security and Safety
MSI	Maritime Safety Information
MTI	Moving Target Indication
MUSIS	Multinational Space-based Imaging System
N	
NASA	National Aeronautics and Space Administration, U.S.
NATO	North Atlantic Treaty Organisation
NAUPLIOS	Navigation and Perilous Goods Input and Output System, project
NGO	Non Governmental Organisation
NOAA	National Oceanic and Atmospheric Administration, US
NPT	Non Proliferation Treaty
NUTS	Nomenclature des Units Territoriales Statistiques
O	
OASIS	Open Advanced System for dISaster & emergency management, project
ODYSSEY	Strategic Pan-European Ballistics Intelligence Platform for Combating Organised Crime and Terrorism, project
OHQ	Operational HeadQuarters
OLAF	Office Europeen de Lutte Anti- Fraude (European Anti-Fraud Office)
OPERAMAR	InterOPERAble Approach to the European Union MARitime Safety & Security Management, project
ORCHESTRA	Open Architecture and Spatial Data Infrastructure for Risk Management, project
ORFEO	Optical and Radar Federated Earth Observation, project
OSC	Operation Safe Commerce
OSCE	Organization for Security and Co-operation in Europe
OSINT	Open Source Intelligence
P	
PAN	Panchromatic
PASR	Preparatory Action Security Research
PDD	Presidential Decision Directive
PNR	Passenger Name Record
POB	Person Over Board, project
POLINSAR	Polarimetric SAR Interferometry, project
PPP	Public-Private Partnership
PRESENSE	Pipeline Remote Sensing for Safety and the Environment, project
PREVIEW	PREVention, Information and Early Warning, project
PRISMA	Hyperspectral Precursor of the Application Mission, satellite
ProATN	Prototype Aeronautical Telecommunication Network, project
PRS	Public Regulated Sevices
PVACS	Institutional Action on Physical Vulnerability of Critical Structures, European Laboratory for Structural Assessment, Joint Research

	Centre, Joint Research Centre
Q	
QUANTUM	Quantum Entanglement in Space Experiments, project
QKD	Quantum Key Distribution
R	
R&D	Research and Development
RADARSAT	RAdio Detection And Ranging SATellite
RCS	Radar Cross Section
RCP	Regional Contingency Plan
RFID	Radio Frequency Identification
RIMS	Reliable Image Management Systems in support of urban and disaster management, project
RIS	River Information Service
RISK-EOS	Project for emergency management
RUNE	Rail User Navigation Equipment
S	
SAFEE	Security of Aircraft in the Future European Environment, project
SAFER	Services and Applications For Emergency Response, project
SALW	Small Arms and Light Weapons
SAR	Synthetic Aperture Radar
SAR-Lupe	satellite family
SARSAT	Search And Rescue Satellite-Aided Tracking
SART	Search And Rescue Transmitter
SATCOM	SATellite COMmunication
ScanSAR	Scanning Synthetic Aperture Radar
SCNI	Institutional Action on Security of Critical Networked Infrastructures, JRC
SDA	Security and Defence Agenda
SDE	Spatial Database Engine
SDI	Spatial Data Infrastructure
SDR	Software Defined Radio
SECTRONIC	Security System for Maritime Infrastructures, Ports and Coastal Zones, project
SEDE	Committee Security and Defence
SENTRE	SEcurity Network for Technological Research in Europe, Project
SEOSAT	Spanish Earth Observation SATellite
SERTIT	Regional Service of Image Treatment and Remote Sensing
SES	Single European Sky initiative
SESAR	Single European Sky ATM Research (former project SESAME)
SEVESEO	Earth Observation Based Services in Support to Industrial and Technological Risk Management
Sicral	Sistema Italiano per Comunicazioni Riservate ed ALlarmi (Italian System for Secret Communications and Emergencies)
SIGINT	SIGnal INTelligence
SitCen	Joint Situation Center
SkyMed	Constellation of Small Satellites for Mediterranean basin Observation
SALW	Small Arms and Light Weapons
S-MAR	Surveillance for Maritime Project conducted by CNES
SMARTregions	Space and Security Knowledge for MARiTime REGIONal Development Project
SME	Small-Medium Enterprice
SOA	Service-Oriented Architecture
SOBCAH	Surveillance of Borders, Coastlines and Harbours
SOLAS	Safety Of Life At Sea
SOTDMA	Self Organizing Time Division Multiple Access
Space-QUEST	Quantum Entanglement in Space Experiments, project
Spainsat	Spain Satellite
SPASEC	Panel of experts on SPAce and SECurity

SPIDER	SPace based Information for Disaster management and Emergency Response, UN
SPIRALE	Système Préparatoire Infra-Rouge pour l´Alerte, satellite family
SPOT	Satellites Pour l´Observation de la Terre, satellite family
SRA	Strategic Research Agenda
SRTM	Shuttle Radar Topography Mission
SSA	Space Situational Awareness
SSAS	Ship Security Alert System
SSN	SafeSeaNet
STACCATO	STAkeholders platform for supply Chain mapping, market Condition Analysis and Technologies Opportunities, Project
STAP	Space Time Adaptive Processing
STIRES	SafeSeaNet Tracking Information Relay and Exchange System
STRAW	Security Technology Active Watch, project
SUMO	Search Unidentified Marine Objects
SVR	Stolen Vehicle Recovery system
T	
T2L2	Transfert de Temps par Lien Laser, project
TAF-TSI	Telematics Applications for Freight
TanDEM-X	Satellite for intelligence gathering
TANGO	Advanced Networks for GMES Operations, project
TAS	Thales Alenia Space
TELESAL	Telemedicine Applications, project
TerraSAR-X	Satellite for intelligence gathering
TEU	Treaty on the European Union
TIES	Tactical Imagery Exploitation System
TIHSI	Thermal Infra-red Hyper Spectral Imaging
TOCHA	Continuous and Operational Tracking with High Autonomous, project
TRACAR	TRAffic & CARgo Supervision System, project
TransControl	Project related to Innovative monitoring and inspection application for the safe transport of dangerous goods
TRVA	TRaceability and Vulnerability Assessement unit, JRC
TSAT	Transformational Satellites Communication Programme
TW	Technology Watch
TWSTFT	Two Way Satellite Time and Frequency Transfert
U	
UAS	Unmanned Aerial System
UAV	Unmanned Aerial Vehicle
UCTE	Union for the Co-ordination of Transmission of Electricity
UK	United Kingdom
UN	United Nations
UN-SPIDER	UN Platform for Space-based Information for Disaster Management and Emergency Response
U.S.	United States
USA	United States of America
USAF	United States Air Force
V	
VASTS	Vulnerability Assessment and Surface Transport Security, project
VATDIS	Vulnerability Assessment in Transport and Distribution System
VDR	Voyage Data Recorder
VDS	Vessel Detection System
VESPO	VEssel Surveillance and POrt security project
VHF	Very High Frequency
VHR	Very High Resolution
VHR-OPT	Very High Resolution Optical
VHR-SAR	Very High Resolution Synthetic Aperture Radar
VIP	Very Important Person
VMS	Vessel Monitoring Service

VTM	Vehicle Traffic Management
VTMIS	Vessel Traffic Monitoring and Information System
VTS	Vessel Traffic Service
W	
WEAG	Western European Armaments Group
WETREP	West European Tanker Reporting System
WIN	Wide Information Network for Risk Management, project
WMD	Weapon of Mass Destruction
WSBC	Western Sea Borders Centre
WTC	World Trade Center
Z	
ZKI	Zentrum für satellitengestützte Kriseninformation (ZKI)/ German Center for satellite-based Crisis Information
9/11	Terrorist attacks on 9 September 2001 on targets in the U.S. including the World Trade Center

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